

# Final

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**CO2:**      [https://daac.ornl.gov/CMS/guides/CMS\\_DARTE\\_V2.html](https://daac.ornl.gov/CMS/guides/CMS_DARTE_V2.html)

**PM2.5:** <https://dataverse.harvard.edu/dataset.xhtml?persistentId=doi:10.7910/DVN/4GDRB1>

## Background

Delete everything in environment

```
rm(list = ls())
```

Load libraries

```
library(arrow)
library(broom)
library(CausalArima)
library(colorspace)
library(ggnewscale)
library(ggpattern)
library(glue)
library(mgcv)
library(moderndiver)
library(moments)
library(npreg)
library(plm)
library(plotly)
library(purrr)
library(scales)
library(scatterplot3d)
library(sf)
```

```
library(spdep)
library(splm)
library(tidyverse)
library(tigris)
library(tseries)
```

Load data for county shapefile (from library(tigris))

```
# Load county shapefile (includes FIPS codes and geometry)
counties_sf <- counties(cb = TRUE, resolution = "20m", year = 2020) %>%
  mutate(fips = paste0(STATEFP, COUNTYFP))
```

```
## |
```

```
# Filter for contiguous U.S.
contiguous_fips <- setdiff(sprintf("%02d", 1:56),
                             c("02", "15", "60", "66", "69", "72", "78"))
counties_sf <- counties_sf %>%
  filter(STATEFP %in% contiguous_fips)
```

Load data for crosswalk (between ZCTA and county)

```
xwalk22 <- read_parquet("us_xwalks__census__zcta2county_yearly__2022.parquet")
xwalk21 <- read_parquet("us_xwalks__census__zcta2county_yearly__2021.parquet")
xwalk20 <- read_parquet("us_xwalks__census__zcta2county_yearly__2020.parquet")
xwalk19 <- read_parquet("us_xwalks__census__zcta2county_yearly__2019.parquet")
xwalk18 <- read_parquet("us_xwalks__census__zcta2county_yearly__2018.parquet")
xwalk17 <- read_parquet("us_xwalks__census__zcta2county_yearly__2017.parquet")
xwalk16 <- read_parquet("us_xwalks__census__zcta2county_yearly__2016.parquet")
xwalk15 <- read_parquet("us_xwalks__census__zcta2county_yearly__2015.parquet")
xwalk14 <- read_parquet("us_xwalks__census__zcta2county_yearly__2014.parquet")
xwalk13 <- read_parquet("us_xwalks__census__zcta2county_yearly__2013.parquet")
xwalk12 <- read_parquet("us_xwalks__census__zcta2county_yearly__2012.parquet")
xwalk11 <- read_parquet("us_xwalks__census__zcta2county_yearly__2011.parquet")
xwalk10 <- read_parquet("us_xwalks__census__zcta2county_yearly__2010.parquet")
xwalk09 <- read_parquet("us_xwalks__census__zcta2county_yearly__2009.parquet")
xwalk08 <- read_parquet("us_xwalks__census__zcta2county_yearly__2008.parquet")
xwalk07 <- read_parquet("us_xwalks__census__zcta2county_yearly__2007.parquet")
xwalk06 <- read_parquet("us_xwalks__census__zcta2county_yearly__2006.parquet")
xwalk05 <- read_parquet("us_xwalks__census__zcta2county_yearly__2005.parquet")
xwalk04 <- read_parquet("us_xwalks__census__zcta2county_yearly__2004.parquet")
xwalk03 <- read_parquet("us_xwalks__census__zcta2county_yearly__2003.parquet")
xwalk02 <- read_parquet("us_xwalks__census__zcta2county_yearly__2002.parquet")
```

```
xwalk01 <- read_parquet("us_xwalks__census__zcta2county_yearly__2001.parquet")
xwalk00 <- read_parquet("us_xwalks__census__zcta2county_yearly__2000.parquet")
```

Load data for total CO2 emissions (at census block group level in kilograms)

```
# Create path to the .gdb directory
co2_path <- "CMS_DARTE_V2_1735/data/DARTE_v2.gdb"
```

```
# List the available layers (feature classes) inside the .gdb
st_layers(dsn = co2_path)
```

```
## Driver: OpenFileGDB
## Available layers:
##               layer_name geometry_type features fields  crs_name
## 1 DARTE_v2_blockgroup_kgco2_1980_2017 Multi Polygon   220333    43 Vulcan_LCC
```

```
# Load the specific layer
co2_cbg <- st_read(dsn = co2_path, layer = "DARTE_v2_blockgroup_kgco2_1980_2017")
```

```
## Reading layer 'DARTE_v2_blockgroup_kgco2_1980_2017' from data source
##   '/n/home10/rtruong/CMS_DARTE_V2_1735/data/DARTE_v2.gdb' using driver 'OpenFileGDB'
## Simple feature collection with 220333 features and 43 fields
## Geometry type: MULTIPOLYGON
## Dimension:      XY
## Bounding box:   xmin: -7034829 ymin: -1957575 xmax: 3488418 ymax: 4595604
## Projected CRS: Vulcan_LCC
```

Load data for average PM2.5 concentration (at ZCTA level in micrograms per cubic meter)

```
# Create path to the .parquet files
pm25_path <- "dataverse_files"
```

```
# List all .parquet files for years 2000-2020
file_list <- list.files(path = pm25_path,
                        pattern = "^pm25__ushap__zcta_yearly__20\\d{2}\\$.parquet$",
                        full.names = TRUE)
```

```
# Read and combine all files into data frame
pm25_zcta <- file_list %>%
  lapply(read_parquet) %>%
  bind_rows()
```

## Data wrangling for co2\_cbg

Convert co2\_cbg to tidy format

```
co2_cbg <- co2_cbg %>%  
  pivot_longer(cols = starts_with("kgco2_"),  
               names_to = "year",  
               names_prefix = "kgco2_",  
               values_to = "value") %>%  
  mutate(year = as.integer(year))
```

Select for variables to avoid large data set

```
co2_cbg <- co2_cbg %>%  
  ungroup() %>%  
  st_drop_geometry() %>%  
  as.data.frame() %>%  
  rename(geoid = GEOID) %>%  
  select(geoid, year, value)
```

## Data wrangling for xwalk

Combine everything into one crosswalk

```
years <- 2000:2022  
file_paths <- paste0("us_xwalks__census__zcta2county_yearly__", years, ".parquet")  
xwalk_list <- map(file_paths, read_parquet)  
xwalk <- bind_rows(xwalk_list)
```

## Aggregation for pm25\_zcta (from ZCTA level to county level)

Join xwalk and pm25\_zcta

```
pm25_county <- inner_join(xwalk, pm25_zcta, by = c("zcta", "year"))
```

Aggregate from ZCTA level to county level

```
pm25_county <- pm25_county %>%
  group_by(year, county) %>%
  mutate(weight = (arealand_pct / 100)) %>%
  mutate(weight = weight / sum(weight, na.rm = TRUE)) %>%
  summarize(pm25 = sum(PM25 * weight, na.rm = TRUE),
    .groups = "drop")
```

## Aggregation for co2\_cbg (from census block group level to county level)

Create variable for county FIPS code

```
co2_cbg <- co2_cbg %>%
  mutate(county = substring(geoid, 1, 5))
```

Aggregate from census block group level to county level

```
co2_county <- co2_cbg %>%
  group_by(year, county) %>%
  summarize(co2 = sum(value, na.rm = TRUE),
    .groups = "drop")
```

## Finalizing files (df has CO2 and PM2.5 at county level from 2000 to 2017)

Filter for contiguous U.S.

```
pm25_county <- pm25_county %>%
  filter(substr(county, 1, 2) %in% contiguous_fips)

co2_county <- co2_county %>%
  filter(substr(county, 1, 2) %in% contiguous_fips)
```

Create df by joining pm25\_county and co2\_county

```
df <- inner_join(pm25_county, co2_county, by = c("county", "year")) %>%
  filter(substr(county, 1, 2) %in% contiguous_fips)
```

## Data visualization

### Aggregate CO2

```
# Aggregate total CO2 for all counties by year
co2_total_year <- co2_county %>%
  group_by(year) %>%
  summarize(total_co2 = sum(co2, na.rm = TRUE))

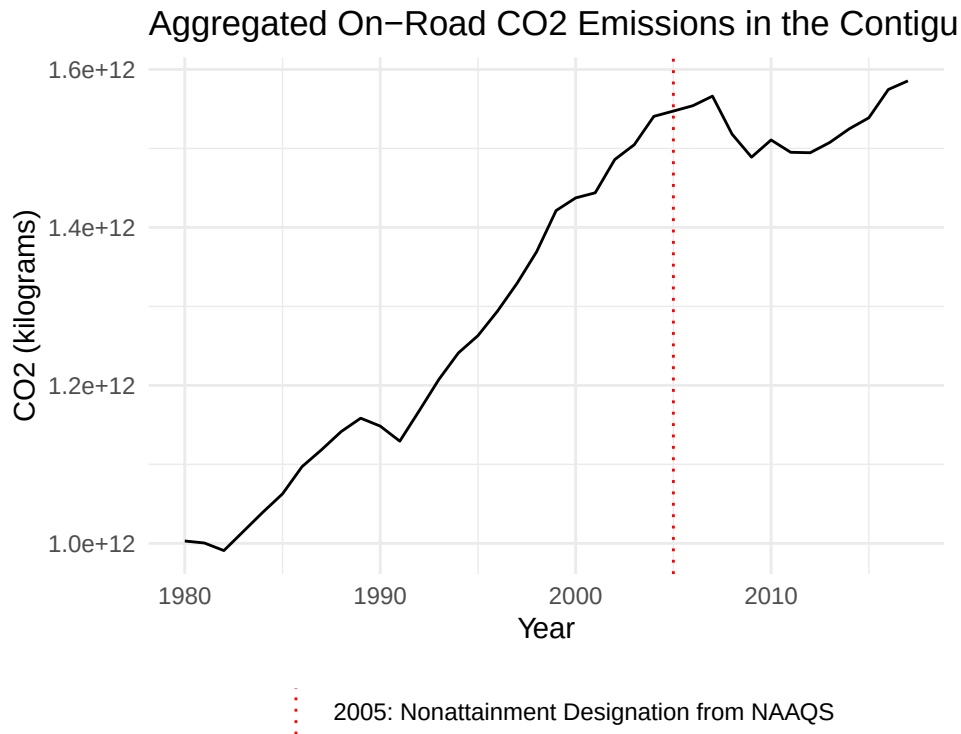
# Check the result
head(co2_total_year)
```

```
## # A tibble: 6 x 2
##   year total_co2
##   <int>   <dbl>
## 1  1980  1.00e12
## 2  1981  1.00e12
## 3  1982  9.91e11
## 4  1983  1.02e12
## 5  1984  1.04e12
## 6  1985  1.06e12
```

### Plot aggregate CO2

```
# Create graph of CO2 data
co2_plot <- co2_total_year %>%
  ggplot() +
  geom_line(aes(x = year, y = total_co2)) +
  geom_vline(aes(xintercept = 2005, linetype = "2005: Nonattainment Designation from NAAQS"),
    color = "red") +
  labs(title = "Aggregated On-Road CO2 Emissions in the Contiguous U.S.",
    x = "Year",
    y = "CO2 (kilograms)",
    fill = "",
    linetype = "") +
  scale_linetype_manual(values = c("2005: Nonattainment Designation from NAAQS" = "dotted")) +
  theme_minimal() +
  theme(legend.position = "bottom")

# Plot and save graph
co2_plot
```



```
ggsave("co2_plot.png",
  plot = co2_plot,
  bg = "white")
```

## Saving 5 x 4 in image

**Remove extreme counties (with CO2 2005-2010 relative changes above +/- 100%)**

Create `df_wide` (in wide format) from `df` (in long/tidy format)

```
df_wide <- df %>%
  pivot_wider(names_from = year,
    values_from = c(co2, pm25))
```

**Calculate 2005-2010 change in CO2 and PM2.5 for each county**

```
df_wide <- df_wide %>%
  mutate(co2_change = co2_2010 - co2_2005,
    co2_rel_change = co2_change / co2_2005,
    co2_rel_change_perc = co2_rel_change * 100,
```

```
pm25_change = pm25_2010 - pm25_2005,
pm25_rel_change = pm25_change / pm25_2005,
pm25_rel_change_perc = pm25_rel_change * 100)
```

Create vector for counties with extreme values of `co2_rel_change_perc`

```
# Filter for extreme counties
extreme_counties <- df_wide %>%
  filter(abs(co2_rel_change_perc) >= 100)

# Get vector of corresponding fips codes
extreme_fips <- extreme_counties$county
```

Create `df_wide_filtered` and `co2_county_filtered` by removing extreme counties

```
# Create df_wide_filtered
df_wide_filtered <- df_wide %>%
  filter(!(county %in% extreme_fips))

# Create co2_county_filtered
co2_county_filtered <- co2_county %>%
  filter(!(county %in% extreme_fips))
```

## Analysis with CausalArima library

TEMPORARY CODE (DELETE LATER): Read `df_causal_arima`

```
df_causal_arima <- read_csv("df_causal_arima.csv")

## Rows: 3052 Columns: 11
## -- Column specification -----
## Delimiter: ","
## chr  (1): county
## dbl (10): ar_order, differencing_order, ma_order, avg_effect, cum_effect, re...
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.
```

Look at most common ARIMA model specifications



```
df_causal_arima %>%
  count(ar_order, differencing_order, ma_order, sort = TRUE)
```

```
## # A tibble: 40 x 4
##   ar_order differencing_order ma_order    n
##   <dbl>         <dbl>      <dbl> <int>
## 1     1           1           1     0   966
## 2     0           1           1     1   604
## 3     0           1           0    251
## 4     0           2           0   213
## 5     2           1           0   200
## 6     2           0           0   192
## 7     0           2           1   117
## 8     0           1           2    62
## 9     1           2           0    58
## 10    2           1           2    43
## # i 30 more rows
```

Combine `df_wide_filtered` with `df_causal_arima` (both at county level)

```
df_wide_filtered <- df_wide_filtered %>%
  left_join(df_causal_arima %>% select(county,
                                     avg_effect,
                                     cum_effect,
                                     rel_effect,
                                     sd,
                                     p_value),
            by = "county")
```

Add polygons to `df_wide_filtered`

```
# Join df_wide_filtered with shapefile
df_wide_filtered <- counties_sf %>%
  inner_join(df_wide_filtered, by = c("fips" = "county"))
```

Plot `co2_rel_change_perc`

```
# Define diverging color palette
div_colors <- c("#2166AC", "#4393C3", "#92C5DE",
               "#F7F7F7",
               "#F4A582", "#D6604D", "#B2182B")
```

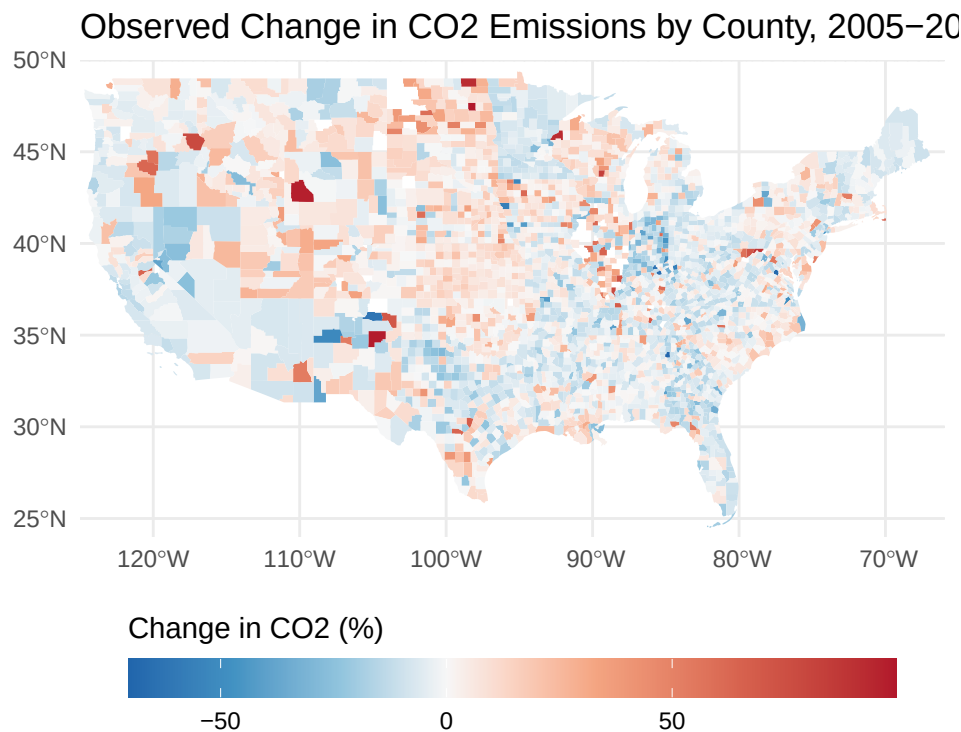
```

# Compute global limits for co2_rel_change_perc
global_min <- min(df_wide_filtered$co2_rel_change_perc, na.rm = TRUE)
global_max <- max(df_wide_filtered$co2_rel_change_perc, na.rm = TRUE)

# Create graph of differences
change_map <- ggplot(data = df_wide_filtered) +
  geom_sf(aes(fill = co2_rel_change_perc), color = NA) +
  scale_fill_gradientn(colors = div_colors,
    na.value = "white",
    limits = c(global_min, global_max),
    values = scales::rescale(c(global_min, 0, global_max)),
    labels = label_number(accuracy = 1),
    guide = guide_colorbar(barwidth = 20,
      barheight = 1,
      title.position = "top",
      title.vjust = 1)) +
  labs(title = glue("Observed Change in CO2 Emissions by County, 2005-2010"),
    fill = "Change in CO2 (%)") +
  theme_minimal() +
  theme(legend.position = "bottom") +
  coord_sf(xlim = c(-125, -66), ylim = c(24, 50), expand = FALSE)

# Plot and save graph
change_map

```



```
ggsave("change_map.png",
  plot = change_map,
  width = 8, height = 6, dpi = 300,
  bg = "white")
```

Plot `rel_effect_perc` and `pm25_rel_change_perc`

```
# Create rel_effect_perc (i.e., rel_effect as a percentage)
df_wide_filtered <- df_wide_filtered %>%
  mutate(rel_effect_perc = rel_effect * 100)

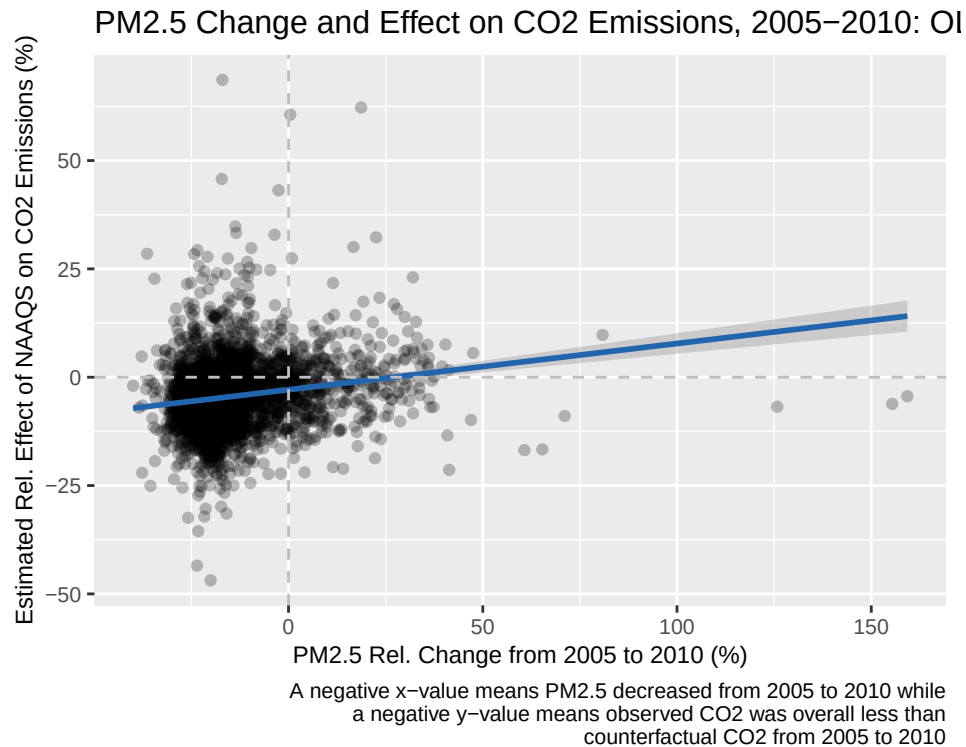
# Plot scatterplot with OLS line
scatterplot_ols <- ggplot(df_wide_filtered,
  aes(y = rel_effect_perc,
      x = pm25_rel_change_perc)) +

  geom_point(alpha = 0.25) +
  geom_smooth(method = "lm", se = TRUE, color = "#2166AC") +
  geom_hline(yintercept = 0, color = "grey", linetype = "dashed") +
  geom_vline(xintercept = 0, color = "grey", linetype = "dashed") +
  labs(x = "PM2.5 Rel. Change from 2005 to 2010 (%)",
    y = "Estimated Rel. Effect of NAAQS on CO2 Emissions (%)",
    title = "PM2.5 Change and Effect on CO2 Emissions, 2005-2010: OLS",
    caption = "A negative x-value means PM2.5 decreased from 2005 to 2010 while
      a negative y-value means observed CO2 was overall less than
      counterfactual CO2 from 2005 to 2010") +

  theme(plot.title = element_text(size = 12),
    plot.caption = element_text(size = 8),
    axis.title.x = element_text(size = 9),
    axis.title.y = element_text(size = 9),
    axis.text = element_text(size = 8))

# Plot and save graph
scatterplot_ols
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```



```
ggsave("scatterplot_ols.png",
  plot = scatterplot_ols,
  width = 8, height = 6, dpi = 300,
  bg = "white")
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```

Model `rel_effect` and `pm25_rel_change` via OLS

```
# Fit linear regression model
model_ols <- lm(rel_effect ~ pm25_rel_change, data = df_wide_filtered)
get_regression_table(model_ols)
```

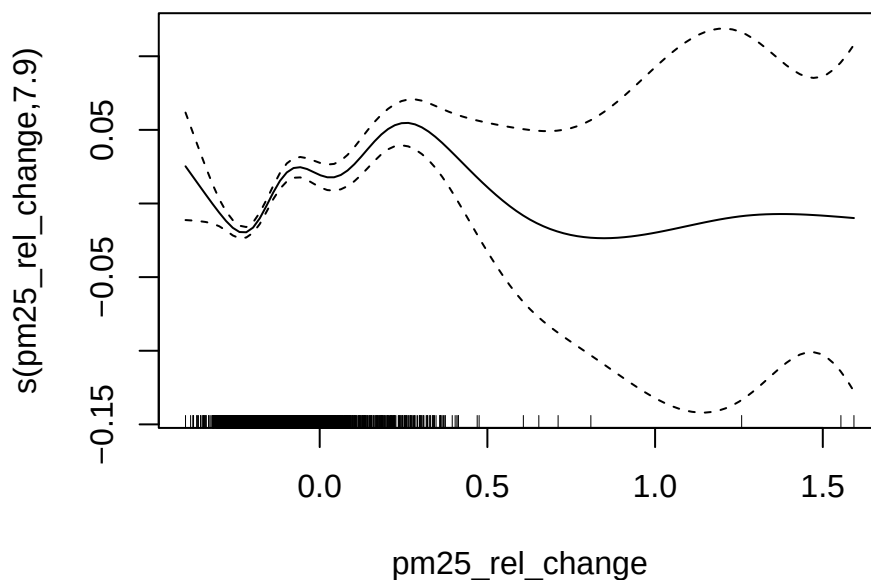
```
## # A tibble: 2 x 7
##   term          estimate std_error statistic p_value lower_ci upper_ci
##   <chr>          <dbl>    <dbl>    <dbl>   <dbl>   <dbl>   <dbl>
## 1 intercept     -0.029    0.002   -13.9     0    -0.033  -0.025
## 2 pm25_rel_change 0.107    0.011    10.1     0     0.086   0.127
```

Model `rel_effect` and `pm25_rel_change` via GAM

```
# Fit generalized additive model
model_gam <- gam(rel_effect ~ s(pm25_rel_change), data = df_wide_filtered)
summary(model_gam)
```

```
##
## Family: gaussian
## Link function: identity
##
## Formula:
## rel_effect ~ s(pm25_rel_change)
##
## Parametric coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) -0.043587  0.001459  -29.86  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Approximate significance of smooth terms:
##             edf Ref.df      F p-value
## s(pm25_rel_change) 7.903  8.695 20.64  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## R-sq.(adj) =  0.0547   Deviance explained = 5.72%
## GCV = 0.00652   Scale est. = 0.006501   n = 3052
```

```
# Plot model
plot(model_gam)
```



## Analysis with plm library

Prepare for plm() with co2\_county\_filtered (i.e., filtered)

```
# Rename variable to fips in co2_county_filtered
co2_county_filtered <- co2_county_filtered %>%
  rename(fips = county)

# Filter for years less than or equal to 2004 (to fit to pre-treatment period)
co2_county_2004_filtered <- co2_county_filtered %>%
  filter(year <= 2004)

# Get shapefile of unique counties in co2_county_2004_filtered
fips_panel <- sort(unique(co2_county_2004_filtered$fips))
counties_sf <- counties(cb = TRUE, resolution = "20m", year = 2020) %>%
  st_as_sf() %>%
  mutate(fips = str_pad(paste0(STATEFP, COUNTYFP), 5, "left", "0")) %>%
  filter(fips %in% fips_panel) %>%
  st_make_valid()

# Join co2_county_2004_filtered with shapefile (to add polygons/geometry)
co2_county_2004_filtered <- counties_sf %>%
  inner_join(co2_county_2004_filtered, by = c("fips" = "fips"))

# Create neighbor list using KNN to ensure each county has 4 neighbors
coords <- st_coordinates(st_centroid(counties_sf))
```

## Warning: st\_centroid assumes attributes are constant over geometries

```
nb_knn <- knn2nb(knearneigh(coords, k = 4))

# Create listw object from nb_knn
listw_object <- nb2listw(nb_knn, style = "W")

# Ensure ordering matches plm()
regions <- counties_sf$fips
co2_county_2004_filtered <- co2_county_2004_filtered %>%
  filter(fips %in% regions) %>%
  mutate(fips = factor(fips, levels = regions)) %>%
  arrange(fips, year)

# Drop 1980 for every county (since there is no lag before)
df_one <- co2_county_2004_filtered %>%
  arrange(fips, year) %>%
  group_by(fips) %>%
  mutate(
```

```

log_co2 = log(co2),
log_lag_co2 = log(lag(co2))
) %>%
filter(!is.na(log_lag_co2)) %>%
ungroup()

```

Run `plm()` with `df_one`: log-log model with 1 year of lag (i.e., most simple model)

```

# Create panel dataframe from df_one
pdf <- pdata.frame(df_one, index = c("fips", "year"))

# Fit model
model_1 <- plm(
  log_co2 ~ log_lag_co2,
  data = pdf,
  model = "within",
  effect = "individual"
)
summary(model_1)

```

```

## Oneway (individual) effect Within Model
##
## Call:
## plm(formula = log_co2 ~ log_lag_co2, data = pdf, effect = "individual",
##      model = "within")
##
## Balanced Panel: n = 3053, T = 24, N = 73272
##
## Residuals:
##      Min.      1st Qu.      Median      3rd Qu.      Max.
## -1.01152650 -0.01954162  0.00083948  0.02046691  0.72662285
##
## Coefficients:
##              Estimate Std. Error t-value Pr(>|t|)
## log_lag_co2 0.98465200 0.00076505   1287 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Total Sum of Squares:    2692
## Residual Sum of Squares: 109.47
## R-Squared:    0.95933
## Adj. R-Squared: 0.95757
## F-statistic: 1656490 on 1 and 70218 DF, p-value: < 2.22e-16

```

```

# Evaluate model with AIC
log_lik_val <- as.numeric(model_1$logLik)
k <- length(coef(model_1))
AIC_val_1 <- -2 * log_lik_val + 2 * k
AIC_val_1

```

```
## numeric(0)
```

This model is  $\log(Y_{i,t}) = \beta_1 \log(y_{i,t-1}) + \mu_i + \varepsilon_{i,t}$ , which implies  $\mathbb{E}[\log(Y_{i,t}) \mid \text{past}] = \beta_1 \log(y_{i,t-1}) + \mu_i$ . We estimate  $\hat{\mathbb{E}}[\log(Y_{i,t}) \mid \text{past}] = \hat{\beta}_1 \log(y_{i,t-1}) + \hat{\mu}_i$ . We thus want  $\hat{\beta}_1$  (which we can extract from the model) and  $\hat{\mu}_i$  (which we can calculate for each  $i$ ).

Let  $y_i = \log(Y_{i,t})$  and  $x_i = \log(y_{i,t-1})$ . Our model is  $y_i = \beta_1 x_i + \mu_i + \varepsilon_{i,t}$ . Averaging over  $t \in \{1, \dots, T\}$ , we have  $\bar{y}_i = \beta_1 \bar{x}_i + \mu_i + \bar{\varepsilon}_{i,t}$ . Since we assume  $\mathbb{E}[\varepsilon_{i,t} \mid \text{past}] = 0$ , for large  $n$ ,  $\bar{\varepsilon}_{i,t} \approx 0$ , so  $\bar{y}_i \approx \beta_1 \bar{x}_i + \mu_i \implies \mu_i \approx \bar{y}_i - \beta_1 \bar{x}_i \implies \hat{\mu}_i \approx \bar{y}_i - \hat{\beta}_1 \bar{x}_i$ .

Use `model_1` to estimate counterfactual CO2 values

```

# Drop geometry
df_one <- df_one %>%
  sf::st_drop_geometry()

# Get beta_1_hat
beta_hat <- coef(model_1)
beta_1_hat <- beta_hat["log_lag_co2"]

# Get mu_i_hat for each county
mu_hat_df <- df_one %>%
  group_by(fips) %>%
  summarize(
    mu_hat = mean(log_co2 - beta_1_hat * log_lag_co2, na.rm = TRUE),
    .groups = "drop"
  )

# Predict 2001 using observed 2000 CO2
pred_2001 <- df_one %>%
  filter(year == 2000) %>%
  left_join(mu_hat_df, by = "fips") %>%
  mutate(
    pred_year = 2001,
    pred_log_co2_2001 = beta_1_hat * log_co2 + mu_hat,
    pred_co2_2001 = exp(pred_log_co2_2001)
  ) %>%
  select(
    fips,

```



```

    pred_year,
    co2_2000 = co2,
    log_co2_2000 = log_co2,
    mu_hat,
    pred_log_co2_2001,
    pred_co2_2001
  )

# Display predicted vs. observed CO2 in 2001 for each county
df_one %>%
  filter(year == 2001) %>%
  rename(co2_2001 = co2) %>%
  left_join(pred_2001, by = "fips") %>%
  select(fips, co2_2000, pred_co2_2001, co2_2001)

```

```

## # A tibble: 3,053 x 4
##   fips    co2_2000 pred_co2_2001  co2_2001
##   <fct>    <dbl>         <dbl>      <dbl>
## 1 01061 136234975.    137431976. 130597273.
## 2 08125  66639622.      66850442.  67927728.
## 3 17177 190152991.    193207695. 184251558.
## 4 28153 133680830.    134458434. 128345273.
## 5 34041  854684648.    863061170. 836189695.
## 6 46051  56747321.      57432813.  57402950.
## 7 51013  855074997.    872676063. 830131352.
## 8 37091 105980776.    106266391. 104644457.
## 9 13057  744331509.    775816674. 772551827.
## 10 13015  955817211.    979376041. 971379827.
## # i 3,043 more rows

```

Run `plm()` with `df_three`: log-log model with 3 years of lag (i.e., more complicated model)

```

# Drop 1980, 1981, 1982 for every county (since we're using three years of lag)
df_three <- co2_county_2004_filtered %>%
  arrange(fips, year) %>%
  group_by(fips) %>%
  mutate(
    log_co2 = log(co2),
    log_lag_co2_one = lag(log_co2, 1),
    log_lag_co2_two = lag(log_co2, 2),
    log_lag_co2_three = lag(log_co2, 3)
  ) %>%
  filter(
    !is.na(log_lag_co2_one),

```

```

    !is.na(log_lag_co2_two),
    !is.na(log_lag_co2_three)
  ) %>%
  ungroup()

# Create panel dataframe from df_three
pdf <- pdata.frame(df_three, index = c("fips", "year"))

# Fit model via plm()
model_2 <- plm(
  log_co2 ~ log_lag_co2_one + log_lag_co2_two + log_lag_co2_three,
  data = pdf,
  model = "within",
  effect = "individual"
)
summary(model_2)

## Oneway (individual) effect Within Model
##
## Call:
## plm(formula = log_co2 ~ log_lag_co2_one + log_lag_co2_two + log_lag_co2_three,
##      data = pdf, effect = "individual", model = "within")
##
## Balanced Panel: n = 3053, T = 22, N = 67166
##
## Residuals:
##      Min.      1st Qu.      Median      3rd Qu.      Max.
## -0.7790162 -0.0154415 -0.0011651  0.0155646  0.4344497
##
## Coefficients:
##              Estimate Std. Error  t-value Pr(>|t|)
## log_lag_co2_one    1.7566735   0.0038807  452.664 < 2.2e-16 ***
## log_lag_co2_two   -0.9399495   0.0069992 -134.293 < 2.2e-16 ***
## log_lag_co2_three  0.1435722   0.0038497   37.294 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Total Sum of Squares:    2270
## Residual Sum of Squares: 46.224
## R-Squared:    0.97964
## Adj. R-Squared: 0.97867
## F-statistic: 1028080 on 3 and 64110 DF, p-value: < 2.22e-16

```

This model is  $\log(Y_{i,t}) = \beta_1 \log(y_{i,t-1}) + \beta_2 \log(y_{i,t-2}) + \beta_3 \log(y_{i,t-3}) + \mu_i + \varepsilon_{i,t}$ , which implies  $E[\log(Y_{i,t}) \mid \text{past}] = \beta_1 \log(y_{i,t-1}) + \beta_2 \log(y_{i,t-2}) + \beta_3 \log(y_{i,t-3}) + \mu_i$ . We estimate  $\hat{E}[\log(Y_{i,t}) \mid \text{past}] = \hat{\beta}_1 \log(y_{i,t-1}) + \hat{\beta}_2 \log(y_{i,t-2}) + \hat{\beta}_3 \log(y_{i,t-3}) + \hat{\mu}_i$ . We thus want  $\hat{\beta}_1$ ,  $\hat{\beta}_2$ , and  $\hat{\beta}_3$  (which we can extract from the model) and  $\hat{\mu}_i$  (which we can calculate for each  $i$ ).

Let  $y_{i,t} = \log(Y_{i,t})$  and  $x_{i,t-j} = \log(y_{i,t-j})$ . Our model is  $y_{i,t} = \beta_1 x_{i,t-1} + \beta_2 x_{i,t-2} + \beta_3 x_{i,t-3} + \mu_i + \varepsilon_{i,t}$ . Averaging over  $t \in \{1, \dots, T\}$ , we have  $\bar{y}_i = \beta_1 \bar{x}_{i-1} + \beta_2 \bar{x}_{i-2} + \beta_3 \bar{x}_{i-3} + \mu_i + \bar{\varepsilon}_{i,t}$ . Since we assume  $\mathbb{E}[\varepsilon_{i,t} \mid \text{past}] = 0$ , for large  $n$ ,  $\bar{\varepsilon}_{i,t} \approx 0$ , so  $\bar{y}_i \approx \beta_1 \bar{x}_{i-1} + \beta_2 \bar{x}_{i-2} + \beta_3 \bar{x}_{i-3} + \mu_i \implies \mu_i \approx \bar{y}_i - \beta_1 \bar{x}_{i-1} - \beta_2 \bar{x}_{i-2} - \beta_3 \bar{x}_{i-3} \implies \hat{\mu}_i \approx \bar{y}_i - \hat{\beta}_1 \bar{x}_{i-1} - \hat{\beta}_2 \bar{x}_{i-2} - \hat{\beta}_3 \bar{x}_{i-3}$ .

Use `model_2` to estimate counterfactual CO2 values

```
# Drop geometry
df_clean <- df_three %>%
  sf::st_drop_geometry()

# Get beta estimates
beta_hat <- coef(model_2)
beta_1_hat <- beta_hat["log_lag_co2_one"]
beta_2_hat <- beta_hat["log_lag_co2_two"]
beta_3_hat <- beta_hat["log_lag_co2_three"]

# Get mu_i_hat for each county
mu_hat_df <- df_clean %>%
  group_by(fips) %>%
  summarize(
    mu_hat = mean(
      log_co2 -
        beta_1_hat * log_lag_co2_one -
        beta_2_hat * log_lag_co2_two -
        beta_3_hat * log_lag_co2_three,
      na.rm = TRUE
    ),
    .groups = "drop"
  )

# Prepare full county-year data without geometry
df_all <- co2_county_filtered %>%
  sf::st_drop_geometry() %>%
  mutate(
    fips = as.character(fips),
    log_co2 = log(co2)
  ) %>%
  arrange(fips, year)

# Create wide data with observed log CO2 values
pred_wide <- df_all %>%
  filter(year >= 2002, year <= 2010) %>%
  select(fips, year, log_co2) %>%
  tidyr::pivot_wider(
    names_from = year,
```

```

    values_from = log_co2,
    names_prefix = "log_co2_"
  ) %>%
  left_join(mu_hat_df, by = "fips")

# For each year in 2005-2010, recursively predict log CO2
for (yr in 2005:2010) {

  # Define lag column names
  lag_1 <- paste0("log_co2_", yr - 1)
  lag_2 <- paste0("log_co2_", yr - 2)
  lag_3 <- paste0("log_co2_", yr - 3)
  pred_col <- paste0("pred_log_co2_", yr)

  # Predict log CO2 for current year
  pred_wide <- pred_wide %>%
    mutate(
      !!pred_col :=
        beta_1_hat * .data[[lag_1]] +
        beta_2_hat * .data[[lag_2]] +
        beta_3_hat * .data[[lag_3]] +
        mu_hat
    )

  # Replace log_co2_yr with predicted value so future years use recursive predictions
  pred_wide <- pred_wide %>%
    mutate(
      !!paste0("log_co2_", yr) := .data[[pred_col]]
    )
}

# Convert predictions to long format
pred_2005_2010 <- pred_wide %>%
  select(fips, starts_with("pred_log_co2_")) %>%
  tidyr::pivot_longer(
    cols = starts_with("pred_log_co2_"),
    names_to = "pred_year",
    values_to = "pred_log_co2"
  ) %>%
  mutate(
    pred_year = as.numeric(stringr::str_remove(pred_year, "pred_log_co2_")),
    pred_co2 = exp(pred_log_co2)
  )

# Join observed CO2 values for 2005-2010
pred_vs_obs <- df_all %>%
  filter(year >= 2005, year <= 2010) %>%

```

```

select(fips, year, observed_co2 = co2, observed_log_co2 = log_co2) %>%
left_join(
  pred_2005_2010,
  by = c("fips" = "fips", "year" = "pred_year")
)

# Display predicted vs. observed CO2 in 2005-2010 for each county
pred_vs_obs %>%
  select(fips, year, observed_co2, pred_co2)

```

```

## # A tibble: 18,318 x 4
##   fips   year observed_co2   pred_co2
##   <chr> <dbl>         <dbl>         <dbl>
## 1 01001  2005   337024676.   341080358.
## 2 01001  2006   333775728.   343027973.
## 3 01001  2007   331416672.   343825522.
## 4 01001  2008   317668472.   343902714.
## 5 01001  2009   320421436.   343568933.
## 6 01001  2010   339279489.   343025362.
## 7 01003  2005   1263603829.  1273381814.
## 8 01003  2006   1260344880.  1285147105.
## 9 01003  2007   1269201522.  1289297656.
## 10 01003  2008   1221088362.  1288656705.
## # i 18,308 more rows

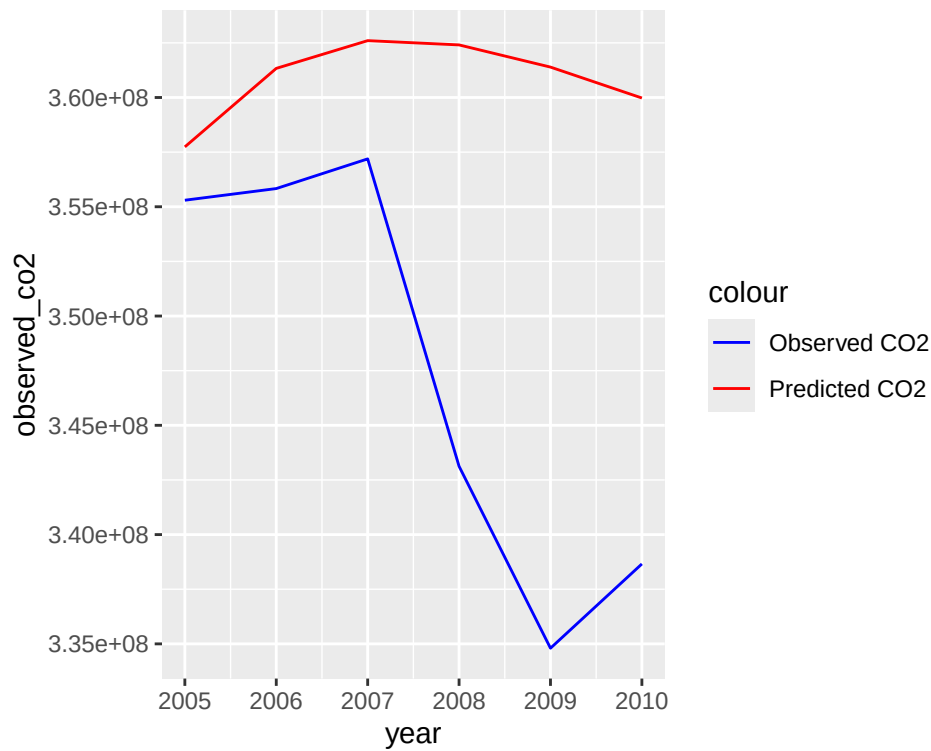
```

Plot predicted vs. observed CO2 for individual county and for total (from model\_2)

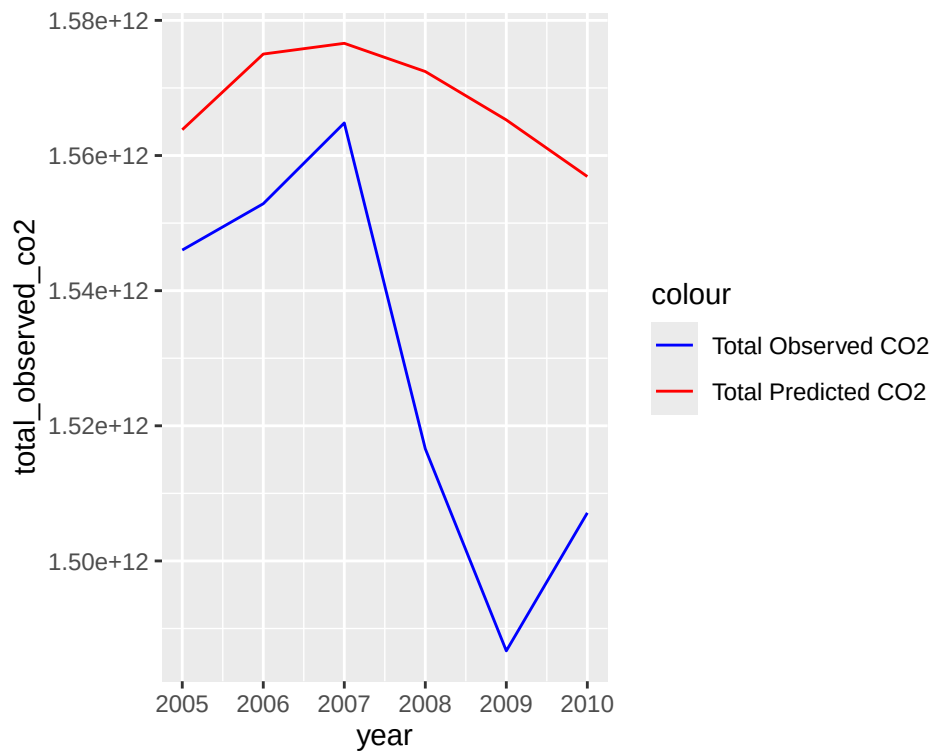
```

# Plot for individual county
pred_vs_obs %>%
  filter(fips == "01053") %>%
  ggplot(aes(x = year)) +
  geom_line(aes(y = observed_co2, color = "Observed CO2")) +
  geom_line(aes(y = pred_co2, color = "Predicted CO2")) +
  scale_color_manual(values = c("Observed CO2" = "blue", "Predicted CO2" = "red"))

```



```
# Plot for total/aggregated
pred_vs_obs %>%
  group_by(year) %>%
  summarize(total_observed_co2 = sum(observed_co2, na.rm = TRUE),
            total_pred_co2 = sum(pred_co2, na.rm = TRUE)) %>%
  ggplot(aes(x = year)) +
  geom_line(aes(y = total_observed_co2, color = "Total Observed CO2")) +
  geom_line(aes(y = total_pred_co2, color = "Total Predicted CO2")) +
  scale_color_manual(values = c("Total Observed CO2" = "blue", "Total Predicted CO2" = "red"))
```



Analyze Normality of residuals (from model\_2) for one county

```
# Define function to get residuals for a county
specified_fips <- "01053"
get_county_residuals_three_lag <- function(
  fips_use = specified_fips,
  co2_train = co2_county_2004_filtered
) {

  # Build 3-lag training data (pre-treatment only, up to 2004)
  df_three <- co2_train %>%
    sf::st_drop_geometry() %>%
    mutate(
      fips = as.character(fips),
      log_co2 = log(co2)
    ) %>%
    arrange(fips, year) %>%
    group_by(fips) %>%
    mutate(
      log_lag_co2_one = lag(log_co2, 1),
      log_lag_co2_two = lag(log_co2, 2),
      log_lag_co2_three = lag(log_co2, 3)
    ) %>%
    filter(
```

```

    !is.na(log_lag_co2_one),
    !is.na(log_lag_co2_two),
    !is.na(log_lag_co2_three)
  ) %>%
  ungroup()

# Convert to panel dataframe
pdf <- pdata.frame(df_three, index = c("fips", "year"))

# Fit within / county fixed effects model
model_2 <- plm(
  log_co2 ~ log_lag_co2_one + log_lag_co2_two + log_lag_co2_three,
  data = pdf,
  model = "within",
  effect = "individual"
)

# Extract beta estimates
beta_hat <- coef(model_2)
beta_1_hat <- beta_hat["log_lag_co2_one"]
beta_2_hat <- beta_hat["log_lag_co2_two"]
beta_3_hat <- beta_hat["log_lag_co2_three"]

# Reconstruct county fixed effects  $\mu_i$ 
mu_hat_df <- df_three %>%
  group_by(fips) %>%
  summarize(
    mu_hat = mean(
      log_co2 -
        beta_1_hat * log_lag_co2_one -
        beta_2_hat * log_lag_co2_two -
        beta_3_hat * log_lag_co2_three,
      na.rm = TRUE
    ),
    .groups = "drop"
  )

# Get fitted values and residuals for county
county_resid_df <- df_three %>%
  filter(fips == fips_use) %>%
  left_join(mu_hat_df, by = "fips") %>%
  mutate(
    fitted_log_co2 =
      beta_1_hat * log_lag_co2_one +
      beta_2_hat * log_lag_co2_two +
      beta_3_hat * log_lag_co2_three +
      mu_hat,

```



```

    residual_log_co2 = log_co2 - fitted_log_co2,

    fitted_co2 = exp(fitted_log_co2),
    residual_co2 = co2 - fitted_co2
  ) %>%
  select(
    fips, year, co2, log_co2,
    fitted_log_co2, residual_log_co2,
    fitted_co2, residual_co2,
    log_lag_co2_one, log_lag_co2_two, log_lag_co2_three,
    mu_hat
  )

  return(list(
    model = model_2,
    beta_hat = beta_hat,
    mu_hat_df = mu_hat_df,
    county_resid_df = county_resid_df
  ))
}

# Get residuals for fips 01053
res_01053 <- get_county_residuais_three_lag(fips_use = "01053")

# Summarize residuals
res_01053$county_residuais

## # A tibble: 22 x 12
##   fips   year      co2 log_co2 fitted_log_co2 residual_log_co2 fitted_co2
##   <chr> <int>    <dbl>  <dbl>         <dbl>         <dbl>      <dbl>
## 1 01053  1983 213502549.   19.2         19.2         0.0159 210126588.
## 2 01053  1984 220159013.   19.2         19.2        -0.0108 222538583.
## 3 01053  1985 221413485.   19.2         19.2        -0.0330 228847352.
## 4 01053  1986 229405499.   19.3         19.2         0.0177 225391295.
## 5 01053  1987 236858288.   19.3         19.3        -0.0117 239653827.
## 6 01053  1988 248886745.   19.3         19.3         0.0142 245388748.
## 7 01053  1989 257007070.   19.4         19.4        -0.0158 261099646.
## 8 01053  1990 259829467.   19.4         19.4        -0.0193 264894093.
## 9 01053  1991 260240502.   19.4         19.4        -0.0138 263868444.
## 10 01053 1992 269343723.   19.4         19.4         0.0234 263109642.
## # i 12 more rows
## # i 5 more variables: residual_co2 <dbl>, log_lag_co2_one <dbl>,
## #   log_lag_co2_two <dbl>, log_lag_co2_three <dbl>, mu_hat <dbl>

summary(res_01053$model)

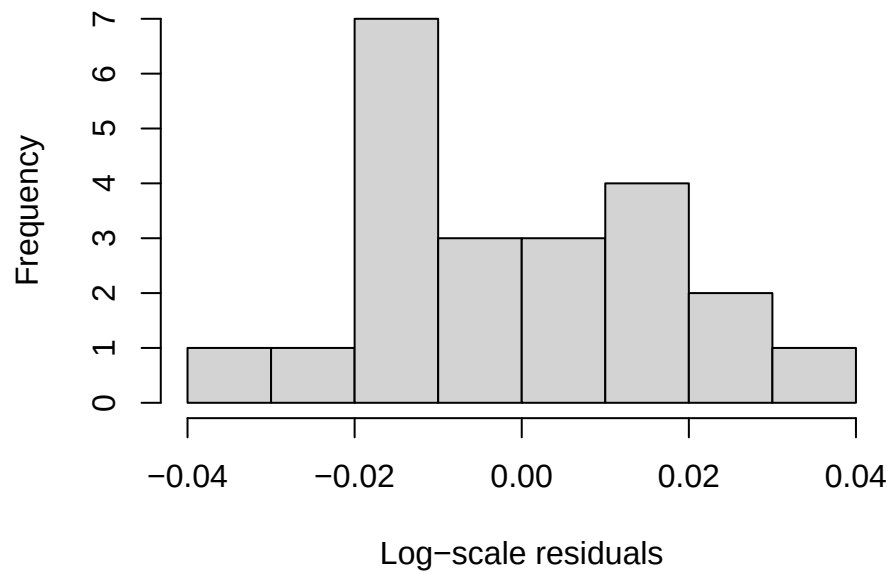
```

```
## Oneway (individual) effect Within Model
```

```
##
## Call:
## plm(formula = log_co2 ~ log_lag_co2_one + log_lag_co2_two + log_lag_co2_three,
##      data = pdf, effect = "individual", model = "within")
##
## Balanced Panel: n = 3053, T = 22, N = 67166
##
## Residuals:
##      Min.      1st Qu.      Median      3rd Qu.      Max.
## -0.7790162 -0.0154415 -0.0011651  0.0155646  0.4344497
##
## Coefficients:
##              Estimate Std. Error  t-value Pr(>|t|)
## log_lag_co2_one      1.7566735   0.0038807  452.664 < 2.2e-16 ***
## log_lag_co2_two     -0.9399495   0.0069992 -134.293 < 2.2e-16 ***
## log_lag_co2_three    0.1435722   0.0038497   37.294 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Total Sum of Squares:      2270
## Residual Sum of Squares: 46.224
## R-Squared:      0.97964
## Adj. R-Squared: 0.97867
## F-statistic: 1028080 on 3 and 64110 DF, p-value: < 2.22e-16

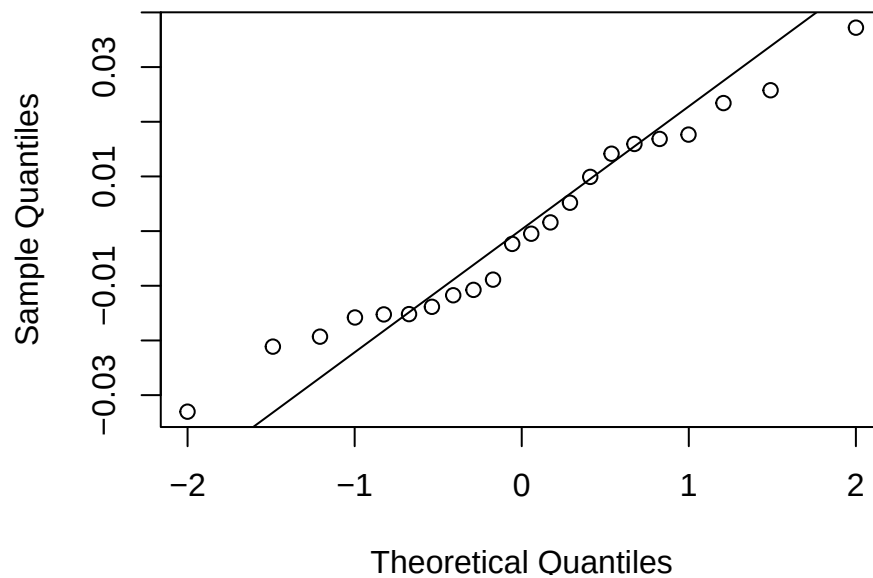
# Plot histogram of residuals
resid_01053 <- res_01053$county_residuals$residual_log_co2
hist(resid_01053, breaks = 10, main = "Residuals for FIPS 01053",
     xlab = "Log-scale residuals")
```

### Residuals for FIPS 01053



```
# Plot Q-Q plot  
qqnorm(resid_01053)  
qqline(resid_01053)
```

### Normal Q-Q Plot

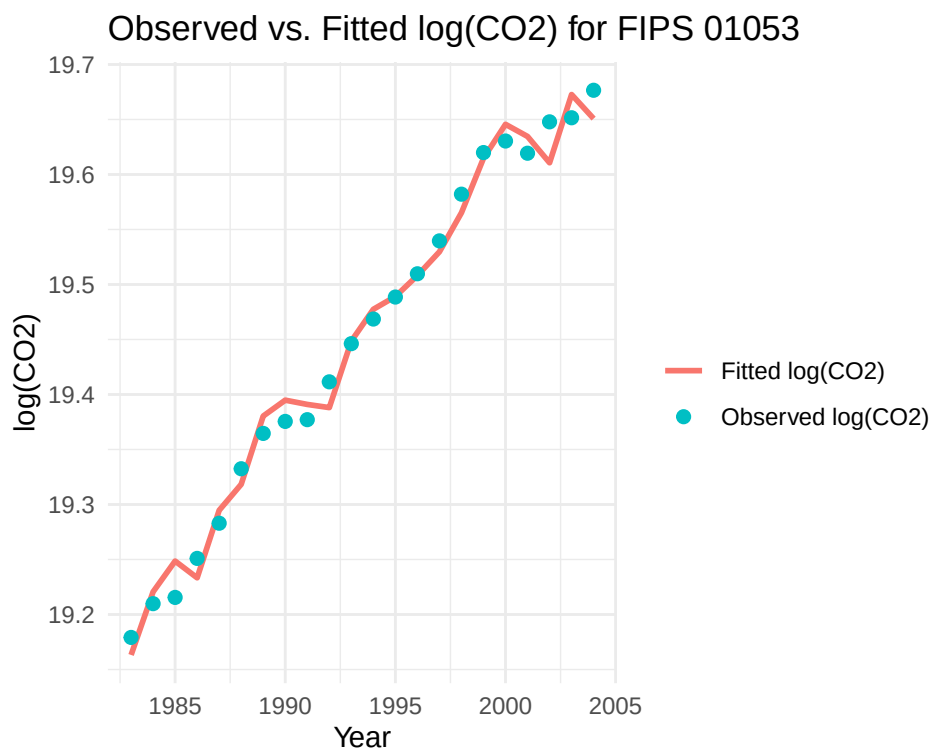


```
# Run Shapiro-Wilk test (high p indicates Normality)  
shapiro.test(resid_01053)
```

##

```
## Shapiro-Wilk normality test
##
## data: resid_01053
## W = 0.96577, p-value = 0.6138
```

```
# Plot observed and fitted values
res_01053$county_residuals %>%
  ggplot(aes(x = year)) +
  geom_line(aes(y = fitted_log_co2, color = "Fitted log(CO2)"), linewidth = 1) +
  geom_point(aes(y = log_co2, color = "Observed log(CO2)"), size = 2) +
  labs(
    title = "Observed vs. Fitted log(CO2) for FIPS 01053",
    x = "Year",
    y = "log(CO2)",
    color = ""
  ) +
  theme_minimal()
```



Analyze Normality of residuals (from model\_2) for all counties

```
# Run on all counties
county_diag_df <- df_three %>%
  group_split(fips) %>%
  map_dfr(function(df_county) {
```

```

# Get county residuals
county_res <- df_county %>%
  left_join(mu_hat_df, by = "fips") %>%
  mutate(
    fitted =
      beta_1_hat * log_lag_co2_one +
      beta_2_hat * log_lag_co2_two +
      beta_3_hat * log_lag_co2_three +
      mu_hat,

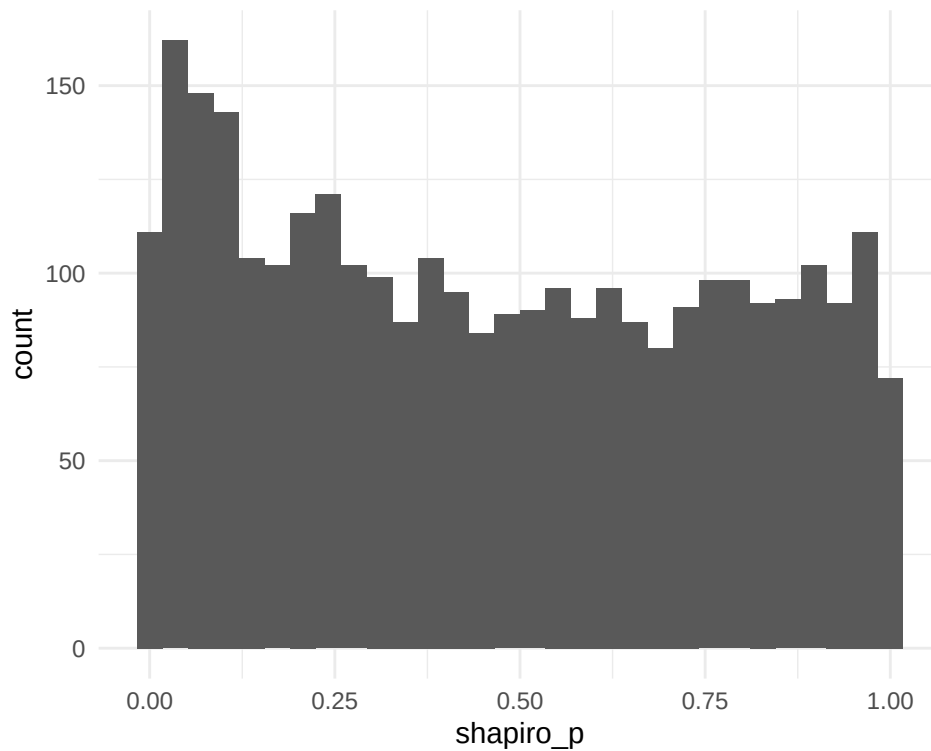
    resid = log_co2 - fitted
  )

r <- county_res$resid

tibble(
  fips = unique(df_county$fips),
  n = length(r),
  mean_resid = mean(r),
  sd_resid = sd(r),
  skewness = moments::skewness(r),
  kurtosis = moments::kurtosis(r),
  shapiro_p = shapiro.test(r)$p.value,
  jb_p = jarque.bera.test(r)$p.value,
  acf1 = acf(r, plot = FALSE)$acf[2]
)
})

# Plot histogram of p-values from Shapiro-Wilk test
ggplot(county_diag_df, aes(x = shapiro_p)) +
  geom_histogram(bins = 30) +
  theme_minimal()

```



```
# Get proportion of counties that have low p-value (non-Normality)
mean(county_diag_df$shapiro_p < 0.01)
```

```
## [1] 0.02129053
```

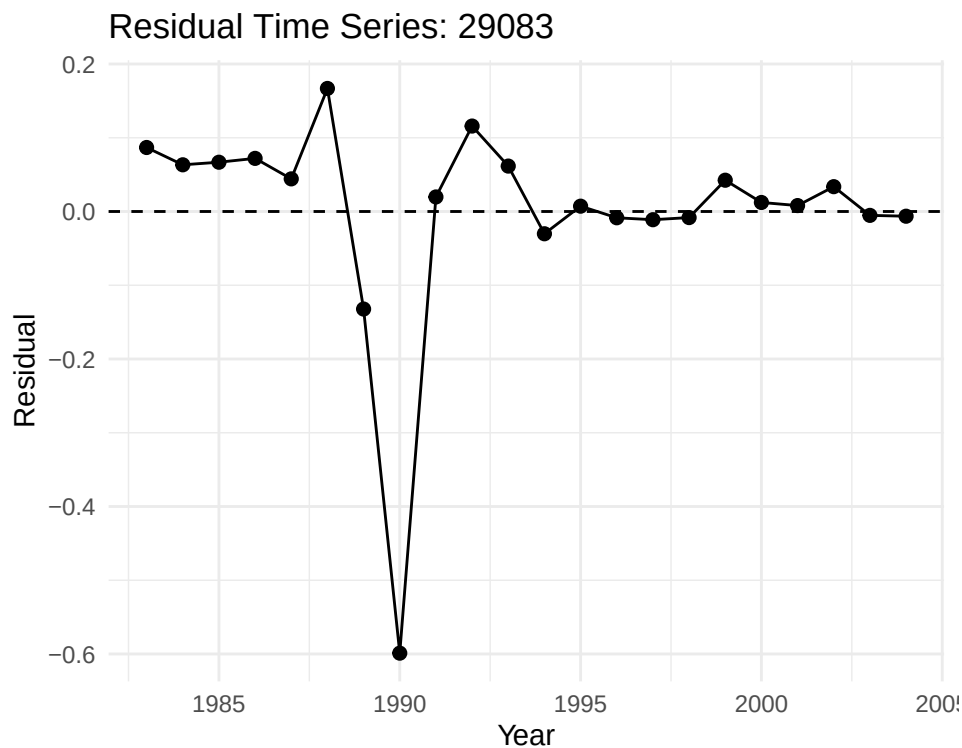
```
# Display counties with low p-value (non-Normality)
county_diag_df %>%
  arrange(shapiro_p) %>%
  slice(1:10)
```

```
## # A tibble: 10 x 9
##   fips      n mean_resid sd_resid skewness kurtosis shapiro_p  jb_p    acf1
##   <fct> <int>      <dbl>    <dbl>    <dbl>    <dbl>      <dbl>    <dbl>    <dbl>
## 1 29083     22 -1.61e-16  0.146    -3.25    14.2  0.00000119 0      0.174
## 2 29007     22 -9.69e-16  0.120    -3.20    13.8  0.00000202 0      0.176
## 3 29079     22 -4.84e-16  0.205    -2.58    10.8  0.0000394 0      0.415
## 4 29121     22 -1.45e-15  0.0990    -2.40     9.66  0.0000671 4.00e-14 0.458
## 5 24001     22  1.94e-15  0.0381    -2.41    10.4  0.0000732 2.22e-16 0.245
## 6 29189     22  6.46e-16  0.0299     2.30     8.88  0.0000788 8.25e-12 0.154
## 7 29055     22  8.07e-16  0.0901    -1.96     7.58  0.000105 6.07e- 8 0.102
## 8 29051     22  9.69e-16  0.0538    -2.31    10.2  0.000131 2.44e-15 0.206
## 9 08061     22 -1.61e-15  0.0413    -2.11     7.90  0.000210 4.68e- 9 -0.201
## 10 54059     22  9.69e-16  0.0375     2.11     8.49  0.000224 2.90e-10 0.0762
```

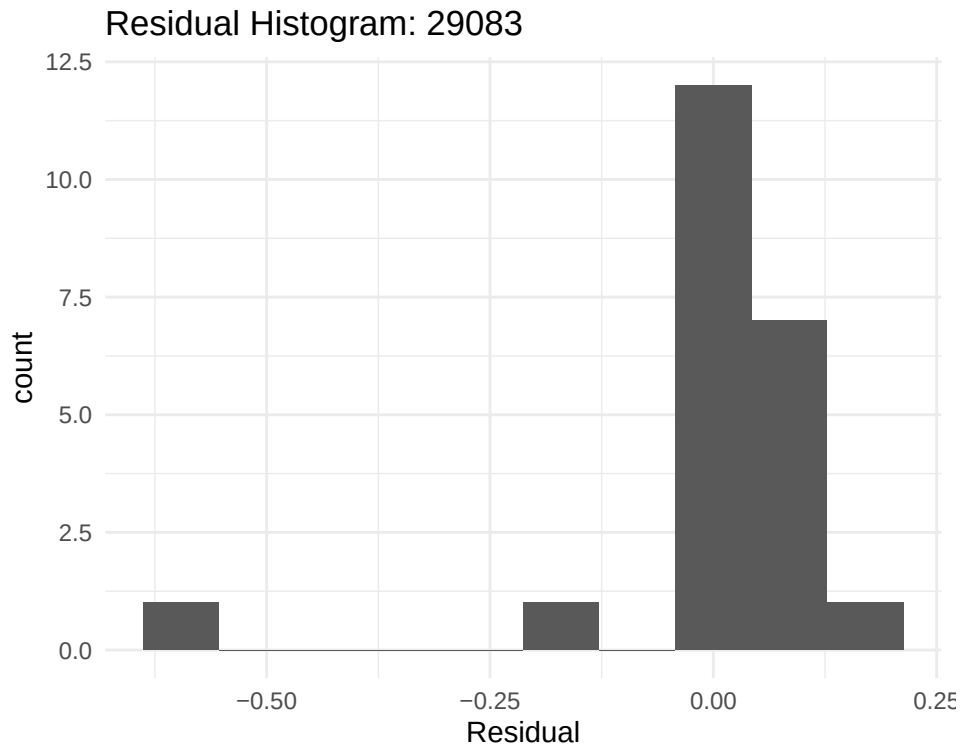
Analyze Normality of residuals (from model\_2) for a “bad” county

```
# Get fips and residuals of a county with low p-value (non-Normality)
bad_fips <- county_diag_df %>%
  arrange(shapiro_p) %>%
  slice(1) %>%
  pull(fips)
bad_res <- get_county_residuals_three_lag(
  fips_use = bad_fips
)$county_residuals

# Plot residuals over time
ggplot(bad_res, aes(x = year, y = residual_log_co2)) +
  geom_line() +
  geom_point(size = 2) +
  geom_hline(yintercept = 0, linetype = "dashed") +
  labs(
    title = paste("Residual Time Series:", bad_fips),
    x = "Year",
    y = "Residual"
  ) +
  theme_minimal()
```

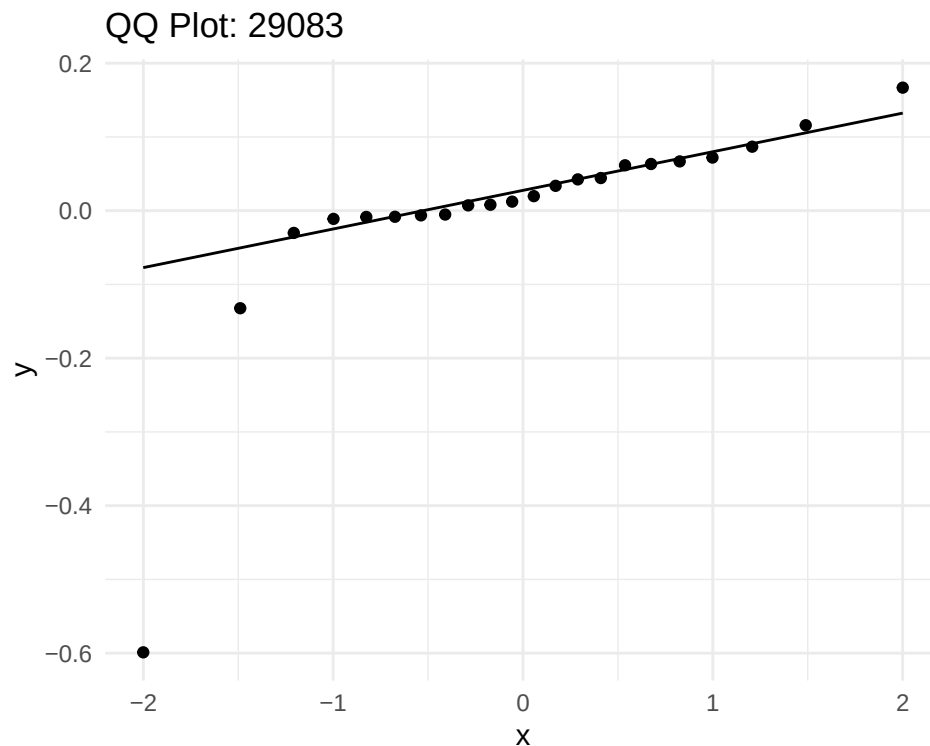


```
# Plot histogram of residuals
ggplot(bad_res, aes(x = residual_log_co2)) +
  geom_histogram(bins = 10) +
  labs(
    title = paste("Residual Histogram:", bad_fips),
    x = "Residual"
  ) +
  theme_minimal()
```



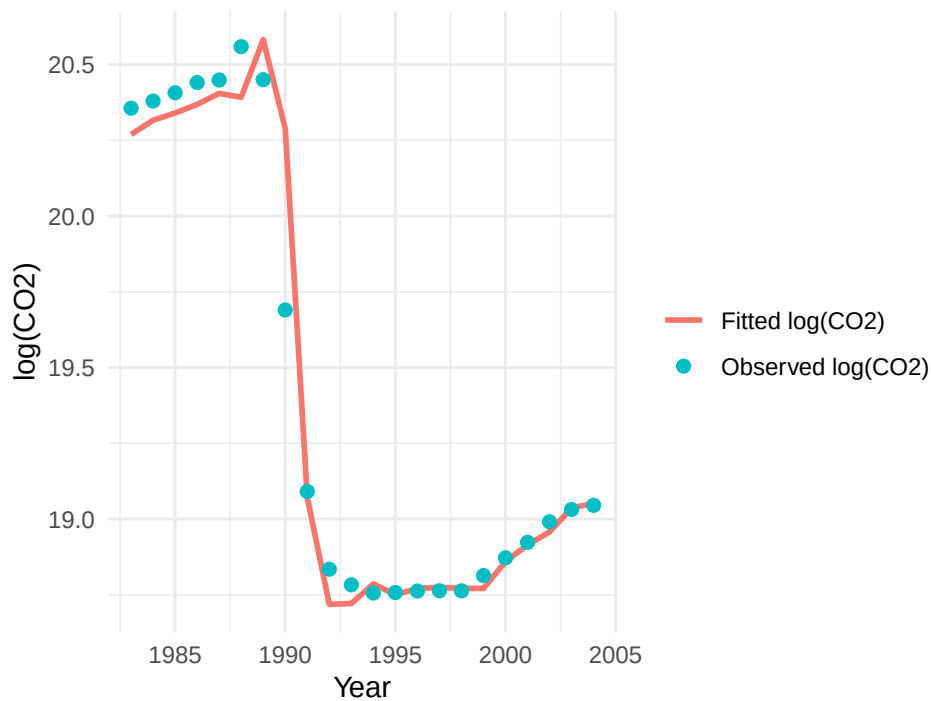
```
# Plot Q-Q plot
ggplot(bad_res, aes(sample = residual_log_co2)) +
  stat_qq() +
  stat_qq_line() +
  labs(
    title = paste("QQ Plot:", bad_fips)
  ) +
  theme_minimal()
```





```
# Plot observed and fitted values
res_29083 <- get_county_residuals_three_lag(fips_use = "29083")
resid_29083 <- res_29083$county_residuals$residual_log_co2
res_29083$county_residuals %>%
  ggplot(aes(x = year)) +
  geom_line(aes(y = fitted_log_co2, color = "Fitted log(CO2)"),
            linewidth = 1) +
  geom_point(aes(y = log_co2, color = "Observed log(CO2)"),
             size = 2) +
  labs(
    title = "Observed vs. Fitted log(CO2) for FIPS 29083",
    x = "Year",
    y = "log(CO2)",
    color = ""
  ) +
  theme_minimal()
```

### Observed vs. Fitted log(CO2) for FIPS 29083



### Analysis with splm library

Recreate model\_2 with `splm()` (i.e., three years of lag and spatial errors)

```
# Create panel dataframe from df_three
pdf <- pdata.frame(df_three, index = c("fips", "year"))

# Fit model via splm()
model_3 <- splm(
  log_co2 ~ log_lag_co2_one + log_lag_co2_two + log_lag_co2_three,
  data = pdf,
  listw = listw_object,
  model = "within",
  effect = "individual"
)
summary(model_3)

## Spatial panel fixed effects error model
##
##
## Call:
## splm(formula = log_co2 ~ log_lag_co2_one + log_lag_co2_two +
##       log_lag_co2_three, data = pdf, listw = listw_object, model = "within",
##       effect = "individual")
```

```
##
## Residuals:
##      Min.      1st Qu.      Median      3rd Qu.      Max.
## -0.6703687 -0.0184503 -0.0015719  0.0184139  0.3735562
##
## Spatial error parameter:
##      Estimate Std. Error t-value Pr(>|t|)
## rho 0.7647068  0.0027124  281.93 < 2.2e-16 ***
##
## Coefficients:
##              Estimate Std. Error t-value Pr(>|t|)
## log_lag_co2_one      2.1187814  0.0034766  609.44 < 2.2e-16 ***
## log_lag_co2_two     -1.5881904  0.0064036 -248.02 < 2.2e-16 ***
## log_lag_co2_three    0.4387239  0.0034194  128.31 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
# Get estimated fixed effects
effects(model_3)
```

```
##
## Intercept:
##              Estimate Std. Error t-value Pr(>|t|)
## (Intercept) 0.5842448  0.0072502  80.583 < 2.2e-16 ***
##
##
## Spatial fixed effects:
##              Estimate Std. Error t-value Pr(>|t|)
## 1      -7.2123e-03  8.0586e-03  -0.8950 0.3707967
## 2     -3.4389e-02  7.8127e-03  -4.4017 1.074e-05 ***
## 3      1.6677e-03  8.1126e-03   0.2056 0.8371311
## 4     -1.2376e-02  8.0023e-03  -1.5466 0.1219701
## 5      4.1036e-02  8.6101e-03   4.7661 1.878e-06 ***
## 6     -3.5421e-02  7.7579e-03  -4.5659 4.975e-06 ***
## 7      4.8785e-02  8.6406e-03   5.6461 1.641e-08 ***
## 8     -1.9433e-02  7.9805e-03  -2.4350 0.0148910 *
## 9      4.3592e-02  8.5083e-03   5.1235 2.999e-07 ***
## 10     5.2593e-02  8.6746e-03   6.0629 1.337e-09 ***
## 11     2.9560e-03  8.1454e-03   0.3629 0.7166774
## 12     -6.6985e-03  8.1171e-03  -0.8252 0.4092411
## 13     1.2030e-03  8.1528e-03   0.1476 0.8826953
## 14     -4.9297e-03  8.0517e-03  -0.6123 0.5403659
## 15     -2.9592e-02  7.9504e-03  -3.7220 0.0001976 ***
## 16     -7.0374e-02  7.5348e-03  -9.3399 < 2.2e-16 ***
## 17      2.1968e-02  8.3935e-03   2.6173 0.0088635 **
## 18     -2.5535e-02  7.9119e-03  -3.2274 0.0012493 **
## 19     -9.8442e-03  8.0460e-03  -1.2235 0.2211442
```

|       |             |            |                    |     |
|-------|-------------|------------|--------------------|-----|
| ## 20 | -8.6134e-02 | 7.1774e-03 | -12.0008 < 2.2e-16 | *** |
| ## 21 | -6.9404e-02 | 7.5479e-03 | -9.1952 < 2.2e-16  | *** |
| ## 22 | -2.5397e-02 | 7.9469e-03 | -3.1958 0.0013945  | **  |
| ## 23 | -2.4727e-02 | 7.9075e-03 | -3.1271 0.0017654  | **  |
| ## 24 | -6.1329e-02 | 7.5218e-03 | -8.1535 3.534e-16  | *** |
| ## 25 | -1.8173e-02 | 7.9162e-03 | -2.2957 0.0216930  | *   |
| ## 26 | 2.9182e-02  | 8.4597e-03 | 3.4495 0.0005617   | *** |
| ## 27 | -9.2729e-02 | 7.1798e-03 | -12.9152 < 2.2e-16 | *** |
| ## 28 | 4.8820e-02  | 8.5622e-03 | 5.7018 1.186e-08   | *** |
| ## 29 | 8.8049e-03  | 8.2077e-03 | 1.0728 0.2833773   |     |
| ## 30 | -6.3606e-02 | 7.5135e-03 | -8.4656 < 2.2e-16  | *** |
| ## 31 | -1.3145e-02 | 7.9673e-03 | -1.6499 0.0989694  | .   |
| ## 32 | -1.4480e-02 | 8.0472e-03 | -1.7994 0.0719625  | .   |
| ## 33 | -5.8385e-02 | 7.6346e-03 | -7.6474 2.050e-14  | *** |
| ## 34 | -5.5912e-03 | 8.0716e-03 | -0.6927 0.4884938  |     |
| ## 35 | -4.1688e-02 | 7.7277e-03 | -5.3946 6.867e-08  | *** |
| ## 36 | -1.2796e-02 | 8.0393e-03 | -1.5917 0.1114560  |     |
| ## 37 | -3.3613e-02 | 7.7780e-03 | -4.3216 1.549e-05  | *** |
| ## 38 | -1.0685e-02 | 8.0985e-03 | -1.3193 0.1870603  |     |
| ## 39 | 5.5747e-03  | 8.2316e-03 | 0.6772 0.4982582   |     |
| ## 40 | -1.1530e-02 | 7.9611e-03 | -1.4482 0.1475506  |     |
| ## 41 | -2.6089e-02 | 7.8645e-03 | -3.3173 0.0009090  | *** |
| ## 42 | 1.3774e-02  | 8.3009e-03 | 1.6594 0.0970407   | .   |
| ## 43 | -2.8436e-02 | 7.8346e-03 | -3.6296 0.0002839  | *** |
| ## 44 | -6.8819e-02 | 7.4556e-03 | -9.2306 < 2.2e-16  | *** |
| ## 45 | 2.1847e-02  | 8.3371e-03 | 2.6204 0.0087818   | **  |
| ## 46 | -4.5111e-02 | 7.7146e-03 | -5.8475 4.990e-09  | *** |
| ## 47 | 4.5836e-02  | 8.6314e-03 | 5.3104 1.094e-07   | *** |
| ## 48 | 1.5964e-02  | 8.3085e-03 | 1.9214 0.0546790   | .   |
| ## 49 | -1.9182e-02 | 7.9330e-03 | -2.4180 0.0156059  | *   |
| ## 50 | 1.4768e-03  | 8.1549e-03 | 0.1811 0.8562910   |     |
| ## 51 | 4.5135e-03  | 8.1987e-03 | 0.5505 0.5819706   |     |
| ## 52 | 2.5905e-02  | 8.4072e-03 | 3.0812 0.0020615   | **  |
| ## 53 | -4.7562e-02 | 7.6631e-03 | -6.2065 5.416e-10  | *** |
| ## 54 | -2.4295e-02 | 7.9128e-03 | -3.0703 0.0021382  | **  |
| ## 55 | -3.6264e-02 | 7.7867e-03 | -4.6572 3.206e-06  | *** |
| ## 56 | -2.0046e-03 | 8.1205e-03 | -0.2469 0.8050147  |     |
| ## 57 | 3.2297e-02  | 8.4550e-03 | 3.8198 0.0001336   | *** |
| ## 58 | 3.4031e-02  | 8.5162e-03 | 3.9961 6.440e-05   | *** |
| ## 59 | 4.3756e-02  | 8.5284e-03 | 5.1307 2.887e-07   | *** |
| ## 60 | -2.9300e-02 | 7.8450e-03 | -3.7349 0.0001878  | *** |
| ## 61 | 1.1385e-02  | 8.2233e-03 | 1.3845 0.1662115   |     |
| ## 62 | 2.5531e-02  | 8.3991e-03 | 3.0398 0.0023676   | **  |
| ## 63 | -3.5453e-02 | 7.7350e-03 | -4.5834 4.574e-06  | *** |
| ## 64 | 7.5463e-02  | 8.9577e-03 | 8.4243 < 2.2e-16   | *** |
| ## 65 | -5.1762e-02 | 7.6340e-03 | -6.7805 1.198e-11  | *** |
| ## 66 | 2.4759e-02  | 8.3784e-03 | 2.9551 0.0031260   | **  |
| ## 67 | -3.4538e-03 | 8.1205e-03 | -0.4253 0.6706014  |     |

|        |             |            |                    |     |
|--------|-------------|------------|--------------------|-----|
| ## 68  | -6.9891e-02 | 7.3697e-03 | -9.4835 < 2.2e-16  | *** |
| ## 69  | 8.9063e-02  | 8.8791e-03 | 10.0307 < 2.2e-16  | *** |
| ## 70  | 2.7449e-02  | 8.5331e-03 | 3.2168 0.0012963   | **  |
| ## 71  | -3.4222e-02 | 7.7922e-03 | -4.3919 1.124e-05  | *** |
| ## 72  | 3.4486e-02  | 8.5235e-03 | 4.0460 5.211e-05   | *** |
| ## 73  | 6.7911e-03  | 8.2031e-03 | 0.8279 0.4077488   |     |
| ## 74  | -5.1683e-03 | 8.0527e-03 | -0.6418 0.5210014  |     |
| ## 75  | 2.1392e-02  | 8.3530e-03 | 2.5610 0.0104386   | *   |
| ## 76  | 1.2342e-02  | 8.4203e-03 | 1.4658 0.1427014   |     |
| ## 77  | -4.6270e-02 | 7.6460e-03 | -6.0516 1.434e-09  | *** |
| ## 78  | -8.5246e-03 | 8.0128e-03 | -1.0639 0.2873860  |     |
| ## 79  | 2.1134e-02  | 8.3280e-03 | 2.5377 0.0111594   | *   |
| ## 80  | -1.7890e-03 | 8.1148e-03 | -0.2205 0.8255115  |     |
| ## 81  | 1.6683e-03  | 8.0700e-03 | 0.2067 0.8362228   |     |
| ## 82  | 1.5000e-02  | 8.2144e-03 | 1.8261 0.0678425   | .   |
| ## 83  | -5.4684e-02 | 7.6821e-03 | -7.1184 1.092e-12  | *** |
| ## 84  | 4.2042e-02  | 8.5857e-03 | 4.8968 9.742e-07   | *** |
| ## 85  | -4.3211e-02 | 7.7050e-03 | -5.6082 2.045e-08  | *** |
| ## 86  | 1.7041e-02  | 8.3295e-03 | 2.0459 0.0407674   | *   |
| ## 87  | -2.1034e-02 | 7.8960e-03 | -2.6639 0.0077231  | **  |
| ## 88  | -4.4412e-02 | 7.8354e-03 | -5.6681 1.444e-08  | *** |
| ## 89  | -6.9744e-02 | 7.4619e-03 | -9.3466 < 2.2e-16  | *** |
| ## 90  | -7.0006e-02 | 7.5961e-03 | -9.2161 < 2.2e-16  | *** |
| ## 91  | -2.5824e-02 | 7.8833e-03 | -3.2758 0.0010538  | **  |
| ## 92  | -7.5850e-03 | 8.0129e-03 | -0.9466 0.3438407  |     |
| ## 93  | -3.8314e-02 | 7.7893e-03 | -4.9189 8.705e-07  | *** |
| ## 94  | 1.5212e-03  | 8.1117e-03 | 0.1875 0.8512482   |     |
| ## 95  | 3.4203e-03  | 8.1822e-03 | 0.4180 0.6759320   |     |
| ## 96  | 2.5375e-02  | 8.4244e-03 | 3.0121 0.0025947   | **  |
| ## 97  | -2.6439e-02 | 7.8566e-03 | -3.3651 0.0007651  | *** |
| ## 98  | 6.9170e-02  | 8.8454e-03 | 7.8198 5.289e-15   | *** |
| ## 99  | 1.7574e-02  | 8.3002e-03 | 2.1174 0.0342291   | *   |
| ## 100 | -4.2371e-03 | 8.1038e-03 | -0.5229 0.6010780  |     |
| ## 101 | -3.6812e-03 | 8.0704e-03 | -0.4561 0.6482962  |     |
| ## 102 | -2.8707e-02 | 7.8977e-03 | -3.6348 0.0002781  | *** |
| ## 103 | 1.1805e-02  | 8.2601e-03 | 1.4292 0.1529541   |     |
| ## 104 | 1.5366e-02  | 8.3081e-03 | 1.8495 0.0643850   | .   |
| ## 105 | 2.9577e-03  | 8.1636e-03 | 0.3623 0.7171296   |     |
| ## 106 | 3.2016e-02  | 8.4320e-03 | 3.7970 0.0001465   | *** |
| ## 107 | 5.8566e-04  | 8.1737e-03 | 0.0717 0.9428790   |     |
| ## 108 | 6.6066e-03  | 8.1923e-03 | 0.8064 0.4199888   |     |
| ## 109 | 5.3626e-04  | 8.1611e-03 | 0.0657 0.9476092   |     |
| ## 110 | -3.6619e-02 | 7.7725e-03 | -4.7113 2.461e-06  | *** |
| ## 111 | -9.8880e-04 | 8.0730e-03 | -0.1225 0.9025165  |     |
| ## 112 | -9.2231e-02 | 7.3223e-03 | -12.5958 < 2.2e-16 | *** |
| ## 113 | -4.1426e-02 | 7.7430e-03 | -5.3502 8.788e-08  | *** |
| ## 114 | -7.2738e-02 | 7.3771e-03 | -9.8599 < 2.2e-16  | *** |
| ## 115 | 1.1084e-02  | 8.2575e-03 | 1.3423 0.1794885   |     |

|        |             |            |         |           |     |
|--------|-------------|------------|---------|-----------|-----|
| ## 116 | -1.0147e-02 | 8.0749e-03 | -1.2566 | 0.2089063 |     |
| ## 117 | -2.5211e-02 | 7.9153e-03 | -3.1852 | 0.0014468 | **  |
| ## 118 | -1.6820e-02 | 8.0194e-03 | -2.0974 | 0.0359577 | *   |
| ## 119 | -1.4679e-02 | 7.9968e-03 | -1.8356 | 0.0664122 | .   |
| ## 120 | 5.7067e-02  | 8.7768e-03 | 6.5020  | 7.926e-11 | *** |
| ## 121 | -4.0466e-02 | 7.7371e-03 | -5.2301 | 1.694e-07 | *** |
| ## 122 | -5.0212e-02 | 7.6689e-03 | -6.5475 | 5.850e-11 | *** |
| ## 123 | 6.2842e-03  | 8.2606e-03 | 0.7607  | 0.4468112 |     |
| ## 124 | 1.5708e-02  | 8.2908e-03 | 1.8946  | 0.0581442 | .   |
| ## 125 | 3.1509e-02  | 8.4455e-03 | 3.7309  | 0.0001908 | *** |
| ## 126 | 4.3360e-02  | 8.5414e-03 | 5.0764  | 3.846e-07 | *** |
| ## 127 | 6.0943e-03  | 8.2266e-03 | 0.7408  | 0.4588142 |     |
| ## 128 | -2.9604e-02 | 7.8926e-03 | -3.7509 | 0.0001762 | *** |
| ## 129 | -8.9585e-03 | 8.0654e-03 | -1.1107 | 0.2666834 |     |
| ## 130 | 2.2787e-02  | 8.3381e-03 | 2.7329  | 0.0062779 | **  |
| ## 131 | -3.8601e-02 | 7.7678e-03 | -4.9694 | 6.717e-07 | *** |
| ## 132 | -1.1254e-02 | 8.0815e-03 | -1.3925 | 0.1637621 |     |
| ## 133 | 5.0951e-02  | 8.7141e-03 | 5.8469  | 5.009e-09 | *** |
| ## 134 | -2.0010e-02 | 7.9522e-03 | -2.5163 | 0.0118607 | *   |
| ## 135 | -1.5598e-02 | 7.9879e-03 | -1.9527 | 0.0508575 | .   |
| ## 136 | -1.4043e-03 | 8.1129e-03 | -0.1731 | 0.8625727 |     |
| ## 137 | -1.3875e-03 | 8.1222e-03 | -0.1708 | 0.8643579 |     |
| ## 138 | 3.8294e-02  | 8.5454e-03 | 4.4812  | 7.423e-06 | *** |
| ## 139 | 2.7645e-04  | 8.1623e-03 | 0.0339  | 0.9729816 |     |
| ## 140 | 1.2585e-02  | 8.3370e-03 | 1.5095  | 0.1311709 |     |
| ## 141 | -3.4413e-02 | 7.7023e-03 | -4.4679 | 7.900e-06 | *** |
| ## 142 | -4.2449e-02 | 7.7245e-03 | -5.4953 | 3.900e-08 | *** |
| ## 143 | -2.8299e-02 | 7.8489e-03 | -3.6055 | 0.0003116 | *** |
| ## 144 | 6.6486e-02  | 8.8831e-03 | 7.4845  | 7.179e-14 | *** |
| ## 145 | -1.6176e-02 | 7.9837e-03 | -2.0261 | 0.0427528 | *   |
| ## 146 | -3.5936e-02 | 7.7022e-03 | -4.6657 | 3.075e-06 | *** |
| ## 147 | 8.2143e-03  | 8.1987e-03 | 1.0019  | 0.3163898 |     |
| ## 148 | -3.1254e-02 | 7.8534e-03 | -3.9796 | 6.902e-05 | *** |
| ## 149 | -6.1304e-03 | 8.1054e-03 | -0.7563 | 0.4494500 |     |
| ## 150 | -1.4207e-02 | 7.9802e-03 | -1.7803 | 0.0750282 | .   |
| ## 151 | 3.7870e-02  | 8.5183e-03 | 4.4457  | 8.759e-06 | *** |
| ## 152 | -2.4677e-02 | 7.8638e-03 | -3.1380 | 0.0017010 | **  |
| ## 153 | -1.1109e-02 | 7.9488e-03 | -1.3976 | 0.1622334 |     |
| ## 154 | -5.1412e-02 | 7.6634e-03 | -6.7087 | 1.964e-11 | *** |
| ## 155 | -5.4821e-02 | 7.5556e-03 | -7.2557 | 3.996e-13 | *** |
| ## 156 | 3.5181e-02  | 8.4333e-03 | 4.1716  | 3.024e-05 | *** |
| ## 157 | 9.3196e-03  | 8.1263e-03 | 1.1468  | 0.2514454 |     |
| ## 158 | -3.2793e-02 | 7.8629e-03 | -4.1707 | 3.037e-05 | *** |
| ## 159 | 3.2669e-02  | 8.4623e-03 | 3.8606  | 0.0001131 | *** |
| ## 160 | -8.7761e-03 | 8.0114e-03 | -1.0955 | 0.2733161 |     |
| ## 161 | 1.1412e-03  | 8.1147e-03 | 0.1406  | 0.8881574 |     |
| ## 162 | -6.2552e-03 | 8.0400e-03 | -0.7780 | 0.4365613 |     |
| ## 163 | 1.1370e-02  | 8.2668e-03 | 1.3753  | 0.1690250 |     |

|        |             |            |         |           |     |
|--------|-------------|------------|---------|-----------|-----|
| ## 164 | -1.6189e-02 | 8.0508e-03 | -2.0108 | 0.0443434 | *   |
| ## 165 | -5.3905e-02 | 7.5565e-03 | -7.1337 | 9.773e-13 | *** |
| ## 166 | -3.1224e-02 | 7.7579e-03 | -4.0248 | 5.702e-05 | *** |
| ## 167 | 1.2184e-02  | 8.2489e-03 | 1.4770  | 0.1396627 |     |
| ## 168 | -1.3355e-02 | 7.9810e-03 | -1.6733 | 0.0942727 | .   |
| ## 169 | 8.6025e-03  | 8.2467e-03 | 1.0431  | 0.2968827 |     |
| ## 170 | 4.7680e-02  | 8.6440e-03 | 5.5160  | 3.468e-08 | *** |
| ## 171 | 6.3504e-02  | 8.8284e-03 | 7.1931  | 6.332e-13 | *** |
| ## 172 | -3.5372e-02 | 7.9823e-03 | -4.4313 | 9.365e-06 | *** |
| ## 173 | -5.4459e-02 | 7.6034e-03 | -7.1624 | 7.929e-13 | *** |
| ## 174 | 8.6017e-03  | 8.2035e-03 | 1.0485  | 0.2943929 |     |
| ## 175 | -4.9664e-03 | 8.0513e-03 | -0.6168 | 0.5373366 |     |
| ## 176 | -1.3660e-02 | 7.9621e-03 | -1.7156 | 0.0862337 | .   |
| ## 177 | -7.2108e-02 | 7.4100e-03 | -9.7313 | < 2.2e-16 | *** |
| ## 178 | -2.7134e-02 | 7.8696e-03 | -3.4479 | 0.0005649 | *** |
| ## 179 | -1.2281e-02 | 7.9257e-03 | -1.5495 | 0.1212542 |     |
| ## 180 | 6.9941e-04  | 8.1264e-03 | 0.0861  | 0.9314137 |     |
| ## 181 | -7.0120e-02 | 7.4399e-03 | -9.4248 | < 2.2e-16 | *** |
| ## 182 | 1.8859e-02  | 8.3157e-03 | 2.2678  | 0.0233399 | *   |
| ## 183 | -2.9068e-02 | 7.7827e-03 | -3.7349 | 0.0001878 | *** |
| ## 184 | 1.0932e-01  | 9.4030e-03 | 11.6265 | < 2.2e-16 | *** |
| ## 185 | 7.1897e-02  | 8.8964e-03 | 8.0816  | 6.393e-16 | *** |
| ## 186 | 6.5440e-02  | 8.8228e-03 | 7.4171  | 1.197e-13 | *** |
| ## 187 | -3.2237e-02 | 7.7428e-03 | -4.1635 | 3.135e-05 | *** |
| ## 188 | 2.9516e-02  | 8.4007e-03 | 3.5135  | 0.0004423 | *** |
| ## 189 | 3.5776e-02  | 8.5501e-03 | 4.1842  | 2.861e-05 | *** |
| ## 190 | -4.9297e-02 | 7.6722e-03 | -6.4254 | 1.315e-10 | *** |
| ## 191 | -4.9207e-02 | 7.6223e-03 | -6.4556 | 1.078e-10 | *** |
| ## 192 | -3.7661e-03 | 8.1354e-03 | -0.4629 | 0.6434182 |     |
| ## 193 | 4.7364e-02  | 8.6292e-03 | 5.4888  | 4.046e-08 | *** |
| ## 194 | 3.4552e-02  | 8.5248e-03 | 4.0532  | 5.053e-05 | *** |
| ## 195 | 1.4149e-02  | 8.3534e-03 | 1.6939  | 0.0902937 | .   |
| ## 196 | -4.9370e-02 | 7.6628e-03 | -6.4429 | 1.172e-10 | *** |
| ## 197 | -4.4472e-02 | 7.6467e-03 | -5.8158 | 6.035e-09 | *** |
| ## 198 | -5.2474e-02 | 7.6182e-03 | -6.8879 | 5.661e-12 | *** |
| ## 199 | 9.3614e-02  | 9.1539e-03 | 10.2266 | < 2.2e-16 | *** |
| ## 200 | 1.6171e-02  | 8.2858e-03 | 1.9517  | 0.0509780 | .   |
| ## 201 | -5.9762e-02 | 7.5295e-03 | -7.9370 | 2.071e-15 | *** |
| ## 202 | 5.0233e-03  | 8.1871e-03 | 0.6136  | 0.5395063 |     |
| ## 203 | 1.8384e-02  | 8.3297e-03 | 2.2070  | 0.0273146 | *   |
| ## 204 | 2.5543e-02  | 8.4241e-03 | 3.0321  | 0.0024283 | **  |
| ## 205 | -3.2588e-02 | 7.8014e-03 | -4.1771 | 2.952e-05 | *** |
| ## 206 | -3.9577e-02 | 7.6942e-03 | -5.1437 | 2.693e-07 | *** |
| ## 207 | -3.7683e-02 | 7.7806e-03 | -4.8431 | 1.278e-06 | *** |
| ## 208 | -1.7446e-02 | 7.9666e-03 | -2.1899 | 0.0285341 | *   |
| ## 209 | -2.3956e-03 | 8.0819e-03 | -0.2964 | 0.7669103 |     |
| ## 210 | 3.4632e-02  | 8.4959e-03 | 4.0763  | 4.576e-05 | *** |
| ## 211 | 3.8689e-02  | 8.4853e-03 | 4.5595  | 5.127e-06 | *** |

|        |             |            |          |           |     |
|--------|-------------|------------|----------|-----------|-----|
| ## 212 | -3.0479e-02 | 7.8417e-03 | -3.8868  | 0.0001016 | *** |
| ## 213 | -2.2161e-02 | 7.8988e-03 | -2.8056  | 0.0050218 | **  |
| ## 214 | 9.3584e-02  | 9.2446e-03 | 10.1232  | < 2.2e-16 | *** |
| ## 215 | 3.4812e-02  | 8.5158e-03 | 4.0879   | 4.353e-05 | *** |
| ## 216 | -7.1745e-02 | 7.3968e-03 | -9.6995  | < 2.2e-16 | *** |
| ## 217 | 2.3929e-02  | 8.3702e-03 | 2.8589   | 0.0042517 | **  |
| ## 218 | -2.6690e-02 | 7.7876e-03 | -3.4273  | 0.0006097 | *** |
| ## 219 | 7.2605e-03  | 8.2027e-03 | 0.8851   | 0.3760854 |     |
| ## 220 | -1.7652e-02 | 7.9518e-03 | -2.2199  | 0.0264238 | *   |
| ## 221 | -4.4914e-02 | 7.6752e-03 | -5.8519  | 4.861e-09 | *** |
| ## 222 | -5.2695e-04 | 8.1426e-03 | -0.0647  | 0.9484014 |     |
| ## 223 | -7.6859e-03 | 8.0298e-03 | -0.9572  | 0.3384774 |     |
| ## 224 | -1.4278e-02 | 7.9497e-03 | -1.7960  | 0.0724932 | .   |
| ## 225 | -9.3508e-02 | 7.1387e-03 | -13.0988 | < 2.2e-16 | *** |
| ## 226 | 1.1118e-01  | 9.3620e-03 | 11.8756  | < 2.2e-16 | *** |
| ## 227 | -5.0888e-02 | 7.5178e-03 | -6.7690  | 1.296e-11 | *** |
| ## 228 | -5.1843e-02 | 7.5200e-03 | -6.8940  | 5.423e-12 | *** |
| ## 229 | 3.6861e-02  | 8.5537e-03 | 4.3094   | 1.637e-05 | *** |
| ## 230 | 3.4531e-02  | 8.4983e-03 | 4.0633   | 4.839e-05 | *** |
| ## 231 | -2.1683e-02 | 7.9002e-03 | -2.7446  | 0.0060586 | **  |
| ## 232 | -2.0269e-02 | 7.9713e-03 | -2.5427  | 0.0109992 | *   |
| ## 233 | -2.6828e-02 | 7.8929e-03 | -3.3991  | 0.0006762 | *** |
| ## 234 | -4.3390e-02 | 7.7026e-03 | -5.6332  | 1.769e-08 | *** |
| ## 235 | -1.5941e-02 | 7.9758e-03 | -1.9987  | 0.0456415 | *   |
| ## 236 | -5.5084e-02 | 7.5406e-03 | -7.3050  | 2.773e-13 | *** |
| ## 237 | -1.2649e-02 | 7.9869e-03 | -1.5837  | 0.1132598 |     |
| ## 238 | 5.4023e-02  | 8.7477e-03 | 6.1757   | 6.587e-10 | *** |
| ## 239 | -3.7934e-02 | 7.7665e-03 | -4.8843  | 1.038e-06 | *** |
| ## 240 | -4.2540e-02 | 7.7278e-03 | -5.5048  | 3.696e-08 | *** |
| ## 241 | 2.4764e-02  | 8.3234e-03 | 2.9752   | 0.0029279 | **  |
| ## 242 | -2.9209e-02 | 7.8484e-03 | -3.7217  | 0.0001979 | *** |
| ## 243 | -2.3035e-02 | 7.9624e-03 | -2.8930  | 0.0038160 | **  |
| ## 244 | -9.2137e-03 | 8.0090e-03 | -1.1504  | 0.2499679 |     |
| ## 245 | -5.0742e-02 | 7.7020e-03 | -6.5882  | 4.451e-11 | *** |
| ## 246 | -2.3185e-02 | 7.9028e-03 | -2.9338  | 0.0033482 | **  |
| ## 247 | 2.9733e-02  | 8.4608e-03 | 3.5142   | 0.0004411 | *** |
| ## 248 | -4.5893e-02 | 7.7424e-03 | -5.9275  | 3.076e-09 | *** |
| ## 249 | -3.7385e-02 | 7.7581e-03 | -4.8188  | 1.444e-06 | *** |
| ## 250 | -5.3546e-02 | 7.6941e-03 | -6.9594  | 3.416e-12 | *** |
| ## 251 | -3.5174e-02 | 7.8042e-03 | -4.5072  | 6.570e-06 | *** |
| ## 252 | -7.5161e-02 | 7.4135e-03 | -10.1383 | < 2.2e-16 | *** |
| ## 253 | -1.4836e-02 | 7.9092e-03 | -1.8758  | 0.0606867 | .   |
| ## 254 | -1.9977e-03 | 8.0629e-03 | -0.2478  | 0.8043214 |     |
| ## 255 | -3.3594e-02 | 7.8456e-03 | -4.2819  | 1.853e-05 | *** |
| ## 256 | 6.6508e-03  | 8.2202e-03 | 0.8091   | 0.4184652 |     |
| ## 257 | -2.1281e-02 | 7.9467e-03 | -2.6780  | 0.0074071 | **  |
| ## 258 | 5.1457e-02  | 8.6531e-03 | 5.9466   | 2.737e-09 | *** |
| ## 259 | -5.7518e-02 | 7.5685e-03 | -7.5996  | 2.970e-14 | *** |



|        |             |            |          |           |     |
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| ## 260 | -8.5796e-03 | 8.1167e-03 | -1.0570  | 0.2905013 |     |
| ## 261 | -6.8219e-02 | 7.4170e-03 | -9.1976  | < 2.2e-16 | *** |
| ## 262 | 1.9123e-02  | 8.2779e-03 | 2.3101   | 0.0208841 | *   |
| ## 263 | 2.3440e-02  | 8.4225e-03 | 2.7831   | 0.0053848 | **  |
| ## 264 | -2.8435e-02 | 7.7105e-03 | -3.6878  | 0.0002262 | *** |
| ## 265 | -4.1899e-02 | 7.7388e-03 | -5.4142  | 6.158e-08 | *** |
| ## 266 | -3.2383e-02 | 7.7508e-03 | -4.1780  | 2.941e-05 | *** |
| ## 267 | -4.2620e-02 | 7.7837e-03 | -5.4755  | 4.362e-08 | *** |
| ## 268 | 2.8853e-02  | 8.4400e-03 | 3.4187   | 0.0006293 | *** |
| ## 269 | 1.8942e-02  | 8.3835e-03 | 2.2594   | 0.0238571 | *   |
| ## 270 | -5.9272e-02 | 7.5449e-03 | -7.8559  | 3.969e-15 | *** |
| ## 271 | -8.7528e-03 | 8.0645e-03 | -1.0853  | 0.2777689 |     |
| ## 272 | 3.7559e-02  | 8.5020e-03 | 4.4177   | 9.976e-06 | *** |
| ## 273 | 2.2900e-02  | 8.3370e-03 | 2.7468   | 0.0060188 | **  |
| ## 274 | 1.6899e-02  | 8.2894e-03 | 2.0386   | 0.0414853 | *   |
| ## 275 | -2.9571e-02 | 7.7865e-03 | -3.7977  | 0.0001460 | *** |
| ## 276 | -2.0321e-02 | 7.9294e-03 | -2.5628  | 0.0103842 | *   |
| ## 277 | -3.1649e-02 | 7.7973e-03 | -4.0590  | 4.929e-05 | *** |
| ## 278 | -3.6201e-02 | 7.8118e-03 | -4.6341  | 3.585e-06 | *** |
| ## 279 | -3.9993e-02 | 7.7218e-03 | -5.1793  | 2.228e-07 | *** |
| ## 280 | -5.5407e-02 | 7.5451e-03 | -7.3435  | 2.082e-13 | *** |
| ## 281 | 5.2227e-02  | 8.6801e-03 | 6.0168   | 1.779e-09 | *** |
| ## 282 | -2.9255e-03 | 8.0912e-03 | -0.3616  | 0.7176705 |     |
| ## 283 | -4.6713e-03 | 8.1128e-03 | -0.5758  | 0.5647558 |     |
| ## 284 | -3.1831e-02 | 7.7907e-03 | -4.0858  | 4.392e-05 | *** |
| ## 285 | -9.2091e-02 | 7.1825e-03 | -12.8216 | < 2.2e-16 | *** |
| ## 286 | 4.7777e-02  | 8.6954e-03 | 5.4946   | 3.917e-08 | *** |
| ## 287 | -3.1977e-02 | 7.8182e-03 | -4.0901  | 4.313e-05 | *** |
| ## 288 | -5.3908e-02 | 7.5701e-03 | -7.1212  | 1.070e-12 | *** |
| ## 289 | -3.6750e-02 | 7.7766e-03 | -4.7256  | 2.294e-06 | *** |
| ## 290 | 1.9119e-02  | 8.3109e-03 | 2.3004   | 0.0214247 | *   |
| ## 291 | -8.9927e-02 | 7.2319e-03 | -12.4348 | < 2.2e-16 | *** |
| ## 292 | -1.5100e-02 | 8.0276e-03 | -1.8810  | 0.0599693 | .   |
| ## 293 | -3.9647e-02 | 7.8145e-03 | -5.0735  | 3.905e-07 | *** |
| ## 294 | 5.8303e-03  | 8.1828e-03 | 0.7125   | 0.4761501 |     |
| ## 295 | 1.7181e-02  | 8.2986e-03 | 2.0704   | 0.0384179 | *   |
| ## 296 | 2.2856e-02  | 8.3562e-03 | 2.7352   | 0.0062341 | **  |
| ## 297 | 2.9925e-03  | 8.1853e-03 | 0.3656   | 0.7146722 |     |
| ## 298 | 9.1898e-03  | 8.2041e-03 | 1.1201   | 0.2626529 |     |
| ## 299 | -2.1539e-02 | 7.8497e-03 | -2.7440  | 0.0060704 | **  |
| ## 300 | 4.0507e-02  | 8.5253e-03 | 4.7514   | 2.020e-06 | *** |
| ## 301 | -5.4611e-02 | 7.6254e-03 | -7.1618  | 7.964e-13 | *** |
| ## 302 | -5.8785e-02 | 7.5144e-03 | -7.8229  | 5.161e-15 | *** |
| ## 303 | 8.2023e-03  | 8.1935e-03 | 1.0011   | 0.3167862 |     |
| ## 304 | -9.5943e-03 | 8.0523e-03 | -1.1915  | 0.2334544 |     |
| ## 305 | 5.0399e-03  | 8.1894e-03 | 0.6154   | 0.5382786 |     |
| ## 306 | 2.6897e-02  | 8.4208e-03 | 3.1941   | 0.0014027 | **  |
| ## 307 | -2.8765e-02 | 7.8539e-03 | -3.6625  | 0.0002498 | *** |

|        |             |            |          |           |     |
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| ## 308 | -4.6070e-02 | 7.6236e-03 | -6.0430  | 1.513e-09 | *** |
| ## 309 | 2.3100e-02  | 8.4060e-03 | 2.7481   | 0.0059946 | **  |
| ## 310 | 7.3486e-02  | 8.9358e-03 | 8.2238   | < 2.2e-16 | *** |
| ## 311 | -4.8286e-02 | 7.6627e-03 | -6.3015  | 2.948e-10 | *** |
| ## 312 | 2.9805e-02  | 8.4581e-03 | 3.5239   | 0.0004253 | *** |
| ## 313 | 1.8731e-02  | 8.3434e-03 | 2.2450   | 0.0247659 | *   |
| ## 314 | 4.8236e-02  | 8.5631e-03 | 5.6330   | 1.771e-08 | *** |
| ## 315 | 5.1989e-04  | 8.1272e-03 | 0.0640   | 0.9489942 |     |
| ## 316 | 2.9915e-02  | 8.4537e-03 | 3.5387   | 0.0004021 | *** |
| ## 317 | -1.6722e-02 | 7.9642e-03 | -2.0996  | 0.0357604 | *   |
| ## 318 | 1.1715e-02  | 8.2596e-03 | 1.4184   | 0.1560730 |     |
| ## 319 | -2.9588e-02 | 7.7807e-03 | -3.8027  | 0.0001431 | *** |
| ## 320 | -2.8117e-02 | 7.8733e-03 | -3.5712  | 0.0003554 | *** |
| ## 321 | -9.2315e-03 | 8.0458e-03 | -1.1474  | 0.2512271 |     |
| ## 322 | 3.4881e-03  | 8.1354e-03 | 0.4288   | 0.6681001 |     |
| ## 323 | 5.2121e-03  | 8.1624e-03 | 0.6386   | 0.5231157 |     |
| ## 324 | 6.6323e-02  | 8.7890e-03 | 7.5461   | 4.485e-14 | *** |
| ## 325 | -1.4754e-02 | 7.9219e-03 | -1.8624  | 0.0625433 | .   |
| ## 326 | 1.8282e-02  | 8.3152e-03 | 2.1986   | 0.0279056 | *   |
| ## 327 | 7.4932e-02  | 8.9036e-03 | 8.4158   | < 2.2e-16 | *** |
| ## 328 | 3.7887e-03  | 8.2041e-03 | 0.4618   | 0.6442208 |     |
| ## 329 | -5.2358e-03 | 8.0315e-03 | -0.6519  | 0.5144579 |     |
| ## 330 | 7.4358e-02  | 8.9298e-03 | 8.3269   | < 2.2e-16 | *** |
| ## 331 | -7.4442e-02 | 7.3690e-03 | -10.1020 | < 2.2e-16 | *** |
| ## 332 | -3.4413e-02 | 7.7504e-03 | -4.4401  | 8.991e-06 | *** |
| ## 333 | -1.9844e-02 | 7.9232e-03 | -2.5046  | 0.0122588 | *   |
| ## 334 | -1.1882e-02 | 7.9750e-03 | -1.4900  | 0.1362362 |     |
| ## 335 | 1.2667e-02  | 8.2127e-03 | 1.5423   | 0.1229945 |     |
| ## 336 | 1.6626e-02  | 8.2395e-03 | 2.0179   | 0.0436031 | *   |
| ## 337 | 9.9464e-03  | 8.2170e-03 | 1.2105   | 0.2260994 |     |
| ## 338 | -1.5054e-03 | 8.0968e-03 | -0.1859  | 0.8525078 |     |
| ## 339 | 1.3615e-02  | 8.2659e-03 | 1.6471   | 0.0995276 | .   |
| ## 340 | 1.2367e-02  | 8.4193e-03 | 1.4689   | 0.1418578 |     |
| ## 341 | -3.1842e-02 | 7.8204e-03 | -4.0716  | 4.669e-05 | *** |
| ## 342 | 1.0450e-02  | 8.1799e-03 | 1.2775   | 0.2014317 |     |
| ## 343 | 2.9579e-02  | 8.4761e-03 | 3.4897   | 0.0004835 | *** |
| ## 344 | -5.7642e-03 | 8.1199e-03 | -0.7099  | 0.4777736 |     |
| ## 345 | 2.9229e-02  | 8.4504e-03 | 3.4589   | 0.0005423 | *** |
| ## 346 | 4.8734e-02  | 8.6506e-03 | 5.6336   | 1.765e-08 | *** |
| ## 347 | -4.4732e-02 | 7.6530e-03 | -5.8450  | 5.065e-09 | *** |
| ## 348 | 2.5799e-02  | 8.4690e-03 | 3.0463   | 0.0023170 | **  |
| ## 349 | -2.8684e-02 | 7.8182e-03 | -3.6689  | 0.0002436 | *** |
| ## 350 | 3.5322e-02  | 8.5082e-03 | 4.1515   | 3.303e-05 | *** |
| ## 351 | -6.7801e-02 | 7.3592e-03 | -9.2131  | < 2.2e-16 | *** |
| ## 352 | 1.8376e-02  | 8.3393e-03 | 2.2035   | 0.0275575 | *   |
| ## 353 | 3.5385e-02  | 8.5164e-03 | 4.1549   | 3.254e-05 | *** |
| ## 354 | -2.3964e-02 | 7.9105e-03 | -3.0294  | 0.0024502 | **  |
| ## 355 | 4.0161e-02  | 8.5687e-03 | 4.6870   | 2.773e-06 | *** |

|        |             |            |          |           |     |
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| ## 356 | 9.3870e-03  | 8.2399e-03 | 1.1392   | 0.2546141 |     |
| ## 357 | -2.9707e-02 | 7.8882e-03 | -3.7660  | 0.0001659 | *** |
| ## 358 | -2.5838e-02 | 7.8360e-03 | -3.2974  | 0.0009760 | *** |
| ## 359 | 1.8194e-02  | 8.3522e-03 | 2.1784   | 0.0293792 | *   |
| ## 360 | -4.2339e-02 | 7.6929e-03 | -5.5037  | 3.719e-08 | *** |
| ## 361 | -4.3778e-02 | 7.7108e-03 | -5.6774  | 1.367e-08 | *** |
| ## 362 | -4.0132e-02 | 7.6781e-03 | -5.2268  | 1.725e-07 | *** |
| ## 363 | -9.2935e-03 | 8.0822e-03 | -1.1499  | 0.2501964 |     |
| ## 364 | -2.3863e-02 | 7.9221e-03 | -3.0122  | 0.0025939 | **  |
| ## 365 | -1.6051e-03 | 8.0855e-03 | -0.1985  | 0.8426405 |     |
| ## 366 | 8.0012e-03  | 8.2579e-03 | 0.9689   | 0.3325924 |     |
| ## 367 | 7.4018e-02  | 8.8644e-03 | 8.3500   | < 2.2e-16 | *** |
| ## 368 | 1.5246e-02  | 8.2632e-03 | 1.8451   | 0.0650221 | .   |
| ## 369 | -9.9403e-03 | 8.0207e-03 | -1.2393  | 0.2152225 |     |
| ## 370 | -3.3610e-02 | 7.8287e-03 | -4.2931  | 1.762e-05 | *** |
| ## 371 | -4.9903e-03 | 8.0320e-03 | -0.6213  | 0.5343997 |     |
| ## 372 | 9.6792e-04  | 8.1215e-03 | 0.1192   | 0.9051332 |     |
| ## 373 | -3.4718e-02 | 7.7701e-03 | -4.4681  | 7.891e-06 | *** |
| ## 374 | -1.1326e-01 | 7.1636e-03 | -15.8106 | < 2.2e-16 | *** |
| ## 375 | -6.0063e-03 | 8.1021e-03 | -0.7413  | 0.4584936 |     |
| ## 376 | 7.3400e-03  | 8.2388e-03 | 0.8909   | 0.3729816 |     |
| ## 377 | -1.3656e-02 | 7.9782e-03 | -1.7117  | 0.0869505 | .   |
| ## 378 | -3.9837e-02 | 7.7382e-03 | -5.1481  | 2.632e-07 | *** |
| ## 379 | -1.9068e-02 | 7.9163e-03 | -2.4087  | 0.0160094 | *   |
| ## 380 | -1.0496e-02 | 8.0205e-03 | -1.3086  | 0.1906747 |     |
| ## 381 | 1.0849e-02  | 8.2184e-03 | 1.3201   | 0.1868119 |     |
| ## 382 | 1.1976e-02  | 8.3077e-03 | 1.4416   | 0.1494264 |     |
| ## 383 | 2.5592e-03  | 8.1819e-03 | 0.3128   | 0.7544416 |     |
| ## 384 | 7.4393e-03  | 8.1802e-03 | 0.9094   | 0.3631269 |     |
| ## 385 | -3.0033e-02 | 7.8852e-03 | -3.8087  | 0.0001397 | *** |
| ## 386 | -1.2367e-02 | 7.9539e-03 | -1.5548  | 0.1199866 |     |
| ## 387 | 1.4746e-02  | 8.2977e-03 | 1.7771   | 0.0755559 | .   |
| ## 388 | -3.9350e-02 | 7.9092e-03 | -4.9751  | 6.520e-07 | *** |
| ## 389 | -3.9099e-03 | 8.1307e-03 | -0.4809  | 0.6306029 |     |
| ## 390 | -1.1169e-01 | 7.0885e-03 | -15.7560 | < 2.2e-16 | *** |
| ## 391 | -7.5512e-03 | 8.0475e-03 | -0.9383  | 0.3480757 |     |
| ## 392 | -2.4639e-02 | 7.9352e-03 | -3.1050  | 0.0019027 | **  |
| ## 393 | -2.7140e-02 | 7.9120e-03 | -3.4302  | 0.0006032 | *** |
| ## 394 | 1.5011e-02  | 8.2968e-03 | 1.8092   | 0.0704187 | .   |
| ## 395 | -2.0823e-02 | 7.9418e-03 | -2.6220  | 0.0087421 | **  |
| ## 396 | -7.1222e-03 | 8.0581e-03 | -0.8839  | 0.3767731 |     |
| ## 397 | 1.0211e-02  | 8.2705e-03 | 1.2346   | 0.2169810 |     |
| ## 398 | -6.2224e-02 | 7.5045e-03 | -8.2916  | < 2.2e-16 | *** |
| ## 399 | 8.2516e-05  | 8.1663e-03 | 0.0101   | 0.9919379 |     |
| ## 400 | -1.6655e-02 | 7.8766e-03 | -2.1144  | 0.0344787 | *   |
| ## 401 | -1.6198e-02 | 7.9934e-03 | -2.0264  | 0.0427234 | *   |
| ## 402 | 3.0316e-02  | 8.4971e-03 | 3.5678   | 0.0003600 | *** |
| ## 403 | -4.6703e-02 | 7.6537e-03 | -6.1020  | 1.047e-09 | *** |

|        |             |            |                    |     |
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| ## 404 | -6.8173e-02 | 7.4287e-03 | -9.1769 < 2.2e-16  | *** |
| ## 405 | 3.4488e-02  | 8.4827e-03 | 4.0657 4.789e-05   | *** |
| ## 406 | -1.6428e-02 | 7.9786e-03 | -2.0589 0.0394995  | *   |
| ## 407 | -6.9072e-02 | 7.4442e-03 | -9.2786 < 2.2e-16  | *** |
| ## 408 | 2.1595e-02  | 8.3986e-03 | 2.5712 0.0101338   | *   |
| ## 409 | -2.3452e-02 | 7.8918e-03 | -2.9717 0.0029620  | **  |
| ## 410 | 1.0938e-02  | 8.2658e-03 | 1.3233 0.1857333   |     |
| ## 411 | -8.8422e-03 | 8.1154e-03 | -1.0896 0.2759062  |     |
| ## 412 | -3.0900e-02 | 7.8731e-03 | -3.9247 8.683e-05  | *** |
| ## 413 | -9.0950e-03 | 8.0808e-03 | -1.1255 0.2603781  |     |
| ## 414 | 1.9930e-02  | 8.4316e-03 | 2.3638 0.0180896   | *   |
| ## 415 | 8.1390e-02  | 9.0118e-03 | 9.0316 < 2.2e-16   | *** |
| ## 416 | -1.8634e-02 | 7.9218e-03 | -2.3522 0.0186620  | *   |
| ## 417 | -9.1763e-02 | 7.0589e-03 | -12.9996 < 2.2e-16 | *** |
| ## 418 | -2.4174e-03 | 8.0198e-03 | -0.3014 0.7630898  |     |
| ## 419 | -4.1911e-02 | 7.6958e-03 | -5.4459 5.153e-08  | *** |
| ## 420 | 5.0829e-03  | 8.1809e-03 | 0.6213 0.5343940   |     |
| ## 421 | 7.1151e-02  | 8.9596e-03 | 7.9413 2.000e-15   | *** |
| ## 422 | -2.9847e-02 | 7.7556e-03 | -3.8484 0.0001189  | *** |
| ## 423 | 2.3218e-02  | 8.3770e-03 | 2.7717 0.0055768   | **  |
| ## 424 | -1.0853e-02 | 7.9984e-03 | -1.3569 0.1748153  |     |
| ## 425 | 6.8766e-03  | 8.1526e-03 | 0.8435 0.3989558   |     |
| ## 426 | 1.8706e-02  | 8.2656e-03 | 2.2632 0.0236253   | *   |
| ## 427 | 3.0338e-02  | 8.4456e-03 | 3.5922 0.0003279   | *** |
| ## 428 | -1.4628e-02 | 7.9594e-03 | -1.8379 0.0660790  | .   |
| ## 429 | 5.4628e-03  | 8.2330e-03 | 0.6635 0.5069932   |     |
| ## 430 | -1.7405e-02 | 8.0179e-03 | -2.1708 0.0299470  | *   |
| ## 431 | -2.2563e-02 | 7.8543e-03 | -2.8727 0.0040702  | **  |
| ## 432 | -6.4061e-02 | 7.4592e-03 | -8.5882 < 2.2e-16  | *** |
| ## 433 | 4.7435e-02  | 8.6963e-03 | 5.4547 4.906e-08   | *** |
| ## 434 | 2.4007e-02  | 8.3632e-03 | 2.8706 0.0040975   | **  |
| ## 435 | 2.1026e-02  | 8.3734e-03 | 2.5111 0.0120372   | *   |
| ## 436 | 7.6956e-03  | 8.2012e-03 | 0.9383 0.3480677   |     |
| ## 437 | 3.5930e-02  | 8.5409e-03 | 4.2068 2.590e-05   | *** |
| ## 438 | 5.3802e-02  | 8.7138e-03 | 6.1744 6.643e-10   | *** |
| ## 439 | 8.2819e-03  | 8.2315e-03 | 1.0061 0.3143565   |     |
| ## 440 | 8.2009e-02  | 9.0464e-03 | 9.0653 < 2.2e-16   | *** |
| ## 441 | 1.8563e-02  | 8.3375e-03 | 2.2264 0.0259881   | *   |
| ## 442 | 1.4244e-03  | 8.1257e-03 | 0.1753 0.8608485   |     |
| ## 443 | 2.3567e-02  | 8.4026e-03 | 2.8047 0.0050358   | **  |
| ## 444 | -3.5488e-02 | 7.7645e-03 | -4.5705 4.865e-06  | *** |
| ## 445 | 3.4535e-02  | 8.5480e-03 | 4.0402 5.341e-05   | *** |
| ## 446 | -6.7882e-02 | 7.4767e-03 | -9.0792 < 2.2e-16  | *** |
| ## 447 | -2.0883e-02 | 7.9688e-03 | -2.6206 0.0087779  | **  |
| ## 448 | 5.7588e-02  | 8.7250e-03 | 6.6004 4.102e-11   | *** |
| ## 449 | 5.0060e-02  | 8.6791e-03 | 5.7679 8.027e-09   | *** |
| ## 450 | -4.1535e-01 | 4.2397e-03 | -97.9684 < 2.2e-16 | *** |
| ## 451 | -3.8000e-02 | 7.7541e-03 | -4.9006 9.554e-07  | *** |

|        |             |            |          |           |     |
|--------|-------------|------------|----------|-----------|-----|
| ## 452 | -3.8882e-03 | 8.0809e-03 | -0.4812  | 0.6304058 |     |
| ## 453 | 2.5109e-02  | 8.4578e-03 | 2.9687   | 0.0029907 | **  |
| ## 454 | -3.2416e-02 | 7.8356e-03 | -4.1370  | 3.519e-05 | *** |
| ## 455 | -2.0733e-02 | 7.9676e-03 | -2.6021  | 0.0092652 | **  |
| ## 456 | 3.1912e-03  | 8.2640e-03 | 0.3862   | 0.6993815 |     |
| ## 457 | -2.9205e-02 | 7.8545e-03 | -3.7183  | 0.0002006 | *** |
| ## 458 | -2.3226e-02 | 7.9111e-03 | -2.9359  | 0.0033259 | **  |
| ## 459 | -2.7727e-02 | 7.8253e-03 | -3.5433  | 0.0003952 | *** |
| ## 460 | 5.5157e-02  | 8.7979e-03 | 6.2694   | 3.625e-10 | *** |
| ## 461 | 1.9552e-02  | 8.2996e-03 | 2.3558   | 0.0184817 | *   |
| ## 462 | -7.6790e-03 | 8.0235e-03 | -0.9571  | 0.3385342 |     |
| ## 463 | -8.8082e-03 | 8.0055e-03 | -1.1003  | 0.2712144 |     |
| ## 464 | 3.3556e-02  | 8.5288e-03 | 3.9345   | 8.339e-05 | *** |
| ## 465 | -1.7481e-03 | 8.1934e-03 | -0.2134  | 0.8310497 |     |
| ## 466 | -9.3325e-03 | 7.9818e-03 | -1.1692  | 0.2423160 |     |
| ## 467 | -6.9415e-03 | 8.0765e-03 | -0.8595  | 0.3900823 |     |
| ## 468 | 5.6050e-02  | 8.7640e-03 | 6.3955   | 1.601e-10 | *** |
| ## 469 | 2.0946e-02  | 8.3786e-03 | 2.5000   | 0.0124199 | *   |
| ## 470 | -1.4219e-02 | 7.9509e-03 | -1.7883  | 0.0737196 | .   |
| ## 471 | -1.1975e-02 | 7.8068e-03 | -1.5339  | 0.1250613 |     |
| ## 472 | 4.2973e-02  | 8.6021e-03 | 4.9956   | 5.866e-07 | *** |
| ## 473 | 2.1846e-02  | 8.2699e-03 | 2.6417   | 0.0082495 | **  |
| ## 474 | -1.7355e-02 | 7.9118e-03 | -2.1936  | 0.0282641 | *   |
| ## 475 | -7.4373e-02 | 7.3629e-03 | -10.1011 | < 2.2e-16 | *** |
| ## 476 | -4.9730e-02 | 7.7149e-03 | -6.4459  | 1.149e-10 | *** |
| ## 477 | -2.9271e-02 | 7.8043e-03 | -3.7506  | 0.0001764 | *** |
| ## 478 | 1.9499e-02  | 8.3403e-03 | 2.3379   | 0.0193914 | *   |
| ## 479 | 4.7455e-02  | 8.7915e-03 | 5.3979   | 6.743e-08 | *** |
| ## 480 | -5.9989e-03 | 8.0687e-03 | -0.7435  | 0.4571939 |     |
| ## 481 | -2.0573e-02 | 7.9512e-03 | -2.5874  | 0.0096710 | **  |
| ## 482 | -3.5451e-02 | 7.8431e-03 | -4.5200  | 6.185e-06 | *** |
| ## 483 | -1.1883e-02 | 8.0485e-03 | -1.4765  | 0.1398145 |     |
| ## 484 | 3.0971e-02  | 8.5217e-03 | 3.6344   | 0.0002786 | *** |
| ## 485 | -1.1823e-01 | 6.9363e-03 | -17.0454 | < 2.2e-16 | *** |
| ## 486 | -2.8033e-02 | 7.8127e-03 | -3.5882  | 0.0003330 | *** |
| ## 487 | -2.6379e-02 | 7.8513e-03 | -3.3599  | 0.0007798 | *** |
| ## 488 | 9.6820e-02  | 9.1230e-03 | 10.6128  | < 2.2e-16 | *** |
| ## 489 | -1.0765e-02 | 8.0418e-03 | -1.3386  | 0.1807062 |     |
| ## 490 | 6.5948e-02  | 8.8982e-03 | 7.4115   | 1.249e-13 | *** |
| ## 491 | -2.3953e-02 | 7.8306e-03 | -3.0589  | 0.0022217 | **  |
| ## 492 | 6.5136e-02  | 8.7648e-03 | 7.4316   | 1.073e-13 | *** |
| ## 493 | 2.3653e-02  | 8.4133e-03 | 2.8114   | 0.0049326 | **  |
| ## 494 | 4.3452e-02  | 8.6241e-03 | 5.0384   | 4.693e-07 | *** |
| ## 495 | -4.1389e-02 | 7.7403e-03 | -5.3472  | 8.931e-08 | *** |
| ## 496 | -1.7639e-02 | 7.9601e-03 | -2.2160  | 0.0266920 | *   |
| ## 497 | 3.7261e-02  | 8.4916e-03 | 4.3880   | 1.144e-05 | *** |
| ## 498 | 1.8350e-02  | 8.2019e-03 | 2.2373   | 0.0252667 | *   |
| ## 499 | -8.6997e-02 | 7.3173e-03 | -11.8891 | < 2.2e-16 | *** |

|        |             |            |          |           |     |
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| ## 500 | -2.6506e-02 | 7.9184e-03 | -3.3474  | 0.0008159 | *** |
| ## 501 | 2.3159e-02  | 8.4158e-03 | 2.7519   | 0.0059252 | **  |
| ## 502 | 1.9872e-02  | 8.2195e-03 | 2.4177   | 0.0156189 | *   |
| ## 503 | 6.0738e-03  | 8.2491e-03 | 0.7363   | 0.4615502 |     |
| ## 504 | -6.1958e-02 | 7.5668e-03 | -8.1882  | 2.652e-16 | *** |
| ## 505 | -6.9512e-03 | 8.0529e-03 | -0.8632  | 0.3880291 |     |
| ## 506 | 4.0175e-03  | 8.2206e-03 | 0.4887   | 0.6250487 |     |
| ## 507 | -2.0854e-02 | 7.9323e-03 | -2.6290  | 0.0085628 | **  |
| ## 508 | -1.4519e-02 | 8.0846e-03 | -1.7959  | 0.0725090 | .   |
| ## 509 | -3.1316e-02 | 7.8673e-03 | -3.9805  | 6.878e-05 | *** |
| ## 510 | 1.3959e-02  | 8.3292e-03 | 1.6759   | 0.0937619 | .   |
| ## 511 | -2.4157e-03 | 8.0766e-03 | -0.2991  | 0.7648641 |     |
| ## 512 | -1.3767e-02 | 7.9889e-03 | -1.7233  | 0.0848369 | .   |
| ## 513 | -2.4490e-02 | 7.8913e-03 | -3.1035  | 0.0019128 | **  |
| ## 514 | 5.1642e-03  | 8.2995e-03 | 0.6222   | 0.5337906 |     |
| ## 515 | 2.6735e-02  | 8.4346e-03 | 3.1697   | 0.0015261 | **  |
| ## 516 | -4.2896e-03 | 8.0528e-03 | -0.5327  | 0.5942544 |     |
| ## 517 | -2.5190e-02 | 7.8524e-03 | -3.2079  | 0.0013369 | **  |
| ## 518 | -8.2821e-02 | 7.2175e-03 | -11.4750 | < 2.2e-16 | *** |
| ## 519 | 1.4419e-03  | 8.0616e-03 | 0.1789   | 0.8580492 |     |
| ## 520 | 2.7205e-02  | 8.4155e-03 | 3.2327   | 0.0012261 | **  |
| ## 521 | -1.5195e-02 | 7.9782e-03 | -1.9045  | 0.0568424 | .   |
| ## 522 | -2.4518e-02 | 7.8771e-03 | -3.1126  | 0.0018543 | **  |
| ## 523 | 2.6711e-02  | 8.4037e-03 | 3.1785   | 0.0014802 | **  |
| ## 524 | 1.9141e-02  | 8.3457e-03 | 2.2935   | 0.0218211 | *   |
| ## 525 | -6.9038e-03 | 8.0678e-03 | -0.8557  | 0.3921548 |     |
| ## 526 | -2.7649e-02 | 7.8303e-03 | -3.5310  | 0.0004139 | *** |
| ## 527 | -2.8302e-02 | 7.8863e-03 | -3.5888  | 0.0003322 | *** |
| ## 528 | -3.4531e-03 | 8.0694e-03 | -0.4279  | 0.6687041 |     |
| ## 529 | -3.4921e-02 | 7.7706e-03 | -4.4940  | 6.990e-06 | *** |
| ## 530 | -1.5051e-01 | 6.5139e-03 | -23.1062 | < 2.2e-16 | *** |
| ## 531 | -2.3697e-02 | 7.8909e-03 | -3.0031  | 0.0026729 | **  |
| ## 532 | -1.9898e-02 | 7.9292e-03 | -2.5094  | 0.0120925 | *   |
| ## 533 | -1.2645e-01 | 6.8296e-03 | -18.5152 | < 2.2e-16 | *** |
| ## 534 | 2.6837e-02  | 8.3969e-03 | 3.1961   | 0.0013932 | **  |
| ## 535 | -2.5845e-02 | 7.7936e-03 | -3.3162  | 0.0009125 | *** |
| ## 536 | -6.8906e-02 | 7.3967e-03 | -9.3157  | < 2.2e-16 | *** |
| ## 537 | 3.2188e-02  | 8.5191e-03 | 3.7783   | 0.0001579 | *** |
| ## 538 | -2.0782e-02 | 7.9143e-03 | -2.6259  | 0.0086422 | **  |
| ## 539 | 2.8058e-04  | 8.1606e-03 | 0.0344   | 0.9725724 |     |
| ## 540 | -8.6348e-03 | 8.1239e-03 | -1.0629  | 0.2878347 |     |
| ## 541 | -5.0870e-02 | 7.6725e-03 | -6.6302  | 3.352e-11 | *** |
| ## 542 | 1.5230e-02  | 8.3122e-03 | 1.8322   | 0.0669186 | .   |
| ## 543 | -4.8522e-02 | 7.3935e-03 | -6.5628  | 5.281e-11 | *** |
| ## 544 | 6.6395e-03  | 8.2453e-03 | 0.8052   | 0.4206783 |     |
| ## 545 | 4.5538e-02  | 8.5709e-03 | 5.3131   | 1.078e-07 | *** |
| ## 546 | -3.2923e-03 | 8.0685e-03 | -0.4080  | 0.6832378 |     |
| ## 547 | -4.4048e-02 | 7.6356e-03 | -5.7689  | 7.981e-09 | *** |

|        |             |            |          |           |     |
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| ## 548 | 4.6542e-02  | 8.6639e-03 | 5.3720   | 7.787e-08 | *** |
| ## 549 | 1.7364e-02  | 8.3535e-03 | 2.0786   | 0.0376540 | *   |
| ## 550 | 1.3010e-02  | 8.2231e-03 | 1.5821   | 0.1136267 |     |
| ## 551 | 9.4774e-03  | 8.2089e-03 | 1.1545   | 0.2482845 |     |
| ## 552 | 2.5073e-02  | 8.4075e-03 | 2.9823   | 0.0028611 | **  |
| ## 553 | -6.7165e-03 | 8.0615e-03 | -0.8332  | 0.4047585 |     |
| ## 554 | -8.8880e-02 | 7.2308e-03 | -12.2919 | < 2.2e-16 | *** |
| ## 555 | 3.0108e-02  | 8.4737e-03 | 3.5532   | 0.0003806 | *** |
| ## 556 | 3.4769e-02  | 8.5036e-03 | 4.0887   | 4.339e-05 | *** |
| ## 557 | -3.8790e-03 | 8.0774e-03 | -0.4802  | 0.6310710 |     |
| ## 558 | 8.6728e-02  | 9.1376e-03 | 9.4913   | < 2.2e-16 | *** |
| ## 559 | 1.8967e-02  | 8.3534e-03 | 2.2705   | 0.0231750 | *   |
| ## 560 | -4.9253e-03 | 8.1159e-03 | -0.6069  | 0.5439393 |     |
| ## 561 | 1.5429e-02  | 8.3080e-03 | 1.8571   | 0.0632970 | .   |
| ## 562 | -9.5534e-04 | 8.1741e-03 | -0.1169  | 0.9069598 |     |
| ## 563 | -2.1502e-02 | 7.8586e-03 | -2.7362  | 0.0062159 | **  |
| ## 564 | 2.6212e-02  | 8.3234e-03 | 3.1492   | 0.0016371 | **  |
| ## 565 | 6.3353e-02  | 8.7786e-03 | 7.2168   | 5.322e-13 | *** |
| ## 566 | -2.1849e-03 | 8.0215e-03 | -0.2724  | 0.7853280 |     |
| ## 567 | -2.5027e-02 | 7.8913e-03 | -3.1715  | 0.0015167 | **  |
| ## 568 | 2.6426e-02  | 8.3879e-03 | 3.1505   | 0.0016297 | **  |
| ## 569 | 4.5160e-02  | 8.6062e-03 | 5.2473   | 1.543e-07 | *** |
| ## 570 | 2.6847e-02  | 8.4241e-03 | 3.1870   | 0.0014376 | **  |
| ## 571 | -6.1816e-03 | 8.0246e-03 | -0.7703  | 0.4411060 |     |
| ## 572 | -4.6905e-02 | 7.6785e-03 | -6.1086  | 1.005e-09 | *** |
| ## 573 | -2.7646e-02 | 7.7308e-03 | -3.5761  | 0.0003488 | *** |
| ## 574 | -4.3051e-02 | 7.6310e-03 | -5.6416  | 1.685e-08 | *** |
| ## 575 | 2.4142e-02  | 8.4057e-03 | 2.8721   | 0.0040778 | **  |
| ## 576 | 5.0447e-03  | 8.1769e-03 | 0.6169   | 0.5372746 |     |
| ## 577 | 3.4403e-02  | 8.5162e-03 | 4.0397   | 5.352e-05 | *** |
| ## 578 | 4.4376e-02  | 8.6107e-03 | 5.1536   | 2.556e-07 | *** |
| ## 579 | -2.7494e-02 | 7.8175e-03 | -3.5170  | 0.0004365 | *** |
| ## 580 | -2.0813e-02 | 7.9954e-03 | -2.6031  | 0.0092380 | **  |
| ## 581 | -7.2453e-02 | 7.4062e-03 | -9.7828  | < 2.2e-16 | *** |
| ## 582 | -5.5707e-02 | 7.5936e-03 | -7.3361  | 2.199e-13 | *** |
| ## 583 | -4.1752e-02 | 7.7636e-03 | -5.3779  | 7.536e-08 | *** |
| ## 584 | -1.6953e-03 | 8.1579e-03 | -0.2078  | 0.8353745 |     |
| ## 585 | 5.8744e-04  | 8.1953e-03 | 0.0717   | 0.9428566 |     |
| ## 586 | -1.1124e-02 | 8.0103e-03 | -1.3887  | 0.1649200 |     |
| ## 587 | -1.5311e-02 | 7.8905e-03 | -1.9404  | 0.0523252 | .   |
| ## 588 | 6.8224e-02  | 8.9076e-03 | 7.6591   | 1.872e-14 | *** |
| ## 589 | 8.2873e-03  | 8.1944e-03 | 1.0113   | 0.3118595 |     |
| ## 590 | -6.2948e-03 | 8.0693e-03 | -0.7801  | 0.4353395 |     |
| ## 591 | -2.8348e-03 | 8.1365e-03 | -0.3484  | 0.7275401 |     |
| ## 592 | 8.9653e-02  | 9.1310e-03 | 9.8185   | < 2.2e-16 | *** |
| ## 593 | 4.7775e-02  | 8.6466e-03 | 5.5253   | 3.289e-08 | *** |
| ## 594 | -6.7482e-03 | 8.0573e-03 | -0.8375  | 0.4022952 |     |
| ## 595 | -1.8993e-04 | 8.0820e-03 | -0.0235  | 0.9812510 |     |

|        |             |            |                    |     |
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| ## 596 | -7.5761e-02 | 7.5058e-03 | -10.0936 < 2.2e-16 | *** |
| ## 597 | -3.8378e-03 | 7.8130e-03 | -0.4912 0.6232766  |     |
| ## 598 | -2.1503e-02 | 7.9093e-03 | -2.7188 0.0065529  | **  |
| ## 599 | -1.7969e-02 | 7.8903e-03 | -2.2774 0.0227651  | *   |
| ## 600 | -5.6993e-02 | 7.5955e-03 | -7.5035 6.216e-14  | *** |
| ## 601 | 2.1139e-02  | 8.4137e-03 | 2.5124 0.0119898   | *   |
| ## 602 | 8.1508e-02  | 9.0113e-03 | 9.0451 < 2.2e-16   | *** |
| ## 603 | -3.1888e-02 | 7.8404e-03 | -4.0671 4.760e-05  | *** |
| ## 604 | -1.6033e-03 | 8.1272e-03 | -0.1973 0.8436122  |     |
| ## 605 | -1.8734e-02 | 7.9729e-03 | -2.3497 0.0187893  | *   |
| ## 606 | -4.3987e-02 | 7.7009e-03 | -5.7120 1.117e-08  | *** |
| ## 607 | -5.7962e-03 | 8.0826e-03 | -0.7171 0.4732998  |     |
| ## 608 | -6.2168e-02 | 7.4490e-03 | -8.3458 < 2.2e-16  | *** |
| ## 609 | -3.2454e-02 | 7.8063e-03 | -4.1574 3.219e-05  | *** |
| ## 610 | -1.5873e-02 | 7.9198e-03 | -2.0042 0.0450505  | *   |
| ## 611 | 6.1134e-02  | 8.8676e-03 | 6.8940 5.425e-12   | *** |
| ## 612 | -1.0752e-02 | 8.0492e-03 | -1.3358 0.1816265  |     |
| ## 613 | -6.1325e-04 | 8.0889e-03 | -0.0758 0.9395679  |     |
| ## 614 | -2.0327e-04 | 8.0995e-03 | -0.0251 0.9799781  |     |
| ## 615 | 2.1352e-02  | 8.3891e-03 | 2.5452 0.0109217   | *   |
| ## 616 | -5.2464e-02 | 7.6244e-03 | -6.8811 5.939e-12  | *** |
| ## 617 | 3.7019e-02  | 8.4869e-03 | 4.3619 1.290e-05   | *** |
| ## 618 | -2.2933e-02 | 7.8952e-03 | -2.9047 0.0036760  | **  |
| ## 619 | -5.6284e-07 | 8.1025e-03 | -0.0001 0.9999446  |     |
| ## 620 | 1.0444e-01  | 9.2251e-03 | 11.3217 < 2.2e-16  | *** |
| ## 621 | -8.6989e-03 | 8.0271e-03 | -1.0837 0.2785009  |     |
| ## 622 | -2.2062e-02 | 7.9249e-03 | -2.7839 0.0053715  | **  |
| ## 623 | -4.5573e-03 | 8.0645e-03 | -0.5651 0.5720028  |     |
| ## 624 | -6.6932e-02 | 7.4667e-03 | -8.9640 < 2.2e-16  | *** |
| ## 625 | 2.9147e-02  | 8.4303e-03 | 3.4573 0.0005455   | *** |
| ## 626 | -1.3981e-02 | 7.9802e-03 | -1.7519 0.0797891  | .   |
| ## 627 | 5.5444e-02  | 8.7142e-03 | 6.3624 1.986e-10   | *** |
| ## 628 | -2.9761e-03 | 8.0597e-03 | -0.3693 0.7119364  |     |
| ## 629 | -8.3909e-03 | 8.0954e-03 | -1.0365 0.2999622  |     |
| ## 630 | 3.1665e-03  | 8.4324e-03 | 0.3755 0.7072744   |     |
| ## 631 | 2.4072e-02  | 8.4024e-03 | 2.8649 0.0041709   | **  |
| ## 632 | -5.2804e-02 | 7.5930e-03 | -6.9544 3.541e-12  | *** |
| ## 633 | -3.4159e-02 | 7.7638e-03 | -4.3998 1.083e-05  | *** |
| ## 634 | -4.8573e-02 | 7.6324e-03 | -6.3640 1.965e-10  | *** |
| ## 635 | -2.7730e-02 | 7.8197e-03 | -3.5462 0.0003909  | *** |
| ## 636 | 2.5586e-02  | 8.4377e-03 | 3.0324 0.0024265   | **  |
| ## 637 | 2.6972e-02  | 8.2749e-03 | 3.2595 0.0011159   | **  |
| ## 638 | -1.9614e-02 | 8.0078e-03 | -2.4494 0.0143089  | *   |
| ## 639 | -3.6844e-03 | 7.9835e-03 | -0.4615 0.6444389  |     |
| ## 640 | -3.8180e-02 | 7.7693e-03 | -4.9143 8.911e-07  | *** |
| ## 641 | -2.0403e-02 | 7.9111e-03 | -2.5790 0.0099077  | **  |
| ## 642 | 3.0437e-02  | 8.4473e-03 | 3.6031 0.0003144   | *** |
| ## 643 | -4.1061e-02 | 8.0624e-03 | -5.0929 3.526e-07  | *** |



|        |             |            |          |           |     |
|--------|-------------|------------|----------|-----------|-----|
| ## 644 | -3.9510e-02 | 7.6427e-03 | -5.1697  | 2.344e-07 | *** |
| ## 645 | -2.7610e-02 | 7.8576e-03 | -3.5138  | 0.0004417 | *** |
| ## 646 | -9.5172e-04 | 8.1214e-03 | -0.1172  | 0.9067126 |     |
| ## 647 | -2.0924e-02 | 7.9422e-03 | -2.6345  | 0.0084261 | **  |
| ## 648 | -6.3030e-02 | 7.6556e-03 | -8.2332  | < 2.2e-16 | *** |
| ## 649 | 4.5034e-02  | 8.6354e-03 | 5.2150   | 1.838e-07 | *** |
| ## 650 | -2.5347e-02 | 7.9247e-03 | -3.1985  | 0.0013815 | **  |
| ## 651 | -1.8782e-02 | 7.8914e-03 | -2.3801  | 0.0173090 | *   |
| ## 652 | -1.1703e-02 | 8.0157e-03 | -1.4600  | 0.1442875 |     |
| ## 653 | 5.2580e-02  | 8.6449e-03 | 6.0822   | 1.186e-09 | *** |
| ## 654 | 2.5542e-03  | 8.1946e-03 | 0.3117   | 0.7552781 |     |
| ## 655 | 5.9007e-03  | 8.2279e-03 | 0.7172   | 0.4732783 |     |
| ## 656 | -3.0987e-02 | 7.6574e-03 | -4.0467  | 5.195e-05 | *** |
| ## 657 | 1.1061e-02  | 8.2203e-03 | 1.3456   | 0.1784347 |     |
| ## 658 | 2.3767e-02  | 8.3075e-03 | 2.8609   | 0.0042240 | **  |
| ## 659 | -8.8246e-03 | 8.2842e-03 | -1.0652  | 0.2867700 |     |
| ## 660 | 1.2795e-02  | 8.3245e-03 | 1.5370   | 0.1242857 |     |
| ## 661 | 3.3895e-02  | 8.4809e-03 | 3.9966   | 6.426e-05 | *** |
| ## 662 | 1.8140e-02  | 8.2898e-03 | 2.1882   | 0.0286519 | *   |
| ## 663 | 3.6847e-02  | 8.5387e-03 | 4.3152   | 1.594e-05 | *** |
| ## 664 | -4.1141e-02 | 7.8148e-03 | -5.2645  | 1.406e-07 | *** |
| ## 665 | -4.0109e-03 | 8.0464e-03 | -0.4985  | 0.6181530 |     |
| ## 666 | -1.1559e-02 | 8.0372e-03 | -1.4382  | 0.1503787 |     |
| ## 667 | -7.0276e-02 | 7.3946e-03 | -9.5037  | < 2.2e-16 | *** |
| ## 668 | -3.4693e-02 | 8.0194e-03 | -4.3261  | 1.518e-05 | *** |
| ## 669 | 5.1224e-02  | 8.6526e-03 | 5.9200   | 3.220e-09 | *** |
| ## 670 | 1.7093e-02  | 8.4811e-03 | 2.0154   | 0.0438594 | *   |
| ## 671 | -1.2121e-02 | 7.9876e-03 | -1.5175  | 0.1291486 |     |
| ## 672 | -4.0155e-02 | 7.7408e-03 | -5.1875  | 2.131e-07 | *** |
| ## 673 | -7.5239e-02 | 7.3965e-03 | -10.1721 | < 2.2e-16 | *** |
| ## 674 | 2.4114e-02  | 8.3591e-03 | 2.8847   | 0.0039174 | **  |
| ## 675 | 3.8752e-03  | 8.1239e-03 | 0.4770   | 0.6333502 |     |
| ## 676 | 4.0657e-02  | 8.5921e-03 | 4.7319   | 2.224e-06 | *** |
| ## 677 | 9.9607e-03  | 8.2284e-03 | 1.2105   | 0.2260757 |     |
| ## 678 | -6.4939e-02 | 7.5100e-03 | -8.6470  | < 2.2e-16 | *** |
| ## 679 | 5.5579e-02  | 8.7607e-03 | 6.3441   | 2.238e-10 | *** |
| ## 680 | 6.3988e-02  | 8.8025e-03 | 7.2693   | 3.614e-13 | *** |
| ## 681 | 1.6020e-02  | 8.3108e-03 | 1.9276   | 0.0539089 | .   |
| ## 682 | -5.4613e-02 | 7.5534e-03 | -7.2303  | 4.819e-13 | *** |
| ## 683 | -9.0584e-03 | 8.0132e-03 | -1.1304  | 0.2582943 |     |
| ## 684 | 4.4213e-03  | 8.2046e-03 | 0.5389   | 0.5899731 |     |
| ## 685 | -8.1689e-06 | 8.0356e-03 | -0.0010  | 0.9991889 |     |
| ## 686 | 3.2661e-02  | 8.4182e-03 | 3.8798   | 0.0001045 | *** |
| ## 687 | -3.6338e-02 | 7.7808e-03 | -4.6702  | 3.009e-06 | *** |
| ## 688 | -1.9020e-02 | 7.9296e-03 | -2.3986  | 0.0164584 | *   |
| ## 689 | -7.6244e-03 | 8.0921e-03 | -0.9422  | 0.3460905 |     |
| ## 690 | 9.8189e-04  | 8.1730e-03 | 0.1201   | 0.9043738 |     |
| ## 691 | 2.5049e-02  | 8.3754e-03 | 2.9908   | 0.0027829 | **  |

|        |             |            |         |           |     |
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| ## 692 | -9.6207e-03 | 8.0237e-03 | -1.1990 | 0.2305099 |     |
| ## 693 | 3.8858e-02  | 8.5983e-03 | 4.5193  | 6.206e-06 | *** |
| ## 694 | -2.0003e-02 | 7.9028e-03 | -2.5311 | 0.0113711 | *   |
| ## 695 | 1.9517e-02  | 8.3273e-03 | 2.3437  | 0.0190914 | *   |
| ## 696 | 1.1904e-02  | 8.2839e-03 | 1.4370  | 0.1507051 |     |
| ## 697 | 2.8343e-02  | 8.4714e-03 | 3.3457  | 0.0008207 | *** |
| ## 698 | -3.2554e-02 | 7.7880e-03 | -4.1800 | 2.916e-05 | *** |
| ## 699 | 3.7675e-02  | 8.5553e-03 | 4.4037  | 1.064e-05 | *** |
| ## 700 | 6.5197e-02  | 8.7847e-03 | 7.4217  | 1.156e-13 | *** |
| ## 701 | 4.4144e-02  | 8.5907e-03 | 5.1385  | 2.770e-07 | *** |
| ## 702 | 1.9468e-02  | 8.2942e-03 | 2.3472  | 0.0189164 | *   |
| ## 703 | -8.0979e-03 | 8.0506e-03 | -1.0059 | 0.3144765 |     |
| ## 704 | -8.4672e-03 | 8.0145e-03 | -1.0565 | 0.2907503 |     |
| ## 705 | -4.3582e-02 | 7.6616e-03 | -5.6884 | 1.283e-08 | *** |
| ## 706 | 7.5753e-02  | 8.8370e-03 | 8.5723  | < 2.2e-16 | *** |
| ## 707 | 1.1041e-02  | 8.1765e-03 | 1.3504  | 0.1768912 |     |
| ## 708 | 2.0361e-02  | 8.3903e-03 | 2.4267  | 0.0152375 | *   |
| ## 709 | 6.7465e-03  | 8.1752e-03 | 0.8252  | 0.4092407 |     |
| ## 710 | -5.3370e-02 | 7.6041e-03 | -7.0185 | 2.243e-12 | *** |
| ## 711 | -3.4087e-04 | 8.1282e-03 | -0.0419 | 0.9665493 |     |
| ## 712 | -3.2856e-02 | 7.8616e-03 | -4.1793 | 2.924e-05 | *** |
| ## 713 | -7.0843e-03 | 8.0452e-03 | -0.8806 | 0.3785563 |     |
| ## 714 | 6.3774e-02  | 8.8802e-03 | 7.1816  | 6.892e-13 | *** |
| ## 715 | -9.2867e-03 | 8.0583e-03 | -1.1524 | 0.2491416 |     |
| ## 716 | -2.0647e-02 | 7.9496e-03 | -2.5973 | 0.0093966 | **  |
| ## 717 | 3.6630e-02  | 8.5420e-03 | 4.2882  | 1.801e-05 | *** |
| ## 718 | 1.8109e-02  | 8.2713e-03 | 2.1894  | 0.0285678 | *   |
| ## 719 | -3.5226e-02 | 7.7527e-03 | -4.5437 | 5.526e-06 | *** |
| ## 720 | 3.0581e-02  | 8.4706e-03 | 3.6102  | 0.0003059 | *** |
| ## 721 | -2.4256e-02 | 7.9168e-03 | -3.0639 | 0.0021848 | **  |
| ## 722 | 2.1023e-02  | 8.3471e-03 | 2.5186  | 0.0117833 | *   |
| ## 723 | 8.9465e-03  | 8.1902e-03 | 1.0923  | 0.2746824 |     |
| ## 724 | -7.8151e-03 | 8.0241e-03 | -0.9740 | 0.3300783 |     |
| ## 725 | 1.2366e-02  | 8.2135e-03 | 1.5056  | 0.1321614 |     |
| ## 726 | 7.6904e-03  | 8.2007e-03 | 0.9378  | 0.3483586 |     |
| ## 727 | -2.9907e-02 | 7.7652e-03 | -3.8514 | 0.0001174 | *** |
| ## 728 | 3.7703e-02  | 8.4856e-03 | 4.4432  | 8.861e-06 | *** |
| ## 729 | 4.0183e-02  | 8.4677e-03 | 4.7455  | 2.080e-06 | *** |
| ## 730 | -4.5517e-02 | 7.6906e-03 | -5.9184 | 3.250e-09 | *** |
| ## 731 | -4.3471e-02 | 7.5989e-03 | -5.7207 | 1.061e-08 | *** |
| ## 732 | 3.1784e-03  | 8.1890e-03 | 0.3881  | 0.6979229 |     |
| ## 733 | -2.0602e-02 | 7.7667e-03 | -2.6526 | 0.0079871 | **  |
| ## 734 | -6.3758e-03 | 8.1371e-03 | -0.7836 | 0.4333019 |     |
| ## 735 | 1.1151e-02  | 8.2657e-03 | 1.3491  | 0.1773111 |     |
| ## 736 | 2.0961e-02  | 8.3452e-03 | 2.5118  | 0.0120125 | *   |
| ## 737 | 7.9514e-03  | 8.2251e-03 | 0.9667  | 0.3336786 |     |
| ## 738 | -4.0306e-03 | 8.3744e-03 | -0.4813 | 0.6303032 |     |
| ## 739 | 2.8767e-02  | 8.4953e-03 | 3.3862  | 0.0007086 | *** |

|        |             |            |          |           |     |
|--------|-------------|------------|----------|-----------|-----|
| ## 740 | -9.1363e-03 | 8.0801e-03 | -1.1307  | 0.2581752 |     |
| ## 741 | -5.5078e-02 | 7.5940e-03 | -7.2528  | 4.083e-13 | *** |
| ## 742 | -1.7771e-02 | 7.9309e-03 | -2.2408  | 0.0250400 | *   |
| ## 743 | 3.2811e-03  | 8.2314e-03 | 0.3986   | 0.6901857 |     |
| ## 744 | 1.8169e-02  | 8.4251e-03 | 2.1566   | 0.0310384 | *   |
| ## 745 | 3.0787e-02  | 8.4559e-03 | 3.6409   | 0.0002717 | *** |
| ## 746 | -2.4306e-02 | 7.9414e-03 | -3.0607  | 0.0022084 | **  |
| ## 747 | -3.6677e-02 | 7.7291e-03 | -4.7452  | 2.083e-06 | *** |
| ## 748 | 2.3478e-02  | 8.4085e-03 | 2.7922   | 0.0052350 | **  |
| ## 749 | 5.6426e-02  | 8.7141e-03 | 6.4753   | 9.463e-11 | *** |
| ## 750 | -2.4554e-02 | 7.8767e-03 | -3.1173  | 0.0018253 | **  |
| ## 751 | -1.9220e-02 | 7.8581e-03 | -2.4459  | 0.0144500 | *   |
| ## 752 | 6.1740e-03  | 8.1657e-03 | 0.7561   | 0.4495935 |     |
| ## 753 | 4.1021e-02  | 8.6034e-03 | 4.7680   | 1.860e-06 | *** |
| ## 754 | -2.0116e-02 | 7.9820e-03 | -2.5202  | 0.0117297 | *   |
| ## 755 | 4.2827e-02  | 8.6010e-03 | 4.9794   | 6.380e-07 | *** |
| ## 756 | 9.1962e-02  | 9.1623e-03 | 10.0370  | < 2.2e-16 | *** |
| ## 757 | 4.3158e-02  | 8.6305e-03 | 5.0006   | 5.715e-07 | *** |
| ## 758 | -5.5422e-02 | 7.6522e-03 | -7.2426  | 4.401e-13 | *** |
| ## 759 | 5.1318e-02  | 8.7340e-03 | 5.8757   | 4.212e-09 | *** |
| ## 760 | 2.6212e-03  | 8.0832e-03 | 0.3243   | 0.7457242 |     |
| ## 761 | 1.2768e-03  | 8.1691e-03 | 0.1563   | 0.8757959 |     |
| ## 762 | -1.0112e-02 | 8.0792e-03 | -1.2516  | 0.2107314 |     |
| ## 763 | -1.0496e-02 | 8.0519e-03 | -1.3036  | 0.1923824 |     |
| ## 764 | -4.4604e-03 | 8.1111e-03 | -0.5499  | 0.5823774 |     |
| ## 765 | -8.4351e-02 | 7.3505e-03 | -11.4756 | < 2.2e-16 | *** |
| ## 766 | 4.3821e-03  | 8.1071e-03 | 0.5405   | 0.5888365 |     |
| ## 767 | -3.1796e-02 | 7.8501e-03 | -4.0504  | 5.113e-05 | *** |
| ## 768 | 4.1297e-03  | 8.2321e-03 | 0.5017   | 0.6159077 |     |
| ## 769 | 3.6965e-04  | 8.0932e-03 | 0.0457   | 0.9635702 |     |
| ## 770 | 2.1801e-02  | 8.3671e-03 | 2.6056   | 0.0091716 | **  |
| ## 771 | -1.2558e-03 | 8.1096e-03 | -0.1549  | 0.8769360 |     |
| ## 772 | -2.8750e-02 | 7.8418e-03 | -3.6663  | 0.0002461 | *** |
| ## 773 | 5.3208e-03  | 8.1646e-03 | 0.6517   | 0.5146007 |     |
| ## 774 | -4.9454e-02 | 7.5719e-03 | -6.5313  | 6.520e-11 | *** |
| ## 775 | 1.3862e-02  | 8.2933e-03 | 1.6715   | 0.0946288 | .   |
| ## 776 | -3.0663e-02 | 7.8456e-03 | -3.9083  | 9.294e-05 | *** |
| ## 777 | -1.1065e-01 | 6.9712e-03 | -15.8722 | < 2.2e-16 | *** |
| ## 778 | 1.3133e-02  | 8.2272e-03 | 1.5963   | 0.1104206 |     |
| ## 779 | -4.5906e-03 | 8.1340e-03 | -0.5644  | 0.5725030 |     |
| ## 780 | -1.4948e-02 | 7.9872e-03 | -1.8716  | 0.0612684 | .   |
| ## 781 | -2.5582e-02 | 7.8059e-03 | -3.2772  | 0.0010483 | **  |
| ## 782 | -8.7579e-02 | 7.2934e-03 | -12.0080 | < 2.2e-16 | *** |
| ## 783 | -3.7933e-02 | 7.7060e-03 | -4.9225  | 8.546e-07 | *** |
| ## 784 | -1.8129e-02 | 7.9549e-03 | -2.2790  | 0.0226669 | *   |
| ## 785 | -6.5394e-02 | 7.5108e-03 | -8.7067  | < 2.2e-16 | *** |
| ## 786 | 6.4375e-02  | 8.7978e-03 | 7.3172   | 2.533e-13 | *** |
| ## 787 | -6.7533e-02 | 7.5219e-03 | -8.9781  | < 2.2e-16 | *** |

|        |             |            |          |           |     |
|--------|-------------|------------|----------|-----------|-----|
| ## 788 | 5.6879e-02  | 8.7523e-03 | 6.4988   | 8.097e-11 | *** |
| ## 789 | -5.3868e-02 | 7.5877e-03 | -7.0994  | 1.253e-12 | *** |
| ## 790 | -7.9452e-02 | 7.3944e-03 | -10.7448 | < 2.2e-16 | *** |
| ## 791 | -5.2818e-02 | 7.5948e-03 | -6.9544  | 3.539e-12 | *** |
| ## 792 | -4.2171e-03 | 8.0416e-03 | -0.5244  | 0.5999919 |     |
| ## 793 | 1.9142e-02  | 8.2231e-03 | 2.3279   | 0.0199196 | *   |
| ## 794 | -1.6763e-02 | 7.9158e-03 | -2.1176  | 0.0342066 | *   |
| ## 795 | 1.3510e-02  | 8.1988e-03 | 1.6479   | 0.0993804 | .   |
| ## 796 | -4.3081e-02 | 7.6819e-03 | -5.6081  | 2.046e-08 | *** |
| ## 797 | -1.7878e-03 | 8.0882e-03 | -0.2210  | 0.8250623 |     |
| ## 798 | -6.6097e-02 | 7.4816e-03 | -8.8346  | < 2.2e-16 | *** |
| ## 799 | 7.5662e-03  | 8.2032e-03 | 0.9223   | 0.3563516 |     |
| ## 800 | 9.6718e-03  | 8.2964e-03 | 1.1658   | 0.2437015 |     |
| ## 801 | -5.1812e-02 | 7.5947e-03 | -6.8221  | 8.975e-12 | *** |
| ## 802 | 2.3989e-02  | 8.3914e-03 | 2.8587   | 0.0042535 | **  |
| ## 803 | 7.9475e-03  | 8.2457e-03 | 0.9638   | 0.3351307 |     |
| ## 804 | 3.0760e-02  | 8.4769e-03 | 3.6287   | 0.0002848 | *** |
| ## 805 | 1.4611e-02  | 8.2594e-03 | 1.7691   | 0.0768828 | .   |
| ## 806 | -1.5356e-02 | 7.9856e-03 | -1.9230  | 0.0544835 | .   |
| ## 807 | -5.9755e-03 | 8.1242e-03 | -0.7355  | 0.4620205 |     |
| ## 808 | -7.5213e-02 | 7.3442e-03 | -10.2412 | < 2.2e-16 | *** |
| ## 809 | 5.9968e-02  | 8.8042e-03 | 6.8113   | 9.669e-12 | *** |
| ## 810 | 9.1793e-03  | 8.2485e-03 | 1.1128   | 0.2657769 |     |
| ## 811 | -6.0550e-02 | 7.5019e-03 | -8.0713  | 6.953e-16 | *** |
| ## 812 | 3.9967e-02  | 8.5659e-03 | 4.6658   | 3.074e-06 | *** |
| ## 813 | 1.0422e-02  | 8.1846e-03 | 1.2733   | 0.2029037 |     |
| ## 814 | 8.4346e-03  | 8.2571e-03 | 1.0215   | 0.3070210 |     |
| ## 815 | 2.0097e-02  | 8.3733e-03 | 2.4001   | 0.0163912 | *   |
| ## 816 | 1.0476e-02  | 8.1739e-03 | 1.2817   | 0.1999578 |     |
| ## 817 | 4.3289e-03  | 8.1766e-03 | 0.5294   | 0.5965094 |     |
| ## 818 | -6.2976e-02 | 7.5286e-03 | -8.3649  | < 2.2e-16 | *** |
| ## 819 | 3.0685e-02  | 8.4772e-03 | 3.6197   | 0.0002950 | *** |
| ## 820 | -3.6380e-02 | 7.7729e-03 | -4.6803  | 2.864e-06 | *** |
| ## 821 | 2.6930e-02  | 8.3285e-03 | 3.2335   | 0.0012228 | **  |
| ## 822 | -2.5647e-02 | 7.8458e-03 | -3.2689  | 0.0010797 | **  |
| ## 823 | -3.8248e-02 | 7.7549e-03 | -4.9321  | 8.136e-07 | *** |
| ## 824 | -6.4190e-02 | 7.5154e-03 | -8.5411  | < 2.2e-16 | *** |
| ## 825 | 3.0166e-03  | 8.1199e-03 | 0.3715   | 0.7102559 |     |
| ## 826 | -1.3107e-01 | 6.6830e-03 | -19.6119 | < 2.2e-16 | *** |
| ## 827 | 3.7316e-02  | 8.4664e-03 | 4.4076   | 1.045e-05 | *** |
| ## 828 | -2.1795e-02 | 7.8898e-03 | -2.7624  | 0.0057377 | **  |
| ## 829 | 9.7283e-03  | 8.2074e-03 | 1.1853   | 0.2358972 |     |
| ## 830 | -1.6839e-02 | 7.9581e-03 | -2.1159  | 0.0343526 | *   |
| ## 831 | -9.3436e-02 | 7.1503e-03 | -13.0673 | < 2.2e-16 | *** |
| ## 832 | -7.0266e-02 | 7.3103e-03 | -9.6119  | < 2.2e-16 | *** |
| ## 833 | -1.5175e-02 | 8.0209e-03 | -1.8919  | 0.0585037 | .   |
| ## 834 | -7.6504e-02 | 7.4355e-03 | -10.2891 | < 2.2e-16 | *** |
| ## 835 | -3.4914e-02 | 7.7227e-03 | -4.5209  | 6.156e-06 | *** |

|        |             |            |                    |     |
|--------|-------------|------------|--------------------|-----|
| ## 836 | 8.5106e-02  | 9.0522e-03 | 9.4017 < 2.2e-16   | *** |
| ## 837 | 1.7306e-02  | 8.2950e-03 | 2.0863 0.0369527   | *   |
| ## 838 | -5.9638e-02 | 7.5280e-03 | -7.9222 2.334e-15  | *** |
| ## 839 | 4.0751e-02  | 8.6022e-03 | 4.7373 2.166e-06   | *** |
| ## 840 | -6.8932e-02 | 7.3585e-03 | -9.3677 < 2.2e-16  | *** |
| ## 841 | 1.8488e-02  | 8.2967e-03 | 2.2284 0.0258555   | *   |
| ## 842 | -2.3620e-02 | 7.9175e-03 | -2.9833 0.0028516  | **  |
| ## 843 | 4.5859e-03  | 8.1957e-03 | 0.5596 0.5757832   |     |
| ## 844 | -2.3013e-02 | 7.9037e-03 | -2.9116 0.0035954  | **  |
| ## 845 | 6.1088e-03  | 8.2455e-03 | 0.7409 0.4587777   |     |
| ## 846 | -3.1028e-02 | 7.8215e-03 | -3.9670 7.279e-05  | *** |
| ## 847 | -7.4723e-02 | 7.3270e-03 | -10.1983 < 2.2e-16 | *** |
| ## 848 | 5.9353e-03  | 8.1611e-03 | 0.7273 0.4670650   |     |
| ## 849 | 3.0711e-02  | 8.5241e-03 | 3.6028 0.0003148   | *** |
| ## 850 | 9.0258e-03  | 8.1496e-03 | 1.1075 0.2680742   |     |
| ## 851 | 1.0239e-01  | 9.3303e-03 | 10.9738 < 2.2e-16  | *** |
| ## 852 | 4.4683e-02  | 8.6346e-03 | 5.1749 2.280e-07   | *** |
| ## 853 | -4.4246e-04 | 8.1517e-03 | -0.0543 0.9567133  |     |
| ## 854 | -6.7973e-02 | 7.4867e-03 | -9.0791 < 2.2e-16  | *** |
| ## 855 | 1.3695e-02  | 8.2740e-03 | 1.6552 0.0978921   | .   |
| ## 856 | 2.0941e-02  | 8.3384e-03 | 2.5114 0.0120270   | *   |
| ## 857 | -6.2767e-03 | 8.0966e-03 | -0.7752 0.4382040  |     |
| ## 858 | -1.7588e-02 | 7.9783e-03 | -2.2045 0.0274922  | *   |
| ## 859 | -2.2242e-02 | 7.9322e-03 | -2.8039 0.0050481  | **  |
| ## 860 | -8.0089e-02 | 7.2671e-03 | -11.0208 < 2.2e-16 | *** |
| ## 861 | -4.0606e-02 | 7.7261e-03 | -5.2556 1.475e-07  | *** |
| ## 862 | 4.1179e-02  | 8.5205e-03 | 4.8330 1.345e-06   | *** |
| ## 863 | -1.4277e-02 | 8.0657e-03 | -1.7701 0.0767068  | .   |
| ## 864 | -6.8730e-02 | 7.3315e-03 | -9.3746 < 2.2e-16  | *** |
| ## 865 | -1.7370e-02 | 7.9667e-03 | -2.1803 0.0292320  | *   |
| ## 866 | 1.6578e-02  | 8.3474e-03 | 1.9860 0.0470287   | *   |
| ## 867 | -1.3755e-02 | 8.0149e-03 | -1.7162 0.0861187  | .   |
| ## 868 | 2.7622e-03  | 8.2245e-03 | 0.3358 0.7369886   |     |
| ## 869 | 2.0633e-02  | 8.3455e-03 | 2.4724 0.0134220   | *   |
| ## 870 | -1.1430e-02 | 7.9624e-03 | -1.4355 0.1511528  |     |
| ## 871 | -7.9174e-03 | 8.0556e-03 | -0.9828 0.3256820  |     |
| ## 872 | -2.8679e-02 | 7.8666e-03 | -3.6457 0.0002667  | *** |
| ## 873 | -2.7584e-02 | 7.7701e-03 | -3.5500 0.0003852  | *** |
| ## 874 | -1.5249e-02 | 7.9937e-03 | -1.9076 0.0564391  | .   |
| ## 875 | -3.1650e-02 | 7.8108e-03 | -4.0521 5.076e-05  | *** |
| ## 876 | -1.7118e-02 | 7.9106e-03 | -2.1640 0.0304658  | *   |
| ## 877 | -3.9952e-02 | 7.6739e-03 | -5.2063 1.927e-07  | *** |
| ## 878 | 3.9457e-02  | 8.5758e-03 | 4.6010 4.205e-06   | *** |
| ## 879 | 2.4847e-02  | 8.3740e-03 | 2.9672 0.0030057   | **  |
| ## 880 | 3.5390e-02  | 8.4912e-03 | 4.1678 3.076e-05   | *** |
| ## 881 | -3.1288e-02 | 7.8162e-03 | -4.0030 6.256e-05  | *** |
| ## 882 | -4.7388e-02 | 7.6810e-03 | -6.1694 6.854e-10  | *** |
| ## 883 | 3.3776e-02  | 8.5716e-03 | 3.9405 8.131e-05   | *** |

|        |             |            |         |           |     |
|--------|-------------|------------|---------|-----------|-----|
| ## 884 | -3.1859e-02 | 7.7875e-03 | -4.0911 | 4.294e-05 | *** |
| ## 885 | -4.0381e-02 | 7.7456e-03 | -5.2134 | 1.854e-07 | *** |
| ## 886 | -7.0058e-02 | 7.5027e-03 | -9.3377 | < 2.2e-16 | *** |
| ## 887 | -4.1079e-02 | 7.7510e-03 | -5.2998 | 1.159e-07 | *** |
| ## 888 | 1.1955e-02  | 8.2304e-03 | 1.4525  | 0.1463519 |     |
| ## 889 | 8.1847e-03  | 8.2083e-03 | 0.9971  | 0.3187055 |     |
| ## 890 | 7.0039e-02  | 8.8640e-03 | 7.9015  | 2.755e-15 | *** |
| ## 891 | -5.0754e-02 | 7.6115e-03 | -6.6681 | 2.591e-11 | *** |
| ## 892 | -2.1431e-02 | 7.9130e-03 | -2.7084 | 0.0067612 | **  |
| ## 893 | -1.7766e-02 | 7.8895e-03 | -2.2519 | 0.0243286 | *   |
| ## 894 | -6.3247e-02 | 7.3600e-03 | -8.5934 | < 2.2e-16 | *** |
| ## 895 | 3.9500e-03  | 8.1559e-03 | 0.4843  | 0.6281697 |     |
| ## 896 | 2.1830e-02  | 8.3772e-03 | 2.6059  | 0.0091631 | **  |
| ## 897 | 2.9638e-02  | 8.5342e-03 | 3.4728  | 0.0005150 | *** |
| ## 898 | 5.6523e-03  | 8.2357e-03 | 0.6863  | 0.4925149 |     |
| ## 899 | -4.8507e-02 | 7.6270e-03 | -6.3599 | 2.019e-10 | *** |
| ## 900 | 4.1966e-02  | 8.5678e-03 | 4.8981  | 9.678e-07 | *** |
| ## 901 | -1.8819e-02 | 7.9355e-03 | -2.3715 | 0.0177182 | *   |
| ## 902 | 4.4526e-02  | 8.6548e-03 | 5.1447  | 2.680e-07 | *** |
| ## 903 | 1.0901e-02  | 8.2623e-03 | 1.3194  | 0.1870335 |     |
| ## 904 | -3.6522e-02 | 7.7502e-03 | -4.7124 | 2.448e-06 | *** |
| ## 905 | -2.4891e-02 | 7.8593e-03 | -3.1670 | 0.0015400 | **  |
| ## 906 | 2.0397e-02  | 8.2748e-03 | 2.4650  | 0.0137024 | *   |
| ## 907 | 1.5010e-03  | 8.1459e-03 | 0.1843  | 0.8538075 |     |
| ## 908 | -8.2701e-03 | 8.0911e-03 | -1.0221 | 0.3067247 |     |
| ## 909 | -2.6384e-02 | 7.8675e-03 | -3.3536 | 0.0007977 | *** |
| ## 910 | -4.8799e-02 | 7.6728e-03 | -6.3600 | 2.017e-10 | *** |
| ## 911 | -7.4528e-03 | 8.0791e-03 | -0.9225 | 0.3562801 |     |
| ## 912 | 2.3211e-02  | 8.3667e-03 | 2.7742  | 0.0055335 | **  |
| ## 913 | -1.3966e-02 | 8.0085e-03 | -1.7439 | 0.0811756 | .   |
| ## 914 | 8.2948e-02  | 9.0587e-03 | 9.1567  | < 2.2e-16 | *** |
| ## 915 | -1.7978e-03 | 8.0563e-03 | -0.2232 | 0.8234145 |     |
| ## 916 | -1.3719e-02 | 8.0018e-03 | -1.7145 | 0.0864288 | .   |
| ## 917 | -1.5480e-02 | 7.9878e-03 | -1.9379 | 0.0526358 | .   |
| ## 918 | 9.3926e-02  | 9.1712e-03 | 10.2414 | < 2.2e-16 | *** |
| ## 919 | -5.3039e-02 | 7.4369e-03 | -7.1318 | 9.903e-13 | *** |
| ## 920 | 1.0312e-02  | 8.2376e-03 | 1.2518  | 0.2106539 |     |
| ## 921 | 1.3282e-02  | 8.2778e-03 | 1.6045  | 0.1085941 |     |
| ## 922 | -1.8128e-02 | 7.9467e-03 | -2.2812 | 0.0225377 | *   |
| ## 923 | -1.1547e-02 | 7.9950e-03 | -1.4443 | 0.1486548 |     |
| ## 924 | -2.7897e-02 | 7.7678e-03 | -3.5914 | 0.0003289 | *** |
| ## 925 | -9.5867e-03 | 8.0579e-03 | -1.1897 | 0.2341585 |     |
| ## 926 | 4.4155e-03  | 8.1417e-03 | 0.5423  | 0.5875871 |     |
| ## 927 | -1.5914e-02 | 8.0104e-03 | -1.9867 | 0.0469551 | *   |
| ## 928 | -2.3573e-02 | 7.8586e-03 | -2.9997 | 0.0027028 | **  |
| ## 929 | 5.1257e-02  | 8.5785e-03 | 5.9751  | 2.299e-09 | *** |
| ## 930 | -1.7846e-02 | 7.9504e-03 | -2.2446 | 0.0247910 | *   |
| ## 931 | -5.0701e-02 | 7.6536e-03 | -6.6245 | 3.484e-11 | *** |

|        |             |            |         |           |     |
|--------|-------------|------------|---------|-----------|-----|
| ## 932 | -3.9364e-03 | 8.0900e-03 | -0.4866 | 0.6265530 |     |
| ## 933 | -4.8133e-02 | 7.5940e-03 | -6.3383 | 2.324e-10 | *** |
| ## 934 | 7.6586e-02  | 8.9576e-03 | 8.5498  | < 2.2e-16 | *** |
| ## 935 | -5.3746e-02 | 7.6365e-03 | -7.0381 | 1.949e-12 | *** |
| ## 936 | -2.5836e-02 | 7.8350e-03 | -3.2975 | 0.0009755 | *** |
| ## 937 | -7.1800e-03 | 8.0365e-03 | -0.8934 | 0.3716286 |     |
| ## 938 | -2.4905e-02 | 7.8324e-03 | -3.1798 | 0.0014740 | **  |
| ## 939 | 1.6337e-03  | 8.1329e-03 | 0.2009  | 0.8407917 |     |
| ## 940 | 7.8584e-03  | 8.2101e-03 | 0.9572  | 0.3384841 |     |
| ## 941 | -2.2809e-02 | 7.9261e-03 | -2.8777 | 0.0040057 | **  |
| ## 942 | 4.7527e-03  | 8.1965e-03 | 0.5798  | 0.5620178 |     |
| ## 943 | -1.1634e-02 | 8.1133e-03 | -1.4339 | 0.1515994 |     |
| ## 944 | -4.0242e-02 | 7.7498e-03 | -5.1927 | 2.073e-07 | *** |
| ## 945 | 1.3136e-02  | 8.3941e-03 | 1.5649  | 0.1176001 |     |
| ## 946 | -2.6030e-03 | 8.0739e-03 | -0.3224 | 0.7471515 |     |
| ## 947 | 2.0377e-03  | 8.1061e-03 | 0.2514  | 0.8015238 |     |
| ## 948 | 8.2910e-03  | 8.1757e-03 | 1.0141  | 0.3105366 |     |
| ## 949 | -1.0329e-02 | 7.9953e-03 | -1.2919 | 0.1963864 |     |
| ## 950 | 1.9544e-02  | 8.3307e-03 | 2.3460  | 0.0189746 | *   |
| ## 951 | 2.1896e-03  | 8.1491e-03 | 0.2687  | 0.7881696 |     |
| ## 952 | -5.6350e-02 | 7.5489e-03 | -7.4647 | 8.347e-14 | *** |
| ## 953 | 1.4726e-02  | 8.3244e-03 | 1.7690  | 0.0768871 | .   |
| ## 954 | -4.6432e-03 | 8.0653e-03 | -0.5757 | 0.5648134 |     |
| ## 955 | -7.1287e-02 | 7.2944e-03 | -9.7728 | < 2.2e-16 | *** |
| ## 956 | 3.1250e-02  | 8.5208e-03 | 3.6675  | 0.0002449 | *** |
| ## 957 | 1.9329e-02  | 8.3166e-03 | 2.3242  | 0.0201157 | *   |
| ## 958 | 1.1469e-02  | 8.2439e-03 | 1.3912  | 0.1641737 |     |
| ## 959 | -5.2411e-02 | 7.6110e-03 | -6.8861 | 5.732e-12 | *** |
| ## 960 | 4.5222e-04  | 8.1013e-03 | 0.0558  | 0.9554850 |     |
| ## 961 | 4.7371e-02  | 8.6528e-03 | 5.4747  | 4.384e-08 | *** |
| ## 962 | -1.1440e-02 | 7.9701e-03 | -1.4354 | 0.1511768 |     |
| ## 963 | -1.0759e-02 | 7.9186e-03 | -1.3588 | 0.1742253 |     |
| ## 964 | -4.0983e-02 | 7.7396e-03 | -5.2952 | 1.189e-07 | *** |
| ## 965 | -1.6446e-02 | 8.0059e-03 | -2.0542 | 0.0399568 | *   |
| ## 966 | -3.7315e-02 | 7.6866e-03 | -4.8546 | 1.207e-06 | *** |
| ## 967 | 6.8675e-03  | 8.1888e-03 | 0.8386  | 0.4016670 |     |
| ## 968 | -2.2395e-03 | 8.0361e-03 | -0.2787 | 0.7804944 |     |
| ## 969 | -2.3567e-03 | 8.1060e-03 | -0.2907 | 0.7712596 |     |
| ## 970 | -1.2965e-02 | 8.0245e-03 | -1.6157 | 0.1061550 |     |
| ## 971 | 5.0987e-03  | 8.2031e-03 | 0.6216  | 0.5342317 |     |
| ## 972 | -3.4646e-02 | 7.7837e-03 | -4.4511 | 8.544e-06 | *** |
| ## 973 | -4.0928e-02 | 7.7471e-03 | -5.2831 | 1.270e-07 | *** |
| ## 974 | 6.4541e-02  | 8.7741e-03 | 7.3558  | 1.898e-13 | *** |
| ## 975 | -5.5255e-02 | 7.7822e-03 | -7.1002 | 1.246e-12 | *** |
| ## 976 | -6.5324e-03 | 8.0989e-03 | -0.8066 | 0.4199069 |     |
| ## 977 | -5.9162e-03 | 8.0811e-03 | -0.7321 | 0.4641045 |     |
| ## 978 | 4.9670e-02  | 8.7417e-03 | 5.6819  | 1.332e-08 | *** |
| ## 979 | 1.7792e-03  | 8.1696e-03 | 0.2178  | 0.8275962 |     |

|         |             |            |           |           |     |
|---------|-------------|------------|-----------|-----------|-----|
| ## 980  | -6.0223e-02 | 7.4830e-03 | -8.0480   | 8.414e-16 | *** |
| ## 981  | -1.5812e-02 | 7.9613e-03 | -1.9861   | 0.0470275 | *   |
| ## 982  | 2.2028e-02  | 8.2935e-03 | 2.6561    | 0.0079051 | **  |
| ## 983  | -1.8582e-02 | 7.9664e-03 | -2.3325   | 0.0196748 | *   |
| ## 984  | -7.6317e-02 | 7.3382e-03 | -10.4000  | < 2.2e-16 | *** |
| ## 985  | -2.1066e-02 | 7.9015e-03 | -2.6660   | 0.0076754 | **  |
| ## 986  | -4.0750e-02 | 7.6902e-03 | -5.2990   | 1.165e-07 | *** |
| ## 987  | 2.3706e-02  | 8.3930e-03 | 2.8245    | 0.0047358 | **  |
| ## 988  | 4.5621e-02  | 8.6020e-03 | 5.3035    | 1.136e-07 | *** |
| ## 989  | -4.7865e-02 | 7.5647e-03 | -6.3274   | 2.493e-10 | *** |
| ## 990  | -2.3109e-02 | 7.8839e-03 | -2.9312   | 0.0033765 | **  |
| ## 991  | -1.3552e-02 | 7.5996e-03 | -1.7832   | 0.0745478 | .   |
| ## 992  | 2.7895e-02  | 8.4091e-03 | 3.3172    | 0.0009093 | *** |
| ## 993  | -6.6037e-02 | 7.4164e-03 | -8.9042   | < 2.2e-16 | *** |
| ## 994  | -7.3878e-02 | 7.3583e-03 | -10.0401  | < 2.2e-16 | *** |
| ## 995  | -2.0687e-02 | 7.9362e-03 | -2.6067   | 0.0091410 | **  |
| ## 996  | -4.3676e-01 | 4.0932e-03 | -106.7034 | < 2.2e-16 | *** |
| ## 997  | -3.0138e-02 | 7.9157e-03 | -3.8073   | 0.0001405 | *** |
| ## 998  | 3.5402e-02  | 8.5021e-03 | 4.1639    | 3.129e-05 | *** |
| ## 999  | -1.0432e-02 | 8.0660e-03 | -1.2934   | 0.1958851 |     |
| ## 1000 | 2.3347e-02  | 8.3992e-03 | 2.7797    | 0.0054405 | **  |
| ## 1001 | -8.0502e-02 | 7.2328e-03 | -11.1301  | < 2.2e-16 | *** |
| ## 1002 | 3.9908e-02  | 8.5964e-03 | 4.6424    | 3.443e-06 | *** |
| ## 1003 | -3.6297e-02 | 7.7932e-03 | -4.6575   | 3.201e-06 | *** |
| ## 1004 | 3.2013e-02  | 8.4746e-03 | 3.7775    | 0.0001584 | *** |
| ## 1005 | -1.8326e-02 | 7.9587e-03 | -2.3026   | 0.0213009 | *   |
| ## 1006 | -2.8170e-02 | 7.8923e-03 | -3.5692   | 0.0003580 | *** |
| ## 1007 | 1.1172e-02  | 8.2117e-03 | 1.3605    | 0.1736603 |     |
| ## 1008 | -7.5571e-02 | 7.4652e-03 | -10.1231  | < 2.2e-16 | *** |
| ## 1009 | -1.3917e-02 | 7.9897e-03 | -1.7418   | 0.0815408 | .   |
| ## 1010 | -4.3982e-02 | 7.7181e-03 | -5.6985   | 1.209e-08 | *** |
| ## 1011 | 4.7985e-02  | 8.6861e-03 | 5.5243    | 3.307e-08 | *** |
| ## 1012 | -8.7205e-03 | 7.9054e-03 | -1.1031   | 0.2699832 |     |
| ## 1013 | -1.2259e-02 | 8.0296e-03 | -1.5268   | 0.1268228 |     |
| ## 1014 | 6.2638e-03  | 8.1802e-03 | 0.7657    | 0.4438392 |     |
| ## 1015 | 9.4766e-04  | 8.1939e-03 | 0.1157    | 0.9079260 |     |
| ## 1016 | -3.9271e-02 | 7.7857e-03 | -5.0439   | 4.561e-07 | *** |
| ## 1017 | -2.3380e-03 | 8.2344e-03 | -0.2839   | 0.7764624 |     |
| ## 1018 | 1.8834e-02  | 8.3243e-03 | 2.2626    | 0.0236631 | *   |
| ## 1019 | -4.1200e-02 | 7.6634e-03 | -5.3761   | 7.610e-08 | *** |
| ## 1020 | 2.1963e-02  | 8.4229e-03 | 2.6076    | 0.0091191 | **  |
| ## 1021 | -9.8232e-03 | 8.3134e-03 | -1.1816   | 0.2373580 |     |
| ## 1022 | -3.7523e-02 | 7.7358e-03 | -4.8506   | 1.231e-06 | *** |
| ## 1023 | 1.9895e-02  | 8.3478e-03 | 2.3832    | 0.0171617 | *   |
| ## 1024 | 2.2509e-02  | 8.3820e-03 | 2.6854    | 0.0072449 | **  |
| ## 1025 | -4.1964e-02 | 7.7002e-03 | -5.4497   | 5.047e-08 | *** |
| ## 1026 | -1.0783e-02 | 8.0006e-03 | -1.3478   | 0.1777387 |     |
| ## 1027 | -6.1309e-02 | 7.4986e-03 | -8.1760   | 2.934e-16 | *** |



|         |             |            |          |           |     |
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| ## 1028 | -8.4160e-02 | 7.2828e-03 | -11.5559 | < 2.2e-16 | *** |
| ## 1029 | 3.1962e-02  | 8.4830e-03 | 3.7678   | 0.0001647 | *** |
| ## 1030 | 3.3043e-02  | 8.4603e-03 | 3.9056   | 9.398e-05 | *** |
| ## 1031 | 6.0430e-03  | 8.1840e-03 | 0.7384   | 0.4602762 |     |
| ## 1032 | -3.1394e-02 | 7.7894e-03 | -4.0304  | 5.569e-05 | *** |
| ## 1033 | -4.3635e-02 | 7.6793e-03 | -5.6821  | 1.331e-08 | *** |
| ## 1034 | -9.7540e-03 | 8.0871e-03 | -1.2061  | 0.2277689 |     |
| ## 1035 | 3.7592e-02  | 8.5288e-03 | 4.4077   | 1.045e-05 | *** |
| ## 1036 | -7.3310e-02 | 7.4168e-03 | -9.8843  | < 2.2e-16 | *** |
| ## 1037 | -2.8466e-02 | 7.8606e-03 | -3.6213  | 0.0002931 | *** |
| ## 1038 | 5.8557e-03  | 8.1829e-03 | 0.7156   | 0.4742355 |     |
| ## 1039 | -3.5663e-02 | 7.8187e-03 | -4.5613  | 5.084e-06 | *** |
| ## 1040 | -1.8242e-02 | 7.9425e-03 | -2.2968  | 0.0216326 | *   |
| ## 1041 | 2.3707e-02  | 8.3788e-03 | 2.8294   | 0.0046631 | **  |
| ## 1042 | -3.5618e-02 | 7.7433e-03 | -4.5998  | 4.229e-06 | *** |
| ## 1043 | 6.6169e-02  | 8.9564e-03 | 7.3879   | 1.492e-13 | *** |
| ## 1044 | -2.2826e-02 | 7.9019e-03 | -2.8887  | 0.0038682 | **  |
| ## 1045 | -4.5433e-03 | 8.1335e-03 | -0.5586  | 0.5764385 |     |
| ## 1046 | -9.0413e-02 | 7.1314e-03 | -12.6782 | < 2.2e-16 | *** |
| ## 1047 | -2.2423e-02 | 7.9502e-03 | -2.8204  | 0.0047965 | **  |
| ## 1048 | 3.0023e-02  | 8.4806e-03 | 3.5402   | 0.0003998 | *** |
| ## 1049 | 2.3157e-02  | 8.3252e-03 | 2.7816   | 0.0054098 | **  |
| ## 1050 | 4.1450e-02  | 8.6353e-03 | 4.8000   | 1.586e-06 | *** |
| ## 1051 | 7.4468e-03  | 8.0656e-03 | 0.9233   | 0.3558620 |     |
| ## 1052 | -3.0915e-02 | 7.7831e-03 | -3.9721  | 7.124e-05 | *** |
| ## 1053 | 4.7389e-02  | 8.6182e-03 | 5.4988   | 3.824e-08 | *** |
| ## 1054 | -3.6106e-02 | 7.6894e-03 | -4.6956  | 2.658e-06 | *** |
| ## 1055 | -4.4439e-02 | 7.7185e-03 | -5.7575  | 8.536e-09 | *** |
| ## 1056 | -6.8309e-02 | 7.4725e-03 | -9.1413  | < 2.2e-16 | *** |
| ## 1057 | -1.3791e-02 | 8.0083e-03 | -1.7221  | 0.0850533 | .   |
| ## 1058 | -1.0354e-02 | 7.9878e-03 | -1.2963  | 0.1948866 |     |
| ## 1059 | 1.5853e-02  | 8.3411e-03 | 1.9006   | 0.0573485 | .   |
| ## 1060 | 1.4493e-02  | 8.3050e-03 | 1.7451   | 0.0809639 | .   |
| ## 1061 | -3.5147e-02 | 7.7978e-03 | -4.5073  | 6.567e-06 | *** |
| ## 1062 | -1.0349e-02 | 7.9639e-03 | -1.2995  | 0.1937712 |     |
| ## 1063 | -1.7440e-02 | 8.0587e-03 | -2.1641  | 0.0304540 | *   |
| ## 1064 | 1.9818e-02  | 8.3080e-03 | 2.3854   | 0.0170598 | *   |
| ## 1065 | -1.0859e-02 | 8.0278e-03 | -1.3527  | 0.1761563 |     |
| ## 1066 | 1.0074e-02  | 8.1623e-03 | 1.2342   | 0.2171443 |     |
| ## 1067 | -2.0540e-02 | 7.9309e-03 | -2.5898  | 0.0096025 | **  |
| ## 1068 | 3.8141e-02  | 8.5325e-03 | 4.4701   | 7.820e-06 | *** |
| ## 1069 | -2.9619e-02 | 7.7342e-03 | -3.8296  | 0.0001283 | *** |
| ## 1070 | -3.5525e-02 | 7.7899e-03 | -4.5605  | 5.104e-06 | *** |
| ## 1071 | -2.5273e-02 | 7.8544e-03 | -3.2177  | 0.0012921 | **  |
| ## 1072 | 1.1108e-02  | 8.1549e-03 | 1.3621   | 0.1731560 |     |
| ## 1073 | 1.9212e-03  | 8.2735e-03 | 0.2322   | 0.8163710 |     |
| ## 1074 | -7.7850e-02 | 7.3860e-03 | -10.5402 | < 2.2e-16 | *** |
| ## 1075 | 2.0763e-02  | 8.3610e-03 | 2.4832   | 0.0130191 | *   |

|         |             |            |          |           |     |
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| ## 1076 | -1.9884e-02 | 7.9550e-03 | -2.4995  | 0.0124357 | *   |
| ## 1077 | 1.2353e-02  | 8.3067e-03 | 1.4871   | 0.1369970 |     |
| ## 1078 | 7.5426e-02  | 8.9193e-03 | 8.4565   | < 2.2e-16 | *** |
| ## 1079 | -4.9949e-02 | 7.6602e-03 | -6.5206  | 7.003e-11 | *** |
| ## 1080 | -1.1316e-02 | 7.9874e-03 | -1.4167  | 0.1565678 |     |
| ## 1081 | 6.0204e-03  | 8.1074e-03 | 0.7426   | 0.4577402 |     |
| ## 1082 | 5.9184e-03  | 8.1801e-03 | 0.7235   | 0.4693673 |     |
| ## 1083 | -2.0894e-02 | 7.9339e-03 | -2.6335  | 0.0084517 | **  |
| ## 1084 | -3.4084e-02 | 7.7488e-03 | -4.3986  | 1.089e-05 | *** |
| ## 1085 | 4.9248e-02  | 8.6772e-03 | 5.6756   | 1.382e-08 | *** |
| ## 1086 | -4.5002e-02 | 7.6892e-03 | -5.8527  | 4.837e-09 | *** |
| ## 1087 | 2.5271e-02  | 8.3573e-03 | 3.0238   | 0.0024960 | **  |
| ## 1088 | -8.4606e-03 | 8.0119e-03 | -1.0560  | 0.2909679 |     |
| ## 1089 | -4.7826e-02 | 7.6168e-03 | -6.2791  | 3.406e-10 | *** |
| ## 1090 | -6.6616e-02 | 7.4196e-03 | -8.9784  | < 2.2e-16 | *** |
| ## 1091 | 1.5843e-02  | 8.2718e-03 | 1.9153   | 0.0554591 | .   |
| ## 1092 | -6.1190e-03 | 7.9908e-03 | -0.7657  | 0.4438278 |     |
| ## 1093 | -2.0690e-04 | 8.1727e-03 | -0.0253  | 0.9798025 |     |
| ## 1094 | 2.6821e-02  | 8.4207e-03 | 3.1851   | 0.0014469 | **  |
| ## 1095 | -4.6761e-02 | 7.6120e-03 | -6.1431  | 8.093e-10 | *** |
| ## 1096 | -1.0189e-01 | 7.0679e-03 | -14.4155 | < 2.2e-16 | *** |
| ## 1097 | -3.6489e-02 | 7.7617e-03 | -4.7012  | 2.587e-06 | *** |
| ## 1098 | -5.8955e-03 | 8.0685e-03 | -0.7307  | 0.4649749 |     |
| ## 1099 | -6.4127e-03 | 8.0966e-03 | -0.7920  | 0.4283429 |     |
| ## 1100 | -2.9227e-02 | 7.8452e-03 | -3.7255  | 0.0001949 | *** |
| ## 1101 | 9.6560e-03  | 8.2374e-03 | 1.1722   | 0.2411078 |     |
| ## 1102 | -1.3115e-02 | 7.9961e-03 | -1.6402  | 0.1009714 |     |
| ## 1103 | -2.0307e-02 | 8.1173e-03 | -2.5017  | 0.0123594 | *   |
| ## 1104 | 6.2979e-02  | 8.8382e-03 | 7.1258   | 1.035e-12 | *** |
| ## 1105 | -7.7252e-02 | 7.3313e-03 | -10.5372 | < 2.2e-16 | *** |
| ## 1106 | -1.1156e-02 | 7.9108e-03 | -1.4102  | 0.1584724 |     |
| ## 1107 | -2.3221e-03 | 8.0772e-03 | -0.2875  | 0.7737417 |     |
| ## 1108 | 1.9318e-03  | 8.1928e-03 | 0.2358   | 0.8135951 |     |
| ## 1109 | -4.4907e-02 | 7.7095e-03 | -5.8249  | 5.716e-09 | *** |
| ## 1110 | -2.6690e-02 | 7.7947e-03 | -3.4241  | 0.0006168 | *** |
| ## 1111 | 1.9852e-02  | 8.3588e-03 | 2.3750   | 0.0175482 | *   |
| ## 1112 | -6.4615e-02 | 7.5327e-03 | -8.5778  | < 2.2e-16 | *** |
| ## 1113 | 3.6030e-02  | 8.4714e-03 | 4.2531   | 2.108e-05 | *** |
| ## 1114 | -3.7490e-02 | 7.7788e-03 | -4.8195  | 1.439e-06 | *** |
| ## 1115 | -5.2846e-02 | 7.6033e-03 | -6.9504  | 3.641e-12 | *** |
| ## 1116 | -9.7894e-02 | 7.1265e-03 | -13.7366 | < 2.2e-16 | *** |
| ## 1117 | 2.7970e-02  | 8.3942e-03 | 3.3321   | 0.0008619 | *** |
| ## 1118 | 2.1142e-03  | 8.0826e-03 | 0.2616   | 0.7936458 |     |
| ## 1119 | -3.6262e-02 | 7.7392e-03 | -4.6856  | 2.792e-06 | *** |
| ## 1120 | 3.8002e-03  | 8.2184e-03 | 0.4624   | 0.6437931 |     |
| ## 1121 | 1.7593e-02  | 8.1898e-03 | 2.1481   | 0.0317036 | *   |
| ## 1122 | -4.5734e-03 | 8.0762e-03 | -0.5663  | 0.5712032 |     |
| ## 1123 | 7.5064e-02  | 8.9566e-03 | 8.3809   | < 2.2e-16 | *** |

|         |             |            |          |           |     |
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| ## 1124 | -9.7925e-02 | 7.1484e-03 | -13.6989 | < 2.2e-16 | *** |
| ## 1125 | -7.0316e-02 | 7.4703e-03 | -9.4128  | < 2.2e-16 | *** |
| ## 1126 | -3.7897e-02 | 7.7312e-03 | -4.9019  | 9.492e-07 | *** |
| ## 1127 | -2.4468e-02 | 7.8814e-03 | -3.1045  | 0.0019058 | **  |
| ## 1128 | 3.5721e-03  | 8.0731e-03 | 0.4425   | 0.6581451 |     |
| ## 1129 | -9.6274e-03 | 8.0512e-03 | -1.1958  | 0.2317832 |     |
| ## 1130 | -1.0539e-02 | 8.0330e-03 | -1.3120  | 0.1895242 |     |
| ## 1131 | -6.3309e-02 | 7.5138e-03 | -8.4256  | < 2.2e-16 | *** |
| ## 1132 | -1.2418e-02 | 7.9956e-03 | -1.5531  | 0.1203885 |     |
| ## 1133 | -2.3284e-02 | 7.9082e-03 | -2.9443  | 0.0032372 | **  |
| ## 1134 | -7.6739e-02 | 7.3464e-03 | -10.4458 | < 2.2e-16 | *** |
| ## 1135 | 4.5775e-02  | 8.6547e-03 | 5.2891   | 1.229e-07 | *** |
| ## 1136 | 2.7456e-03  | 8.2069e-03 | 0.3346   | 0.7379625 |     |
| ## 1137 | -5.3774e-02 | 7.6167e-03 | -7.0600  | 1.665e-12 | *** |
| ## 1138 | 1.3612e-03  | 8.1808e-03 | 0.1664   | 0.8678507 |     |
| ## 1139 | -2.9315e-02 | 7.8110e-03 | -3.7531  | 0.0001747 | *** |
| ## 1140 | -2.2498e-02 | 7.8368e-03 | -2.8708  | 0.0040949 | **  |
| ## 1141 | -6.1751e-03 | 8.1253e-03 | -0.7600  | 0.4472684 |     |
| ## 1142 | -3.2385e-02 | 7.7727e-03 | -4.1665  | 3.094e-05 | *** |
| ## 1143 | -1.5602e-02 | 7.9800e-03 | -1.9551  | 0.0505738 | .   |
| ## 1144 | 2.7953e-02  | 8.4047e-03 | 3.3258   | 0.0008816 | *** |
| ## 1145 | 2.5646e-02  | 8.1942e-03 | 3.1298   | 0.0017493 | **  |
| ## 1146 | -1.9328e-02 | 7.9599e-03 | -2.4281  | 0.0151767 | *   |
| ## 1147 | -3.2371e-02 | 7.8258e-03 | -4.1364  | 3.527e-05 | *** |
| ## 1148 | -1.6434e-02 | 7.9489e-03 | -2.0674  | 0.0386927 | *   |
| ## 1149 | 1.2023e-02  | 8.2419e-03 | 1.4588   | 0.1446222 |     |
| ## 1150 | -2.0088e-02 | 7.9498e-03 | -2.5269  | 0.0115076 | *   |
| ## 1151 | -1.6568e-02 | 7.8721e-03 | -2.1047  | 0.0353205 | *   |
| ## 1152 | -9.3404e-03 | 8.0102e-03 | -1.1661  | 0.2435886 |     |
| ## 1153 | 6.9339e-03  | 8.2207e-03 | 0.8435   | 0.3989670 |     |
| ## 1154 | -7.7374e-03 | 8.1086e-03 | -0.9542  | 0.3399730 |     |
| ## 1155 | -8.6812e-03 | 8.0159e-03 | -1.0830  | 0.2788108 |     |
| ## 1156 | -1.6335e-02 | 7.9717e-03 | -2.0491  | 0.0404477 | *   |
| ## 1157 | -5.0780e-03 | 8.1197e-03 | -0.6254  | 0.5317154 |     |
| ## 1158 | 3.3456e-02  | 8.5246e-03 | 3.9246   | 8.687e-05 | *** |
| ## 1159 | 1.0550e-01  | 9.2534e-03 | 11.4010  | < 2.2e-16 | *** |
| ## 1160 | 9.4161e-03  | 8.2708e-03 | 1.1385   | 0.2549211 |     |
| ## 1161 | -4.2986e-02 | 7.6133e-03 | -5.6461  | 1.641e-08 | *** |
| ## 1162 | 2.1466e-02  | 8.3824e-03 | 2.5609   | 0.0104405 | *   |
| ## 1163 | 2.8454e-02  | 8.4687e-03 | 3.3599   | 0.0007798 | *** |
| ## 1164 | -5.4017e-02 | 7.6287e-03 | -7.0807  | 1.434e-12 | *** |
| ## 1165 | -2.2112e-02 | 7.8915e-03 | -2.8021  | 0.0050778 | **  |
| ## 1166 | -5.2258e-02 | 7.6850e-03 | -6.8001  | 1.046e-11 | *** |
| ## 1167 | 8.5208e-02  | 9.1512e-03 | 9.3111   | < 2.2e-16 | *** |
| ## 1168 | 8.6755e-02  | 9.2083e-03 | 9.4213   | < 2.2e-16 | *** |
| ## 1169 | 5.0665e-02  | 8.6574e-03 | 5.8523   | 4.849e-09 | *** |
| ## 1170 | 4.2075e-02  | 8.5135e-03 | 4.9422   | 7.724e-07 | *** |
| ## 1171 | 4.1614e-02  | 8.5521e-03 | 4.8659   | 1.139e-06 | *** |

|         |             |            |          |           |     |
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| ## 1172 | 2.9157e-02  | 8.4690e-03 | 3.4428   | 0.0005758 | *** |
| ## 1173 | 3.1829e-04  | 8.1214e-03 | 0.0392   | 0.9687380 |     |
| ## 1174 | 7.0841e-02  | 8.9316e-03 | 7.9315   | 2.165e-15 | *** |
| ## 1175 | 7.2568e-03  | 8.2575e-03 | 0.8788   | 0.3795039 |     |
| ## 1176 | 8.2801e-02  | 9.0052e-03 | 9.1947   | < 2.2e-16 | *** |
| ## 1177 | 3.2155e-03  | 8.1910e-03 | 0.3926   | 0.6946456 |     |
| ## 1178 | 9.2249e-02  | 9.1939e-03 | 10.0337  | < 2.2e-16 | *** |
| ## 1179 | 1.1295e-01  | 9.2496e-03 | 12.2111  | < 2.2e-16 | *** |
| ## 1180 | 8.7520e-02  | 9.0937e-03 | 9.6242   | < 2.2e-16 | *** |
| ## 1181 | -7.4795e-04 | 8.1340e-03 | -0.0920  | 0.9267349 |     |
| ## 1182 | 1.3588e-02  | 8.3403e-03 | 1.6292   | 0.1032649 |     |
| ## 1183 | -5.1244e-03 | 8.0618e-03 | -0.6356  | 0.5250123 |     |
| ## 1184 | -1.5714e-02 | 7.9668e-03 | -1.9724  | 0.0485645 | *   |
| ## 1185 | -6.7917e-02 | 7.3924e-03 | -9.1874  | < 2.2e-16 | *** |
| ## 1186 | -8.5278e-02 | 6.9830e-03 | -12.2122 | < 2.2e-16 | *** |
| ## 1187 | 4.7178e-02  | 8.5906e-03 | 5.4918   | 3.978e-08 | *** |
| ## 1188 | 2.0935e-02  | 8.3251e-03 | 2.5147   | 0.0119121 | *   |
| ## 1189 | 8.9819e-02  | 9.0429e-03 | 9.9325   | < 2.2e-16 | *** |
| ## 1190 | -5.5859e-03 | 8.0484e-03 | -0.6940  | 0.4876574 |     |
| ## 1191 | 4.1156e-02  | 8.5027e-03 | 4.8403   | 1.296e-06 | *** |
| ## 1192 | 1.5067e-02  | 8.2436e-03 | 1.8277   | 0.0676006 | .   |
| ## 1193 | 1.1720e-02  | 8.3109e-03 | 1.4102   | 0.1584860 |     |
| ## 1194 | 4.9749e-02  | 8.5956e-03 | 5.7877   | 7.137e-09 | *** |
| ## 1195 | -8.0894e-03 | 8.0307e-03 | -1.0073  | 0.3137867 |     |
| ## 1196 | -1.7993e-02 | 8.0092e-03 | -2.2465  | 0.0246700 | *   |
| ## 1197 | 6.4810e-02  | 8.8793e-03 | 7.2989   | 2.901e-13 | *** |
| ## 1198 | 4.8078e-02  | 8.6848e-03 | 5.5358   | 3.098e-08 | *** |
| ## 1199 | -8.9122e-03 | 7.8282e-03 | -1.1385  | 0.2549204 |     |
| ## 1200 | -7.4996e-03 | 8.0739e-03 | -0.9289  | 0.3529527 |     |
| ## 1201 | -3.6554e-02 | 7.7401e-03 | -4.7226  | 2.328e-06 | *** |
| ## 1202 | -2.1862e-02 | 7.9187e-03 | -2.7608  | 0.0057666 | **  |
| ## 1203 | 4.7634e-02  | 8.6059e-03 | 5.5350   | 3.112e-08 | *** |
| ## 1204 | -1.3192e-02 | 7.9841e-03 | -1.6523  | 0.0984786 | .   |
| ## 1205 | 1.2849e-02  | 8.2697e-03 | 1.5537   | 0.1202532 |     |
| ## 1206 | 1.2172e-02  | 8.2026e-03 | 1.4839   | 0.1378253 |     |
| ## 1207 | -3.0844e-03 | 8.1464e-03 | -0.3786  | 0.7049710 |     |
| ## 1208 | -1.7285e-02 | 7.9397e-03 | -2.1771  | 0.0294748 | *   |
| ## 1209 | 1.3903e-03  | 8.1284e-03 | 0.1710   | 0.8641865 |     |
| ## 1210 | 2.0484e-02  | 8.2941e-03 | 2.4697   | 0.0135240 | *   |
| ## 1211 | 1.8637e-02  | 8.3074e-03 | 2.2434   | 0.0248703 | *   |
| ## 1212 | -5.1278e-02 | 7.6243e-03 | -6.7256  | 1.749e-11 | *** |
| ## 1213 | -3.4384e-02 | 7.7403e-03 | -4.4422  | 8.904e-06 | *** |
| ## 1214 | -3.4245e-02 | 7.7915e-03 | -4.3952  | 1.107e-05 | *** |
| ## 1215 | -1.0094e-02 | 7.9913e-03 | -1.2631  | 0.2065495 |     |
| ## 1216 | -4.1386e-02 | 7.7331e-03 | -5.3518  | 8.706e-08 | *** |
| ## 1217 | -3.9815e-02 | 7.7066e-03 | -5.1664  | 2.387e-07 | *** |
| ## 1218 | -1.4090e-02 | 7.9218e-03 | -1.7787  | 0.0752965 | .   |
| ## 1219 | -3.3321e-01 | 4.7071e-03 | -70.7898 | < 2.2e-16 | *** |

|         |             |            |          |           |     |
|---------|-------------|------------|----------|-----------|-----|
| ## 1220 | -4.6652e-02 | 7.6953e-03 | -6.0624  | 1.341e-09 | *** |
| ## 1221 | -3.0995e-02 | 7.7767e-03 | -3.9856  | 6.731e-05 | *** |
| ## 1222 | 9.6883e-03  | 8.3660e-03 | 1.1581   | 0.2468433 |     |
| ## 1223 | -2.8654e-02 | 7.7780e-03 | -3.6839  | 0.0002297 | *** |
| ## 1224 | -8.1449e-02 | 7.3509e-03 | -11.0800 | < 2.2e-16 | *** |
| ## 1225 | 4.0154e-02  | 8.4984e-03 | 4.7248   | 2.303e-06 | *** |
| ## 1226 | 7.2993e-02  | 8.9794e-03 | 8.1290   | 4.329e-16 | *** |
| ## 1227 | -2.0105e-02 | 7.9274e-03 | -2.5361  | 0.0112085 | *   |
| ## 1228 | 2.8791e-02  | 8.4349e-03 | 3.4134   | 0.0006417 | *** |
| ## 1229 | -2.5090e-01 | 5.3115e-03 | -47.2378 | < 2.2e-16 | *** |
| ## 1230 | 1.2652e-02  | 8.2433e-03 | 1.5348   | 0.1248364 |     |
| ## 1231 | -3.6746e-02 | 7.7009e-03 | -4.7717  | 1.827e-06 | *** |
| ## 1232 | -5.4301e-02 | 7.5958e-03 | -7.1488  | 8.753e-13 | *** |
| ## 1233 | -1.3950e-02 | 7.9937e-03 | -1.7451  | 0.0809713 | .   |
| ## 1234 | -4.9181e-02 | 7.6618e-03 | -6.4189  | 1.372e-10 | *** |
| ## 1235 | 1.3126e-02  | 8.2903e-03 | 1.5833   | 0.1133541 |     |
| ## 1236 | 3.9321e-02  | 8.5721e-03 | 4.5870   | 4.496e-06 | *** |
| ## 1237 | 7.2785e-04  | 8.1550e-03 | 0.0893   | 0.9288815 |     |
| ## 1238 | 1.1692e-02  | 8.3168e-03 | 1.4058   | 0.1597741 |     |
| ## 1239 | -1.8911e-02 | 7.9548e-03 | -2.3774  | 0.0174360 | *   |
| ## 1240 | -4.5383e-03 | 8.1260e-03 | -0.5585  | 0.5765071 |     |
| ## 1241 | -8.0452e-02 | 7.2823e-03 | -11.0475 | < 2.2e-16 | *** |
| ## 1242 | 2.2970e-02  | 8.3687e-03 | 2.7447   | 0.0060566 | **  |
| ## 1243 | 2.6811e-02  | 8.4591e-03 | 3.1695   | 0.0015272 | **  |
| ## 1244 | -2.4743e-02 | 7.8897e-03 | -3.1361  | 0.0017119 | **  |
| ## 1245 | -6.3299e-02 | 7.5420e-03 | -8.3929  | < 2.2e-16 | *** |
| ## 1246 | -8.5164e-02 | 7.3071e-03 | -11.6549 | < 2.2e-16 | *** |
| ## 1247 | 5.9513e-03  | 8.1760e-03 | 0.7279   | 0.4666739 |     |
| ## 1248 | 8.7318e-03  | 8.1945e-03 | 1.0656   | 0.2866215 |     |
| ## 1249 | -4.0487e-02 | 7.6158e-03 | -5.3162  | 1.060e-07 | *** |
| ## 1250 | 1.9497e-03  | 8.1365e-03 | 0.2396   | 0.8106244 |     |
| ## 1251 | 1.2936e-02  | 8.2553e-03 | 1.5670   | 0.1171054 |     |
| ## 1252 | -2.4307e-02 | 7.9344e-03 | -3.0635  | 0.0021878 | **  |
| ## 1253 | -1.6917e-02 | 7.9252e-03 | -2.1346  | 0.0327929 | *   |
| ## 1254 | -6.0541e-02 | 7.4901e-03 | -8.0828  | 6.331e-16 | *** |
| ## 1255 | -8.2637e-03 | 8.0654e-03 | -1.0246  | 0.3055584 |     |
| ## 1256 | -6.5015e-02 | 7.4931e-03 | -8.6767  | < 2.2e-16 | *** |
| ## 1257 | -5.1226e-02 | 7.5869e-03 | -6.7519  | 1.459e-11 | *** |
| ## 1258 | -2.1522e-02 | 7.9502e-03 | -2.7070  | 0.0067887 | **  |
| ## 1259 | -2.9116e-02 | 7.8213e-03 | -3.7227  | 0.0001971 | *** |
| ## 1260 | -3.0588e-02 | 7.7429e-03 | -3.9504  | 7.801e-05 | *** |
| ## 1261 | 4.8620e-04  | 8.0912e-03 | 0.0601   | 0.9520843 |     |
| ## 1262 | -9.6137e-02 | 7.1550e-03 | -13.4364 | < 2.2e-16 | *** |
| ## 1263 | 6.7297e-02  | 8.8879e-03 | 7.5717   | 3.684e-14 | *** |
| ## 1264 | 9.4675e-02  | 9.1279e-03 | 10.3720  | < 2.2e-16 | *** |
| ## 1265 | -8.9868e-03 | 8.0403e-03 | -1.1177  | 0.2636881 |     |
| ## 1266 | 3.0340e-02  | 8.4353e-03 | 3.5968   | 0.0003222 | *** |
| ## 1267 | -1.5821e-02 | 8.0168e-03 | -1.9734  | 0.0484458 | *   |

|         |             |            |          |           |     |
|---------|-------------|------------|----------|-----------|-----|
| ## 1268 | -7.4534e-03 | 8.0391e-03 | -0.9271  | 0.3538522 |     |
| ## 1269 | 6.3691e-02  | 8.6780e-03 | 7.3393   | 2.147e-13 | *** |
| ## 1270 | -2.9149e-03 | 8.0618e-03 | -0.3616  | 0.7176765 |     |
| ## 1271 | -2.1931e-02 | 7.8334e-03 | -2.7996  | 0.0051164 | **  |
| ## 1272 | 2.3191e-02  | 8.3771e-03 | 2.7684   | 0.0056334 | **  |
| ## 1273 | -4.2593e-02 | 7.6677e-03 | -5.5548  | 2.779e-08 | *** |
| ## 1274 | -3.4503e-02 | 7.7791e-03 | -4.4354  | 9.192e-06 | *** |
| ## 1275 | 2.1586e-02  | 8.3231e-03 | 2.5934   | 0.0095018 | **  |
| ## 1276 | -2.6838e-02 | 7.8676e-03 | -3.4112  | 0.0006467 | *** |
| ## 1277 | -3.8549e-02 | 7.7326e-03 | -4.9852  | 6.190e-07 | *** |
| ## 1278 | -2.3556e-02 | 7.9031e-03 | -2.9806  | 0.0028771 | **  |
| ## 1279 | 1.7563e-02  | 8.3595e-03 | 2.1010   | 0.0356432 | *   |
| ## 1280 | 6.7057e-02  | 8.9211e-03 | 7.5167   | 5.618e-14 | *** |
| ## 1281 | -8.2037e-02 | 7.3741e-03 | -11.1250 | < 2.2e-16 | *** |
| ## 1282 | 1.0727e-02  | 8.2706e-03 | 1.2971   | 0.1946103 |     |
| ## 1283 | 4.7569e-02  | 8.6247e-03 | 5.5154   | 3.479e-08 | *** |
| ## 1284 | 1.9637e-02  | 8.3311e-03 | 2.3570   | 0.0184218 | *   |
| ## 1285 | -4.6507e-02 | 7.6516e-03 | -6.0781  | 1.216e-09 | *** |
| ## 1286 | -9.6657e-03 | 8.0555e-03 | -1.1999  | 0.2301804 |     |
| ## 1287 | -2.8234e-03 | 8.0599e-03 | -0.3503  | 0.7261114 |     |
| ## 1288 | 2.8180e-02  | 8.3892e-03 | 3.3591   | 0.0007820 | *** |
| ## 1289 | 1.3224e-02  | 8.3285e-03 | 1.5878   | 0.1123385 |     |
| ## 1290 | -3.3894e-02 | 7.7817e-03 | -4.3556  | 1.327e-05 | *** |
| ## 1291 | -8.4504e-04 | 8.0109e-03 | -0.1055  | 0.9159908 |     |
| ## 1292 | -1.9821e-03 | 8.1246e-03 | -0.2440  | 0.8072647 |     |
| ## 1293 | -2.9734e-02 | 7.8141e-03 | -3.8052  | 0.0001417 | *** |
| ## 1294 | -6.2752e-02 | 7.4863e-03 | -8.3822  | < 2.2e-16 | *** |
| ## 1295 | 3.8883e-02  | 8.5806e-03 | 4.5315   | 5.856e-06 | *** |
| ## 1296 | 8.4579e-02  | 9.0207e-03 | 9.3761   | < 2.2e-16 | *** |
| ## 1297 | -3.6909e-02 | 7.7481e-03 | -4.7636  | 1.901e-06 | *** |
| ## 1298 | 3.6897e-02  | 8.5601e-03 | 4.3104   | 1.630e-05 | *** |
| ## 1299 | 1.7749e-03  | 8.2123e-03 | 0.2161   | 0.8288899 |     |
| ## 1300 | -2.9608e-02 | 7.8211e-03 | -3.7856  | 0.0001533 | *** |
| ## 1301 | 4.6965e-03  | 8.0839e-03 | 0.5810   | 0.5612597 |     |
| ## 1302 | 5.3118e-03  | 8.2344e-03 | 0.6451   | 0.5188800 |     |
| ## 1303 | -2.1986e-02 | 7.9604e-03 | -2.7620  | 0.0057456 | **  |
| ## 1304 | 9.2061e-02  | 9.1945e-03 | 10.0126  | < 2.2e-16 | *** |
| ## 1305 | -3.6417e-02 | 7.8910e-03 | -4.6150  | 3.931e-06 | *** |
| ## 1306 | -3.7145e-02 | 7.7737e-03 | -4.7783  | 1.768e-06 | *** |
| ## 1307 | -1.9376e-02 | 8.0056e-03 | -2.4203  | 0.0155080 | *   |
| ## 1308 | 7.9527e-03  | 8.3343e-03 | 0.9542   | 0.3399742 |     |
| ## 1309 | -3.0112e-02 | 7.8689e-03 | -3.8268  | 0.0001298 | *** |
| ## 1310 | 9.3954e-02  | 9.1812e-03 | 10.2333  | < 2.2e-16 | *** |
| ## 1311 | 6.1933e-02  | 8.8246e-03 | 7.0183   | 2.246e-12 | *** |
| ## 1312 | 8.8648e-02  | 9.1486e-03 | 9.6898   | < 2.2e-16 | *** |
| ## 1313 | 6.8497e-02  | 8.9621e-03 | 7.6430   | 2.123e-14 | *** |
| ## 1314 | 1.7089e-02  | 8.3687e-03 | 2.0420   | 0.0411473 | *   |
| ## 1315 | 4.1235e-02  | 8.5577e-03 | 4.8185   | 1.447e-06 | *** |

|         |             |            |          |           |     |
|---------|-------------|------------|----------|-----------|-----|
| ## 1316 | -6.6515e-02 | 7.4628e-03 | -8.9129  | < 2.2e-16 | *** |
| ## 1317 | 3.7388e-02  | 8.5764e-03 | 4.3594   | 1.304e-05 | *** |
| ## 1318 | 6.6879e-03  | 8.1599e-03 | 0.8196   | 0.4124418 |     |
| ## 1319 | 2.3745e-02  | 8.3769e-03 | 2.8345   | 0.0045892 | **  |
| ## 1320 | 6.2366e-02  | 8.8074e-03 | 7.0811   | 1.430e-12 | *** |
| ## 1321 | 6.7135e-02  | 8.9323e-03 | 7.5160   | 5.649e-14 | *** |
| ## 1322 | -1.8863e-02 | 7.9314e-03 | -2.3783  | 0.0173922 | *   |
| ## 1323 | 2.1723e-03  | 8.1841e-03 | 0.2654   | 0.7906773 |     |
| ## 1324 | 1.6997e-02  | 8.3395e-03 | 2.0381   | 0.0415419 | *   |
| ## 1325 | 1.0485e-02  | 8.2978e-03 | 1.2636   | 0.2063882 |     |
| ## 1326 | 1.6775e-02  | 8.3154e-03 | 2.0174   | 0.0436549 | *   |
| ## 1327 | 2.7548e-02  | 8.3717e-03 | 3.2906   | 0.0009998 | *** |
| ## 1328 | 4.6127e-02  | 8.6094e-03 | 5.3577   | 8.429e-08 | *** |
| ## 1329 | -3.2750e-02 | 8.2708e-03 | -3.9597  | 7.503e-05 | *** |
| ## 1330 | -6.9035e-02 | 7.3865e-03 | -9.3462  | < 2.2e-16 | *** |
| ## 1331 | 6.7793e-03  | 8.1824e-03 | 0.8285   | 0.4073732 |     |
| ## 1332 | 2.8464e-02  | 8.4481e-03 | 3.3693   | 0.0007535 | *** |
| ## 1333 | 3.9447e-03  | 8.1784e-03 | 0.4823   | 0.6295722 |     |
| ## 1334 | -7.3884e-03 | 8.0321e-03 | -0.9198  | 0.3576516 |     |
| ## 1335 | 3.6385e-02  | 8.5545e-03 | 4.2534   | 2.106e-05 | *** |
| ## 1336 | 1.0399e-02  | 8.2301e-03 | 1.2635   | 0.2063977 |     |
| ## 1337 | -5.4862e-02 | 7.6025e-03 | -7.2164  | 5.339e-13 | *** |
| ## 1338 | 1.7591e-02  | 8.3227e-03 | 2.1136   | 0.0345462 | *   |
| ## 1339 | 7.8619e-02  | 9.0051e-03 | 8.7305   | < 2.2e-16 | *** |
| ## 1340 | 2.8808e-02  | 8.4767e-03 | 3.3985   | 0.0006776 | *** |
| ## 1341 | -3.4024e-02 | 7.8131e-03 | -4.3547  | 1.332e-05 | *** |
| ## 1342 | 3.0634e-02  | 8.5329e-03 | 3.5901   | 0.0003305 | *** |
| ## 1343 | -5.9839e-03 | 8.1185e-03 | -0.7371  | 0.4610830 |     |
| ## 1344 | 2.1412e-04  | 8.0805e-03 | 0.0265   | 0.9788598 |     |
| ## 1345 | -8.7653e-03 | 8.0246e-03 | -1.0923  | 0.2747008 |     |
| ## 1346 | 1.8147e-02  | 8.3389e-03 | 2.1762   | 0.0295423 | *   |
| ## 1347 | -3.5867e-02 | 7.7551e-03 | -4.6249  | 3.748e-06 | *** |
| ## 1348 | -1.1364e-02 | 7.8966e-03 | -1.4391  | 0.1501262 |     |
| ## 1349 | -4.3730e-02 | 7.6559e-03 | -5.7120  | 1.117e-08 | *** |
| ## 1350 | -4.0070e-02 | 7.7039e-03 | -5.2012  | 1.980e-07 | *** |
| ## 1351 | -8.3975e-02 | 7.3253e-03 | -11.4637 | < 2.2e-16 | *** |
| ## 1352 | -4.1747e-02 | 7.7400e-03 | -5.3938  | 6.900e-08 | *** |
| ## 1353 | 1.0370e-02  | 8.2662e-03 | 1.2545   | 0.2096618 |     |
| ## 1354 | -6.9715e-02 | 7.5067e-03 | -9.2870  | < 2.2e-16 | *** |
| ## 1355 | -4.5738e-02 | 7.7186e-03 | -5.9257  | 3.110e-09 | *** |
| ## 1356 | -5.2663e-02 | 7.6714e-03 | -6.8649  | 6.655e-12 | *** |
| ## 1357 | 1.3056e-02  | 8.2524e-03 | 1.5820   | 0.1136397 |     |
| ## 1358 | 4.2929e-03  | 8.1415e-03 | 0.5273   | 0.5979904 |     |
| ## 1359 | -4.1954e-04 | 8.1744e-03 | -0.0513  | 0.9590677 |     |
| ## 1360 | -4.6537e-02 | 7.6305e-03 | -6.0988  | 1.069e-09 | *** |
| ## 1361 | -1.8306e-02 | 7.9748e-03 | -2.2954  | 0.0217079 | *   |
| ## 1362 | -2.5514e-02 | 7.8859e-03 | -3.2353  | 0.0012150 | **  |
| ## 1363 | 3.6149e-02  | 8.4225e-03 | 4.2920   | 1.771e-05 | *** |

|         |             |            |          |           |     |
|---------|-------------|------------|----------|-----------|-----|
| ## 1364 | 3.1651e-02  | 8.4877e-03 | 3.7291   | 0.0001922 | *** |
| ## 1365 | -7.2142e-02 | 7.4607e-03 | -9.6696  | < 2.2e-16 | *** |
| ## 1366 | 2.2793e-02  | 8.4502e-03 | 2.6974   | 0.0069891 | **  |
| ## 1367 | -1.3811e-03 | 8.1238e-03 | -0.1700  | 0.8650026 |     |
| ## 1368 | -1.0018e-03 | 8.1174e-03 | -0.1234  | 0.9017835 |     |
| ## 1369 | -3.4093e-02 | 7.7752e-03 | -4.3849  | 1.161e-05 | *** |
| ## 1370 | 9.0997e-02  | 9.1956e-03 | 9.8957   | < 2.2e-16 | *** |
| ## 1371 | -3.4575e-03 | 8.1511e-03 | -0.4242  | 0.6714336 |     |
| ## 1372 | 9.2932e-02  | 9.2195e-03 | 10.0799  | < 2.2e-16 | *** |
| ## 1373 | 3.6649e-02  | 8.5144e-03 | 4.3043   | 1.675e-05 | *** |
| ## 1374 | 1.8652e-02  | 8.3015e-03 | 2.2468   | 0.0246535 | *   |
| ## 1375 | -5.0705e-02 | 7.6196e-03 | -6.6545  | 2.843e-11 | *** |
| ## 1376 | -1.0309e-02 | 8.0206e-03 | -1.2854  | 0.1986679 |     |
| ## 1377 | -2.9683e-02 | 7.6832e-03 | -3.8633  | 0.0001119 | *** |
| ## 1378 | 2.7070e-02  | 8.3474e-03 | 3.2429   | 0.0011832 | **  |
| ## 1379 | 8.0410e-02  | 8.9996e-03 | 8.9348   | < 2.2e-16 | *** |
| ## 1380 | 2.1901e-02  | 8.3546e-03 | 2.6214   | 0.0087562 | **  |
| ## 1381 | 5.6258e-02  | 8.7234e-03 | 6.4491   | 1.125e-10 | *** |
| ## 1382 | -2.1519e-05 | 8.0058e-03 | -0.0027  | 0.9978554 |     |
| ## 1383 | 5.2967e-02  | 8.8254e-03 | 6.0016   | 1.954e-09 | *** |
| ## 1384 | 3.5893e-02  | 8.5230e-03 | 4.2113   | 2.539e-05 | *** |
| ## 1385 | 1.6012e-02  | 8.3488e-03 | 1.9178   | 0.0551328 | .   |
| ## 1386 | 7.6566e-03  | 8.1702e-03 | 0.9371   | 0.3486874 |     |
| ## 1387 | 5.3061e-02  | 8.6784e-03 | 6.1141   | 9.708e-10 | *** |
| ## 1388 | -1.2668e-02 | 7.9706e-03 | -1.5893  | 0.1119891 |     |
| ## 1389 | -3.8038e-02 | 7.7236e-03 | -4.9248  | 8.443e-07 | *** |
| ## 1390 | -8.1157e-02 | 7.3153e-03 | -11.0942 | < 2.2e-16 | *** |
| ## 1391 | 9.3254e-02  | 9.2051e-03 | 10.1307  | < 2.2e-16 | *** |
| ## 1392 | 7.6614e-02  | 8.9172e-03 | 8.5917   | < 2.2e-16 | *** |
| ## 1393 | -5.8406e-03 | 8.0667e-03 | -0.7240  | 0.4690448 |     |
| ## 1394 | -4.2081e-03 | 8.1089e-03 | -0.5190  | 0.6037944 |     |
| ## 1395 | -2.4943e-02 | 7.8392e-03 | -3.1819  | 0.0014631 | **  |
| ## 1396 | 2.3334e-02  | 8.4147e-03 | 2.7730   | 0.0055543 | **  |
| ## 1397 | 1.6413e-02  | 8.2719e-03 | 1.9842   | 0.0472292 | *   |
| ## 1398 | 1.5844e-03  | 8.1697e-03 | 0.1939   | 0.8462298 |     |
| ## 1399 | -1.1109e-02 | 8.1136e-03 | -1.3692  | 0.1709303 |     |
| ## 1400 | 2.9217e-02  | 8.4735e-03 | 3.4480   | 0.0005647 | *** |
| ## 1401 | 1.0652e-02  | 8.2878e-03 | 1.2853   | 0.1986806 |     |
| ## 1402 | 9.5611e-02  | 9.0701e-03 | 10.5413  | < 2.2e-16 | *** |
| ## 1403 | -4.7374e-02 | 7.6980e-03 | -6.1541  | 7.549e-10 | *** |
| ## 1404 | -8.3621e-03 | 7.9985e-03 | -1.0455  | 0.2958101 |     |
| ## 1405 | -2.5614e-02 | 7.8808e-03 | -3.2501  | 0.0011536 | **  |
| ## 1406 | 1.4847e-04  | 8.1222e-03 | 0.0183   | 0.9854155 |     |
| ## 1407 | -3.3000e-02 | 7.8101e-03 | -4.2253  | 2.386e-05 | *** |
| ## 1408 | 6.8790e-02  | 8.8216e-03 | 7.7979   | 6.294e-15 | *** |
| ## 1409 | 3.2355e-02  | 8.5119e-03 | 3.8012   | 0.0001440 | *** |
| ## 1410 | 1.4774e-02  | 8.1660e-03 | 1.8093   | 0.0704094 | .   |
| ## 1411 | -4.5847e-02 | 7.6774e-03 | -5.9716  | 2.349e-09 | *** |



|         |             |            |          |           |     |
|---------|-------------|------------|----------|-----------|-----|
| ## 1412 | 6.3074e-03  | 8.2338e-03 | 0.7660   | 0.4436582 |     |
| ## 1413 | -1.5863e-02 | 7.9319e-03 | -1.9999  | 0.0455115 | *   |
| ## 1414 | 3.4153e-02  | 8.4545e-03 | 4.0397   | 5.353e-05 | *** |
| ## 1415 | 1.9105e-02  | 8.3070e-03 | 2.2999   | 0.0214541 | *   |
| ## 1416 | 9.7013e-02  | 9.1960e-03 | 10.5494  | < 2.2e-16 | *** |
| ## 1417 | -1.9732e-02 | 7.9648e-03 | -2.4774  | 0.0132353 | *   |
| ## 1418 | 2.6400e-02  | 8.4304e-03 | 3.1315   | 0.0017390 | **  |
| ## 1419 | 4.1997e-03  | 8.2474e-03 | 0.5092   | 0.6106047 |     |
| ## 1420 | -4.3233e-02 | 7.6916e-03 | -5.6208  | 1.901e-08 | *** |
| ## 1421 | 1.1098e-02  | 8.1742e-03 | 1.3577   | 0.1745675 |     |
| ## 1422 | -4.5823e-02 | 7.6707e-03 | -5.9738  | 2.318e-09 | *** |
| ## 1423 | -2.9376e-02 | 7.8315e-03 | -3.7511  | 0.0001761 | *** |
| ## 1424 | -3.6056e-02 | 7.8027e-03 | -4.6209  | 3.820e-06 | *** |
| ## 1425 | -1.5722e-02 | 7.9570e-03 | -1.9758  | 0.0481721 | *   |
| ## 1426 | 1.3173e-02  | 8.2826e-03 | 1.5904   | 0.1117345 |     |
| ## 1427 | 2.4920e-02  | 8.4204e-03 | 2.9595   | 0.0030812 | **  |
| ## 1428 | 3.1691e-02  | 8.4453e-03 | 3.7525   | 0.0001751 | *** |
| ## 1429 | -7.9778e-02 | 7.2860e-03 | -10.9494 | < 2.2e-16 | *** |
| ## 1430 | 3.8340e-02  | 8.4851e-03 | 4.5185   | 6.227e-06 | *** |
| ## 1431 | -3.5884e-01 | 4.6291e-03 | -77.5179 | < 2.2e-16 | *** |
| ## 1432 | -1.5267e-02 | 8.0425e-03 | -1.8983  | 0.0576588 | .   |
| ## 1433 | 3.0174e-02  | 8.4571e-03 | 3.5679   | 0.0003598 | *** |
| ## 1434 | -2.6625e-02 | 7.8456e-03 | -3.3936  | 0.0006897 | *** |
| ## 1435 | -1.3893e-02 | 7.9382e-03 | -1.7502  | 0.0800914 | .   |
| ## 1436 | -7.6812e-02 | 7.3191e-03 | -10.4946 | < 2.2e-16 | *** |
| ## 1437 | -6.7479e-02 | 7.4745e-03 | -9.0278  | < 2.2e-16 | *** |
| ## 1438 | 5.2160e-03  | 8.2274e-03 | 0.6340   | 0.5260924 |     |
| ## 1439 | -5.9079e-03 | 8.0423e-03 | -0.7346  | 0.4625798 |     |
| ## 1440 | 6.6370e-03  | 8.1331e-03 | 0.8161   | 0.4144703 |     |
| ## 1441 | 1.3472e-02  | 8.2652e-03 | 1.6300   | 0.1031112 |     |
| ## 1442 | -2.2873e-02 | 7.7931e-03 | -2.9350  | 0.0033351 | **  |
| ## 1443 | -4.2344e-03 | 8.1133e-03 | -0.5219  | 0.6017345 |     |
| ## 1444 | -4.0920e-02 | 7.7367e-03 | -5.2891  | 1.229e-07 | *** |
| ## 1445 | 4.5129e-02  | 8.6661e-03 | 5.2076   | 1.913e-07 | *** |
| ## 1446 | 2.6608e-02  | 8.4265e-03 | 3.1577   | 0.0015901 | **  |
| ## 1447 | -4.1198e-02 | 7.5303e-03 | -5.4709  | 4.476e-08 | *** |
| ## 1448 | 4.8520e-02  | 8.6873e-03 | 5.5852   | 2.334e-08 | *** |
| ## 1449 | 1.1144e-02  | 8.2328e-03 | 1.3536   | 0.1758730 |     |
| ## 1450 | -1.9230e-02 | 7.9461e-03 | -2.4200  | 0.0155200 | *   |
| ## 1451 | 1.1074e-02  | 8.2722e-03 | 1.3387   | 0.1806557 |     |
| ## 1452 | 3.4852e-03  | 8.1448e-03 | 0.4279   | 0.6687199 |     |
| ## 1453 | -6.1247e-02 | 7.5351e-03 | -8.1282  | 4.357e-16 | *** |
| ## 1454 | -3.1898e-02 | 7.8066e-03 | -4.0861  | 4.388e-05 | *** |
| ## 1455 | -3.7167e-02 | 7.7585e-03 | -4.7904  | 1.664e-06 | *** |
| ## 1456 | 1.3676e-02  | 8.2137e-03 | 1.6650   | 0.0959182 | .   |
| ## 1457 | -7.5160e-02 | 7.3599e-03 | -10.2122 | < 2.2e-16 | *** |
| ## 1458 | -2.9032e-03 | 8.1474e-03 | -0.3563  | 0.7215853 |     |
| ## 1459 | -1.1223e-02 | 7.9546e-03 | -1.4109  | 0.1582644 |     |

|         |             |            |         |           |     |
|---------|-------------|------------|---------|-----------|-----|
| ## 1460 | 4.5668e-02  | 8.6301e-03 | 5.2917  | 1.212e-07 | *** |
| ## 1461 | -4.5244e-03 | 8.0543e-03 | -0.5617 | 0.5742940 |     |
| ## 1462 | -9.0585e-04 | 8.1840e-03 | -0.1107 | 0.9118656 |     |
| ## 1463 | 5.5170e-02  | 8.7651e-03 | 6.2943  | 3.088e-10 | *** |
| ## 1464 | 1.7379e-02  | 8.3772e-03 | 2.0745  | 0.0380298 | *   |
| ## 1465 | -6.0182e-02 | 7.5770e-03 | -7.9427 | 1.978e-15 | *** |
| ## 1466 | 6.0160e-02  | 8.8252e-03 | 6.8168  | 9.310e-12 | *** |
| ## 1467 | 1.4921e-02  | 8.3055e-03 | 1.7965  | 0.0724071 | .   |
| ## 1468 | 3.4430e-02  | 8.4858e-03 | 4.0573  | 4.964e-05 | *** |
| ## 1469 | 2.0935e-02  | 8.2987e-03 | 2.5227  | 0.0116467 | *   |
| ## 1470 | 6.9752e-02  | 8.8241e-03 | 7.9047  | 2.685e-15 | *** |
| ## 1471 | 1.9113e-02  | 8.3469e-03 | 2.2898  | 0.0220314 | *   |
| ## 1472 | 6.8402e-03  | 8.2666e-03 | 0.8275  | 0.4079792 |     |
| ## 1473 | 6.9964e-02  | 8.9532e-03 | 7.8145  | 5.519e-15 | *** |
| ## 1474 | -1.2002e-03 | 8.1138e-03 | -0.1479 | 0.8824034 |     |
| ## 1475 | -4.7273e-02 | 7.6754e-03 | -6.1590 | 7.318e-10 | *** |
| ## 1476 | 5.5205e-03  | 8.1591e-03 | 0.6766  | 0.4986569 |     |
| ## 1477 | -2.6624e-02 | 7.8510e-03 | -3.3911 | 0.0006960 | *** |
| ## 1478 | 3.7818e-02  | 8.5585e-03 | 4.4187  | 9.929e-06 | *** |
| ## 1479 | 3.0559e-02  | 8.4872e-03 | 3.6007  | 0.0003174 | *** |
| ## 1480 | 2.5480e-02  | 8.3189e-03 | 3.0629  | 0.0021923 | **  |
| ## 1481 | 1.1950e-03  | 8.1772e-03 | 0.1461  | 0.8838154 |     |
| ## 1482 | -9.2688e-03 | 7.9634e-03 | -1.1639 | 0.2444553 |     |
| ## 1483 | 2.7978e-02  | 8.4794e-03 | 3.2996  | 0.0009683 | *** |
| ## 1484 | 1.5156e-02  | 8.3134e-03 | 1.8231  | 0.0682861 | .   |
| ## 1485 | -2.8175e-02 | 7.8595e-03 | -3.5848 | 0.0003373 | *** |
| ## 1486 | 2.0453e-02  | 8.3575e-03 | 2.4473  | 0.0143928 | *   |
| ## 1487 | 8.4671e-02  | 9.0512e-03 | 9.3547  | < 2.2e-16 | *** |
| ## 1488 | 5.2198e-02  | 8.7071e-03 | 5.9949  | 2.036e-09 | *** |
| ## 1489 | 4.2297e-02  | 8.6281e-03 | 4.9022  | 9.476e-07 | *** |
| ## 1490 | 2.3711e-02  | 8.4050e-03 | 2.8210  | 0.0047869 | **  |
| ## 1491 | 4.9522e-02  | 8.6579e-03 | 5.7198  | 1.066e-08 | *** |
| ## 1492 | -4.3495e-02 | 7.6749e-03 | -5.6672 | 1.452e-08 | *** |
| ## 1493 | -3.2586e-03 | 8.1098e-03 | -0.4018 | 0.6878243 |     |
| ## 1494 | -2.1230e-02 | 7.9236e-03 | -2.6793 | 0.0073778 | **  |
| ## 1495 | 5.9350e-02  | 8.7746e-03 | 6.7638  | 1.344e-11 | *** |
| ## 1496 | -9.0125e-03 | 7.9871e-03 | -1.1284 | 0.2591600 |     |
| ## 1497 | 2.3550e-02  | 8.3784e-03 | 2.8108  | 0.0049421 | **  |
| ## 1498 | -6.3106e-02 | 7.4814e-03 | -8.4350 | < 2.2e-16 | *** |
| ## 1499 | -9.8762e-03 | 8.0760e-03 | -1.2229 | 0.2213613 |     |
| ## 1500 | -5.2625e-02 | 7.6948e-03 | -6.8391 | 7.970e-12 | *** |
| ## 1501 | 3.2918e-03  | 8.1818e-03 | 0.4023  | 0.6874364 |     |
| ## 1502 | -3.4060e-02 | 7.8766e-03 | -4.3242 | 1.531e-05 | *** |
| ## 1503 | -6.9989e-02 | 7.4576e-03 | -9.3849 | < 2.2e-16 | *** |
| ## 1504 | -2.3976e-02 | 7.8543e-03 | -3.0526 | 0.0022687 | **  |
| ## 1505 | 5.8905e-03  | 8.1720e-03 | 0.7208  | 0.4710226 |     |
| ## 1506 | -7.6855e-03 | 8.0296e-03 | -0.9571 | 0.3384930 |     |
| ## 1507 | 1.9132e-03  | 8.1121e-03 | 0.2359  | 0.8135479 |     |

|         |             |            |         |           |     |
|---------|-------------|------------|---------|-----------|-----|
| ## 1508 | 4.5739e-02  | 8.6285e-03 | 5.3010  | 1.152e-07 | *** |
| ## 1509 | 2.8320e-02  | 8.4916e-03 | 3.3350  | 0.0008529 | *** |
| ## 1510 | 1.1525e-02  | 8.2450e-03 | 1.3978  | 0.1621815 |     |
| ## 1511 | 2.6410e-02  | 8.4561e-03 | 3.1231  | 0.0017895 | **  |
| ## 1512 | -6.4874e-03 | 8.0601e-03 | -0.8049 | 0.4208956 |     |
| ## 1513 | 4.8631e-02  | 8.6744e-03 | 5.6063  | 2.067e-08 | *** |
| ## 1514 | 1.6699e-01  | 1.0045e-02 | 16.6240 | < 2.2e-16 | *** |
| ## 1515 | -4.8746e-02 | 7.5986e-03 | -6.4151 | 1.408e-10 | *** |
| ## 1516 | 1.9230e-02  | 8.1663e-03 | 2.3548  | 0.0185323 | *   |
| ## 1517 | -5.7003e-02 | 7.6709e-03 | -7.4311 | 1.077e-13 | *** |
| ## 1518 | 5.4309e-02  | 8.7685e-03 | 6.1936  | 5.879e-10 | *** |
| ## 1519 | 4.1828e-02  | 8.5250e-03 | 4.9066  | 9.269e-07 | *** |
| ## 1520 | -1.6457e-02 | 7.9621e-03 | -2.0669 | 0.0387397 | *   |
| ## 1521 | -2.8274e-02 | 7.8102e-03 | -3.6202 | 0.0002944 | *** |
| ## 1522 | 1.4968e-02  | 8.3549e-03 | 1.7915  | 0.0732162 | .   |
| ## 1523 | 1.3643e-02  | 8.3245e-03 | 1.6389  | 0.1012413 |     |
| ## 1524 | 6.9208e-02  | 8.9165e-03 | 7.7618  | 8.372e-15 | *** |
| ## 1525 | 2.7171e-05  | 8.1360e-03 | 0.0033  | 0.9973354 |     |
| ## 1526 | 2.1679e-02  | 8.3180e-03 | 2.6063  | 0.0091521 | **  |
| ## 1527 | 5.9530e-02  | 8.7689e-03 | 6.7887  | 1.131e-11 | *** |
| ## 1528 | 5.0724e-02  | 8.6686e-03 | 5.8515  | 4.871e-09 | *** |
| ## 1529 | 7.8901e-02  | 9.0195e-03 | 8.7478  | < 2.2e-16 | *** |
| ## 1530 | -4.3367e-02 | 7.7392e-03 | -5.6036 | 2.100e-08 | *** |
| ## 1531 | -3.1800e-02 | 7.7794e-03 | -4.0877 | 4.357e-05 | *** |
| ## 1532 | -2.0310e-02 | 7.8496e-03 | -2.5873 | 0.0096722 | **  |
| ## 1533 | -1.3482e-02 | 8.0276e-03 | -1.6795 | 0.0930581 | .   |
| ## 1534 | -9.3939e-03 | 8.0276e-03 | -1.1702 | 0.2419201 |     |
| ## 1535 | 2.0011e-02  | 8.4220e-03 | 2.3761  | 0.0174976 | *   |
| ## 1536 | -7.0479e-02 | 7.2907e-03 | -9.6670 | < 2.2e-16 | *** |
| ## 1537 | -3.8942e-02 | 7.7206e-03 | -5.0439 | 4.561e-07 | *** |
| ## 1538 | 6.3208e-02  | 8.7817e-03 | 7.1977  | 6.122e-13 | *** |
| ## 1539 | -1.5166e-03 | 8.1604e-03 | -0.1858 | 0.8525673 |     |
| ## 1540 | -3.7672e-02 | 7.8992e-03 | -4.7691 | 1.851e-06 | *** |
| ## 1541 | 4.1133e-02  | 8.5752e-03 | 4.7968  | 1.612e-06 | *** |
| ## 1542 | 1.8594e-02  | 8.2844e-03 | 2.2445  | 0.0248004 | *   |
| ## 1543 | 6.0199e-02  | 8.7065e-03 | 6.9143  | 4.702e-12 | *** |
| ## 1544 | 2.3236e-02  | 8.3189e-03 | 2.7931  | 0.0052203 | **  |
| ## 1545 | 5.7381e-02  | 8.7592e-03 | 6.5509  | 5.719e-11 | *** |
| ## 1546 | 3.4759e-02  | 8.4945e-03 | 4.0919  | 4.278e-05 | *** |
| ## 1547 | -1.7076e-02 | 7.9549e-03 | -2.1466 | 0.0318225 | *   |
| ## 1548 | 6.3064e-02  | 8.7963e-03 | 7.1694  | 7.532e-13 | *** |
| ## 1549 | -3.9353e-02 | 7.7148e-03 | -5.1009 | 3.380e-07 | *** |
| ## 1550 | 6.9724e-02  | 8.8296e-03 | 7.8967  | 2.864e-15 | *** |
| ## 1551 | -1.9519e-02 | 7.9523e-03 | -2.4545 | 0.0141097 | *   |
| ## 1552 | 9.2691e-02  | 9.1575e-03 | 10.1219 | < 2.2e-16 | *** |
| ## 1553 | 6.4815e-02  | 8.8306e-03 | 7.3399  | 2.138e-13 | *** |
| ## 1554 | -1.4803e-02 | 7.9720e-03 | -1.8568 | 0.0633368 | .   |
| ## 1555 | 1.0734e-01  | 9.2902e-03 | 11.5545 | < 2.2e-16 | *** |

|         |             |            |          |           |     |
|---------|-------------|------------|----------|-----------|-----|
| ## 1556 | 2.8790e-02  | 8.4252e-03 | 3.4171   | 0.0006329 | *** |
| ## 1557 | -5.0874e-02 | 7.6614e-03 | -6.6403  | 3.131e-11 | *** |
| ## 1558 | -4.3763e-02 | 7.7096e-03 | -5.6764  | 1.376e-08 | *** |
| ## 1559 | -4.2556e-02 | 7.7050e-03 | -5.5231  | 3.330e-08 | *** |
| ## 1560 | -1.9508e-02 | 7.9609e-03 | -2.4505  | 0.0142674 | *   |
| ## 1561 | -8.7200e-02 | 7.2619e-03 | -12.0078 | < 2.2e-16 | *** |
| ## 1562 | -6.2839e-03 | 8.1085e-03 | -0.7750  | 0.4383523 |     |
| ## 1563 | -2.2484e-02 | 7.8767e-03 | -2.8545  | 0.0043103 | **  |
| ## 1564 | -4.1114e-03 | 8.0989e-03 | -0.5077  | 0.6116949 |     |
| ## 1565 | 1.9711e-02  | 8.3019e-03 | 2.3743   | 0.0175844 | *   |
| ## 1566 | -6.0006e-02 | 7.5648e-03 | -7.9323  | 2.150e-15 | *** |
| ## 1567 | 3.0255e-02  | 8.5281e-03 | 3.5477   | 0.0003886 | *** |
| ## 1568 | 1.5866e-02  | 8.3261e-03 | 1.9055   | 0.0567127 | .   |
| ## 1569 | 5.1764e-03  | 8.1856e-03 | 0.6324   | 0.5271387 |     |
| ## 1570 | -6.1859e-02 | 7.5347e-03 | -8.2098  | < 2.2e-16 | *** |
| ## 1571 | -3.1272e-02 | 7.7431e-03 | -4.0387  | 5.375e-05 | *** |
| ## 1572 | 5.3392e-02  | 8.6950e-03 | 6.1405   | 8.227e-10 | *** |
| ## 1573 | 5.4908e-02  | 8.7488e-03 | 6.2761   | 3.473e-10 | *** |
| ## 1574 | 2.7663e-02  | 8.4696e-03 | 3.2661   | 0.0010903 | **  |
| ## 1575 | 4.3043e-02  | 8.6508e-03 | 4.9756   | 6.504e-07 | *** |
| ## 1576 | 6.8329e-02  | 8.9083e-03 | 7.6704   | 1.715e-14 | *** |
| ## 1577 | -9.8173e-03 | 8.0307e-03 | -1.2225  | 0.2215322 |     |
| ## 1578 | 3.0976e-02  | 8.4349e-03 | 3.6724   | 0.0002403 | *** |
| ## 1579 | -8.0596e-03 | 8.0772e-03 | -0.9978  | 0.3183665 |     |
| ## 1580 | -2.2047e-02 | 7.9789e-03 | -2.7632  | 0.0057236 | **  |
| ## 1581 | -6.8792e-02 | 7.4159e-03 | -9.2763  | < 2.2e-16 | *** |
| ## 1582 | 3.9711e-03  | 8.1096e-03 | 0.4897   | 0.6243565 |     |
| ## 1583 | 8.8202e-02  | 9.1075e-03 | 9.6846   | < 2.2e-16 | *** |
| ## 1584 | 3.6968e-02  | 8.5587e-03 | 4.3193   | 1.565e-05 | *** |
| ## 1585 | 9.5405e-02  | 9.1953e-03 | 10.3754  | < 2.2e-16 | *** |
| ## 1586 | -7.9504e-03 | 8.0733e-03 | -0.9848  | 0.3247370 |     |
| ## 1587 | 1.3133e-02  | 8.2662e-03 | 1.5887   | 0.1121205 |     |
| ## 1588 | 8.6893e-03  | 8.2182e-03 | 1.0573   | 0.2903648 |     |
| ## 1589 | 9.6649e-02  | 9.2101e-03 | 10.4938  | < 2.2e-16 | *** |
| ## 1590 | -4.3935e-03 | 8.1026e-03 | -0.5422  | 0.5876575 |     |
| ## 1591 | 8.9901e-02  | 9.1221e-03 | 9.8553   | < 2.2e-16 | *** |
| ## 1592 | 2.6010e-02  | 8.4184e-03 | 3.0896   | 0.0020042 | **  |
| ## 1593 | 6.8051e-03  | 8.1698e-03 | 0.8329   | 0.4048740 |     |
| ## 1594 | -8.2513e-02 | 7.3911e-03 | -11.1638 | < 2.2e-16 | *** |
| ## 1595 | 3.6468e-02  | 8.5154e-03 | 4.2826   | 1.847e-05 | *** |
| ## 1596 | 9.1424e-03  | 8.2262e-03 | 1.1114   | 0.2664084 |     |
| ## 1597 | -2.4969e-02 | 7.9034e-03 | -3.1593  | 0.0015814 | **  |
| ## 1598 | -2.6889e-02 | 7.9032e-03 | -3.4023  | 0.0006683 | *** |
| ## 1599 | -4.7057e-02 | 7.6358e-03 | -6.1626  | 7.156e-10 | *** |
| ## 1600 | 1.4819e-02  | 8.2303e-03 | 1.8006   | 0.0717694 | .   |
| ## 1601 | -3.0687e-02 | 7.8750e-03 | -3.8968  | 9.748e-05 | *** |
| ## 1602 | 1.7902e-02  | 8.2603e-03 | 2.1672   | 0.0302171 | *   |
| ## 1603 | 3.9470e-02  | 8.5400e-03 | 4.6218   | 3.805e-06 | *** |

|         |             |            |          |           |     |
|---------|-------------|------------|----------|-----------|-----|
| ## 1604 | -4.6068e-02 | 7.6916e-03 | -5.9894  | 2.106e-09 | *** |
| ## 1605 | -1.4193e-02 | 7.8406e-03 | -1.8102  | 0.0702602 | .   |
| ## 1606 | -2.4292e-02 | 7.9375e-03 | -3.0604  | 0.0022106 | **  |
| ## 1607 | -1.9291e-02 | 7.9069e-03 | -2.4398  | 0.0146964 | *   |
| ## 1608 | -3.2453e-02 | 7.7364e-03 | -4.1948  | 2.731e-05 | *** |
| ## 1609 | -4.2457e-02 | 7.7376e-03 | -5.4871  | 4.086e-08 | *** |
| ## 1610 | -2.0142e-02 | 7.9326e-03 | -2.5391  | 0.0111142 | *   |
| ## 1611 | 1.6641e-02  | 8.3418e-03 | 1.9950   | 0.0460478 | *   |
| ## 1612 | -1.7856e-02 | 7.9292e-03 | -2.2519  | 0.0243283 | *   |
| ## 1613 | 1.9803e-02  | 8.3248e-03 | 2.3788   | 0.0173682 | *   |
| ## 1614 | 2.8802e-02  | 8.3732e-03 | 3.4398   | 0.0005821 | *** |
| ## 1615 | 6.6660e-03  | 8.2184e-03 | 0.8111   | 0.4173039 |     |
| ## 1616 | 7.1939e-02  | 9.0116e-03 | 7.9829   | 1.429e-15 | *** |
| ## 1617 | -3.9760e-02 | 7.7373e-03 | -5.1387  | 2.766e-07 | *** |
| ## 1618 | -2.9581e-03 | 8.0995e-03 | -0.3652  | 0.7149515 |     |
| ## 1619 | -9.2491e-02 | 7.2291e-03 | -12.7943 | < 2.2e-16 | *** |
| ## 1620 | -3.8505e-02 | 7.7563e-03 | -4.9644  | 6.892e-07 | *** |
| ## 1621 | 2.0228e-02  | 8.3529e-03 | 2.4217   | 0.0154472 | *   |
| ## 1622 | 5.7322e-02  | 8.8008e-03 | 6.5132   | 7.355e-11 | *** |
| ## 1623 | -1.7834e-02 | 8.2251e-03 | -2.1683  | 0.0301368 | *   |
| ## 1624 | 4.5695e-02  | 8.6413e-03 | 5.2880   | 1.237e-07 | *** |
| ## 1625 | -4.8076e-02 | 7.7240e-03 | -6.2242  | 4.840e-10 | *** |
| ## 1626 | -3.9693e-02 | 7.7454e-03 | -5.1247  | 2.980e-07 | *** |
| ## 1627 | 4.0095e-02  | 8.5695e-03 | 4.6788   | 2.886e-06 | *** |
| ## 1628 | -2.2251e-02 | 7.9141e-03 | -2.8116  | 0.0049292 | **  |
| ## 1629 | -1.2968e-03 | 8.0680e-03 | -0.1607  | 0.8723060 |     |
| ## 1630 | 1.5753e-03  | 8.0438e-03 | 0.1958   | 0.8447294 |     |
| ## 1631 | 6.0898e-02  | 8.7273e-03 | 6.9779   | 2.997e-12 | *** |
| ## 1632 | 2.4303e-02  | 8.3280e-03 | 2.9182   | 0.0035207 | **  |
| ## 1633 | 1.3409e-02  | 8.2804e-03 | 1.6193   | 0.1053756 |     |
| ## 1634 | -8.0821e-03 | 8.0955e-03 | -0.9983  | 0.3181123 |     |
| ## 1635 | -7.1722e-02 | 7.3352e-03 | -9.7777  | < 2.2e-16 | *** |
| ## 1636 | -4.9132e-02 | 7.6073e-03 | -6.4585  | 1.057e-10 | *** |
| ## 1637 | 5.3725e-02  | 8.7449e-03 | 6.1436   | 8.068e-10 | *** |
| ## 1638 | 2.4905e-02  | 8.3736e-03 | 2.9742   | 0.0029372 | **  |
| ## 1639 | -4.8736e-03 | 8.0933e-03 | -0.6022  | 0.5470521 |     |
| ## 1640 | 8.5209e-03  | 8.2219e-03 | 1.0364   | 0.3000295 |     |
| ## 1641 | -7.6137e-02 | 7.3080e-03 | -10.4183 | < 2.2e-16 | *** |
| ## 1642 | 7.3439e-02  | 8.8769e-03 | 8.2730   | < 2.2e-16 | *** |
| ## 1643 | 5.9192e-03  | 8.2215e-03 | 0.7200   | 0.4715413 |     |
| ## 1644 | 1.5623e-02  | 8.2924e-03 | 1.8840   | 0.0595584 | .   |
| ## 1645 | 1.3159e-02  | 8.2499e-03 | 1.5951   | 0.1106914 |     |
| ## 1646 | -1.6492e-02 | 8.0071e-03 | -2.0597  | 0.0394251 | *   |
| ## 1647 | -3.7602e-03 | 8.0733e-03 | -0.4658  | 0.6413899 |     |
| ## 1648 | -8.3093e-03 | 7.9958e-03 | -1.0392  | 0.2987087 |     |
| ## 1649 | -5.8226e-02 | 7.4776e-03 | -7.7867  | 6.876e-15 | *** |
| ## 1650 | -4.4761e-02 | 7.6763e-03 | -5.8310  | 5.509e-09 | *** |
| ## 1651 | -2.0660e-02 | 7.9336e-03 | -2.6041  | 0.0092108 | **  |

|         |             |            |          |           |     |
|---------|-------------|------------|----------|-----------|-----|
| ## 1652 | -1.7425e-03 | 8.1638e-03 | -0.2134  | 0.8309821 |     |
| ## 1653 | 2.1596e-02  | 8.4269e-03 | 2.5627   | 0.0103864 | *   |
| ## 1654 | 6.7301e-03  | 8.1529e-03 | 0.8255   | 0.4090931 |     |
| ## 1655 | 7.8147e-02  | 8.9983e-03 | 8.6846   | < 2.2e-16 | *** |
| ## 1656 | -1.7063e-02 | 8.0039e-03 | -2.1318  | 0.0330253 | *   |
| ## 1657 | 4.3090e-02  | 8.5582e-03 | 5.0349   | 4.781e-07 | *** |
| ## 1658 | 8.8161e-03  | 8.2001e-03 | 1.0751   | 0.2823173 |     |
| ## 1659 | -2.0412e-02 | 7.9356e-03 | -2.5722  | 0.0101067 | *   |
| ## 1660 | -6.3346e-02 | 7.5340e-03 | -8.4079  | < 2.2e-16 | *** |
| ## 1661 | 3.4696e-02  | 8.4941e-03 | 4.0848   | 4.412e-05 | *** |
| ## 1662 | -1.8649e-02 | 7.8474e-03 | -2.3764  | 0.0174808 | *   |
| ## 1663 | 7.6365e-02  | 8.9490e-03 | 8.5333   | < 2.2e-16 | *** |
| ## 1664 | -6.2940e-05 | 8.1467e-03 | -0.0077  | 0.9938357 |     |
| ## 1665 | 1.6274e-02  | 8.3091e-03 | 1.9586   | 0.0501548 | .   |
| ## 1666 | -3.0177e-02 | 7.8152e-03 | -3.8613  | 0.0001128 | *** |
| ## 1667 | 1.5337e-02  | 8.3119e-03 | 1.8452   | 0.0650083 | .   |
| ## 1668 | 6.5215e-03  | 8.1844e-03 | 0.7968   | 0.4255515 |     |
| ## 1669 | -2.7130e-02 | 7.8848e-03 | -3.4408  | 0.0005799 | *** |
| ## 1670 | 3.3116e-02  | 8.5715e-03 | 3.8635   | 0.0001118 | *** |
| ## 1671 | 3.7949e-02  | 8.6514e-03 | 4.3864   | 1.152e-05 | *** |
| ## 1672 | 2.0985e-02  | 8.3734e-03 | 2.5062   | 0.0122038 | *   |
| ## 1673 | -3.6336e-02 | 7.8100e-03 | -4.6525  | 3.279e-06 | *** |
| ## 1674 | 1.2004e-02  | 8.2426e-03 | 1.4563   | 0.1453085 |     |
| ## 1675 | -3.0252e-04 | 8.1368e-03 | -0.0372  | 0.9703417 |     |
| ## 1676 | -2.1243e-02 | 7.8412e-03 | -2.7092  | 0.0067453 | **  |
| ## 1677 | -2.0574e-02 | 7.9283e-03 | -2.5950  | 0.0094581 | **  |
| ## 1678 | 7.1710e-02  | 8.8923e-03 | 8.0642   | 7.371e-16 | *** |
| ## 1679 | 3.6238e-02  | 8.5223e-03 | 4.2522   | 2.117e-05 | *** |
| ## 1680 | 8.2136e-03  | 8.2451e-03 | 0.9962   | 0.3191631 |     |
| ## 1681 | -7.5053e-02 | 7.6600e-03 | -9.7981  | < 2.2e-16 | *** |
| ## 1682 | -3.1294e-02 | 7.8404e-03 | -3.9914  | 6.567e-05 | *** |
| ## 1683 | -7.1715e-02 | 7.4112e-03 | -9.6766  | < 2.2e-16 | *** |
| ## 1684 | 1.1436e-02  | 8.2551e-03 | 1.3854   | 0.1659423 |     |
| ## 1685 | 3.7149e-02  | 8.5115e-03 | 4.3646   | 1.273e-05 | *** |
| ## 1686 | -1.5272e-02 | 7.9235e-03 | -1.9274  | 0.0539278 | .   |
| ## 1687 | -1.1486e-02 | 7.9451e-03 | -1.4457  | 0.1482570 |     |
| ## 1688 | 2.8209e-02  | 8.3900e-03 | 3.3623   | 0.0007730 | *** |
| ## 1689 | 5.2799e-02  | 8.6409e-03 | 6.1104   | 9.939e-10 | *** |
| ## 1690 | 2.3113e-02  | 8.4210e-03 | 2.7448   | 0.0060557 | **  |
| ## 1691 | 6.2727e-03  | 8.2450e-03 | 0.7608   | 0.4467805 |     |
| ## 1692 | -4.5037e-02 | 7.7723e-03 | -5.7946  | 6.848e-09 | *** |
| ## 1693 | -8.3758e-02 | 7.1751e-03 | -11.6734 | < 2.2e-16 | *** |
| ## 1694 | 7.2793e-05  | 8.1124e-03 | 0.0090   | 0.9928407 |     |
| ## 1695 | -2.7754e-02 | 7.8695e-03 | -3.5268  | 0.0004206 | *** |
| ## 1696 | 9.6191e-02  | 9.1694e-03 | 10.4904  | < 2.2e-16 | *** |
| ## 1697 | 1.6949e-02  | 8.3237e-03 | 2.0362   | 0.0417285 | *   |
| ## 1698 | 2.8990e-02  | 8.4022e-03 | 3.4503   | 0.0005600 | *** |
| ## 1699 | 2.1457e-02  | 8.3957e-03 | 2.5557   | 0.0105965 | *   |

|         |             |            |         |           |     |
|---------|-------------|------------|---------|-----------|-----|
| ## 1700 | 8.6430e-03  | 8.2229e-03 | 1.0511  | 0.2932209 |     |
| ## 1701 | -2.8835e-02 | 7.8003e-03 | -3.6966 | 0.0002185 | *** |
| ## 1702 | 1.8651e-02  | 8.3507e-03 | 2.2334  | 0.0255220 | *   |
| ## 1703 | -2.0483e-04 | 8.2010e-03 | -0.0250 | 0.9800738 |     |
| ## 1704 | -5.2050e-02 | 7.5806e-03 | -6.8662 | 6.594e-12 | *** |
| ## 1705 | 8.8302e-03  | 8.2874e-03 | 1.0655  | 0.2866496 |     |
| ## 1706 | 1.7668e-02  | 8.3302e-03 | 2.1210  | 0.0339246 | *   |
| ## 1707 | 2.2281e-02  | 8.3666e-03 | 2.6631  | 0.0077429 | **  |
| ## 1708 | -1.9746e-02 | 7.8999e-03 | -2.4996 | 0.0124343 | *   |
| ## 1709 | 6.8372e-03  | 8.2226e-03 | 0.8315  | 0.4056818 |     |
| ## 1710 | -4.6733e-03 | 8.0756e-03 | -0.5787 | 0.5627954 |     |
| ## 1711 | -1.7817e-02 | 7.9481e-03 | -2.2417 | 0.0249808 | *   |
| ## 1712 | -2.7970e-02 | 7.8166e-03 | -3.5783 | 0.0003458 | *** |
| ## 1713 | 2.0217e-02  | 8.3656e-03 | 2.4167  | 0.0156608 | *   |
| ## 1714 | 2.8005e-02  | 8.4383e-03 | 3.3188  | 0.0009040 | *** |
| ## 1715 | 2.4376e-02  | 8.4015e-03 | 2.9014  | 0.0037146 | **  |
| ## 1716 | -2.1710e-02 | 7.8763e-03 | -2.7564 | 0.0058450 | **  |
| ## 1717 | 3.9133e-02  | 8.4986e-03 | 4.6046  | 4.133e-06 | *** |
| ## 1718 | -2.2988e-02 | 7.9296e-03 | -2.8991 | 0.0037429 | **  |
| ## 1719 | -2.7459e-03 | 8.1599e-03 | -0.3365 | 0.7364831 |     |
| ## 1720 | -3.8641e-03 | 8.0522e-03 | -0.4799 | 0.6313103 |     |
| ## 1721 | -4.6796e-02 | 7.6371e-03 | -6.1274 | 8.931e-10 | *** |
| ## 1722 | 1.3758e-02  | 8.2761e-03 | 1.6624  | 0.0964273 | .   |
| ## 1723 | 1.5401e-02  | 8.2455e-03 | 1.8678  | 0.0617928 | .   |
| ## 1724 | 1.2627e-02  | 8.2468e-03 | 1.5312  | 0.1257262 |     |
| ## 1725 | -1.6864e-02 | 7.9425e-03 | -2.1233 | 0.0337289 | *   |
| ## 1726 | -3.1813e-04 | 8.1140e-03 | -0.0392 | 0.9687250 |     |
| ## 1727 | 6.0041e-04  | 8.1588e-03 | 0.0736  | 0.9413362 |     |
| ## 1728 | -4.9230e-02 | 7.6782e-03 | -6.4118 | 1.439e-10 | *** |
| ## 1729 | 3.4898e-02  | 8.5293e-03 | 4.0915  | 4.286e-05 | *** |
| ## 1730 | 9.1246e-02  | 9.1385e-03 | 9.9848  | < 2.2e-16 | *** |
| ## 1731 | 4.6542e-02  | 8.5784e-03 | 5.4255  | 5.781e-08 | *** |
| ## 1732 | -6.4455e-03 | 8.1251e-03 | -0.7933 | 0.4276146 |     |
| ## 1733 | 1.1601e-02  | 8.2042e-03 | 1.4140  | 0.1573595 |     |
| ## 1734 | -3.5549e-02 | 7.8028e-03 | -4.5559 | 5.215e-06 | *** |
| ## 1735 | -1.7582e-02 | 8.0098e-03 | -2.1951 | 0.0281541 | *   |
| ## 1736 | -2.8373e-02 | 7.8684e-03 | -3.6060 | 0.0003110 | *** |
| ## 1737 | 2.5103e-02  | 8.4013e-03 | 2.9880  | 0.0028082 | **  |
| ## 1738 | -2.8293e-02 | 7.8747e-03 | -3.5929 | 0.0003270 | *** |
| ## 1739 | -1.5313e-02 | 8.0502e-03 | -1.9021 | 0.0571538 | .   |
| ## 1740 | 2.3250e-02  | 8.4061e-03 | 2.7659  | 0.0056769 | **  |
| ## 1741 | 8.1796e-02  | 9.0963e-03 | 8.9922  | < 2.2e-16 | *** |
| ## 1742 | 4.9925e-03  | 8.2265e-03 | 0.6069  | 0.5439277 |     |
| ## 1743 | 3.4148e-02  | 8.4754e-03 | 4.0291  | 5.600e-05 | *** |
| ## 1744 | 1.1405e-02  | 8.2398e-03 | 1.3841  | 0.1663257 |     |
| ## 1745 | 4.3897e-02  | 8.6282e-03 | 5.0876  | 3.627e-07 | *** |
| ## 1746 | -5.9302e-02 | 7.5179e-03 | -7.8881 | 3.068e-15 | *** |
| ## 1747 | -3.6307e-03 | 8.1144e-03 | -0.4474 | 0.6545527 |     |

|         |             |            |          |           |     |
|---------|-------------|------------|----------|-----------|-----|
| ## 1748 | 2.8352e-02  | 8.5171e-03 | 3.3288   | 0.0008721 | *** |
| ## 1749 | 2.6124e-02  | 8.3282e-03 | 3.1368   | 0.0017080 | **  |
| ## 1750 | -3.1561e-02 | 7.8262e-03 | -4.0328  | 5.512e-05 | *** |
| ## 1751 | -1.1721e-02 | 7.9792e-03 | -1.4690  | 0.1418456 |     |
| ## 1752 | -2.4465e-03 | 8.1066e-03 | -0.3018  | 0.7628121 |     |
| ## 1753 | -3.9210e-02 | 7.6734e-03 | -5.1099  | 3.223e-07 | *** |
| ## 1754 | 9.3949e-03  | 8.2237e-03 | 1.1424   | 0.2532806 |     |
| ## 1755 | 2.4208e-02  | 8.4397e-03 | 2.8684   | 0.0041260 | **  |
| ## 1756 | 7.7836e-03  | 8.1907e-03 | 0.9503   | 0.3419607 |     |
| ## 1757 | -5.9849e-03 | 8.0454e-03 | -0.7439  | 0.4569432 |     |
| ## 1758 | -2.1764e-02 | 7.9138e-03 | -2.7502  | 0.0059559 | **  |
| ## 1759 | -1.5669e-02 | 8.0051e-03 | -1.9573  | 0.0503113 | .   |
| ## 1760 | 8.5056e-02  | 9.0423e-03 | 9.4065   | < 2.2e-16 | *** |
| ## 1761 | 6.3894e-02  | 8.8360e-03 | 7.2311   | 4.792e-13 | *** |
| ## 1762 | -2.1681e-02 | 7.9207e-03 | -2.7373  | 0.0061945 | **  |
| ## 1763 | 3.0858e-02  | 8.4533e-03 | 3.6504   | 0.0002619 | *** |
| ## 1764 | -4.2735e-02 | 7.7099e-03 | -5.5429  | 2.975e-08 | *** |
| ## 1765 | -3.4373e-02 | 7.7806e-03 | -4.4177  | 9.974e-06 | *** |
| ## 1766 | 3.0573e-02  | 8.4188e-03 | 3.6316   | 0.0002817 | *** |
| ## 1767 | 2.6957e-02  | 8.3874e-03 | 3.2140   | 0.0013091 | **  |
| ## 1768 | -4.2399e-02 | 7.6300e-03 | -5.5569  | 2.746e-08 | *** |
| ## 1769 | 5.2948e-02  | 8.7243e-03 | 6.0691   | 1.287e-09 | *** |
| ## 1770 | -5.5313e-02 | 7.6144e-03 | -7.2644  | 3.748e-13 | *** |
| ## 1771 | 6.0867e-02  | 8.8221e-03 | 6.8995   | 5.220e-12 | *** |
| ## 1772 | -2.3460e-02 | 7.8672e-03 | -2.9820  | 0.0028633 | **  |
| ## 1773 | 1.9139e-02  | 8.3601e-03 | 2.2893   | 0.0220592 | *   |
| ## 1774 | 1.9787e-03  | 8.1483e-03 | 0.2428   | 0.8081295 |     |
| ## 1775 | -8.3459e-03 | 8.1418e-03 | -1.0251  | 0.3053290 |     |
| ## 1776 | -1.0746e-02 | 8.0627e-03 | -1.3328  | 0.1826028 |     |
| ## 1777 | 9.8774e-03  | 8.2233e-03 | 1.2012   | 0.2296926 |     |
| ## 1778 | 6.6780e-03  | 8.1734e-03 | 0.8170   | 0.4139070 |     |
| ## 1779 | -1.0217e-02 | 7.9810e-03 | -1.2801  | 0.2005084 |     |
| ## 1780 | 1.8127e-03  | 8.1841e-03 | 0.2215   | 0.8247086 |     |
| ## 1781 | -1.3120e-02 | 7.9584e-03 | -1.6486  | 0.0992329 | .   |
| ## 1782 | 1.4344e-03  | 8.1347e-03 | 0.1763   | 0.8600367 |     |
| ## 1783 | 4.2307e-02  | 8.5434e-03 | 4.9520   | 7.345e-07 | *** |
| ## 1784 | 1.9701e-02  | 8.3647e-03 | 2.3552   | 0.0185114 | *   |
| ## 1785 | 4.6481e-02  | 8.5656e-03 | 5.4265   | 5.748e-08 | *** |
| ## 1786 | 7.4315e-02  | 8.9604e-03 | 8.2937   | < 2.2e-16 | *** |
| ## 1787 | -2.5122e-02 | 7.8498e-03 | -3.2003  | 0.0013728 | **  |
| ## 1788 | -3.5781e-02 | 7.7750e-03 | -4.6021  | 4.183e-06 | *** |
| ## 1789 | 6.9238e-05  | 8.0963e-03 | 0.0086   | 0.9931767 |     |
| ## 1790 | 4.5779e-02  | 8.6870e-03 | 5.2698   | 1.366e-07 | *** |
| ## 1791 | -9.0040e-02 | 7.2340e-03 | -12.4468 | < 2.2e-16 | *** |
| ## 1792 | -1.5534e-03 | 8.1721e-03 | -0.1901  | 0.8492468 |     |
| ## 1793 | -6.7928e-02 | 7.4362e-03 | -9.1347  | < 2.2e-16 | *** |
| ## 1794 | 3.3160e-03  | 8.4794e-03 | 0.3911   | 0.6957468 |     |
| ## 1795 | -2.9853e-02 | 7.9644e-03 | -3.7482  | 0.0001781 | *** |



|         |             |            |         |           |     |
|---------|-------------|------------|---------|-----------|-----|
| ## 1796 | -4.1328e-03 | 8.1950e-03 | -0.5043 | 0.6140444 |     |
| ## 1797 | 1.1465e-02  | 8.2234e-03 | 1.3942  | 0.1632445 |     |
| ## 1798 | -5.2829e-02 | 7.6624e-03 | -6.8946 | 5.403e-12 | *** |
| ## 1799 | -1.6342e-02 | 7.9763e-03 | -2.0488 | 0.0404847 | *   |
| ## 1800 | -2.3595e-03 | 7.9904e-03 | -0.2953 | 0.7677702 |     |
| ## 1801 | 4.9783e-02  | 8.6572e-03 | 5.7505  | 8.898e-09 | *** |
| ## 1802 | 3.3964e-02  | 8.5266e-03 | 3.9833  | 6.798e-05 | *** |
| ## 1803 | -7.6889e-03 | 8.0106e-03 | -0.9598 | 0.3371347 |     |
| ## 1804 | -3.5707e-02 | 7.7978e-03 | -4.5791 | 4.669e-06 | *** |
| ## 1805 | -1.2642e-02 | 8.0225e-03 | -1.5758 | 0.1150804 |     |
| ## 1806 | -2.5785e-02 | 7.8793e-03 | -3.2725 | 0.0010658 | **  |
| ## 1807 | 2.2320e-02  | 8.3912e-03 | 2.6600  | 0.0078151 | **  |
| ## 1808 | 4.5935e-02  | 8.6094e-03 | 5.3354  | 9.534e-08 | *** |
| ## 1809 | 6.1066e-02  | 8.7627e-03 | 6.9688  | 3.195e-12 | *** |
| ## 1810 | 8.2815e-03  | 8.2903e-03 | 0.9989  | 0.3178268 |     |
| ## 1811 | 6.0974e-03  | 8.2167e-03 | 0.7421  | 0.4580431 |     |
| ## 1812 | 2.8050e-02  | 8.4922e-03 | 3.3031  | 0.0009563 | *** |
| ## 1813 | -5.7957e-02 | 7.5488e-03 | -7.6776 | 1.621e-14 | *** |
| ## 1814 | 3.0495e-02  | 8.4543e-03 | 3.6070  | 0.0003098 | *** |
| ## 1815 | -2.4779e-03 | 8.1916e-03 | -0.3025 | 0.7622802 |     |
| ## 1816 | 9.9953e-02  | 9.2346e-03 | 10.8238 | < 2.2e-16 | *** |
| ## 1817 | -2.0786e-02 | 7.9699e-03 | -2.6081 | 0.0091048 | **  |
| ## 1818 | 8.3246e-02  | 9.0398e-03 | 9.2088  | < 2.2e-16 | *** |
| ## 1819 | -2.2771e-02 | 7.9177e-03 | -2.8760 | 0.0040275 | **  |
| ## 1820 | 8.0804e-03  | 8.1820e-03 | 0.9876  | 0.3233596 |     |
| ## 1821 | -2.7389e-02 | 7.8620e-03 | -3.4838 | 0.0004944 | *** |
| ## 1822 | -4.7971e-02 | 7.6619e-03 | -6.2609 | 3.826e-10 | *** |
| ## 1823 | 2.9783e-02  | 8.4937e-03 | 3.5065  | 0.0004540 | *** |
| ## 1824 | -1.4617e-02 | 8.0165e-03 | -1.8234 | 0.0682458 | .   |
| ## 1825 | 1.6793e-02  | 8.3523e-03 | 2.0106  | 0.0443680 | *   |
| ## 1826 | 2.1538e-02  | 8.4076e-03 | 2.5617  | 0.0104169 | *   |
| ## 1827 | -6.0313e-03 | 8.0222e-03 | -0.7518 | 0.4521578 |     |
| ## 1828 | 1.1665e-01  | 9.4334e-03 | 12.3654 | < 2.2e-16 | *** |
| ## 1829 | -3.9574e-02 | 7.7628e-03 | -5.0979 | 3.435e-07 | *** |
| ## 1830 | 4.8180e-03  | 8.1904e-03 | 0.5883  | 0.5563591 |     |
| ## 1831 | 3.4276e-02  | 8.4933e-03 | 4.0357  | 5.445e-05 | *** |
| ## 1832 | 5.7506e-02  | 8.7297e-03 | 6.5874  | 4.475e-11 | *** |
| ## 1833 | -5.0459e-02 | 7.6667e-03 | -6.5816 | 4.654e-11 | *** |
| ## 1834 | -3.6858e-02 | 7.7692e-03 | -4.7441 | 2.094e-06 | *** |
| ## 1835 | 5.2457e-02  | 8.7863e-03 | 5.9703  | 2.368e-09 | *** |
| ## 1836 | 5.5587e-02  | 8.6041e-03 | 6.4605  | 1.043e-10 | *** |
| ## 1837 | 4.2325e-02  | 8.5421e-03 | 4.9549  | 7.237e-07 | *** |
| ## 1838 | -6.9120e-02 | 7.4358e-03 | -9.2956 | < 2.2e-16 | *** |
| ## 1839 | -3.2708e-02 | 7.8197e-03 | -4.1828 | 2.880e-05 | *** |
| ## 1840 | 6.0976e-02  | 8.8250e-03 | 6.9094  | 4.866e-12 | *** |
| ## 1841 | 1.0394e-01  | 9.2726e-03 | 11.2095 | < 2.2e-16 | *** |
| ## 1842 | 9.7088e-02  | 9.2290e-03 | 10.5198 | < 2.2e-16 | *** |
| ## 1843 | -1.9416e-02 | 7.9385e-03 | -2.4458 | 0.0144540 | *   |

|         |             |            |                   |     |
|---------|-------------|------------|-------------------|-----|
| ## 1844 | 1.1209e-01  | 9.3784e-03 | 11.9519 < 2.2e-16 | *** |
| ## 1845 | 7.2764e-02  | 8.9463e-03 | 8.1334 4.176e-16  | *** |
| ## 1846 | -5.2369e-03 | 8.0942e-03 | -0.6470 0.5176373 |     |
| ## 1847 | -6.5211e-02 | 7.4483e-03 | -8.7552 < 2.2e-16 | *** |
| ## 1848 | 6.8346e-02  | 8.7883e-03 | 7.7769 7.432e-15  | *** |
| ## 1849 | 1.1358e-02  | 8.3441e-03 | 1.3613 0.1734335  |     |
| ## 1850 | -2.8901e-02 | 7.8731e-03 | -3.6708 0.0002418 | *** |
| ## 1851 | 6.6026e-02  | 8.8334e-03 | 7.4745 7.747e-14  | *** |
| ## 1852 | 1.7689e-02  | 8.3350e-03 | 2.1222 0.0338196  | *   |
| ## 1853 | 3.0004e-02  | 8.4768e-03 | 3.5396 0.0004008  | *** |
| ## 1854 | -2.9735e-02 | 7.8286e-03 | -3.7983 0.0001457 | *** |
| ## 1855 | -2.7338e-02 | 7.7982e-03 | -3.5057 0.0004554 | *** |
| ## 1856 | -4.9191e-02 | 7.5864e-03 | -6.4840 8.930e-11 | *** |
| ## 1857 | 1.3228e-01  | 9.6265e-03 | 13.7416 < 2.2e-16 | *** |
| ## 1858 | 7.4436e-02  | 8.8962e-03 | 8.3673 < 2.2e-16  | *** |
| ## 1859 | 2.0786e-03  | 8.1253e-03 | 0.2558 0.7980934  |     |
| ## 1860 | -2.4693e-02 | 7.8512e-03 | -3.1451 0.0016603 | **  |
| ## 1861 | 6.3482e-02  | 8.7738e-03 | 7.2354 4.641e-13  | *** |
| ## 1862 | -2.9792e-03 | 8.1049e-03 | -0.3676 0.7131897 |     |
| ## 1863 | 1.3300e-01  | 9.6081e-03 | 13.8423 < 2.2e-16 | *** |
| ## 1864 | 1.0959e-01  | 9.3835e-03 | 11.6793 < 2.2e-16 | *** |
| ## 1865 | -2.5856e-02 | 7.8334e-03 | -3.3008 0.0009641 | *** |
| ## 1866 | 5.9343e-02  | 8.8683e-03 | 6.6916 2.208e-11  | *** |
| ## 1867 | 1.6686e-02  | 8.3555e-03 | 1.9970 0.0458230  | *   |
| ## 1868 | 6.8451e-02  | 8.9167e-03 | 7.6767 1.632e-14  | *** |
| ## 1869 | 6.0702e-02  | 8.7095e-03 | 6.9696 3.178e-12  | *** |
| ## 1870 | -3.4264e-02 | 7.6360e-03 | -4.4873 7.215e-06 | *** |
| ## 1871 | -2.4977e-02 | 7.9017e-03 | -3.1609 0.0015727 | **  |
| ## 1872 | 9.2314e-03  | 8.2772e-03 | 1.1153 0.2647290  |     |
| ## 1873 | 5.4174e-02  | 8.4897e-03 | 6.3812 1.757e-10  | *** |
| ## 1874 | 9.1736e-04  | 8.1549e-03 | 0.1125 0.9104335  |     |
| ## 1875 | -6.8110e-02 | 7.4900e-03 | -9.0934 < 2.2e-16 | *** |
| ## 1876 | 1.9921e-02  | 8.3414e-03 | 2.3882 0.0169310  | *   |
| ## 1877 | -5.2044e-02 | 7.6544e-03 | -6.7993 1.052e-11 | *** |
| ## 1878 | -4.3922e-02 | 7.7533e-03 | -5.6649 1.471e-08 | *** |
| ## 1879 | 2.2117e-03  | 8.1396e-03 | 0.2717 0.7858398  |     |
| ## 1880 | 8.0622e-03  | 8.2139e-03 | 0.9815 0.3263317  |     |
| ## 1881 | 7.5385e-03  | 8.2363e-03 | 0.9153 0.3600460  |     |
| ## 1882 | -3.3200e-02 | 7.8151e-03 | -4.2483 2.154e-05 | *** |
| ## 1883 | 8.0540e-03  | 8.1798e-03 | 0.9846 0.3248074  |     |
| ## 1884 | 2.8584e-02  | 8.4617e-03 | 3.3781 0.0007299  | *** |
| ## 1885 | 1.4844e-02  | 8.3062e-03 | 1.7871 0.0739281  | .   |
| ## 1886 | -1.3422e-02 | 8.0152e-03 | -1.6746 0.0940199 | .   |
| ## 1887 | -1.0457e-02 | 7.9389e-03 | -1.3172 0.1877830 |     |
| ## 1888 | 1.0992e-01  | 9.3140e-03 | 11.8015 < 2.2e-16 | *** |
| ## 1889 | 2.8857e-03  | 8.0987e-03 | 0.3563 0.7216011  |     |
| ## 1890 | 5.3957e-02  | 8.6923e-03 | 6.2075 5.385e-10  | *** |
| ## 1891 | 8.2181e-02  | 9.0643e-03 | 9.0664 < 2.2e-16  | *** |

|         |             |            |         |           |     |
|---------|-------------|------------|---------|-----------|-----|
| ## 1892 | 8.8515e-04  | 8.1262e-03 | 0.1089  | 0.9132614 |     |
| ## 1893 | -2.0468e-02 | 7.8665e-03 | -2.6019 | 0.0092714 | **  |
| ## 1894 | -5.4425e-02 | 7.5168e-03 | -7.2405 | 4.470e-13 | *** |
| ## 1895 | 4.2282e-02  | 8.5687e-03 | 4.9344  | 8.038e-07 | *** |
| ## 1896 | 5.7206e-02  | 8.9080e-03 | 6.4219  | 1.346e-10 | *** |
| ## 1897 | 5.4025e-02  | 8.7530e-03 | 6.1722  | 6.736e-10 | *** |
| ## 1898 | 8.3984e-02  | 9.1103e-03 | 9.2186  | < 2.2e-16 | *** |
| ## 1899 | 3.3764e-02  | 8.5329e-03 | 3.9570  | 7.591e-05 | *** |
| ## 1900 | -4.1485e-03 | 8.1181e-03 | -0.5110 | 0.6093353 |     |
| ## 1901 | 1.1656e-01  | 9.3855e-03 | 12.4194 | < 2.2e-16 | *** |
| ## 1902 | 4.9777e-02  | 8.7243e-03 | 5.7056  | 1.160e-08 | *** |
| ## 1903 | -4.6744e-02 | 7.5792e-03 | -6.1674 | 6.942e-10 | *** |
| ## 1904 | -1.9575e-02 | 7.8135e-03 | -2.5053 | 0.0122354 | *   |
| ## 1905 | -2.0971e-02 | 7.8647e-03 | -2.6665 | 0.0076647 | **  |
| ## 1906 | 4.3019e-04  | 8.1406e-03 | 0.0528  | 0.9578555 |     |
| ## 1907 | 4.4484e-02  | 8.6261e-03 | 5.1568  | 2.512e-07 | *** |
| ## 1908 | -4.4945e-02 | 7.6489e-03 | -5.8760 | 4.204e-09 | *** |
| ## 1909 | -5.6855e-02 | 7.4779e-03 | -7.6031 | 2.891e-14 | *** |
| ## 1910 | -1.9562e-02 | 7.8962e-03 | -2.4774 | 0.0132345 | *   |
| ## 1911 | 2.6941e-02  | 8.4626e-03 | 3.1836  | 0.0014546 | **  |
| ## 1912 | 5.9443e-02  | 8.7538e-03 | 6.7906  | 1.117e-11 | *** |
| ## 1913 | -4.3370e-02 | 7.6666e-03 | -5.6570 | 1.540e-08 | *** |
| ## 1914 | 2.3076e-02  | 8.3732e-03 | 2.7560  | 0.0058520 | **  |
| ## 1915 | 4.5523e-02  | 8.5910e-03 | 5.2989  | 1.165e-07 | *** |
| ## 1916 | 4.8598e-03  | 8.1979e-03 | 0.5928  | 0.5533112 |     |
| ## 1917 | -4.2737e-03 | 8.1275e-03 | -0.5258 | 0.5990005 |     |
| ## 1918 | 6.9315e-02  | 8.9539e-03 | 7.7413  | 9.840e-15 | *** |
| ## 1919 | 3.2154e-02  | 8.4784e-03 | 3.7924  | 0.0001492 | *** |
| ## 1920 | -1.5195e-02 | 8.0563e-03 | -1.8861 | 0.0592751 | .   |
| ## 1921 | 2.1581e-02  | 8.4984e-03 | 2.5394  | 0.0111043 | *   |
| ## 1922 | 1.0969e-02  | 8.2672e-03 | 1.3268  | 0.1845819 |     |
| ## 1923 | 4.4671e-02  | 8.7485e-03 | 5.1061  | 3.289e-07 | *** |
| ## 1924 | 1.7048e-02  | 8.3698e-03 | 2.0368  | 0.0416696 | *   |
| ## 1925 | -1.8773e-02 | 7.9414e-03 | -2.3639 | 0.0180846 | *   |
| ## 1926 | -4.9727e-02 | 7.6176e-03 | -6.5279 | 6.669e-11 | *** |
| ## 1927 | -1.7341e-02 | 7.9609e-03 | -2.1782 | 0.0293892 | *   |
| ## 1928 | -1.4691e-02 | 7.9436e-03 | -1.8494 | 0.0644052 | .   |
| ## 1929 | -2.7429e-02 | 7.8026e-03 | -3.5153 | 0.0004392 | *** |
| ## 1930 | 1.2140e-01  | 9.4411e-03 | 12.8583 | < 2.2e-16 | *** |
| ## 1931 | 1.7779e-02  | 8.5279e-03 | 2.0848  | 0.0370907 | *   |
| ## 1932 | 9.3950e-02  | 9.2156e-03 | 10.1947 | < 2.2e-16 | *** |
| ## 1933 | 3.3720e-02  | 8.4441e-03 | 3.9933  | 6.515e-05 | *** |
| ## 1934 | 6.7088e-02  | 8.7553e-03 | 7.6626  | 1.823e-14 | *** |
| ## 1935 | 3.0796e-02  | 8.5390e-03 | 3.6065  | 0.0003104 | *** |
| ## 1936 | 2.1073e-02  | 8.3548e-03 | 2.5222  | 0.0116610 | *   |
| ## 1937 | -3.6980e-02 | 7.6996e-03 | -4.8029 | 1.564e-06 | *** |
| ## 1938 | 1.5623e-02  | 8.4206e-03 | 1.8554  | 0.0635448 | .   |
| ## 1939 | -4.3434e-02 | 7.9487e-03 | -5.4643 | 4.648e-08 | *** |

|         |             |            |          |           |     |
|---------|-------------|------------|----------|-----------|-----|
| ## 1940 | -5.7447e-02 | 7.5574e-03 | -7.6015  | 2.927e-14 | *** |
| ## 1941 | -4.8879e-02 | 7.6241e-03 | -6.4111  | 1.445e-10 | *** |
| ## 1942 | -2.0092e-02 | 7.9416e-03 | -2.5300  | 0.0114054 | *   |
| ## 1943 | 5.7980e-02  | 8.7288e-03 | 6.6424   | 3.087e-11 | *** |
| ## 1944 | 1.6890e-02  | 8.3232e-03 | 2.0293   | 0.0424264 | *   |
| ## 1945 | 1.3746e-02  | 8.2243e-03 | 1.6714   | 0.0946518 | .   |
| ## 1946 | 2.1501e-02  | 8.3742e-03 | 2.5675   | 0.0102438 | *   |
| ## 1947 | -2.7446e-02 | 7.8749e-03 | -3.4852  | 0.0004918 | *** |
| ## 1948 | 3.9018e-02  | 8.5776e-03 | 4.5488   | 5.395e-06 | *** |
| ## 1949 | 1.5580e-02  | 8.2745e-03 | 1.8830   | 0.0597065 | .   |
| ## 1950 | -5.0795e-03 | 8.0855e-03 | -0.6282  | 0.5298566 |     |
| ## 1951 | -5.2325e-02 | 7.6049e-03 | -6.8804  | 5.968e-12 | *** |
| ## 1952 | 1.8744e-02  | 8.3558e-03 | 2.2432   | 0.0248838 | *   |
| ## 1953 | 1.4095e-01  | 9.7307e-03 | 14.4851  | < 2.2e-16 | *** |
| ## 1954 | 9.0781e-04  | 8.1619e-03 | 0.1112   | 0.9114380 |     |
| ## 1955 | 2.3637e-02  | 8.3553e-03 | 2.8290   | 0.0046699 | **  |
| ## 1956 | 5.7244e-03  | 8.2695e-03 | 0.6922   | 0.4887895 |     |
| ## 1957 | 3.8020e-02  | 8.5166e-03 | 4.4642   | 8.037e-06 | *** |
| ## 1958 | 4.0870e-02  | 8.5129e-03 | 4.8009   | 1.579e-06 | *** |
| ## 1959 | 1.7342e-02  | 8.3170e-03 | 2.0851   | 0.0370604 | *   |
| ## 1960 | 5.7867e-02  | 8.8636e-03 | 6.5286   | 6.639e-11 | *** |
| ## 1961 | 6.2294e-03  | 8.2653e-03 | 0.7537   | 0.4510420 |     |
| ## 1962 | 1.2161e-02  | 8.2938e-03 | 1.4663   | 0.1425570 |     |
| ## 1963 | 4.8136e-02  | 8.6313e-03 | 5.5769   | 2.448e-08 | *** |
| ## 1964 | 6.6188e-02  | 8.7993e-03 | 7.5220   | 5.395e-14 | *** |
| ## 1965 | 9.2826e-02  | 9.1270e-03 | 10.1705  | < 2.2e-16 | *** |
| ## 1966 | 2.1311e-02  | 8.4616e-03 | 2.5185   | 0.0117848 | *   |
| ## 1967 | -4.9730e-02 | 7.5921e-03 | -6.5503  | 5.744e-11 | *** |
| ## 1968 | 3.5416e-02  | 8.5187e-03 | 4.1575   | 3.218e-05 | *** |
| ## 1969 | -1.3175e-02 | 7.9973e-03 | -1.6474  | 0.0994781 | .   |
| ## 1970 | 5.1408e-02  | 8.6955e-03 | 5.9120   | 3.379e-09 | *** |
| ## 1971 | -7.4832e-02 | 7.4241e-03 | -10.0795 | < 2.2e-16 | *** |
| ## 1972 | -5.2944e-02 | 7.5649e-03 | -6.9987  | 2.584e-12 | *** |
| ## 1973 | 5.9733e-03  | 8.2500e-03 | 0.7240   | 0.4690474 |     |
| ## 1974 | -9.7608e-03 | 8.1235e-03 | -1.2016  | 0.2295366 |     |
| ## 1975 | -7.7319e-02 | 7.3988e-03 | -10.4502 | < 2.2e-16 | *** |
| ## 1976 | 3.1513e-02  | 8.4537e-03 | 3.7277   | 0.0001932 | *** |
| ## 1977 | -1.9650e-02 | 7.9453e-03 | -2.4732  | 0.0133905 | *   |
| ## 1978 | -6.1585e-02 | 7.4876e-03 | -8.2249  | < 2.2e-16 | *** |
| ## 1979 | -5.5617e-02 | 7.5477e-03 | -7.3688  | 1.722e-13 | *** |
| ## 1980 | 7.3829e-02  | 8.9285e-03 | 8.2689   | < 2.2e-16 | *** |
| ## 1981 | -9.7053e-03 | 8.0521e-03 | -1.2053  | 0.2280816 |     |
| ## 1982 | -1.2325e-02 | 7.9760e-03 | -1.5452  | 0.1222880 |     |
| ## 1983 | 4.5782e-02  | 8.6163e-03 | 5.3134   | 1.076e-07 | *** |
| ## 1984 | -7.5038e-02 | 7.4318e-03 | -10.0969 | < 2.2e-16 | *** |
| ## 1985 | 2.1429e-02  | 8.2667e-03 | 2.5922   | 0.0095356 | **  |
| ## 1986 | -1.3792e-03 | 8.1152e-03 | -0.1700  | 0.8650493 |     |
| ## 1987 | 3.3935e-02  | 8.4781e-03 | 4.0027   | 6.262e-05 | *** |

|         |             |            |          |           |     |
|---------|-------------|------------|----------|-----------|-----|
| ## 1988 | 1.6536e-02  | 8.2799e-03 | 1.9971   | 0.0458121 | *   |
| ## 1989 | -3.4364e-02 | 7.7953e-03 | -4.4083  | 1.042e-05 | *** |
| ## 1990 | 1.2390e-02  | 8.2302e-03 | 1.5054   | 0.1322205 |     |
| ## 1991 | -1.1636e-02 | 7.9914e-03 | -1.4561  | 0.1453735 |     |
| ## 1992 | 6.7263e-02  | 8.8980e-03 | 7.5593   | 4.051e-14 | *** |
| ## 1993 | -4.9630e-02 | 7.6288e-03 | -6.5056  | 7.741e-11 | *** |
| ## 1994 | -2.6861e-03 | 8.1014e-03 | -0.3316  | 0.7402205 |     |
| ## 1995 | 4.6174e-03  | 8.0513e-03 | 0.5735   | 0.5663057 |     |
| ## 1996 | -2.5545e-03 | 8.0765e-03 | -0.3163  | 0.7517822 |     |
| ## 1997 | -1.5329e-02 | 7.9290e-03 | -1.9333  | 0.0532006 | .   |
| ## 1998 | -4.3714e-02 | 7.7272e-03 | -5.6572  | 1.539e-08 | *** |
| ## 1999 | 2.3871e-02  | 8.3597e-03 | 2.8555   | 0.0042964 | **  |
| ## 2000 | 6.8303e-03  | 8.2247e-03 | 0.8305   | 0.4062776 |     |
| ## 2001 | -4.6915e-02 | 7.6967e-03 | -6.0954  | 1.091e-09 | *** |
| ## 2002 | 1.1613e-01  | 9.4474e-03 | 12.2924  | < 2.2e-16 | *** |
| ## 2003 | -1.8591e-02 | 8.0239e-03 | -2.3170  | 0.0205025 | *   |
| ## 2004 | -5.6918e-02 | 7.5631e-03 | -7.5257  | 5.245e-14 | *** |
| ## 2005 | -7.9828e-02 | 7.2733e-03 | -10.9755 | < 2.2e-16 | *** |
| ## 2006 | -3.3349e-02 | 7.5558e-03 | -4.4137  | 1.016e-05 | *** |
| ## 2007 | 3.1677e-05  | 8.1125e-03 | 0.0039   | 0.9968845 |     |
| ## 2008 | 2.2728e-02  | 8.4292e-03 | 2.6963   | 0.0070113 | **  |
| ## 2009 | 1.1903e-02  | 8.3074e-03 | 1.4328   | 0.1519233 |     |
| ## 2010 | 2.9894e-02  | 8.4680e-03 | 3.5302   | 0.0004153 | *** |
| ## 2011 | -3.0923e-02 | 7.8544e-03 | -3.9371  | 8.247e-05 | *** |
| ## 2012 | 1.1054e-04  | 8.1242e-03 | 0.0136   | 0.9891443 |     |
| ## 2013 | 2.9915e-02  | 8.4677e-03 | 3.5329   | 0.0004111 | *** |
| ## 2014 | 6.5319e-02  | 8.8876e-03 | 7.3495   | 1.990e-13 | *** |
| ## 2015 | -4.5160e-03 | 8.1004e-03 | -0.5575  | 0.5771867 |     |
| ## 2016 | 6.0153e-02  | 8.8158e-03 | 6.8234   | 8.891e-12 | *** |
| ## 2017 | 2.0209e-02  | 8.3241e-03 | 2.4277   | 0.0151934 | *   |
| ## 2018 | 5.5211e-02  | 8.7228e-03 | 6.3296   | 2.459e-10 | *** |
| ## 2019 | 1.0105e-01  | 9.2598e-03 | 10.9123  | < 2.2e-16 | *** |
| ## 2020 | -2.5133e-02 | 7.8692e-03 | -3.1939  | 0.0014038 | **  |
| ## 2021 | 5.2796e-02  | 8.7857e-03 | 6.0093   | 1.864e-09 | *** |
| ## 2022 | 2.0809e-03  | 8.0671e-03 | 0.2580   | 0.7964430 |     |
| ## 2023 | 1.4062e-01  | 9.7014e-03 | 14.4948  | < 2.2e-16 | *** |
| ## 2024 | -2.3173e-02 | 7.8899e-03 | -2.9370  | 0.0033137 | **  |
| ## 2025 | 7.8077e-02  | 9.0199e-03 | 8.6561   | < 2.2e-16 | *** |
| ## 2026 | -4.5781e-02 | 7.7088e-03 | -5.9388  | 2.871e-09 | *** |
| ## 2027 | -2.3875e-02 | 7.8452e-03 | -3.0433  | 0.0023400 | **  |
| ## 2028 | 4.1588e-02  | 8.6312e-03 | 4.8184   | 1.447e-06 | *** |
| ## 2029 | -2.9241e-02 | 7.7867e-03 | -3.7553  | 0.0001732 | *** |
| ## 2030 | 2.9200e-02  | 8.3526e-03 | 3.4959   | 0.0004725 | *** |
| ## 2031 | 4.0614e-02  | 8.5535e-03 | 4.7482   | 2.052e-06 | *** |
| ## 2032 | 6.4442e-02  | 8.8497e-03 | 7.2818   | 3.294e-13 | *** |
| ## 2033 | -1.7952e-02 | 7.9894e-03 | -2.2470  | 0.0246402 | *   |
| ## 2034 | -3.4037e-02 | 7.8827e-03 | -4.3179  | 1.575e-05 | *** |
| ## 2035 | -2.1045e-02 | 7.9156e-03 | -2.6587  | 0.0078438 | **  |

|    |      |             |            |          |           |     |
|----|------|-------------|------------|----------|-----------|-----|
| ## | 2036 | -1.1387e-02 | 7.9574e-03 | -1.4310  | 0.1524234 |     |
| ## | 2037 | -1.9232e-02 | 8.0027e-03 | -2.4032  | 0.0162511 | *   |
| ## | 2038 | -6.3469e-02 | 7.4167e-03 | -8.5576  | < 2.2e-16 | *** |
| ## | 2039 | 6.0232e-02  | 8.7459e-03 | 6.8869   | 5.703e-12 | *** |
| ## | 2040 | 3.7419e-02  | 8.6626e-03 | 4.3196   | 1.563e-05 | *** |
| ## | 2041 | 8.1046e-02  | 9.0475e-03 | 8.9578   | < 2.2e-16 | *** |
| ## | 2042 | -8.0060e-04 | 8.1679e-03 | -0.0980  | 0.9219172 |     |
| ## | 2043 | 4.1930e-03  | 8.1283e-03 | 0.5159   | 0.6059522 |     |
| ## | 2044 | -4.3926e-04 | 8.1109e-03 | -0.0542  | 0.9568099 |     |
| ## | 2045 | -5.4453e-02 | 7.6359e-03 | -7.1311  | 9.955e-13 | *** |
| ## | 2046 | 5.0935e-02  | 8.6573e-03 | 5.8834   | 4.018e-09 | *** |
| ## | 2047 | -1.0286e-02 | 8.0442e-03 | -1.2787  | 0.2009926 |     |
| ## | 2048 | 6.4873e-02  | 8.8391e-03 | 7.3393   | 2.146e-13 | *** |
| ## | 2049 | 5.5641e-02  | 8.7493e-03 | 6.3595   | 2.025e-10 | *** |
| ## | 2050 | -5.2711e-02 | 7.6339e-03 | -6.9048  | 5.029e-12 | *** |
| ## | 2051 | -9.8678e-03 | 7.9728e-03 | -1.2377  | 0.2158348 |     |
| ## | 2052 | -9.5351e-03 | 8.0703e-03 | -1.1815  | 0.2374025 |     |
| ## | 2053 | 9.4419e-02  | 9.2518e-03 | 10.2055  | < 2.2e-16 | *** |
| ## | 2054 | 4.4524e-02  | 8.6035e-03 | 5.1751   | 2.278e-07 | *** |
| ## | 2055 | -3.9958e-02 | 7.7148e-03 | -5.1794  | 2.226e-07 | *** |
| ## | 2056 | -4.6256e-02 | 7.6963e-03 | -6.0102  | 1.853e-09 | *** |
| ## | 2057 | -5.0485e-03 | 8.1554e-03 | -0.6190  | 0.5358904 |     |
| ## | 2058 | -1.7770e-02 | 7.9542e-03 | -2.2340  | 0.0254827 | *   |
| ## | 2059 | 5.1906e-02  | 8.7581e-03 | 5.9266   | 3.093e-09 | *** |
| ## | 2060 | 4.1155e-02  | 8.5315e-03 | 4.8239   | 1.408e-06 | *** |
| ## | 2061 | 2.6722e-02  | 8.4480e-03 | 3.1631   | 0.0015610 | **  |
| ## | 2062 | 1.8616e-02  | 8.3397e-03 | 2.2322   | 0.0256043 | *   |
| ## | 2063 | -1.1463e-02 | 8.0574e-03 | -1.4226  | 0.1548455 |     |
| ## | 2064 | 3.6814e-03  | 8.1565e-03 | 0.4513   | 0.6517438 |     |
| ## | 2065 | 4.0321e-02  | 8.5081e-03 | 4.7391   | 2.147e-06 | *** |
| ## | 2066 | -1.5845e-02 | 7.9507e-03 | -1.9929  | 0.0462712 | *   |
| ## | 2067 | 6.2622e-04  | 8.1371e-03 | 0.0770   | 0.9386563 |     |
| ## | 2068 | -6.4304e-02 | 7.4852e-03 | -8.5907  | < 2.2e-16 | *** |
| ## | 2069 | -3.6901e-02 | 7.7306e-03 | -4.7734  | 1.812e-06 | *** |
| ## | 2070 | 5.6663e-02  | 8.7337e-03 | 6.4879   | 8.706e-11 | *** |
| ## | 2071 | 2.6332e-03  | 8.3540e-03 | 0.3152   | 0.7526089 |     |
| ## | 2072 | -1.2790e-02 | 8.0200e-03 | -1.5947  | 0.1107714 |     |
| ## | 2073 | -8.1256e-02 | 7.3247e-03 | -11.0935 | < 2.2e-16 | *** |
| ## | 2074 | -2.5662e-02 | 7.9023e-03 | -3.2475  | 0.0011643 | **  |
| ## | 2075 | -2.1580e-02 | 8.0262e-03 | -2.6887  | 0.0071737 | **  |
| ## | 2076 | 4.7134e-02  | 8.5690e-03 | 5.5005   | 3.787e-08 | *** |
| ## | 2077 | -7.0545e-02 | 7.4094e-03 | -9.5210  | < 2.2e-16 | *** |
| ## | 2078 | 3.3710e-02  | 8.5763e-03 | 3.9306   | 8.473e-05 | *** |
| ## | 2079 | 5.9542e-02  | 8.7504e-03 | 6.8046   | 1.014e-11 | *** |
| ## | 2080 | 4.3238e-02  | 8.6030e-03 | 5.0259   | 5.012e-07 | *** |
| ## | 2081 | 3.7821e-02  | 8.4898e-03 | 4.4548   | 8.397e-06 | *** |
| ## | 2082 | 7.1651e-02  | 8.9757e-03 | 7.9827   | 1.431e-15 | *** |
| ## | 2083 | 7.2999e-03  | 8.2042e-03 | 0.8898   | 0.3735858 |     |

|         |             |            |         |           |     |
|---------|-------------|------------|---------|-----------|-----|
| ## 2084 | -3.6287e-02 | 7.8066e-03 | -4.6482 | 3.349e-06 | *** |
| ## 2085 | 5.2054e-02  | 8.6793e-03 | 5.9974  | 2.005e-09 | *** |
| ## 2086 | 3.5309e-02  | 8.5695e-03 | 4.1203  | 3.784e-05 | *** |
| ## 2087 | 7.0067e-02  | 8.9083e-03 | 7.8653  | 3.682e-15 | *** |
| ## 2088 | 8.4210e-02  | 9.0448e-03 | 9.3103  | < 2.2e-16 | *** |
| ## 2089 | -3.2694e-02 | 7.7999e-03 | -4.1917 | 2.769e-05 | *** |
| ## 2090 | -3.3576e-02 | 7.8295e-03 | -4.2884 | 1.800e-05 | *** |
| ## 2091 | -6.4839e-02 | 7.4199e-03 | -8.7386 | < 2.2e-16 | *** |
| ## 2092 | -1.6029e-02 | 8.0468e-03 | -1.9920 | 0.0463698 | *   |
| ## 2093 | -3.9823e-02 | 7.7292e-03 | -5.1522 | 2.574e-07 | *** |
| ## 2094 | 1.0958e-02  | 8.2003e-03 | 1.3362  | 0.1814722 |     |
| ## 2095 | 1.9505e-03  | 8.1171e-03 | 0.2403  | 0.8100992 |     |
| ## 2096 | 2.9940e-02  | 8.4763e-03 | 3.5322  | 0.0004122 | *** |
| ## 2097 | 3.3951e-02  | 8.4901e-03 | 3.9989  | 6.363e-05 | *** |
| ## 2098 | -2.0487e-03 | 7.9193e-03 | -0.2587 | 0.7958695 |     |
| ## 2099 | -2.7383e-03 | 8.1464e-03 | -0.3361 | 0.7367694 |     |
| ## 2100 | 2.5051e-02  | 8.4389e-03 | 2.9685  | 0.0029926 | **  |
| ## 2101 | 2.2154e-02  | 8.3361e-03 | 2.6576  | 0.0078701 | **  |
| ## 2102 | 6.7811e-02  | 8.8118e-03 | 7.6955  | 1.409e-14 | *** |
| ## 2103 | 1.5148e-02  | 8.2833e-03 | 1.8287  | 0.0674407 | .   |
| ## 2104 | 1.0432e-01  | 9.3003e-03 | 11.2165 | < 2.2e-16 | *** |
| ## 2105 | -8.9194e-03 | 8.0088e-03 | -1.1137 | 0.2654113 |     |
| ## 2106 | -3.6932e-02 | 7.7522e-03 | -4.7641 | 1.897e-06 | *** |
| ## 2107 | 2.9560e-02  | 8.5117e-03 | 3.4728  | 0.0005150 | *** |
| ## 2108 | 8.5518e-03  | 8.1870e-03 | 1.0446  | 0.2962267 |     |
| ## 2109 | -1.2791e-02 | 7.9415e-03 | -1.6107 | 0.1072544 |     |
| ## 2110 | 7.3247e-03  | 8.2015e-03 | 0.8931  | 0.3718073 |     |
| ## 2111 | -3.2191e-02 | 7.8136e-03 | -4.1198 | 3.791e-05 | *** |
| ## 2112 | 5.0120e-02  | 8.6437e-03 | 5.7985  | 6.690e-09 | *** |
| ## 2113 | -1.0814e-02 | 8.0355e-03 | -1.3457 | 0.1783898 |     |
| ## 2114 | -2.7958e-02 | 7.7609e-03 | -3.6025 | 0.0003152 | *** |
| ## 2115 | 1.0008e-02  | 8.1726e-03 | 1.2246  | 0.2207347 |     |
| ## 2116 | -2.7924e-03 | 8.1421e-03 | -0.3430 | 0.7316299 |     |
| ## 2117 | 2.0235e-02  | 8.3833e-03 | 2.4138  | 0.0157885 | *   |
| ## 2118 | 1.2046e-01  | 9.4704e-03 | 12.7198 | < 2.2e-16 | *** |
| ## 2119 | -7.0062e-02 | 7.3607e-03 | -9.5184 | < 2.2e-16 | *** |
| ## 2120 | -3.5347e-02 | 7.7597e-03 | -4.5552 | 5.234e-06 | *** |
| ## 2121 | 9.1956e-03  | 8.2001e-03 | 1.1214  | 0.2621185 |     |
| ## 2122 | -9.8557e-03 | 7.9355e-03 | -1.2420 | 0.2142457 |     |
| ## 2123 | -2.3396e-03 | 8.1296e-03 | -0.2878 | 0.7735122 |     |
| ## 2124 | -2.5348e-02 | 7.9425e-03 | -3.1914 | 0.0014159 | **  |
| ## 2125 | -5.8935e-03 | 8.0650e-03 | -0.7307 | 0.4649331 |     |
| ## 2126 | -2.0797e-03 | 8.0810e-03 | -0.2574 | 0.7969061 |     |
| ## 2127 | 6.7776e-02  | 8.8553e-03 | 7.6537  | 1.953e-14 | *** |
| ## 2128 | 3.0737e-03  | 8.1362e-03 | 0.3778  | 0.7055937 |     |
| ## 2129 | 3.0767e-02  | 8.4617e-03 | 3.6360  | 0.0002769 | *** |
| ## 2130 | -6.3988e-02 | 7.4887e-03 | -8.5447 | < 2.2e-16 | *** |
| ## 2131 | -4.4918e-02 | 7.6716e-03 | -5.8551 | 4.767e-09 | *** |

|         |             |            |         |           |     |
|---------|-------------|------------|---------|-----------|-----|
| ## 2132 | -5.0816e-02 | 7.4043e-03 | -6.8630 | 6.743e-12 | *** |
| ## 2133 | 2.5558e-02  | 8.3956e-03 | 3.0442  | 0.0023330 | **  |
| ## 2134 | 2.7506e-02  | 8.4678e-03 | 3.2484  | 0.0011607 | **  |
| ## 2135 | -1.6947e-02 | 7.9522e-03 | -2.1311 | 0.0330840 | *   |
| ## 2136 | 4.1738e-02  | 8.5681e-03 | 4.8713  | 1.109e-06 | *** |
| ## 2137 | -3.3471e-02 | 7.7810e-03 | -4.3016 | 1.695e-05 | *** |
| ## 2138 | -2.6186e-02 | 7.9056e-03 | -3.3123 | 0.0009254 | *** |
| ## 2139 | -1.0658e-02 | 8.0201e-03 | -1.3289 | 0.1838889 |     |
| ## 2140 | -2.4558e-02 | 7.8685e-03 | -3.1210 | 0.0018023 | **  |
| ## 2141 | 4.3991e-02  | 8.5654e-03 | 5.1359  | 2.808e-07 | *** |
| ## 2142 | 3.3940e-02  | 8.4904e-03 | 3.9975  | 6.402e-05 | *** |
| ## 2143 | -3.0862e-02 | 7.7840e-03 | -3.9648 | 7.344e-05 | *** |
| ## 2144 | 1.7135e-03  | 8.1502e-03 | 0.2102  | 0.8334764 |     |
| ## 2145 | 3.7722e-02  | 8.5013e-03 | 4.4372  | 9.113e-06 | *** |
| ## 2146 | 2.1100e-02  | 8.3963e-03 | 2.5130  | 0.0119721 | *   |
| ## 2147 | 1.0800e-02  | 8.2708e-03 | 1.3058  | 0.1916123 |     |
| ## 2148 | -2.8952e-02 | 7.8459e-03 | -3.6901 | 0.0002242 | *** |
| ## 2149 | 4.6236e-02  | 8.6809e-03 | 5.3262  | 1.003e-07 | *** |
| ## 2150 | 8.5744e-03  | 8.2568e-03 | 1.0385  | 0.2990547 |     |
| ## 2151 | 1.6285e-02  | 8.3218e-03 | 1.9569  | 0.0503541 | .   |
| ## 2152 | 1.2113e-01  | 9.5452e-03 | 12.6902 | < 2.2e-16 | *** |
| ## 2153 | -5.4576e-02 | 7.6444e-03 | -7.1394 | 9.374e-13 | *** |
| ## 2154 | 1.9091e-02  | 8.2695e-03 | 2.3086  | 0.0209669 | *   |
| ## 2155 | 2.9745e-02  | 8.4795e-03 | 3.5079  | 0.0004517 | *** |
| ## 2156 | 3.4455e-02  | 8.4981e-03 | 4.0545  | 5.024e-05 | *** |
| ## 2157 | -1.9480e-02 | 7.8952e-03 | -2.4673 | 0.0136118 | *   |
| ## 2158 | -4.5185e-02 | 7.6149e-03 | -5.9338 | 2.961e-09 | *** |
| ## 2159 | -1.1992e-02 | 8.0133e-03 | -1.4965 | 0.1345243 |     |
| ## 2160 | -2.1329e-02 | 7.9337e-03 | -2.6884 | 0.0071785 | **  |
| ## 2161 | -4.5053e-03 | 8.1005e-03 | -0.5562 | 0.5780941 |     |
| ## 2162 | -4.4799e-02 | 7.5952e-03 | -5.8983 | 3.673e-09 | *** |
| ## 2163 | 4.6926e-02  | 8.6805e-03 | 5.4060  | 6.446e-08 | *** |
| ## 2164 | -8.2898e-04 | 8.1501e-03 | -0.1017 | 0.9189835 |     |
| ## 2165 | 2.4807e-02  | 8.3889e-03 | 2.9571  | 0.0031057 | **  |
| ## 2166 | -3.1075e-02 | 7.8008e-03 | -3.9836 | 6.788e-05 | *** |
| ## 2167 | 3.6693e-02  | 8.5515e-03 | 4.2908  | 1.781e-05 | *** |
| ## 2168 | 4.4024e-02  | 8.5949e-03 | 5.1221  | 3.022e-07 | *** |
| ## 2169 | -1.0431e-02 | 7.9628e-03 | -1.3100 | 0.1902044 |     |
| ## 2170 | 2.9005e-02  | 8.4061e-03 | 3.4505  | 0.0005595 | *** |
| ## 2171 | -3.9422e-02 | 7.7043e-03 | -5.1169 | 3.106e-07 | *** |
| ## 2172 | -1.6274e-02 | 7.9671e-03 | -2.0426 | 0.0410920 | *   |
| ## 2173 | 1.0749e-01  | 9.2870e-03 | 11.5741 | < 2.2e-16 | *** |
| ## 2174 | 4.7348e-03  | 8.2219e-03 | 0.5759  | 0.5647004 |     |
| ## 2175 | 9.3404e-03  | 8.2733e-03 | 1.1290  | 0.2589073 |     |
| ## 2176 | -4.1994e-02 | 7.7206e-03 | -5.4393 | 5.350e-08 | *** |
| ## 2177 | -1.2550e-02 | 8.0420e-03 | -1.5605 | 0.1186380 |     |
| ## 2178 | -2.9404e-02 | 7.7879e-03 | -3.7756 | 0.0001597 | *** |
| ## 2179 | 2.1452e-03  | 8.2238e-03 | 0.2609  | 0.7942031 |     |



|         |             |            |         |           |     |
|---------|-------------|------------|---------|-----------|-----|
| ## 2180 | -2.2726e-03 | 8.0613e-03 | -0.2819 | 0.7780124 |     |
| ## 2181 | 6.5798e-02  | 8.8530e-03 | 7.4322  | 1.068e-13 | *** |
| ## 2182 | -4.5219e-02 | 7.6844e-03 | -5.8845 | 3.992e-09 | *** |
| ## 2183 | 2.2367e-03  | 8.0963e-03 | 0.2763  | 0.7823481 |     |
| ## 2184 | 2.8573e-02  | 8.4405e-03 | 3.3852  | 0.0007113 | *** |
| ## 2185 | 2.7882e-02  | 8.3645e-03 | 3.3333  | 0.0008581 | *** |
| ## 2186 | -1.6279e-03 | 8.1344e-03 | -0.2001 | 0.8413849 |     |
| ## 2187 | -7.2870e-03 | 8.0865e-03 | -0.9011 | 0.3675213 |     |
| ## 2188 | 1.0104e-01  | 9.2414e-03 | 10.9330 | < 2.2e-16 | *** |
| ## 2189 | -3.9863e-02 | 7.7559e-03 | -5.1397 | 2.752e-07 | *** |
| ## 2190 | 6.0002e-02  | 8.8486e-03 | 6.7810  | 1.193e-11 | *** |
| ## 2191 | -1.3494e-02 | 8.0134e-03 | -1.6839 | 0.0922051 | .   |
| ## 2192 | 1.1793e-01  | 9.3928e-03 | 12.5554 | < 2.2e-16 | *** |
| ## 2193 | -8.2146e-03 | 8.0504e-03 | -1.0204 | 0.3075388 |     |
| ## 2194 | 7.2244e-02  | 8.9420e-03 | 8.0792  | 6.518e-16 | *** |
| ## 2195 | 2.8739e-02  | 8.4451e-03 | 3.4031  | 0.0006663 | *** |
| ## 2196 | 3.8898e-02  | 8.5303e-03 | 4.5600  | 5.116e-06 | *** |
| ## 2197 | 4.9129e-02  | 8.6243e-03 | 5.6966  | 1.223e-08 | *** |
| ## 2198 | -4.6509e-02 | 7.6330e-03 | -6.0931 | 1.107e-09 | *** |
| ## 2199 | 1.6427e-02  | 8.3532e-03 | 1.9665  | 0.0492400 | *   |
| ## 2200 | 3.6157e-02  | 8.5133e-03 | 4.2471  | 2.166e-05 | *** |
| ## 2201 | 6.8843e-02  | 8.9170e-03 | 7.7204  | 1.160e-14 | *** |
| ## 2202 | -6.5756e-02 | 7.4444e-03 | -8.8330 | < 2.2e-16 | *** |
| ## 2203 | -6.5976e-02 | 7.4322e-03 | -8.8770 | < 2.2e-16 | *** |
| ## 2204 | -3.3040e-02 | 7.7676e-03 | -4.2536 | 2.104e-05 | *** |
| ## 2205 | -4.8863e-02 | 7.6176e-03 | -6.4145 | 1.413e-10 | *** |
| ## 2206 | 2.5793e-02  | 8.4489e-03 | 3.0528  | 0.0022674 | **  |
| ## 2207 | 2.9813e-03  | 8.1741e-03 | 0.3647  | 0.7153191 |     |
| ## 2208 | 6.3081e-02  | 8.7821e-03 | 7.1830  | 6.821e-13 | *** |
| ## 2209 | 4.9263e-02  | 8.6715e-03 | 5.6810  | 1.339e-08 | *** |
| ## 2210 | -3.9294e-02 | 7.7806e-03 | -5.0503 | 4.412e-07 | *** |
| ## 2211 | -2.6310e-02 | 7.8970e-03 | -3.3317 | 0.0008633 | *** |
| ## 2212 | 2.0511e-02  | 8.4059e-03 | 2.4401  | 0.0146824 | *   |
| ## 2213 | 2.5918e-02  | 8.4333e-03 | 3.0733  | 0.0021167 | **  |
| ## 2214 | 7.4514e-03  | 8.1162e-03 | 0.9181  | 0.3585687 |     |
| ## 2215 | 2.3520e-03  | 8.1001e-03 | 0.2904  | 0.7715314 |     |
| ## 2216 | -1.0042e-02 | 8.0285e-03 | -1.2507 | 0.2110337 |     |
| ## 2217 | 5.5615e-02  | 8.5650e-03 | 6.4933  | 8.400e-11 | *** |
| ## 2218 | -9.4203e-03 | 8.0160e-03 | -1.1752 | 0.2399173 |     |
| ## 2219 | -3.8524e-02 | 7.6171e-03 | -5.0576 | 4.246e-07 | *** |
| ## 2220 | -5.9676e-02 | 7.5639e-03 | -7.8896 | 3.033e-15 | *** |
| ## 2221 | 1.1117e-02  | 8.3120e-03 | 1.3374  | 0.1810800 |     |
| ## 2222 | 1.6491e-02  | 8.3384e-03 | 1.9777  | 0.0479578 | *   |
| ## 2223 | -9.3174e-03 | 8.0337e-03 | -1.1598 | 0.2461330 |     |
| ## 2224 | -5.3900e-02 | 7.6184e-03 | -7.0749 | 1.495e-12 | *** |
| ## 2225 | -6.0699e-03 | 8.0441e-03 | -0.7546 | 0.4505071 |     |
| ## 2226 | 1.4679e-02  | 8.3093e-03 | 1.7665  | 0.0773063 | .   |
| ## 2227 | -2.0239e-02 | 7.8874e-03 | -2.5659 | 0.0102897 | *   |

|         |             |            |         |           |     |
|---------|-------------|------------|---------|-----------|-----|
| ## 2228 | 2.1091e-02  | 8.3383e-03 | 2.5295  | 0.0114238 | *   |
| ## 2229 | 7.0572e-04  | 8.1711e-03 | 0.0864  | 0.9311737 |     |
| ## 2230 | -3.9629e-03 | 8.0902e-03 | -0.4898 | 0.6242446 |     |
| ## 2231 | 9.1018e-04  | 8.1166e-03 | 0.1121  | 0.9107142 |     |
| ## 2232 | -8.4024e-03 | 8.1051e-03 | -1.0367 | 0.2998863 |     |
| ## 2233 | 4.7831e-02  | 8.6066e-03 | 5.5575  | 2.736e-08 | *** |
| ## 2234 | 3.8055e-02  | 8.5687e-03 | 4.4412  | 8.948e-06 | *** |
| ## 2235 | -1.3270e-02 | 7.9281e-03 | -1.6738 | 0.0941680 | .   |
| ## 2236 | 1.0250e-03  | 8.0137e-03 | 0.1279  | 0.8982239 |     |
| ## 2237 | -2.9084e-02 | 7.8183e-03 | -3.7199 | 0.0001993 | *** |
| ## 2238 | -2.9665e-02 | 7.9110e-03 | -3.7498 | 0.0001770 | *** |
| ## 2239 | -1.4651e-02 | 7.9721e-03 | -1.8378 | 0.0660973 | .   |
| ## 2240 | 4.6640e-02  | 8.6513e-03 | 5.3910  | 7.006e-08 | *** |
| ## 2241 | 1.2743e-02  | 8.3057e-03 | 1.5342  | 0.1249825 |     |
| ## 2242 | 2.5622e-03  | 8.1772e-03 | 0.3133  | 0.7540297 |     |
| ## 2243 | -3.5274e-02 | 7.8044e-03 | -4.5197 | 6.192e-06 | *** |
| ## 2244 | -5.8539e-02 | 7.4443e-03 | -7.8636 | 3.733e-15 | *** |
| ## 2245 | -6.2590e-02 | 7.5492e-03 | -8.2909 | < 2.2e-16 | *** |
| ## 2246 | -5.3943e-02 | 7.5757e-03 | -7.1205 | 1.075e-12 | *** |
| ## 2247 | -7.2564e-03 | 8.1749e-03 | -0.8876 | 0.3747296 |     |
| ## 2248 | -2.5079e-02 | 7.9054e-03 | -3.1724 | 0.0015117 | **  |
| ## 2249 | 7.0146e-02  | 9.0146e-03 | 7.7813  | 7.176e-15 | *** |
| ## 2250 | 1.1520e-03  | 8.1703e-03 | 0.1410  | 0.8878691 |     |
| ## 2251 | 4.5481e-02  | 8.6486e-03 | 5.2587  | 1.450e-07 | *** |
| ## 2252 | 1.4737e-03  | 8.1286e-03 | 0.1813  | 0.8561313 |     |
| ## 2253 | 2.7744e-02  | 8.3668e-03 | 3.3159  | 0.0009135 | *** |
| ## 2254 | 8.2572e-03  | 8.1898e-03 | 1.0082  | 0.3133413 |     |
| ## 2255 | 6.5886e-03  | 8.2242e-03 | 0.8011  | 0.4230649 |     |
| ## 2256 | 1.0964e-02  | 8.2643e-03 | 1.3267  | 0.1846223 |     |
| ## 2257 | 5.6853e-02  | 8.7314e-03 | 6.5113  | 7.450e-11 | *** |
| ## 2258 | -1.9104e-02 | 7.9574e-03 | -2.4007 | 0.0163624 | *   |
| ## 2259 | -3.1677e-02 | 7.7920e-03 | -4.0654 | 4.796e-05 | *** |
| ## 2260 | -8.6759e-03 | 8.0538e-03 | -1.0772 | 0.2813716 |     |
| ## 2261 | -2.9518e-02 | 7.8818e-03 | -3.7451 | 0.0001803 | *** |
| ## 2262 | 1.2100e-02  | 8.3110e-03 | 1.4558  | 0.1454342 |     |
| ## 2263 | 1.8437e-02  | 8.2891e-03 | 2.2242  | 0.0261349 | *   |
| ## 2264 | 5.0071e-02  | 8.6352e-03 | 5.7985  | 6.693e-09 | *** |
| ## 2265 | 2.7841e-02  | 8.4170e-03 | 3.3078  | 0.0009405 | *** |
| ## 2266 | 4.1504e-02  | 8.5284e-03 | 4.8666  | 1.136e-06 | *** |
| ## 2267 | -3.5295e-02 | 7.8248e-03 | -4.5106 | 6.464e-06 | *** |
| ## 2268 | -3.4786e-02 | 7.8038e-03 | -4.4576 | 8.288e-06 | *** |
| ## 2269 | 6.6937e-02  | 8.7772e-03 | 7.6263  | 2.416e-14 | *** |
| ## 2270 | -3.2099e-02 | 7.7887e-03 | -4.1212 | 3.770e-05 | *** |
| ## 2271 | 3.8962e-02  | 8.6118e-03 | 4.5243  | 6.061e-06 | *** |
| ## 2272 | 3.9874e-03  | 8.1975e-03 | 0.4864  | 0.6266710 |     |
| ## 2273 | 3.5002e-02  | 8.4767e-03 | 4.1292  | 3.641e-05 | *** |
| ## 2274 | 4.2422e-02  | 8.5495e-03 | 4.9619  | 6.981e-07 | *** |
| ## 2275 | -3.0584e-02 | 7.8059e-03 | -3.9181 | 8.924e-05 | *** |

|    |      |             |            |         |           |     |
|----|------|-------------|------------|---------|-----------|-----|
| ## | 2276 | -3.8763e-02 | 7.7166e-03 | -5.0232 | 5.080e-07 | *** |
| ## | 2277 | -3.4162e-02 | 7.7945e-03 | -4.3828 | 1.172e-05 | *** |
| ## | 2278 | 8.3080e-02  | 9.0618e-03 | 9.1681  | < 2.2e-16 | *** |
| ## | 2279 | -6.4698e-02 | 7.5268e-03 | -8.5957 | < 2.2e-16 | *** |
| ## | 2280 | 7.3602e-03  | 8.2159e-03 | 0.8958  | 0.3703338 |     |
| ## | 2281 | -2.7639e-02 | 7.8686e-03 | -3.5126 | 0.0004437 | *** |
| ## | 2282 | -2.1765e-02 | 7.8688e-03 | -2.7659 | 0.0056759 | **  |
| ## | 2283 | -1.8214e-02 | 7.9886e-03 | -2.2800 | 0.0226081 | *   |
| ## | 2284 | 9.9679e-03  | 8.2795e-03 | 1.2039  | 0.2286231 |     |
| ## | 2285 | -1.6170e-02 | 8.0240e-03 | -2.0152 | 0.0438853 | *   |
| ## | 2286 | 2.5464e-02  | 8.3921e-03 | 3.0343  | 0.0024110 | **  |
| ## | 2287 | -6.4163e-02 | 7.3602e-03 | -8.7176 | < 2.2e-16 | *** |
| ## | 2288 | 1.5702e-02  | 8.3896e-03 | 1.8717  | 0.0612537 | .   |
| ## | 2289 | -3.7417e-02 | 7.7413e-03 | -4.8334 | 1.342e-06 | *** |
| ## | 2290 | -1.9147e-03 | 8.1022e-03 | -0.2363 | 0.8131808 |     |
| ## | 2291 | 6.0828e-02  | 8.8573e-03 | 6.8676  | 6.529e-12 | *** |
| ## | 2292 | -1.7138e-02 | 7.9515e-03 | -2.1554 | 0.0311329 | *   |
| ## | 2293 | 6.2502e-02  | 8.8188e-03 | 7.0873  | 1.368e-12 | *** |
| ## | 2294 | 3.3683e-03  | 8.1487e-03 | 0.4134  | 0.6793416 |     |
| ## | 2295 | -8.9870e-03 | 8.0031e-03 | -1.1229 | 0.2614634 |     |
| ## | 2296 | 4.2265e-02  | 8.5724e-03 | 4.9304  | 8.206e-07 | *** |
| ## | 2297 | 7.4945e-03  | 8.0874e-03 | 0.9267  | 0.3540902 |     |
| ## | 2298 | 1.8446e-02  | 8.3778e-03 | 2.2018  | 0.0276780 | *   |
| ## | 2299 | 3.8859e-03  | 8.2513e-03 | 0.4709  | 0.6376765 |     |
| ## | 2300 | 1.9135e-03  | 8.1322e-03 | 0.2353  | 0.8139723 |     |
| ## | 2301 | 1.4900e-02  | 8.3029e-03 | 1.7945  | 0.0727328 | .   |
| ## | 2302 | 2.9162e-02  | 8.4725e-03 | 3.4419  | 0.0005776 | *** |
| ## | 2303 | -6.2702e-02 | 7.7454e-03 | -8.0953 | 5.711e-16 | *** |
| ## | 2304 | -5.7462e-02 | 7.5948e-03 | -7.5659 | 3.851e-14 | *** |
| ## | 2305 | -4.4416e-02 | 7.6035e-03 | -5.8415 | 5.174e-09 | *** |
| ## | 2306 | -1.2013e-02 | 8.0235e-03 | -1.4972 | 0.1343332 |     |
| ## | 2307 | 2.5074e-02  | 8.4143e-03 | 2.9799  | 0.0028835 | **  |
| ## | 2308 | -1.6643e-02 | 8.0628e-03 | -2.0642 | 0.0389960 | *   |
| ## | 2309 | -3.8104e-03 | 8.1140e-03 | -0.4696 | 0.6386345 |     |
| ## | 2310 | 4.6956e-02  | 8.7302e-03 | 5.3786  | 7.507e-08 | *** |
| ## | 2311 | 4.6410e-02  | 8.7670e-03 | 5.2937  | 1.199e-07 | *** |
| ## | 2312 | -5.1707e-03 | 8.0829e-03 | -0.6397 | 0.5223653 |     |
| ## | 2313 | 7.4918e-03  | 8.2186e-03 | 0.9116  | 0.3619937 |     |
| ## | 2314 | -9.7279e-05 | 8.0959e-03 | -0.0120 | 0.9904129 |     |
| ## | 2315 | 1.1412e-02  | 8.2637e-03 | 1.3810  | 0.1672726 |     |
| ## | 2316 | 4.3911e-02  | 8.5533e-03 | 5.1338  | 2.840e-07 | *** |
| ## | 2317 | -3.2146e-02 | 7.8441e-03 | -4.0981 | 4.165e-05 | *** |
| ## | 2318 | 2.2174e-02  | 8.3228e-03 | 2.6643  | 0.0077153 | **  |
| ## | 2319 | -4.4012e-03 | 8.1212e-03 | -0.5419 | 0.5878586 |     |
| ## | 2320 | -1.8750e-02 | 7.9726e-03 | -2.3519 | 0.0186796 | *   |
| ## | 2321 | -4.0532e-02 | 7.3873e-03 | -5.4867 | 4.096e-08 | *** |
| ## | 2322 | 1.6603e-02  | 8.3533e-03 | 1.9877  | 0.0468501 | *   |
| ## | 2323 | -2.0782e-03 | 8.1331e-03 | -0.2555 | 0.7983188 |     |

|         |             |            |          |           |     |
|---------|-------------|------------|----------|-----------|-----|
| ## 2324 | 5.2311e-02  | 8.7717e-03 | 5.9636   | 2.467e-09 | *** |
| ## 2325 | 4.1465e-03  | 8.2026e-03 | 0.5055   | 0.6131972 |     |
| ## 2326 | 3.8343e-02  | 8.6006e-03 | 4.4582   | 8.264e-06 | *** |
| ## 2327 | 3.7390e-02  | 8.5525e-03 | 4.3718   | 1.232e-05 | *** |
| ## 2328 | 1.6659e-02  | 8.3440e-03 | 1.9965   | 0.0458813 | *   |
| ## 2329 | 8.0279e-02  | 9.0201e-03 | 8.9000   | < 2.2e-16 | *** |
| ## 2330 | 5.3171e-02  | 8.7241e-03 | 6.0947   | 1.096e-09 | *** |
| ## 2331 | 1.6966e-02  | 8.3328e-03 | 2.0361   | 0.0417442 | *   |
| ## 2332 | 3.9046e-02  | 8.5512e-03 | 4.5662   | 4.967e-06 | *** |
| ## 2333 | 2.8386e-02  | 8.3903e-03 | 3.3832   | 0.0007164 | *** |
| ## 2334 | -2.6460e-02 | 7.8567e-03 | -3.3678  | 0.0007577 | *** |
| ## 2335 | 3.4576e-02  | 8.4902e-03 | 4.0724   | 4.653e-05 | *** |
| ## 2336 | 1.5861e-04  | 8.1287e-03 | 0.0195   | 0.9844322 |     |
| ## 2337 | -2.5304e-02 | 7.9047e-03 | -3.2011  | 0.0013688 | **  |
| ## 2338 | 1.5297e-02  | 8.3060e-03 | 1.8417   | 0.0655254 | .   |
| ## 2339 | -3.1732e-02 | 7.7540e-03 | -4.0923  | 4.271e-05 | *** |
| ## 2340 | -3.4893e-02 | 7.8187e-03 | -4.4628  | 8.089e-06 | *** |
| ## 2341 | 4.0740e-02  | 8.5481e-03 | 4.7659   | 1.880e-06 | *** |
| ## 2342 | -4.2681e-02 | 7.6439e-03 | -5.5837  | 2.355e-08 | *** |
| ## 2343 | -3.3867e-02 | 7.8805e-03 | -4.2976  | 1.727e-05 | *** |
| ## 2344 | -2.1660e-02 | 7.7585e-03 | -2.7918  | 0.0052414 | **  |
| ## 2345 | 1.6913e-02  | 8.3243e-03 | 2.0318   | 0.0421753 | *   |
| ## 2346 | 6.1138e-03  | 8.1462e-03 | 0.7505   | 0.4529482 |     |
| ## 2347 | -3.6428e-02 | 7.7438e-03 | -4.7042  | 2.549e-06 | *** |
| ## 2348 | 2.7036e-02  | 8.4360e-03 | 3.2048   | 0.0013516 | **  |
| ## 2349 | 3.7228e-02  | 8.3900e-03 | 4.4372   | 9.112e-06 | *** |
| ## 2350 | 1.2067e-02  | 8.2954e-03 | 1.4547   | 0.1457601 |     |
| ## 2351 | 7.4213e-02  | 8.8894e-03 | 8.3485   | < 2.2e-16 | *** |
| ## 2352 | 3.6114e-02  | 8.5942e-03 | 4.2021   | 2.644e-05 | *** |
| ## 2353 | -3.4371e-02 | 7.8038e-03 | -4.4044  | 1.061e-05 | *** |
| ## 2354 | 2.7656e-02  | 8.4305e-03 | 3.2805   | 0.0010363 | **  |
| ## 2355 | 3.3432e-02  | 8.4919e-03 | 3.9369   | 8.255e-05 | *** |
| ## 2356 | -7.5784e-02 | 7.3657e-03 | -10.2887 | < 2.2e-16 | *** |
| ## 2357 | 5.4228e-02  | 8.6618e-03 | 6.2607   | 3.834e-10 | *** |
| ## 2358 | 3.3381e-02  | 8.5679e-03 | 3.8961   | 9.776e-05 | *** |
| ## 2359 | 5.3757e-02  | 8.7255e-03 | 6.1609   | 7.233e-10 | *** |
| ## 2360 | 6.1625e-02  | 8.8463e-03 | 6.9661   | 3.258e-12 | *** |
| ## 2361 | 1.8112e-02  | 8.2991e-03 | 2.1824   | 0.0290832 | *   |
| ## 2362 | -1.6956e-02 | 7.9123e-03 | -2.1429  | 0.0321174 | *   |
| ## 2363 | -7.1962e-02 | 7.3730e-03 | -9.7601  | < 2.2e-16 | *** |
| ## 2364 | 2.8080e-02  | 8.4219e-03 | 3.3341   | 0.0008557 | *** |
| ## 2365 | -2.6056e-02 | 7.8474e-03 | -3.3204  | 0.0008990 | *** |
| ## 2366 | -3.3285e-02 | 7.7968e-03 | -4.2690  | 1.963e-05 | *** |
| ## 2367 | 1.4064e-02  | 8.3247e-03 | 1.6894   | 0.0911354 | .   |
| ## 2368 | 1.9865e-03  | 8.1634e-03 | 0.2433   | 0.8077448 |     |
| ## 2369 | -1.7577e-03 | 8.0939e-03 | -0.2172  | 0.8280774 |     |
| ## 2370 | 1.7398e-02  | 8.2279e-03 | 2.1145   | 0.0344734 | *   |
| ## 2371 | -2.1403e-02 | 7.9047e-03 | -2.7077  | 0.0067757 | **  |

|         |             |            |          |           |     |
|---------|-------------|------------|----------|-----------|-----|
| ## 2372 | -1.3296e-02 | 8.1201e-03 | -1.6374  | 0.1015488 |     |
| ## 2373 | 2.0562e-02  | 8.3291e-03 | 2.4687   | 0.0135618 | *   |
| ## 2374 | 7.1906e-02  | 8.9486e-03 | 8.0354   | 9.327e-16 | *** |
| ## 2375 | -1.1009e-03 | 8.0579e-03 | -0.1366  | 0.8913276 |     |
| ## 2376 | -3.4393e-02 | 7.6289e-03 | -4.5082  | 6.537e-06 | *** |
| ## 2377 | 5.1851e-02  | 8.6776e-03 | 5.9753   | 2.297e-09 | *** |
| ## 2378 | 6.4421e-04  | 8.1554e-03 | 0.0790   | 0.9370391 |     |
| ## 2379 | -9.5496e-02 | 7.1835e-03 | -13.2938 | < 2.2e-16 | *** |
| ## 2380 | 6.4389e-02  | 8.8237e-03 | 7.2973   | 2.937e-13 | *** |
| ## 2381 | -1.5638e-02 | 8.0125e-03 | -1.9517  | 0.0509758 | .   |
| ## 2382 | -4.6321e-02 | 7.6506e-03 | -6.0546  | 1.408e-09 | *** |
| ## 2383 | 4.9900e-02  | 8.6376e-03 | 5.7771   | 7.601e-09 | *** |
| ## 2384 | -1.6759e-02 | 7.9631e-03 | -2.1046  | 0.0353287 | *   |
| ## 2385 | 7.6835e-02  | 8.9974e-03 | 8.5397   | < 2.2e-16 | *** |
| ## 2386 | -1.7151e-02 | 7.9500e-03 | -2.1573  | 0.0309803 | *   |
| ## 2387 | 4.1473e-02  | 8.6076e-03 | 4.8182   | 1.449e-06 | *** |
| ## 2388 | -5.6458e-02 | 7.5855e-03 | -7.4428  | 9.856e-14 | *** |
| ## 2389 | -9.1349e-03 | 7.9426e-03 | -1.1501  | 0.2501005 |     |
| ## 2390 | -1.6107e-02 | 7.9672e-03 | -2.0216  | 0.0432134 | *   |
| ## 2391 | 3.3756e-02  | 8.4835e-03 | 3.9790   | 6.920e-05 | *** |
| ## 2392 | -2.1322e-02 | 7.9158e-03 | -2.6936  | 0.0070677 | **  |
| ## 2393 | 3.9177e-02  | 8.4642e-03 | 4.6286   | 3.682e-06 | *** |
| ## 2394 | -9.4568e-03 | 8.0540e-03 | -1.1742  | 0.2403260 |     |
| ## 2395 | 2.5113e-02  | 8.3957e-03 | 2.9911   | 0.0027794 | **  |
| ## 2396 | -6.4656e-03 | 8.0893e-03 | -0.7993  | 0.4241299 |     |
| ## 2397 | 5.3180e-03  | 8.1682e-03 | 0.6511   | 0.5150084 |     |
| ## 2398 | -5.5344e-02 | 7.5426e-03 | -7.3376  | 2.175e-13 | *** |
| ## 2399 | -4.5376e-03 | 8.1228e-03 | -0.5586  | 0.5764166 |     |
| ## 2400 | -3.5831e-03 | 8.0371e-03 | -0.4458  | 0.6557265 |     |
| ## 2401 | 5.7243e-03  | 8.2294e-03 | 0.6956   | 0.4866839 |     |
| ## 2402 | 3.8308e-02  | 8.5674e-03 | 4.4713   | 7.773e-06 | *** |
| ## 2403 | 4.0418e-02  | 8.6286e-03 | 4.6842   | 2.811e-06 | *** |
| ## 2404 | 1.6155e-02  | 8.2936e-03 | 1.9479   | 0.0514219 | .   |
| ## 2405 | 2.9510e-02  | 8.3935e-03 | 3.5158   | 0.0004385 | *** |
| ## 2406 | -7.2215e-03 | 8.0286e-03 | -0.8995  | 0.3683975 |     |
| ## 2407 | 4.1562e-02  | 8.6067e-03 | 4.8291   | 1.372e-06 | *** |
| ## 2408 | -4.1475e-02 | 7.7465e-03 | -5.3540  | 8.604e-08 | *** |
| ## 2409 | 5.2030e-02  | 8.6311e-03 | 6.0282   | 1.658e-09 | *** |
| ## 2410 | 8.8924e-02  | 9.1427e-03 | 9.7263   | < 2.2e-16 | *** |
| ## 2411 | 6.1687e-03  | 8.2296e-03 | 0.7496   | 0.4535104 |     |
| ## 2412 | 7.8605e-03  | 8.2056e-03 | 0.9580   | 0.3380873 |     |
| ## 2413 | 6.8642e-02  | 8.8742e-03 | 7.7350   | 1.034e-14 | *** |
| ## 2414 | -2.7512e-02 | 7.8402e-03 | -3.5091  | 0.0004496 | *** |
| ## 2415 | 7.4743e-02  | 8.9826e-03 | 8.3208   | < 2.2e-16 | *** |
| ## 2416 | 1.1815e-02  | 8.1879e-03 | 1.4430   | 0.1490184 |     |
| ## 2417 | 3.1991e-02  | 8.3801e-03 | 3.8174   | 0.0001349 | *** |
| ## 2418 | -2.7600e-02 | 7.8422e-03 | -3.5194  | 0.0004325 | *** |
| ## 2419 | -7.9126e-02 | 7.4056e-03 | -10.6846 | < 2.2e-16 | *** |

|         |             |            |          |           |     |
|---------|-------------|------------|----------|-----------|-----|
| ## 2420 | -1.2863e-02 | 7.9845e-03 | -1.6109  | 0.1071916 |     |
| ## 2421 | 1.7353e-02  | 8.3289e-03 | 2.0835   | 0.0372044 | *   |
| ## 2422 | 8.8321e-05  | 8.0945e-03 | 0.0109   | 0.9912943 |     |
| ## 2423 | -1.3980e-02 | 8.0163e-03 | -1.7440  | 0.0811648 | .   |
| ## 2424 | 1.6620e-02  | 8.3620e-03 | 1.9876   | 0.0468570 | *   |
| ## 2425 | -1.2422e-02 | 7.9681e-03 | -1.5590  | 0.1190026 |     |
| ## 2426 | -2.1776e-02 | 7.9501e-03 | -2.7391  | 0.0061611 | **  |
| ## 2427 | -3.1278e-02 | 7.8191e-03 | -4.0002  | 6.328e-05 | *** |
| ## 2428 | 4.2160e-04  | 8.1280e-03 | 0.0519   | 0.9586320 |     |
| ## 2429 | -8.0908e-03 | 8.0438e-03 | -1.0058  | 0.3144905 |     |
| ## 2430 | 6.2359e-02  | 8.8390e-03 | 7.0550   | 1.726e-12 | *** |
| ## 2431 | 5.0569e-02  | 8.6903e-03 | 5.8190   | 5.920e-09 | *** |
| ## 2432 | 5.8241e-03  | 8.2085e-03 | 0.7095   | 0.4779987 |     |
| ## 2433 | 1.6346e-02  | 8.2828e-03 | 1.9735   | 0.0484364 | *   |
| ## 2434 | 3.5913e-02  | 8.5416e-03 | 4.2045   | 2.617e-05 | *** |
| ## 2435 | 2.7789e-02  | 8.4403e-03 | 3.2924   | 0.0009935 | *** |
| ## 2436 | 2.4046e-02  | 8.4089e-03 | 2.8595   | 0.0042425 | **  |
| ## 2437 | 1.7920e-03  | 8.1449e-03 | 0.2200   | 0.8258585 |     |
| ## 2438 | -6.6864e-02 | 7.4821e-03 | -8.9366  | < 2.2e-16 | *** |
| ## 2439 | 9.7102e-02  | 9.2038e-03 | 10.5502  | < 2.2e-16 | *** |
| ## 2440 | 1.0746e-01  | 9.2796e-03 | 11.5797  | < 2.2e-16 | *** |
| ## 2441 | 1.7379e-02  | 8.3887e-03 | 2.0717   | 0.0382913 | *   |
| ## 2442 | -3.2276e-02 | 7.7452e-03 | -4.1672  | 3.084e-05 | *** |
| ## 2443 | -5.2360e-02 | 7.5937e-03 | -6.8951  | 5.383e-12 | *** |
| ## 2444 | -4.5402e-03 | 8.1185e-03 | -0.5592  | 0.5759994 |     |
| ## 2445 | -1.4583e-02 | 8.0117e-03 | -1.8203  | 0.0687168 | .   |
| ## 2446 | 4.1366e-02  | 8.5005e-03 | 4.8663   | 1.137e-06 | *** |
| ## 2447 | -7.9290e-02 | 7.3945e-03 | -10.7229 | < 2.2e-16 | *** |
| ## 2448 | 1.2842e-02  | 8.2419e-03 | 1.5582   | 0.1191958 |     |
| ## 2449 | 3.5785e-02  | 8.4954e-03 | 4.2123   | 2.527e-05 | *** |
| ## 2450 | -2.9913e-02 | 7.8024e-03 | -3.8338  | 0.0001262 | *** |
| ## 2451 | 1.5517e-02  | 8.2692e-03 | 1.8765   | 0.0605871 | .   |
| ## 2452 | 1.6556e-02  | 8.3142e-03 | 1.9913   | 0.0464439 | *   |
| ## 2453 | 1.3482e-02  | 8.2414e-03 | 1.6359   | 0.1018542 |     |
| ## 2454 | 3.0092e-02  | 8.4561e-03 | 3.5587   | 0.0003727 | *** |
| ## 2455 | -1.0740e-02 | 8.0020e-03 | -1.3421  | 0.1795626 |     |
| ## 2456 | -7.1744e-02 | 7.4045e-03 | -9.6892  | < 2.2e-16 | *** |
| ## 2457 | -4.0564e-02 | 7.7104e-03 | -5.2609  | 1.433e-07 | *** |
| ## 2458 | 3.4152e-02  | 8.4977e-03 | 4.0189   | 5.846e-05 | *** |
| ## 2459 | 1.8253e-02  | 8.3684e-03 | 2.1811   | 0.0291732 | *   |
| ## 2460 | 5.4438e-02  | 8.6952e-03 | 6.2607   | 3.832e-10 | *** |
| ## 2461 | 1.0125e-02  | 8.3007e-03 | 1.2197   | 0.2225613 |     |
| ## 2462 | 3.6048e-02  | 8.4605e-03 | 4.2607   | 2.038e-05 | *** |
| ## 2463 | 1.0747e-02  | 8.2741e-03 | 1.2988   | 0.1939965 |     |
| ## 2464 | 1.0320e-02  | 8.2678e-03 | 1.2482   | 0.2119700 |     |
| ## 2465 | 2.4947e-02  | 8.4412e-03 | 2.9554   | 0.0031231 | **  |
| ## 2466 | 2.6153e-04  | 8.0066e-03 | 0.0327   | 0.9739425 |     |
| ## 2467 | -2.6297e-02 | 7.9287e-03 | -3.3167  | 0.0009108 | *** |

|         |             |            |          |           |     |
|---------|-------------|------------|----------|-----------|-----|
| ## 2468 | 1.1650e-02  | 8.2181e-03 | 1.4176   | 0.1563018 |     |
| ## 2469 | -7.5822e-02 | 7.3948e-03 | -10.2533 | < 2.2e-16 | *** |
| ## 2470 | 1.1206e-02  | 8.2350e-03 | 1.3607   | 0.1736019 |     |
| ## 2471 | 2.4969e-02  | 8.3871e-03 | 2.9771   | 0.0029100 | **  |
| ## 2472 | 6.2105e-02  | 8.7518e-03 | 7.0962   | 1.283e-12 | *** |
| ## 2473 | -7.9581e-02 | 7.3450e-03 | -10.8346 | < 2.2e-16 | *** |
| ## 2474 | -1.8088e-02 | 7.9677e-03 | -2.2702  | 0.0231954 | *   |
| ## 2475 | -3.3624e-02 | 7.8287e-03 | -4.2949  | 1.748e-05 | *** |
| ## 2476 | 2.2304e-02  | 8.3872e-03 | 2.6594   | 0.0078290 | **  |
| ## 2477 | -3.3711e-02 | 7.7202e-03 | -4.3666  | 1.262e-05 | *** |
| ## 2478 | -3.5692e-02 | 7.7083e-03 | -4.6303  | 3.651e-06 | *** |
| ## 2479 | 4.3527e-02  | 8.6064e-03 | 5.0574   | 4.249e-07 | *** |
| ## 2480 | 9.1555e-02  | 9.2245e-03 | 9.9251   | < 2.2e-16 | *** |
| ## 2481 | 2.4426e-02  | 8.4491e-03 | 2.8910   | 0.0038401 | **  |
| ## 2482 | -2.9920e-02 | 7.8030e-03 | -3.8345  | 0.0001258 | *** |
| ## 2483 | 2.5601e-02  | 8.3995e-03 | 3.0479   | 0.0023042 | **  |
| ## 2484 | 8.1188e-02  | 9.0426e-03 | 8.9784   | < 2.2e-16 | *** |
| ## 2485 | 8.8308e-02  | 9.0891e-03 | 9.7159   | < 2.2e-16 | *** |
| ## 2486 | 1.5571e-02  | 8.2833e-03 | 1.8798   | 0.0601381 | .   |
| ## 2487 | 3.2001e-02  | 8.4928e-03 | 3.7680   | 0.0001645 | *** |
| ## 2488 | 2.2723e-02  | 8.3849e-03 | 2.7100   | 0.0067277 | **  |
| ## 2489 | -1.9308e-02 | 8.0493e-03 | -2.3987  | 0.0164524 | *   |
| ## 2490 | 5.0984e-02  | 8.6438e-03 | 5.8983   | 3.672e-09 | *** |
| ## 2491 | 8.9668e-02  | 9.1399e-03 | 9.8106   | < 2.2e-16 | *** |
| ## 2492 | 8.1843e-03  | 8.1129e-03 | 1.0088   | 0.3130648 |     |
| ## 2493 | -4.5295e-02 | 7.6042e-03 | -5.9567  | 2.574e-09 | *** |
| ## 2494 | -6.9234e-02 | 7.4526e-03 | -9.2899  | < 2.2e-16 | *** |
| ## 2495 | 8.4014e-02  | 9.0734e-03 | 9.2593   | < 2.2e-16 | *** |
| ## 2496 | 3.7535e-02  | 8.4478e-03 | 4.4431   | 8.865e-06 | *** |
| ## 2497 | -3.4520e-03 | 8.0797e-03 | -0.4273  | 0.6691965 |     |
| ## 2498 | -1.9638e-03 | 8.0812e-03 | -0.2430  | 0.8080034 |     |
| ## 2499 | 1.4719e-02  | 8.2555e-03 | 1.7829   | 0.0745991 | .   |
| ## 2500 | -1.1780e-02 | 8.0634e-03 | -1.4609  | 0.1440551 |     |
| ## 2501 | 1.0096e-03  | 8.2156e-03 | 0.1229   | 0.9021991 |     |
| ## 2502 | 3.1204e-02  | 8.4877e-03 | 3.6763   | 0.0002366 | *** |
| ## 2503 | 9.9405e-02  | 9.2706e-03 | 10.7226  | < 2.2e-16 | *** |
| ## 2504 | 1.2823e-02  | 8.2692e-03 | 1.5507   | 0.1209854 |     |
| ## 2505 | 3.0182e-02  | 8.4913e-03 | 3.5544   | 0.0003788 | *** |
| ## 2506 | 5.5900e-02  | 8.7316e-03 | 6.4021   | 1.533e-10 | *** |
| ## 2507 | 2.0193e-02  | 8.3137e-03 | 2.4289   | 0.0151456 | *   |
| ## 2508 | -4.3666e-02 | 7.6667e-03 | -5.6955  | 1.230e-08 | *** |
| ## 2509 | -2.2517e-02 | 7.9557e-03 | -2.8302  | 0.0046512 | **  |
| ## 2510 | 2.0807e-02  | 8.3465e-03 | 2.4929   | 0.0126704 | *   |
| ## 2511 | -5.4440e-03 | 8.0743e-03 | -0.6742  | 0.5001546 |     |
| ## 2512 | 7.7198e-02  | 8.9935e-03 | 8.5838   | < 2.2e-16 | *** |
| ## 2513 | 2.0639e-02  | 8.3502e-03 | 2.4716   | 0.0134502 | *   |
| ## 2514 | 3.4498e-02  | 8.4612e-03 | 4.0773   | 4.557e-05 | *** |
| ## 2515 | -3.3816e-02 | 7.6840e-03 | -4.4009  | 1.078e-05 | *** |

|         |             |            |         |           |     |
|---------|-------------|------------|---------|-----------|-----|
| ## 2516 | 3.3500e-02  | 8.5064e-03 | 3.9383  | 8.208e-05 | *** |
| ## 2517 | 9.7053e-03  | 8.2921e-03 | 1.1704  | 0.2418296 |     |
| ## 2518 | 5.3609e-02  | 8.7311e-03 | 6.1400  | 8.252e-10 | *** |
| ## 2519 | -9.6565e-03 | 8.0772e-03 | -1.1955 | 0.2318846 |     |
| ## 2520 | -1.9043e-02 | 7.9187e-03 | -2.4049 | 0.0161790 | *   |
| ## 2521 | 6.1158e-02  | 8.8313e-03 | 6.9251  | 4.356e-12 | *** |
| ## 2522 | -4.2276e-03 | 8.1008e-03 | -0.5219 | 0.6017611 |     |
| ## 2523 | 7.5644e-02  | 8.9599e-03 | 8.4425  | < 2.2e-16 | *** |
| ## 2524 | 6.3473e-03  | 8.1766e-03 | 0.7763  | 0.4375846 |     |
| ## 2525 | -3.2200e-02 | 7.8318e-03 | -4.1114 | 3.932e-05 | *** |
| ## 2526 | -2.3197e-03 | 8.1405e-03 | -0.2850 | 0.7756786 |     |
| ## 2527 | 6.4113e-02  | 8.9042e-03 | 7.2003  | 6.008e-13 | *** |
| ## 2528 | -1.5240e-02 | 7.9914e-03 | -1.9070 | 0.0565211 | .   |
| ## 2529 | -1.7870e-02 | 7.9399e-03 | -2.2506 | 0.0244108 | *   |
| ## 2530 | 4.6984e-02  | 8.6470e-03 | 5.4335  | 5.526e-08 | *** |
| ## 2531 | -2.6277e-02 | 7.8997e-03 | -3.3263 | 0.0008800 | *** |
| ## 2532 | -5.1968e-02 | 7.6327e-03 | -6.8086 | 9.857e-12 | *** |
| ## 2533 | -3.5007e-02 | 7.7186e-03 | -4.5353 | 5.751e-06 | *** |
| ## 2534 | 1.3986e-03  | 8.1344e-03 | 0.1719  | 0.8634894 |     |
| ## 2535 | -4.7278e-02 | 7.6452e-03 | -6.1840 | 6.250e-10 | *** |
| ## 2536 | 1.3061e-02  | 8.2667e-03 | 1.5799  | 0.1141197 |     |
| ## 2537 | -5.1455e-03 | 8.0414e-03 | -0.6399 | 0.5222505 |     |
| ## 2538 | -4.7703e-02 | 7.6611e-03 | -6.2267 | 4.764e-10 | *** |
| ## 2539 | 2.2452e-02  | 8.3949e-03 | 2.6745  | 0.0074844 | **  |
| ## 2540 | -1.5092e-02 | 8.0160e-03 | -1.8827 | 0.0597426 | .   |
| ## 2541 | 1.3579e-02  | 8.2572e-03 | 1.6445  | 0.1000635 |     |
| ## 2542 | 1.3468e-01  | 9.4534e-03 | 14.2465 | < 2.2e-16 | *** |
| ## 2543 | -5.7658e-03 | 8.0731e-03 | -0.7142 | 0.4751008 |     |
| ## 2544 | -4.4711e-02 | 7.6364e-03 | -5.8549 | 4.772e-09 | *** |
| ## 2545 | 1.5491e-02  | 8.3036e-03 | 1.8656  | 0.0620985 | .   |
| ## 2546 | 9.4201e-02  | 9.0955e-03 | 10.3569 | < 2.2e-16 | *** |
| ## 2547 | 7.0306e-02  | 8.9450e-03 | 7.8598  | 3.846e-15 | *** |
| ## 2548 | -1.4799e-02 | 7.9865e-03 | -1.8531 | 0.0638736 | .   |
| ## 2549 | -1.6945e-02 | 7.9743e-03 | -2.1249 | 0.0335940 | *   |
| ## 2550 | 3.2486e-02  | 8.5520e-03 | 3.7987  | 0.0001455 | *** |
| ## 2551 | -1.0403e-02 | 8.0053e-03 | -1.2995 | 0.1937566 |     |
| ## 2552 | 6.6873e-02  | 8.8641e-03 | 7.5443  | 4.546e-14 | *** |
| ## 2553 | -4.5703e-02 | 7.6786e-03 | -5.9519 | 2.650e-09 | *** |
| ## 2554 | 2.6138e-02  | 8.3745e-03 | 3.1211  | 0.0018017 | **  |
| ## 2555 | 1.9723e-02  | 8.3927e-03 | 2.3501  | 0.0187708 | *   |
| ## 2556 | -1.7613e-02 | 7.9773e-03 | -2.2079 | 0.0272480 | *   |
| ## 2557 | 2.5920e-02  | 8.4143e-03 | 3.0805  | 0.0020668 | **  |
| ## 2558 | 6.4229e-04  | 8.0615e-03 | 0.0797  | 0.9364974 |     |
| ## 2559 | -2.3349e-02 | 7.8798e-03 | -2.9631 | 0.0030456 | **  |
| ## 2560 | 6.6132e-02  | 8.7932e-03 | 7.5208  | 5.443e-14 | *** |
| ## 2561 | -4.3824e-02 | 7.8191e-03 | -5.6048 | 2.085e-08 | *** |
| ## 2562 | 6.0970e-02  | 8.8552e-03 | 6.8852  | 5.770e-12 | *** |
| ## 2563 | 4.3973e-02  | 8.6358e-03 | 5.0920  | 3.543e-07 | *** |



|         |             |            |          |           |     |
|---------|-------------|------------|----------|-----------|-----|
| ## 2564 | 5.0837e-02  | 8.7418e-03 | 5.8153   | 6.051e-09 | *** |
| ## 2565 | -1.0026e-02 | 8.0400e-03 | -1.2470  | 0.2124006 |     |
| ## 2566 | -3.5950e-03 | 8.0577e-03 | -0.4462  | 0.6554821 |     |
| ## 2567 | 3.6222e-02  | 8.5789e-03 | 4.2222   | 2.419e-05 | *** |
| ## 2568 | -2.0555e-02 | 7.8702e-03 | -2.6117  | 0.0090085 | **  |
| ## 2569 | 5.8491e-02  | 8.7410e-03 | 6.6916   | 2.208e-11 | *** |
| ## 2570 | 1.1472e-02  | 8.3830e-03 | 1.3685   | 0.1711628 |     |
| ## 2571 | 6.0376e-02  | 8.8452e-03 | 6.8258   | 8.742e-12 | *** |
| ## 2572 | -1.7750e-02 | 7.9577e-03 | -2.2306  | 0.0257096 | *   |
| ## 2573 | 1.0647e-01  | 9.2327e-03 | 11.5321  | < 2.2e-16 | *** |
| ## 2574 | -5.6909e-02 | 7.5701e-03 | -7.5176  | 5.580e-14 | *** |
| ## 2575 | 7.3472e-02  | 8.9997e-03 | 8.1638   | 3.247e-16 | *** |
| ## 2576 | 3.7951e-02  | 8.5058e-03 | 4.4618   | 8.129e-06 | *** |
| ## 2577 | 3.7393e-02  | 8.3844e-03 | 4.4599   | 8.201e-06 | *** |
| ## 2578 | -2.4922e-02 | 7.9337e-03 | -3.1413  | 0.0016818 | **  |
| ## 2579 | 4.8488e-02  | 8.5970e-03 | 5.6401   | 1.700e-08 | *** |
| ## 2580 | 2.9281e-03  | 8.0533e-03 | 0.3636   | 0.7161657 |     |
| ## 2581 | 9.1190e-03  | 8.2393e-03 | 1.1068   | 0.2683933 |     |
| ## 2582 | 1.0896e-02  | 8.1850e-03 | 1.3312   | 0.1831352 |     |
| ## 2583 | 7.7495e-03  | 8.3069e-03 | 0.9329   | 0.3508729 |     |
| ## 2584 | -1.5166e-02 | 7.9787e-03 | -1.9008  | 0.0573226 | .   |
| ## 2585 | 2.0535e-03  | 8.1483e-03 | 0.2520   | 0.8010278 |     |
| ## 2586 | -2.5404e-02 | 7.8166e-03 | -3.2500  | 0.0011539 | **  |
| ## 2587 | 4.6357e-02  | 8.6459e-03 | 5.3617   | 8.245e-08 | *** |
| ## 2588 | -3.3098e-02 | 7.7781e-03 | -4.2553  | 2.088e-05 | *** |
| ## 2589 | -1.8479e-02 | 7.9078e-03 | -2.3368  | 0.0194485 | *   |
| ## 2590 | 2.3087e-02  | 8.3637e-03 | 2.7604   | 0.0057738 | **  |
| ## 2591 | 2.6280e-02  | 8.4208e-03 | 3.1208   | 0.0018036 | **  |
| ## 2592 | -1.2958e-02 | 8.0434e-03 | -1.6110  | 0.1071797 |     |
| ## 2593 | -6.7817e-02 | 7.4287e-03 | -9.1291  | < 2.2e-16 | *** |
| ## 2594 | -3.9046e-02 | 7.7544e-03 | -5.0353  | 4.770e-07 | *** |
| ## 2595 | 1.1853e-02  | 8.2264e-03 | 1.4409   | 0.1496228 |     |
| ## 2596 | -1.1438e-02 | 7.9859e-03 | -1.4323  | 0.1520474 |     |
| ## 2597 | -1.4889e-02 | 7.9346e-03 | -1.8765  | 0.0605822 | .   |
| ## 2598 | -2.7679e-02 | 7.8656e-03 | -3.5190  | 0.0004331 | *** |
| ## 2599 | 4.4979e-02  | 8.7219e-03 | 5.1570   | 2.509e-07 | *** |
| ## 2600 | -1.0960e-02 | 7.9940e-03 | -1.3711  | 0.1703454 |     |
| ## 2601 | -3.5788e-03 | 8.0728e-03 | -0.4433  | 0.6575410 |     |
| ## 2602 | -3.4300e-03 | 8.1471e-03 | -0.4210  | 0.6737453 |     |
| ## 2603 | 2.8401e-02  | 8.3818e-03 | 3.3885   | 0.0007029 | *** |
| ## 2604 | 9.3677e-02  | 9.2593e-03 | 10.1171  | < 2.2e-16 | *** |
| ## 2605 | 3.3404e-02  | 8.4440e-03 | 3.9559   | 7.624e-05 | *** |
| ## 2606 | 7.1849e-02  | 8.8752e-03 | 8.0954   | 5.707e-16 | *** |
| ## 2607 | -4.3333e-02 | 7.7208e-03 | -5.6124  | 1.995e-08 | *** |
| ## 2608 | -4.4999e-02 | 7.7353e-03 | -5.8174  | 5.976e-09 | *** |
| ## 2609 | 3.9301e-02  | 8.5114e-03 | 4.6175   | 3.885e-06 | *** |
| ## 2610 | -7.6414e-02 | 7.3159e-03 | -10.4448 | < 2.2e-16 | *** |
| ## 2611 | 1.1283e-03  | 8.1047e-03 | 0.1392   | 0.8892846 |     |

|    |      |             |            |          |           |     |
|----|------|-------------|------------|----------|-----------|-----|
| ## | 2612 | -7.9286e-02 | 7.3302e-03 | -10.8165 | < 2.2e-16 | *** |
| ## | 2613 | -3.8578e-04 | 8.2721e-03 | -0.0466  | 0.9628034 |     |
| ## | 2614 | 3.3312e-02  | 8.4649e-03 | 3.9354   | 8.307e-05 | *** |
| ## | 2615 | -2.0272e-02 | 7.9487e-03 | -2.5503  | 0.0107618 | *   |
| ## | 2616 | -2.3358e-02 | 7.8968e-03 | -2.9579  | 0.0030978 | **  |
| ## | 2617 | -1.9736e-02 | 7.8343e-03 | -2.5192  | 0.0117606 | *   |
| ## | 2618 | -3.4196e-02 | 7.6728e-03 | -4.4568  | 8.319e-06 | *** |
| ## | 2619 | 4.5547e-02  | 8.6471e-03 | 5.2672   | 1.385e-07 | *** |
| ## | 2620 | -1.7315e-02 | 7.9865e-03 | -2.1681  | 0.0301540 | *   |
| ## | 2621 | 6.3615e-02  | 8.8306e-03 | 7.2040   | 5.849e-13 | *** |
| ## | 2622 | -7.7107e-03 | 8.0680e-03 | -0.9557  | 0.3392138 |     |
| ## | 2623 | -5.6989e-02 | 7.7230e-03 | -7.3791  | 1.593e-13 | *** |
| ## | 2624 | 3.5815e-02  | 8.5053e-03 | 4.2109   | 2.544e-05 | *** |
| ## | 2625 | 9.6264e-03  | 8.2597e-03 | 1.1655   | 0.2438298 |     |
| ## | 2626 | -1.5075e-02 | 8.0407e-03 | -1.8749  | 0.0608125 | .   |
| ## | 2627 | 9.8395e-04  | 8.1161e-03 | 0.1212   | 0.9035054 |     |
| ## | 2628 | 3.0895e-02  | 8.4102e-03 | 3.6735   | 0.0002393 | *** |
| ## | 2629 | -7.8698e-03 | 8.0863e-03 | -0.9732  | 0.3304451 |     |
| ## | 2630 | 1.4403e-02  | 8.2853e-03 | 1.7384   | 0.0821382 | .   |
| ## | 2631 | 4.0890e-02  | 8.6462e-03 | 4.7292   | 2.254e-06 | *** |
| ## | 2632 | 3.4212e-03  | 8.1610e-03 | 0.4192   | 0.6750626 |     |
| ## | 2633 | -1.9071e-02 | 7.9052e-03 | -2.4125  | 0.0158422 | *   |
| ## | 2634 | -7.4603e-02 | 7.3576e-03 | -10.1395 | < 2.2e-16 | *** |
| ## | 2635 | 9.6104e-03  | 8.2986e-03 | 1.1581   | 0.2468306 |     |
| ## | 2636 | 1.7376e-02  | 8.3744e-03 | 2.0749   | 0.0379958 | *   |
| ## | 2637 | 9.7379e-03  | 8.2531e-03 | 1.1799   | 0.2380365 |     |
| ## | 2638 | 2.4765e-02  | 8.3911e-03 | 2.9514   | 0.0031635 | **  |
| ## | 2639 | -5.1554e-02 | 7.3296e-03 | -7.0337  | 2.012e-12 | *** |
| ## | 2640 | 1.8486e-03  | 8.1777e-03 | 0.2261   | 0.8211585 |     |
| ## | 2641 | -3.6102e-04 | 8.1214e-03 | -0.0445  | 0.9645427 |     |
| ## | 2642 | 1.3784e-03  | 8.1772e-03 | 0.1686   | 0.8661393 |     |
| ## | 2643 | 3.0569e-02  | 8.4120e-03 | 3.6340   | 0.0002790 | *** |
| ## | 2644 | 9.9061e-02  | 9.2278e-03 | 10.7350  | < 2.2e-16 | *** |
| ## | 2645 | -3.3376e-02 | 7.7971e-03 | -4.2806  | 1.864e-05 | *** |
| ## | 2646 | -3.3667e-03 | 8.0537e-03 | -0.4180  | 0.6759261 |     |
| ## | 2647 | -1.8609e-02 | 7.9682e-03 | -2.3355  | 0.0195198 | *   |
| ## | 2648 | 1.7458e-02  | 8.3374e-03 | 2.0940   | 0.0362599 | *   |
| ## | 2649 | 7.2803e-02  | 8.9844e-03 | 8.1033   | 5.349e-16 | *** |
| ## | 2650 | -7.6062e-02 | 7.3467e-03 | -10.3532 | < 2.2e-16 | *** |
| ## | 2651 | -5.9093e-02 | 7.5861e-03 | -7.7895  | 6.725e-15 | *** |
| ## | 2652 | 1.0240e-02  | 8.2210e-03 | 1.2456   | 0.2129145 |     |
| ## | 2653 | -4.0385e-03 | 8.1004e-03 | -0.4986  | 0.6180892 |     |
| ## | 2654 | 4.3485e-02  | 8.5948e-03 | 5.0595   | 4.204e-07 | *** |
| ## | 2655 | 6.8526e-02  | 8.8858e-03 | 7.7118   | 1.240e-14 | *** |
| ## | 2656 | -2.5311e-02 | 7.8503e-03 | -3.2242  | 0.0012635 | **  |
| ## | 2657 | -7.8376e-02 | 7.4272e-03 | -10.5526 | < 2.2e-16 | *** |
| ## | 2658 | -5.0919e-02 | 7.6534e-03 | -6.6531  | 2.869e-11 | *** |
| ## | 2659 | 2.6908e-03  | 8.0602e-03 | 0.3338   | 0.7385006 |     |

|         |             |            |          |           |     |
|---------|-------------|------------|----------|-----------|-----|
| ## 2660 | -2.3184e-02 | 7.9201e-03 | -2.9273  | 0.0034197 | **  |
| ## 2661 | -4.1731e-03 | 8.0674e-03 | -0.5173  | 0.6049629 |     |
| ## 2662 | -2.0422e-02 | 7.9392e-03 | -2.5723  | 0.0101034 | *   |
| ## 2663 | 2.6770e-02  | 8.3817e-03 | 3.1938   | 0.0014039 | **  |
| ## 2664 | 7.2275e-02  | 8.8985e-03 | 8.1222   | 4.578e-16 | *** |
| ## 2665 | 3.5250e-02  | 8.4950e-03 | 4.1495   | 3.333e-05 | *** |
| ## 2666 | -1.4615e-03 | 8.0582e-03 | -0.1814  | 0.8560815 |     |
| ## 2667 | -4.0493e-02 | 7.7341e-03 | -5.2356  | 1.644e-07 | *** |
| ## 2668 | -1.7820e-02 | 7.9949e-03 | -2.2290  | 0.0258169 | *   |
| ## 2669 | -9.2117e-02 | 7.1534e-03 | -12.8773 | < 2.2e-16 | *** |
| ## 2670 | -3.3138e-03 | 8.1297e-03 | -0.4076  | 0.6835542 |     |
| ## 2671 | 1.3306e-02  | 8.1626e-03 | 1.6301   | 0.1030763 |     |
| ## 2672 | 1.2750e-02  | 8.2751e-03 | 1.5408   | 0.1233651 |     |
| ## 2673 | -1.4923e-02 | 7.9807e-03 | -1.8699  | 0.0614966 | .   |
| ## 2674 | 1.6275e-02  | 8.3978e-03 | 1.9380   | 0.0526172 | .   |
| ## 2675 | -3.0494e-02 | 7.7850e-03 | -3.9170  | 8.966e-05 | *** |
| ## 2676 | -2.2592e-02 | 7.9253e-03 | -2.8506  | 0.0043634 | **  |
| ## 2677 | 3.2424e-03  | 8.0963e-03 | 0.4005   | 0.6888075 |     |
| ## 2678 | 2.5508e-02  | 8.3627e-03 | 3.0502   | 0.0022871 | **  |
| ## 2679 | 2.4097e-02  | 8.4228e-03 | 2.8610   | 0.0042237 | **  |
| ## 2680 | -2.9782e-02 | 7.8256e-03 | -3.8057  | 0.0001414 | *** |
| ## 2681 | 1.4329e-02  | 8.2790e-03 | 1.7308   | 0.0834952 | .   |
| ## 2682 | 1.7351e-02  | 8.2396e-03 | 2.1058   | 0.0352183 | *   |
| ## 2683 | 4.2982e-03  | 8.2433e-03 | 0.5214   | 0.6020797 |     |
| ## 2684 | -3.9691e-03 | 8.0767e-03 | -0.4914  | 0.6231217 |     |
| ## 2685 | -2.7167e-02 | 7.8475e-03 | -3.4619  | 0.0005363 | *** |
| ## 2686 | -1.5363e-02 | 7.9839e-03 | -1.9242  | 0.0543279 | .   |
| ## 2687 | -4.8356e-03 | 7.9998e-03 | -0.6045  | 0.5455372 |     |
| ## 2688 | -1.0798e-02 | 8.0353e-03 | -1.3439  | 0.1789957 |     |
| ## 2689 | 1.3549e-02  | 8.2796e-03 | 1.6364   | 0.1017577 |     |
| ## 2690 | -5.1389e-02 | 7.6517e-03 | -6.7160  | 1.867e-11 | *** |
| ## 2691 | -2.5454e-02 | 7.8272e-03 | -3.2519  | 0.0011462 | **  |
| ## 2692 | 4.4127e-02  | 8.6569e-03 | 5.0973   | 3.444e-07 | *** |
| ## 2693 | -2.6322e-02 | 7.8546e-03 | -3.3512  | 0.0008047 | *** |
| ## 2694 | 5.5912e-02  | 8.7416e-03 | 6.3961   | 1.594e-10 | *** |
| ## 2695 | 2.7684e-02  | 8.4182e-03 | 3.2886   | 0.0010069 | **  |
| ## 2696 | 2.0636e-02  | 8.3385e-03 | 2.4748   | 0.0133312 | *   |
| ## 2697 | -2.5142e-02 | 7.8235e-03 | -3.2136  | 0.0013108 | **  |
| ## 2698 | -2.0972e-02 | 7.9213e-03 | -2.6475  | 0.0081093 | **  |
| ## 2699 | 1.1966e-02  | 8.2667e-03 | 1.4475   | 0.1477524 |     |
| ## 2700 | -6.4923e-02 | 7.4967e-03 | -8.6602  | < 2.2e-16 | *** |
| ## 2701 | 4.4773e-02  | 8.5853e-03 | 5.2151   | 1.837e-07 | *** |
| ## 2702 | -4.2632e-02 | 7.7291e-03 | -5.5158  | 3.472e-08 | *** |
| ## 2703 | 3.6797e-04  | 8.1658e-03 | 0.0451   | 0.9640574 |     |
| ## 2704 | 9.4195e-02  | 9.1762e-03 | 10.2651  | < 2.2e-16 | *** |
| ## 2705 | -8.0696e-02 | 7.2692e-03 | -11.1011 | < 2.2e-16 | *** |
| ## 2706 | 8.0407e-03  | 8.1901e-03 | 0.9818   | 0.3262159 |     |
| ## 2707 | 4.5488e-02  | 8.6215e-03 | 5.2761   | 1.320e-07 | *** |

|         |             |            |          |           |     |
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| ## 2708 | 1.4300e-02  | 8.2814e-03 | 1.7267   | 0.0842140 | .   |
| ## 2709 | 1.3762e-02  | 8.2776e-03 | 1.6625   | 0.0964108 | .   |
| ## 2710 | 2.2909e-03  | 8.2260e-03 | 0.2785   | 0.7806355 |     |
| ## 2711 | 7.4153e-02  | 9.0115e-03 | 8.2287   | < 2.2e-16 | *** |
| ## 2712 | -2.1264e-02 | 7.9962e-03 | -2.6592  | 0.0078322 | **  |
| ## 2713 | -6.3984e-02 | 7.4629e-03 | -8.5736  | < 2.2e-16 | *** |
| ## 2714 | -4.3984e-02 | 7.7337e-03 | -5.6873  | 1.291e-08 | *** |
| ## 2715 | 6.7358e-03  | 8.1728e-03 | 0.8242   | 0.4098406 |     |
| ## 2716 | 1.8635e-02  | 8.3144e-03 | 2.2413   | 0.0250040 | *   |
| ## 2717 | 3.2593e-03  | 8.1204e-03 | 0.4014   | 0.6881421 |     |
| ## 2718 | 5.3311e-02  | 8.7436e-03 | 6.0971   | 1.080e-09 | *** |
| ## 2719 | -3.0959e-02 | 7.5950e-03 | -4.0762  | 4.577e-05 | *** |
| ## 2720 | -3.2615e-03 | 8.1006e-03 | -0.4026  | 0.6872275 |     |
| ## 2721 | -8.0140e-02 | 7.4035e-03 | -10.8246 | < 2.2e-16 | *** |
| ## 2722 | -1.9034e-02 | 7.9702e-03 | -2.3882  | 0.0169328 | *   |
| ## 2723 | -1.1519e-02 | 8.0585e-03 | -1.4294  | 0.1528932 |     |
| ## 2724 | -2.9836e-02 | 7.7844e-03 | -3.8328  | 0.0001267 | *** |
| ## 2725 | 1.4838e-02  | 8.3319e-03 | 1.7809   | 0.0749299 | .   |
| ## 2726 | 4.3115e-02  | 8.6141e-03 | 5.0051   | 5.583e-07 | *** |
| ## 2727 | 1.5628e-02  | 8.2712e-03 | 1.8895   | 0.0588299 | .   |
| ## 2728 | -1.2491e-02 | 7.9804e-03 | -1.5652  | 0.1175251 |     |
| ## 2729 | -1.3259e-02 | 7.9303e-03 | -1.6719  | 0.0945366 | .   |
| ## 2730 | -6.7049e-02 | 7.3934e-03 | -9.0688  | < 2.2e-16 | *** |
| ## 2731 | -4.5759e-02 | 7.5993e-03 | -6.0215  | 1.728e-09 | *** |
| ## 2732 | -2.0851e-02 | 7.9414e-03 | -2.6256  | 0.0086488 | **  |
| ## 2733 | 2.4841e-02  | 8.3789e-03 | 2.9647   | 0.0030295 | **  |
| ## 2734 | 4.6706e-02  | 8.6387e-03 | 5.4066   | 6.423e-08 | *** |
| ## 2735 | -1.5312e-03 | 8.0985e-03 | -0.1891  | 0.8500397 |     |
| ## 2736 | -3.9865e-02 | 7.6968e-03 | -5.1795  | 2.225e-07 | *** |
| ## 2737 | 6.5308e-02  | 8.8921e-03 | 7.3445   | 2.066e-13 | *** |
| ## 2738 | -2.7730e-02 | 7.8289e-03 | -3.5420  | 0.0003971 | *** |
| ## 2739 | 2.1912e-02  | 8.3469e-03 | 2.6252   | 0.0086602 | **  |
| ## 2740 | 3.3753e-02  | 8.4821e-03 | 3.9793   | 6.911e-05 | *** |
| ## 2741 | 7.1403e-03  | 8.1398e-03 | 0.8772   | 0.3803701 |     |
| ## 2742 | -5.3188e-04 | 8.1178e-03 | -0.0655  | 0.9477600 |     |
| ## 2743 | 8.5937e-03  | 8.2028e-03 | 1.0477   | 0.2947989 |     |
| ## 2744 | -1.5155e-02 | 7.9110e-03 | -1.9157  | 0.0554085 | .   |
| ## 2745 | 5.5474e-03  | 8.2211e-03 | 0.6748   | 0.4998149 |     |
| ## 2746 | -1.0303e-02 | 8.0248e-03 | -1.2840  | 0.1991593 |     |
| ## 2747 | -2.6483e-02 | 7.8413e-03 | -3.3774  | 0.0007319 | *** |
| ## 2748 | -1.7077e-02 | 7.8982e-03 | -2.1621  | 0.0306102 | *   |
| ## 2749 | 9.7012e-02  | 9.2246e-03 | 10.5166  | < 2.2e-16 | *** |
| ## 2750 | 1.1751e-03  | 8.1278e-03 | 0.1446   | 0.8850433 |     |
| ## 2751 | -2.4825e-03 | 8.1476e-03 | -0.3047  | 0.7605979 |     |
| ## 2752 | 3.9120e-02  | 8.5937e-03 | 4.5522   | 5.310e-06 | *** |
| ## 2753 | -1.7391e-02 | 7.9757e-03 | -2.1805  | 0.0292192 | *   |
| ## 2754 | -2.8833e-02 | 7.8747e-03 | -3.6615  | 0.0002508 | *** |
| ## 2755 | 7.4156e-02  | 8.8983e-03 | 8.3337   | < 2.2e-16 | *** |

|         |             |            |          |           |     |
|---------|-------------|------------|----------|-----------|-----|
| ## 2756 | 1.2676e-02  | 8.2968e-03 | 1.5278   | 0.1265708 |     |
| ## 2757 | -2.0254e-02 | 7.9576e-03 | -2.5453  | 0.0109193 | *   |
| ## 2758 | 1.1218e-01  | 9.3970e-03 | 11.9375  | < 2.2e-16 | *** |
| ## 2759 | 6.0823e-02  | 8.7907e-03 | 6.9190   | 4.549e-12 | *** |
| ## 2760 | -2.7620e-03 | 8.1507e-03 | -0.3389  | 0.7347113 |     |
| ## 2761 | -7.5404e-03 | 8.0423e-03 | -0.9376  | 0.3484553 |     |
| ## 2762 | 3.3727e-02  | 8.4779e-03 | 3.9782   | 6.943e-05 | *** |
| ## 2763 | 6.1523e-02  | 8.7632e-03 | 7.0206   | 2.209e-12 | *** |
| ## 2764 | -4.6802e-02 | 7.6899e-03 | -6.0861  | 1.157e-09 | *** |
| ## 2765 | 2.3967e-02  | 8.3433e-03 | 2.8725   | 0.0040718 | **  |
| ## 2766 | 6.7916e-02  | 8.8519e-03 | 7.6725   | 1.687e-14 | *** |
| ## 2767 | -1.9166e-02 | 7.9259e-03 | -2.4182  | 0.0155992 | *   |
| ## 2768 | 3.3485e-02  | 8.5062e-03 | 3.9366   | 8.266e-05 | *** |
| ## 2769 | 8.8285e-03  | 8.2435e-03 | 1.0710   | 0.2841856 |     |
| ## 2770 | -6.1841e-02 | 7.5957e-03 | -8.1415  | 3.903e-16 | *** |
| ## 2771 | -2.5031e-02 | 7.8977e-03 | -3.1694  | 0.0015276 | **  |
| ## 2772 | 1.4799e-02  | 8.2462e-03 | 1.7947   | 0.0727088 | .   |
| ## 2773 | 5.6105e-02  | 8.7855e-03 | 6.3861   | 1.702e-10 | *** |
| ## 2774 | -2.1043e-02 | 7.9177e-03 | -2.6577  | 0.0078673 | **  |
| ## 2775 | -3.9349e-02 | 7.6436e-03 | -5.1479  | 2.634e-07 | *** |
| ## 2776 | 2.5422e-03  | 8.1401e-03 | 0.3123   | 0.7548050 |     |
| ## 2777 | 2.9010e-02  | 8.5170e-03 | 3.4061   | 0.0006590 | *** |
| ## 2778 | -2.6061e-02 | 7.8140e-03 | -3.3351  | 0.0008526 | *** |
| ## 2779 | -9.8604e-03 | 8.0460e-03 | -1.2255  | 0.2203843 |     |
| ## 2780 | 2.8367e-02  | 8.5069e-03 | 3.3346   | 0.0008543 | *** |
| ## 2781 | -2.3219e-02 | 7.9096e-03 | -2.9355  | 0.0033302 | **  |
| ## 2782 | -2.2389e-02 | 7.9206e-03 | -2.8266  | 0.0047039 | **  |
| ## 2783 | -1.5251e-02 | 7.9835e-03 | -1.9103  | 0.0560976 | .   |
| ## 2784 | 2.0042e-02  | 8.3129e-03 | 2.4109   | 0.0159126 | *   |
| ## 2785 | -5.5259e-02 | 7.5265e-03 | -7.3420  | 2.105e-13 | *** |
| ## 2786 | 1.0415e-01  | 9.3138e-03 | 11.1825  | < 2.2e-16 | *** |
| ## 2787 | 1.6197e-02  | 8.3209e-03 | 1.9466   | 0.0515863 | .   |
| ## 2788 | 4.9493e-02  | 8.6615e-03 | 5.7141   | 1.103e-08 | *** |
| ## 2789 | 3.7597e-02  | 8.5727e-03 | 4.3857   | 1.156e-05 | *** |
| ## 2790 | 2.7548e-02  | 8.4434e-03 | 3.2627   | 0.0011037 | **  |
| ## 2791 | -4.3180e-03 | 8.0759e-03 | -0.5347  | 0.5928741 |     |
| ## 2792 | 1.3077e-02  | 8.2912e-03 | 1.5772   | 0.1147578 |     |
| ## 2793 | 1.6254e-02  | 8.3342e-03 | 1.9503   | 0.0511445 | .   |
| ## 2794 | -3.4056e-03 | 8.0590e-03 | -0.4226  | 0.6726049 |     |
| ## 2795 | -6.4962e-03 | 8.0736e-03 | -0.8046  | 0.4210351 |     |
| ## 2796 | -3.5881e-02 | 7.7993e-03 | -4.6005  | 4.215e-06 | *** |
| ## 2797 | -4.8630e-03 | 8.1232e-03 | -0.5987  | 0.5494030 |     |
| ## 2798 | -1.0660e-02 | 8.0562e-03 | -1.3232  | 0.1857762 |     |
| ## 2799 | -8.1736e-02 | 7.3037e-03 | -11.1911 | < 2.2e-16 | *** |
| ## 2800 | 2.1854e-02  | 8.4413e-03 | 2.5889   | 0.0096273 | **  |
| ## 2801 | -8.9032e-02 | 7.3564e-03 | -12.1026 | < 2.2e-16 | *** |
| ## 2802 | 6.1855e-02  | 8.7555e-03 | 7.0646   | 1.610e-12 | *** |
| ## 2803 | -2.3341e-02 | 7.8692e-03 | -2.9661  | 0.0030164 | **  |

|         |             |            |          |           |     |
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| ## 2804 | 3.0460e-02  | 8.4674e-03 | 3.5973   | 0.0003216 | *** |
| ## 2805 | 3.3661e-02  | 8.4796e-03 | 3.9696   | 7.199e-05 | *** |
| ## 2806 | 1.9994e-02  | 8.3544e-03 | 2.3933   | 0.0166993 | *   |
| ## 2807 | -9.2182e-03 | 8.0487e-03 | -1.1453  | 0.2520843 |     |
| ## 2808 | -4.6291e-02 | 7.6843e-03 | -6.0241  | 1.700e-09 | *** |
| ## 2809 | 2.5423e-02  | 8.3752e-03 | 3.0355   | 0.0024015 | **  |
| ## 2810 | -7.3557e-02 | 7.3236e-03 | -10.0439 | < 2.2e-16 | *** |
| ## 2811 | -3.4693e-03 | 8.0891e-03 | -0.4289  | 0.6680067 |     |
| ## 2812 | 1.2992e-02  | 8.3124e-03 | 1.5630   | 0.1180554 |     |
| ## 2813 | -3.6791e-02 | 7.7826e-03 | -4.7273  | 2.276e-06 | *** |
| ## 2814 | 7.0395e-03  | 8.2168e-03 | 0.8567   | 0.3916022 |     |
| ## 2815 | 3.0547e-02  | 8.4517e-03 | 3.6143   | 0.0003012 | *** |
| ## 2816 | 5.8147e-02  | 8.7724e-03 | 6.6284   | 3.394e-11 | *** |
| ## 2817 | -4.3454e-02 | 7.8139e-03 | -5.5611  | 2.680e-08 | *** |
| ## 2818 | 1.2847e-02  | 8.2841e-03 | 1.5508   | 0.1209420 |     |
| ## 2819 | 1.6101e-02  | 8.3174e-03 | 1.9358   | 0.0528897 | .   |
| ## 2820 | -2.7106e-02 | 7.9018e-03 | -3.4303  | 0.0006029 | *** |
| ## 2821 | 1.0541e-02  | 8.2066e-03 | 1.2844   | 0.1989897 |     |
| ## 2822 | 2.6661e-02  | 8.4132e-03 | 3.1689   | 0.0015301 | **  |
| ## 2823 | 6.0831e-02  | 8.8023e-03 | 6.9109   | 4.817e-12 | *** |
| ## 2824 | 2.6775e-02  | 8.4439e-03 | 3.1709   | 0.0015196 | **  |
| ## 2825 | 2.8854e-02  | 8.4285e-03 | 3.4234   | 0.0006185 | *** |
| ## 2826 | 1.8124e-02  | 8.4188e-03 | 2.1528   | 0.0313350 | *   |
| ## 2827 | 3.7939e-02  | 8.5330e-03 | 4.4462   | 8.741e-06 | *** |
| ## 2828 | -7.8325e-03 | 8.0784e-03 | -0.9696  | 0.3322644 |     |
| ## 2829 | -1.8284e-02 | 7.7848e-03 | -2.3486  | 0.0188428 | *   |
| ## 2830 | 2.5940e-02  | 8.4918e-03 | 3.0547   | 0.0022529 | **  |
| ## 2831 | 7.9327e-03  | 8.2044e-03 | 0.9669   | 0.3336043 |     |
| ## 2832 | 3.4653e-02  | 8.5550e-03 | 4.0505   | 5.110e-05 | *** |
| ## 2833 | -8.0273e-04 | 8.1144e-03 | -0.0989  | 0.9211968 |     |
| ## 2834 | 1.2609e-02  | 8.1562e-03 | 1.5459   | 0.1221343 |     |
| ## 2835 | -1.8040e-02 | 7.9478e-03 | -2.2698  | 0.0232179 | *   |
| ## 2836 | -1.1066e-02 | 8.1205e-03 | -1.3627  | 0.1729853 |     |
| ## 2837 | 6.9070e-02  | 8.8424e-03 | 7.8112   | 5.664e-15 | *** |
| ## 2838 | -8.4818e-03 | 8.0568e-03 | -1.0527  | 0.2924578 |     |
| ## 2839 | -4.5990e-02 | 7.7271e-03 | -5.9518  | 2.653e-09 | *** |
| ## 2840 | 2.3460e-03  | 8.1442e-03 | 0.2881   | 0.7733056 |     |
| ## 2841 | 2.2667e-02  | 8.3596e-03 | 2.7115   | 0.0066974 | **  |
| ## 2842 | -1.8688e-02 | 7.9460e-03 | -2.3519  | 0.0186790 | *   |
| ## 2843 | -9.8374e-03 | 8.0260e-03 | -1.2257  | 0.2203115 |     |
| ## 2844 | -5.5476e-03 | 8.1249e-03 | -0.6828  | 0.4947370 |     |
| ## 2845 | 4.2112e-02  | 8.5705e-03 | 4.9136   | 8.941e-07 | *** |
| ## 2846 | 1.2100e-02  | 8.2153e-03 | 1.4728   | 0.1408012 |     |
| ## 2847 | 7.3424e-02  | 8.9583e-03 | 8.1961   | 2.483e-16 | *** |
| ## 2848 | 3.5298e-02  | 8.5087e-03 | 4.1484   | 3.347e-05 | *** |
| ## 2849 | 2.9278e-03  | 8.1386e-03 | 0.3597   | 0.7190449 |     |
| ## 2850 | 1.4831e-02  | 8.2717e-03 | 1.7930   | 0.0729704 | .   |
| ## 2851 | 5.3135e-03  | 8.2334e-03 | 0.6454   | 0.5186968 |     |

|         |             |            |          |           |     |
|---------|-------------|------------|----------|-----------|-----|
| ## 2852 | 1.3132e-02  | 8.2445e-03 | 1.5928   | 0.1111979 |     |
| ## 2853 | 1.5584e-02  | 8.2640e-03 | 1.8858   | 0.0593228 | .   |
| ## 2854 | 2.8818e-02  | 8.4369e-03 | 3.4158   | 0.0006360 | *** |
| ## 2855 | -4.0779e-03 | 8.1261e-03 | -0.5018  | 0.6157861 |     |
| ## 2856 | 3.9996e-02  | 8.5710e-03 | 4.6665   | 3.064e-06 | *** |
| ## 2857 | -5.2768e-02 | 7.6082e-03 | -6.9358  | 4.040e-12 | *** |
| ## 2858 | 2.8086e-02  | 8.4173e-03 | 3.3367   | 0.0008479 | *** |
| ## 2859 | 4.0179e-02  | 8.5145e-03 | 4.7189   | 2.371e-06 | *** |
| ## 2860 | -3.7102e-02 | 7.7552e-03 | -4.7842  | 1.717e-06 | *** |
| ## 2861 | 2.9531e-02  | 8.4145e-03 | 3.5096   | 0.0004488 | *** |
| ## 2862 | -9.0109e-03 | 8.0168e-03 | -1.1240  | 0.2610109 |     |
| ## 2863 | 2.7570e-02  | 8.4480e-03 | 3.2635   | 0.0011006 | **  |
| ## 2864 | 8.2690e-03  | 8.2077e-03 | 1.0075   | 0.3137082 |     |
| ## 2865 | 5.8781e-02  | 8.6994e-03 | 6.7569   | 1.410e-11 | *** |
| ## 2866 | -2.5636e-02 | 7.7695e-03 | -3.2995  | 0.0009684 | *** |
| ## 2867 | 3.3252e-03  | 8.1510e-03 | 0.4079   | 0.6833111 |     |
| ## 2868 | 3.1230e-02  | 8.4659e-03 | 3.6890   | 0.0002252 | *** |
| ## 2869 | -6.9529e-03 | 8.0508e-03 | -0.8636  | 0.3877945 |     |
| ## 2870 | -1.5486e-02 | 8.0016e-03 | -1.9354  | 0.0529466 | .   |
| ## 2871 | -5.6318e-02 | 7.4992e-03 | -7.5099  | 5.917e-14 | *** |
| ## 2872 | -2.8099e-03 | 8.1702e-03 | -0.3439  | 0.7309087 |     |
| ## 2873 | -7.6013e-03 | 8.0319e-03 | -0.9464  | 0.3439526 |     |
| ## 2874 | -2.2095e-02 | 7.6999e-03 | -2.8695  | 0.0041112 | **  |
| ## 2875 | 8.8453e-02  | 9.1483e-03 | 9.6688   | < 2.2e-16 | *** |
| ## 2876 | -4.5019e-02 | 7.6469e-03 | -5.8872  | 3.927e-09 | *** |
| ## 2877 | -7.9245e-02 | 7.2243e-03 | -10.9692 | < 2.2e-16 | *** |
| ## 2878 | 1.1548e-01  | 9.3751e-03 | 12.3177  | < 2.2e-16 | *** |
| ## 2879 | 2.4522e-02  | 8.3568e-03 | 2.9344   | 0.0033424 | **  |
| ## 2880 | -8.4529e-04 | 8.1722e-03 | -0.1034  | 0.9176181 |     |
| ## 2881 | -7.1489e-02 | 7.4655e-03 | -9.5759  | < 2.2e-16 | *** |
| ## 2882 | 6.5449e-02  | 8.8054e-03 | 7.4328   | 1.063e-13 | *** |
| ## 2883 | 5.6145e-02  | 8.7170e-03 | 6.4409   | 1.188e-10 | *** |
| ## 2884 | 7.6283e-02  | 8.9759e-03 | 8.4986   | < 2.2e-16 | *** |
| ## 2885 | -4.0107e-02 | 7.6497e-03 | -5.2429  | 1.580e-07 | *** |
| ## 2886 | 1.3037e-01  | 9.6155e-03 | 13.5588  | < 2.2e-16 | *** |
| ## 2887 | 2.5047e-02  | 8.4116e-03 | 2.9777   | 0.0029046 | **  |
| ## 2888 | -6.5779e-02 | 7.5323e-03 | -8.7330  | < 2.2e-16 | *** |
| ## 2889 | -5.3303e-02 | 7.6584e-03 | -6.9600  | 3.402e-12 | *** |
| ## 2890 | -5.8667e-02 | 7.5947e-03 | -7.7248  | 1.121e-14 | *** |
| ## 2891 | 4.0969e-02  | 8.6140e-03 | 4.7561   | 1.974e-06 | *** |
| ## 2892 | 5.1877e-02  | 8.7092e-03 | 5.9566   | 2.576e-09 | *** |
| ## 2893 | 7.3282e-03  | 8.2644e-03 | 0.8867   | 0.3752348 |     |
| ## 2894 | 1.4481e-02  | 8.2740e-03 | 1.7502   | 0.0800814 | .   |
| ## 2895 | 7.1957e-02  | 8.9254e-03 | 8.0620   | 7.506e-16 | *** |
| ## 2896 | 8.3577e-02  | 9.0262e-03 | 9.2594   | < 2.2e-16 | *** |
| ## 2897 | -1.8278e-02 | 7.9563e-03 | -2.2973  | 0.0216024 | *   |
| ## 2898 | 5.4343e-03  | 8.1769e-03 | 0.6646   | 0.5063114 |     |
| ## 2899 | -3.2005e-02 | 7.7983e-03 | -4.1040  | 4.060e-05 | *** |

|         |             |            |          |           |     |
|---------|-------------|------------|----------|-----------|-----|
| ## 2900 | 3.4862e-02  | 8.4707e-03 | 4.1156   | 3.861e-05 | *** |
| ## 2901 | 3.3396e-02  | 8.4989e-03 | 3.9295   | 8.514e-05 | *** |
| ## 2902 | 2.0472e-02  | 8.3430e-03 | 2.4538   | 0.0141364 | *   |
| ## 2903 | 1.8934e-03  | 8.1743e-03 | 0.2316   | 0.8168296 |     |
| ## 2904 | -1.0317e-02 | 7.9998e-03 | -1.2897  | 0.1971584 |     |
| ## 2905 | 4.7970e-02  | 8.6741e-03 | 5.5303   | 3.196e-08 | *** |
| ## 2906 | -7.9277e-03 | 8.0639e-03 | -0.9831  | 0.3255526 |     |
| ## 2907 | -6.3071e-02 | 7.4787e-03 | -8.4334  | < 2.2e-16 | *** |
| ## 2908 | 5.2140e-02  | 8.6417e-03 | 6.0336   | 1.604e-09 | *** |
| ## 2909 | -2.8419e-02 | 7.8534e-03 | -3.6187  | 0.0002961 | *** |
| ## 2910 | -6.1842e-02 | 7.4345e-03 | -8.3182  | < 2.2e-16 | *** |
| ## 2911 | -9.2210e-05 | 8.0892e-03 | -0.0114  | 0.9909049 |     |
| ## 2912 | 4.9019e-03  | 8.1688e-03 | 0.6001   | 0.5484593 |     |
| ## 2913 | 1.0945e-02  | 8.2765e-03 | 1.3224   | 0.1860318 |     |
| ## 2914 | 2.5926e-02  | 8.4383e-03 | 3.0724   | 0.0021232 | **  |
| ## 2915 | -2.0640e-04 | 8.1190e-03 | -0.0254  | 0.9797184 |     |
| ## 2916 | 3.6085e-03  | 8.1466e-03 | 0.4429   | 0.6578028 |     |
| ## 2917 | -6.8665e-02 | 7.3594e-03 | -9.3301  | < 2.2e-16 | *** |
| ## 2918 | 1.1887e-03  | 8.1061e-03 | 0.1466   | 0.8834142 |     |
| ## 2919 | 1.0401e-01  | 9.3062e-03 | 11.1760  | < 2.2e-16 | *** |
| ## 2920 | -3.4267e-02 | 7.8481e-03 | -4.3662  | 1.264e-05 | *** |
| ## 2921 | 1.2146e-02  | 8.2336e-03 | 1.4751   | 0.1401795 |     |
| ## 2922 | -3.5542e-02 | 7.7630e-03 | -4.5784  | 4.687e-06 | *** |
| ## 2923 | -1.6179e-02 | 8.0096e-03 | -2.0200  | 0.0433837 | *   |
| ## 2924 | 2.0498e-02  | 8.3538e-03 | 2.4537   | 0.0141387 | *   |
| ## 2925 | 5.0107e-02  | 8.7095e-03 | 5.7531   | 8.760e-09 | *** |
| ## 2926 | 2.8415e-02  | 8.3342e-03 | 3.4094   | 0.0006509 | *** |
| ## 2927 | 1.1327e-02  | 8.2318e-03 | 1.3759   | 0.1688375 |     |
| ## 2928 | 2.0728e-02  | 8.3519e-03 | 2.4819   | 0.0130695 | *   |
| ## 2929 | 1.5852e-02  | 8.3223e-03 | 1.9048   | 0.0568053 | .   |
| ## 2930 | -4.6538e-03 | 8.0642e-03 | -0.5771  | 0.5638751 |     |
| ## 2931 | -8.1819e-03 | 8.0519e-03 | -1.0161  | 0.3095600 |     |
| ## 2932 | 1.0328e-02  | 8.2096e-03 | 1.2580   | 0.2083862 |     |
| ## 2933 | 8.1800e-02  | 9.0751e-03 | 9.0137   | < 2.2e-16 | *** |
| ## 2934 | -2.3637e-02 | 7.8716e-03 | -3.0028  | 0.0026751 | **  |
| ## 2935 | -1.4546e-02 | 8.0051e-03 | -1.8171  | 0.0692020 | .   |
| ## 2936 | 1.4014e-02  | 8.2915e-03 | 1.6902   | 0.0909864 | .   |
| ## 2937 | -5.2745e-02 | 7.5805e-03 | -6.9580  | 3.451e-12 | *** |
| ## 2938 | -2.7793e-02 | 7.8860e-03 | -3.5244  | 0.0004245 | *** |
| ## 2939 | -3.4091e-02 | 7.6973e-03 | -4.4289  | 9.471e-06 | *** |
| ## 2940 | 3.7730e-02  | 8.5063e-03 | 4.4356   | 9.182e-06 | *** |
| ## 2941 | -9.6063e-03 | 8.0449e-03 | -1.1941  | 0.2324447 |     |
| ## 2942 | 6.5784e-03  | 8.2257e-03 | 0.7997   | 0.4238609 |     |
| ## 2943 | 1.5959e-02  | 8.3218e-03 | 1.9177   | 0.0551526 | .   |
| ## 2944 | -2.5179e-02 | 7.9008e-03 | -3.1868  | 0.0014385 | **  |
| ## 2945 | -8.3085e-02 | 7.2428e-03 | -11.4714 | < 2.2e-16 | *** |
| ## 2946 | -2.1156e-02 | 7.9025e-03 | -2.6771  | 0.0074257 | **  |
| ## 2947 | 1.0330e-02  | 8.2403e-03 | 1.2536   | 0.2099947 |     |



|    |      |             |            |          |           |     |
|----|------|-------------|------------|----------|-----------|-----|
| ## | 2948 | -1.4584e-02 | 7.9513e-03 | -1.8341  | 0.0666331 | .   |
| ## | 2949 | -3.0668e-02 | 7.8166e-03 | -3.9234  | 8.731e-05 | *** |
| ## | 2950 | -4.4028e-02 | 7.8056e-03 | -5.6405  | 1.695e-08 | *** |
| ## | 2951 | 1.0022e-01  | 9.0867e-03 | 11.0290  | < 2.2e-16 | *** |
| ## | 2952 | -8.4890e-02 | 7.3273e-03 | -11.5853 | < 2.2e-16 | *** |
| ## | 2953 | -1.6002e-02 | 7.9496e-03 | -2.0130  | 0.0441183 | *   |
| ## | 2954 | 6.8058e-03  | 8.2175e-03 | 0.8282   | 0.4075529 |     |
| ## | 2955 | 1.0090e-02  | 8.2629e-03 | 1.2211   | 0.2220349 |     |
| ## | 2956 | 6.6664e-02  | 8.8810e-03 | 7.5063   | 6.081e-14 | *** |
| ## | 2957 | -1.3714e-02 | 7.9674e-03 | -1.7213  | 0.0851991 | .   |
| ## | 2958 | 2.4700e-02  | 8.4004e-03 | 2.9403   | 0.0032792 | **  |
| ## | 2959 | 1.4796e-02  | 8.3643e-03 | 1.7690   | 0.0768942 | .   |
| ## | 2960 | -4.1187e-03 | 7.8994e-03 | -0.5214  | 0.6020944 |     |
| ## | 2961 | -4.2228e-04 | 8.1661e-03 | -0.0517  | 0.9587581 |     |
| ## | 2962 | 3.4384e-02  | 8.4905e-03 | 4.0497   | 5.128e-05 | *** |
| ## | 2963 | -7.6748e-02 | 7.3552e-03 | -10.4346 | < 2.2e-16 | *** |
| ## | 2964 | -3.0440e-02 | 7.9173e-03 | -3.8448  | 0.0001207 | *** |
| ## | 2965 | -5.9742e-02 | 7.5587e-03 | -7.9037  | 2.707e-15 | *** |
| ## | 2966 | -6.8245e-02 | 7.4195e-03 | -9.1980  | < 2.2e-16 | *** |
| ## | 2967 | 1.3977e-02  | 8.2772e-03 | 1.6887   | 0.0912847 | .   |
| ## | 2968 | 6.8341e-04  | 8.1471e-03 | 0.0839   | 0.9331492 |     |
| ## | 2969 | -4.2874e-02 | 7.7215e-03 | -5.5526  | 2.814e-08 | *** |
| ## | 2970 | -1.4755e-02 | 8.0024e-03 | -1.8438  | 0.0652090 | .   |
| ## | 2971 | 1.6211e-02  | 8.3261e-03 | 1.9470   | 0.0515331 | .   |
| ## | 2972 | 4.6424e-03  | 8.1483e-03 | 0.5697   | 0.5688508 |     |
| ## | 2973 | -9.5597e-03 | 8.0588e-03 | -1.1862  | 0.2355252 |     |
| ## | 2974 | 2.0225e-02  | 8.3786e-03 | 2.4139   | 0.0157808 | *   |
| ## | 2975 | -2.4832e-03 | 8.1174e-03 | -0.3059  | 0.7596756 |     |
| ## | 2976 | 1.3496e-02  | 8.2709e-03 | 1.6317   | 0.1027363 |     |
| ## | 2977 | -4.4123e-02 | 7.6836e-03 | -5.7425  | 9.326e-09 | *** |
| ## | 2978 | -4.1688e-02 | 7.7609e-03 | -5.3715  | 7.807e-08 | *** |
| ## | 2979 | 4.6829e-03  | 8.2084e-03 | 0.5705   | 0.5683434 |     |
| ## | 2980 | -3.4839e-03 | 8.0561e-03 | -0.4325  | 0.6654091 |     |
| ## | 2981 | 8.4802e-02  | 8.9907e-03 | 9.4322   | < 2.2e-16 | *** |
| ## | 2982 | -1.8089e-02 | 7.9875e-03 | -2.2647  | 0.0235305 | *   |
| ## | 2983 | 1.9697e-03  | 8.1309e-03 | 0.2422   | 0.8085902 |     |
| ## | 2984 | 1.7177e-04  | 8.1349e-03 | 0.0211   | 0.9831537 |     |
| ## | 2985 | -3.3278e-02 | 7.8351e-03 | -4.2473  | 2.164e-05 | *** |
| ## | 2986 | 5.8840e-03  | 8.2006e-03 | 0.7175   | 0.4730610 |     |
| ## | 2987 | -7.2450e-02 | 7.3633e-03 | -9.8393  | < 2.2e-16 | *** |
| ## | 2988 | 6.0997e-03  | 8.1910e-03 | 0.7447   | 0.4564644 |     |
| ## | 2989 | 6.2193e-02  | 8.8120e-03 | 7.0577   | 1.693e-12 | *** |
| ## | 2990 | -6.3304e-02 | 7.4114e-03 | -8.5414  | < 2.2e-16 | *** |
| ## | 2991 | -4.6031e-02 | 7.6475e-03 | -6.0192  | 1.753e-09 | *** |
| ## | 2992 | 5.5727e-02  | 8.7855e-03 | 6.3431   | 2.252e-10 | *** |
| ## | 2993 | -2.7121e-02 | 7.8274e-03 | -3.4649  | 0.0005303 | *** |
| ## | 2994 | 2.4173e-02  | 8.3948e-03 | 2.8796   | 0.0039824 | **  |
| ## | 2995 | -3.2837e-02 | 7.8327e-03 | -4.1923  | 2.762e-05 | *** |

|         |             |            |         |           |     |
|---------|-------------|------------|---------|-----------|-----|
| ## 2996 | 5.5194e-02  | 8.7507e-03 | 6.3074  | 2.837e-10 | *** |
| ## 2997 | 8.6761e-02  | 9.1318e-03 | 9.5009  | < 2.2e-16 | *** |
| ## 2998 | -4.3797e-02 | 7.6478e-03 | -5.7268 | 1.024e-08 | *** |
| ## 2999 | 6.4645e-03  | 8.2381e-03 | 0.7847  | 0.4326243 |     |
| ## 3000 | -4.4419e-02 | 7.6270e-03 | -5.8240 | 5.746e-09 | *** |
| ## 3001 | -1.6205e-02 | 7.9560e-03 | -2.0369 | 0.0416629 | *   |
| ## 3002 | 6.9242e-02  | 8.9236e-03 | 7.7594  | 8.535e-15 | *** |
| ## 3003 | 3.5117e-02  | 8.5285e-03 | 4.1176  | 3.829e-05 | *** |
| ## 3004 | -1.7284e-02 | 7.9506e-03 | -2.1739 | 0.0297098 | *   |
| ## 3005 | 1.5564e-02  | 8.2872e-03 | 1.8781  | 0.0603695 | .   |
| ## 3006 | 2.4137e-02  | 8.3942e-03 | 2.8755  | 0.0040345 | **  |
| ## 3007 | 3.2836e-02  | 8.5401e-03 | 3.8449  | 0.0001206 | *** |
| ## 3008 | -3.8193e-02 | 7.7970e-03 | -4.8984 | 9.663e-07 | *** |
| ## 3009 | -5.0471e-02 | 7.7076e-03 | -6.5481 | 5.826e-11 | *** |
| ## 3010 | 6.0643e-02  | 8.8340e-03 | 6.8648  | 6.659e-12 | *** |
| ## 3011 | -9.5285e-03 | 8.0081e-03 | -1.1899 | 0.2341012 |     |
| ## 3012 | 9.4346e-03  | 8.2291e-03 | 1.1465  | 0.2515928 |     |
| ## 3013 | -2.5680e-02 | 7.8878e-03 | -3.2557 | 0.0011312 | **  |
| ## 3014 | -2.1559e-02 | 7.9223e-03 | -2.7212 | 0.0065038 | **  |
| ## 3015 | -1.9434e-02 | 7.9505e-03 | -2.4443 | 0.0145127 | *   |
| ## 3016 | -2.0585e-02 | 7.9363e-03 | -2.5938 | 0.0094935 | **  |
| ## 3017 | -9.6201e-03 | 8.0582e-03 | -1.1938 | 0.2325486 |     |
| ## 3018 | -2.1332e-02 | 7.9330e-03 | -2.6890 | 0.0071661 | **  |
| ## 3019 | 2.0656e-02  | 8.4211e-03 | 2.4529  | 0.0141716 | *   |
| ## 3020 | 1.7801e-02  | 8.3059e-03 | 2.1431  | 0.0321013 | *   |
| ## 3021 | 2.6855e-02  | 8.4411e-03 | 3.1814  | 0.0014655 | **  |
| ## 3022 | 1.0402e-02  | 8.2126e-03 | 1.2666  | 0.2052875 |     |
| ## 3023 | -4.4918e-02 | 7.6513e-03 | -5.8706 | 4.341e-09 | *** |
| ## 3024 | 5.5775e-02  | 8.7464e-03 | 6.3769  | 1.807e-10 | *** |
| ## 3025 | -1.5389e-02 | 7.9623e-03 | -1.9328 | 0.0532621 | .   |
| ## 3026 | 5.8756e-02  | 8.7518e-03 | 6.7136  | 1.899e-11 | *** |
| ## 3027 | 2.5130e-03  | 8.1562e-03 | 0.3081  | 0.7579977 |     |
| ## 3028 | -1.5979e-02 | 7.9794e-03 | -2.0026 | 0.0452241 | *   |
| ## 3029 | 2.9060e-02  | 8.4944e-03 | 3.4210  | 0.0006239 | *** |
| ## 3030 | -3.6882e-02 | 7.7712e-03 | -4.7460 | 2.075e-06 | *** |
| ## 3031 | -8.6192e-03 | 8.0680e-03 | -1.0683 | 0.2853761 |     |
| ## 3032 | 2.8385e-02  | 8.4758e-03 | 3.3489  | 0.0008112 | *** |
| ## 3033 | 6.4421e-02  | 8.7473e-03 | 7.3647  | 1.775e-13 | *** |
| ## 3034 | 4.3098e-02  | 8.8192e-03 | 4.8869  | 1.024e-06 | *** |
| ## 3035 | 4.1073e-03  | 8.1669e-03 | 0.5029  | 0.6150183 |     |
| ## 3036 | -2.3930e-02 | 7.8763e-03 | -3.0382 | 0.0023800 | **  |
| ## 3037 | -5.1468e-02 | 7.5818e-03 | -6.7884 | 1.134e-11 | *** |
| ## 3038 | -7.7955e-03 | 7.9745e-03 | -0.9776 | 0.3282928 |     |
| ## 3039 | -1.5112e-02 | 7.9891e-03 | -1.8916 | 0.0585479 | .   |
| ## 3040 | -5.7483e-02 | 7.5554e-03 | -7.6082 | 2.780e-14 | *** |
| ## 3041 | -2.2869e-02 | 7.9312e-03 | -2.8835 | 0.0039333 | **  |
| ## 3042 | 2.4907e-02  | 8.3990e-03 | 2.9654  | 0.0030226 | **  |
| ## 3043 | 3.4245e-02  | 8.4279e-03 | 4.0633  | 4.838e-05 | *** |

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## 3044 6.7821e-04 8.1615e-03 0.0831 0.9337732
## 3045 -9.4483e-03 7.9270e-03 -1.1919 0.2332938
## 3046 -5.6326e-02 7.5517e-03 -7.4587 8.738e-14 ***
## 3047 -5.3385e-02 7.5853e-03 -7.0380 1.951e-12 ***
## 3048 -3.2066e-02 7.7781e-03 -4.1226 3.746e-05 ***
## 3049 -5.8807e-02 7.5244e-03 -7.8155 5.477e-15 ***
## 3050 -3.6555e-02 7.9754e-03 -4.5834 4.575e-06 ***
## 3051 -3.6525e-02 7.7938e-03 -4.6864 2.780e-06 ***
## 3052 -3.7813e-02 7.7587e-03 -4.8736 1.096e-06 ***
## 3053 -6.3516e-02 7.4755e-03 -8.4966 < 2.2e-16 ***
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## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

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